

Futures of the past

The diachrony of future constructions across languages

Edited by

Stefan Hartmann

Lena Schnee

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Chapter 1

Introduction

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This introduction gives a very brief overview of the state of the art in research on the linguistic expression of future reference, as well as on the diachronic development of future constructions across languages. It also offers a brief summary of the individual chapters to the present edited volume, discussing their contributions to the overarching research questions addressed in this book, and outlines potential avenues for further contrastive research on the diachrony of future constructions.

Thinking and talking about future events is part of our everyday lives. As such, it is not surprising that languages have developed a variety of markers for futurity. According to Dahl & Velupillai (2013), “[i]t is relatively rare for a language to totally lack any grammatical means for marking the future. Most languages have at least one or more weakly grammaticalized devices for doing so.” In Bybee et al.’s (1994) broad typological sample, 70 out of 90 languages had at least one grammatical marker for future time reference, and among these, 49 had more than one form to refer to the future (also see Wiemer et al. 2024: 2). The fact that many languages have more than one way of expressing future time reference can be seen as a reflection of the diachronic tendency of futures to be unstable over time and to be formed anew in polygenesis (Wiemer et al. 2024: 2). This instability makes them an even more interesting and promising object of study from a diachronic and contrastive point of view. Crucially, there are vast typological differences between languages, both in the degree to which future markers are grammaticalized and in the diachronic paths of development that these markers have taken. The present volume focuses on the latter aspect: Its goal is to offer in-depth studies on the diachronic development of future markers across different languages.

In doing so, it connects to a rich research tradition that has addressed the evolution of grammatical markers for future time reference, which in turn is



part of a broader literature on tense and aspect. Fleischman (1982: 8, 1990: 26) describes the temporal structure of future tenses as either $S > R/E$ or $S/R > E$, taking up Reichenbach's (1947) tense model that distinguishes between the point in time when an utterance is made (point of speech, S), the point in time when an event occurs (point of the event, E), and the point of reference (R), which can coincide with S or E but can also be independent, e.g. in the case of the future perfect (*I will have seen John*: $S > E > R$; Reichenbach 1947: 3). In his broad typological overview of future tenses, Ultan (1978) argues that most languages show either a prospective or a retrospective tense system: in the former, the present tense can generally be used to refer to the future; in the latter, the present tense can generally be used to refer to the past. Regardless of whether a language has a prospective or a retrospective tense system, he argues that "future tenses are generally more marked than either past or present tenses" (Ultan 1978: 116), which is reflected in the markedness of the grammatical features of future tenses. For instance, if a language has a periphrastic past tense (which he considers more marked than bound suffixes used for tense marking), it will also have a periphrastic future, but not vice versa (Ultan 1978: 90–91).

Dahl (2000: 309–310) points out that the future has a very different epistemological status compared to the present or the past: "We cannot perceive or remember future states of affairs, and it has been disputed whether statements about the future can be said to have a determinate truth value." However, we do talk a lot about intentions on the one hand and predictions on the other, which leads Dahl (2000: 310) to the distinction between prediction-based and intention-based future time reference. He observes that in European languages, markers that are originally restricted to intention-based future time reference tend to develop into general future makers; at the same time, a grammatical opposition between intention- and prediction-based future time reference "is less common than one would perhaps think in view of the apparent cognitive salience of that distinction" (Dahl 2000: 310).

Bybee et al. (1994: 244) propose a distinction between primary futures, which evolve from a fairly restricted set of lexical sources (constructions involving movement verbs; markers of obligation, desire, and ability; temporal adverbs) on the one hand and what they call aspectual futures on the other, which "use a form whose principal function is the marking of present tense or perfective or imperfective aspect". The latter category would include, for example, cases of futurate present such as English *The Yankees play the Red Sox tomorrow*. (Hilpert 2008: 158), which would not typically be considered an instance of a future construction, let alone a future *tense*. This is why some terminological remarks are in order. When we talk about future *constructions* in the subtitle of the present

book, we mean linguistic patterns that can be used for future time reference. We use the term in a fairly theory-neutral sense here, although it is clearly inspired by the notion of constructions in the sense of form-meaning pairs in constructionist approaches to language (Ungerer & Hartmann 2023). Note, however, that the term is used in different senses in the literature, including the contributions to the present volume. For example, Wiemer et al. (2024: 1) use it interchangeably with *marker* or *gram*. Especially from a diachronic point of view, the distinction between a pattern that is a fully-fledged grammatical marker for future time reference and one that can be *used* for referring to future event even though this is not its primary purpose is of course blurred, which is why the scope of the present volume is not limited to cases that are clearly identifiable as future tenses – in fact, there have been controversial debates whether the German *werden* + infinitive construction, which several of the papers deal with, is actually a future construction or not (e.g. Saltveit 1962, Vater 1975). What counts for our purposes is, first and foremost, that a pattern is used (more or less regularly) for future time reference.

The present volume does not aim to cover the full typological breadth of languages, or of grammaticalization scenarios of future constructions. Instead, it presents some case studies from a diverse set of languages that can be considered representative for some of the recurrent questions that are discussed with regard to the diachronic development of linguistic markers of future reference.

A contrastive perspective on the diachrony of future constructions can prove insightful for a variety of reasons. Firstly, the developmental patterns of future constructions constitute prime examples of grammaticalization (Hopper & Traugott 2003). But future constructions differ in their grammaticalization paths: While some derive from modal verbs (e.g. English *will/shall*), others recruit change-of-state verbs (e.g. German *werden* ‘become’) or forms of *come* and *go* (e.g. English *BE going to*; Duala *-ya* ‘go’, Bambara *nà* ‘come’, among many others, see Kuteva et al. 2019) as future auxiliaries. Secondly, the competition patterns between future constructions merit further investigation from a cross-linguistic perspective. The examples mentioned above also show that in many cases, future constructions going back to verbs of movement compete with future constructions that have developed from modal verbs (see e.g. Kuteva et al. 2019). In other languages, change-of-state verbs such as German *werden* ‘become’ have prevailed as future markers (Dahl 1985, 2000). Crucially, languages differ not only in the ways in which they mark future time reference, but also in the extent to which speakers can choose between future tense forms and the so-called futurate present when referring to future events (see e.g. Dahl & Velupillai 2013).

A third aspect is equally important: The diachronic development of future constructions, as well as their synchronic diversity, can potentially give important clues to our cognitive conceptualizations of time, as well as to cultural factors that may play a role in how time is conceptualized and expressed linguistically (Fleischman 1982). In fact, Coseriu (1957) puts forward the far-reaching hypothesis that the rise of Christianity was a major factor in the emergence of modal futures in Romance languages. While this idea has to be taken with some scepticism (see Fleischman 1982: 49–50 for critical remarks), it does seem plausible to assume that socio-historical and cultural phenomena can have an influence on the use of grammatical forms. In addition, the influence of writing and literacy should probably not be underestimated, as differences between oral and literal contexts have often been found to be crucial for tense uses, especially when it comes to the use of future constructions (see e.g. Fleischman 1990, Nau & Sprau-niené 2022).

The goal of this volume is thus to take a contrastive perspective on the evolution of future constructions by bringing together scholars working on different languages and language families. In particular, we want to address the following questions:

- Which cross-linguistic tendencies can be observed in the grammaticalization of future constructions?
- To what extent can we compare the grammaticalization paths of future constructions with different origins?
- How do different future constructions compete in different languages, and can these patterns of competition be compared cross-linguistically?
- Which cognitive implications do findings on the diachronic development of future constructions entail?

As for the first question, a number of contributions re-address common hypotheses about the main grammaticalization paths of future constructions, while others, like Rhee and Khammee, delve deeper into the intricacies of grammaticalization paths of future constructions in two languages, Thai and Korean. In addition, grammaticalization processes can be intertwined with other processes of language change that are not traditionally considered part of grammaticalization. For instance, hypoanalysis is identified as a key process with respect to both the development of the Middle Armenian Future (Meyer) as well as the futurate

present of perfective Verbs in North Slavic languages (Wiemer). These contributions also address the third question mentioned above, i.e. parallel patterns of competition between different future constructions in different languages.

Importantly, the contributions to our volume also address languages and language stages for which no systematic analysis of future tenses exists so far, e.g. Middle Low German (Ihden & Schnee). They also address understudied topics such as the encoding of ‘future-in-the-past’ (Petrova & Chen) or “failed” grammaticalization paths in the development of future constructions (Skrzypek, Smirnova), which arguably have implications for the fourth question raised above, i.e. for a deeper understanding of the cognitive mechanisms underlying the grammaticalization of future reference markers.

Another aspect that the contributions to this volume add to the existing literature is an in-depth discussion of data types that are suitable for investigating the development of future markers. Bible translations are a widely-used resource in this domain as they are often based on the Latin version of the Bible, the Vulgata, and as Latin has a morphological future marker, it can be revealing to check how Latin future forms are translated (Ihden & Schnee).

In sum, the edited volume opens up new perspectives on long-standing questions regarding the diachronic development of future constructions across languages, with implications not only for research on tense-aspect-mood systems but also for research on grammaticalization and language change more generally. It brings together eleven papers presenting empirical case studies on future constructions in High and Low German, Norwegian, Swedish, West Polish, Korean, Thai, as well as in multiple Slavic languages. The articles focus on a variety of research questions, some offering a broad-scope overview of previously under-investigated cases of grammaticalization in lesser studied languages, others zooming in on specific aspects pertaining to the development of well-investigated patterns.

Three papers in the present volume deal with the German future construction *werden* + Infinitive. One of them, by Sarah Ihden and Lena Schnee, focuses on Low German. They present an onomasiological study that investigates the factors that influence the use of different grammatical structures expressing future reference in Middle Low German. In addition to modal verb constructions, which are considered the most common choices in the old Middle Low German grammars, futurate usage of the present tense is also possible, as well as verbal constructions with *werden* with an infinitive or a present participle. Drawing on selected texts from the reference corpus of Middle Low German / Low Rhenisch, Ihden and Schnee check how future constructions are translated in Middle Low German Bible texts, showing that both the translator and the semantic context

have a major influence on the choice between different variants. Furthermore, parallels and differences are identified in comparison to the development in High German.

Lena Schnee and **Stefan Hartmann** investigate the relation between *werden* + Infinitive and *werden* + Present Participle based on data from historical corpora of German. *werden* + Infinitive is the most important overt future marker of present-day German, but there are many different theories about its emergence and development, in many of which *werden* + Present Participle plays a crucial role. While it is fairly uncontroversial by now that *werden* + Infinitive did not emerge from some kind of reanalysis of *werden* + Present Participle, as assumed in very early approaches (e.g. *werde lachend* ‘become laughing’ > *werde lachen* ‘will laugh’), it still seems plausible to assume that the two constructions influenced each other in their development. These patterns of competition are investigated using association and dispersion measures. The results provide tentative evidence that over the time span taken into account in the study, the two constructions first become more similar and then start occupying different functional niches, with *werden* + Present Participle largely falling out of use except in a few dialects.

The diachronic change of the German *werden* + Infinitive construction is also the topic of **Elena Smirnova**’s paper, which investigates its development in the early stages of New High German. Drawing on data from the German Text Archive and the DWDS Core Corpus of the 20th century, she shows that the construction has undergone a decline in frequency, being replaced by the futurate present in many contexts, and that the construction has established a clear preference for the verb *sein* ‘to be’, which indicates that it has come to occupy a fairly narrow functional niche, and that two tendencies that are typically considered characteristic of grammaticalization cannot be observed any more in the case of this pattern: context expansion (Himmelmann 2004) on the one hand, and obligatorification (Lehmann 2015) on the other. On this account, then, the grammaticalization of *werden* + Infinitive has run into a dead end.

Dominika Skrzypek examines constructions with a *werden*-cognate, *varda*, in Swedish, which did not embark on the future grammaticalisation path after all. Similarly to the situation in German, *varda* was found in a passive construction with the past participle as well as with an infinitive, in future or inchoative contexts. Her corpus study traces the development of *varda* usages from 1300 to 1750, comparing the overall conditions with those in German. Several reasons for the “demise” are discussed, shedding light on the more general question of promoting and hindering factors of grammaticalisation. One main hindering factor for *varda* is the rise of the Middle Low German loan *bliva* ‘remain’, which developed

a new meaning ‘become’ in Swedish and replaced *varda* in all constructions except the infinitive one, leading to its near disappearance.

While in Swedish, Middle Low German *bliva* did not start taking an infinitive, it did so in Norwegian: **Olaf Mikkelsen** and **Dylan Glynn** investigate the competition between the Norwegian future constructions *blir å* + Infinitive and *vil* + Infinitive. While the former used to be considered a feature of Northern Norwegian dialects, the authors present evidence that it has spread over the last decade, showing considerable functional overlap with *vil* + Infinitive as well as with other competing future constructions. Mikkelsen and Glynn present a corpus study based on a Norwegian blog corpus, showing that “*blir å INF* is encroaching on the semantic territory of *vil INF*” and deeming it “a future that may still be”.

Two contributions deal with the relation between grammaticalization and hypoanalysis. Hypoanalysis refers to the phenomenon that a contextual semantic or functional property is reanalyzed as an inherent property of the syntactic unit (see Croft 2000: 126). The paper by **Björn Wiemer** discusses the interaction of grammaticalization and hypoanalysis in the development of future tenses in Slavic languages. Wiemer shows that in North Slavic languages, the present tense of perfective stems comes to mark perfective future by default. Wiemer argues that this is an instance of hypoanalysis. He slightly adapts Croft’s definition of hypoanalysis, allowing for it to be applied not only to cases where a new function was previously only implied but also to cases where a function was always possible but turns into the most prominent one. In South Slavic, by contrast, ‘become’- and ‘want’-based futures emerged, which can be seen as results of grammaticalization.

Robin Meyer traces the development of future constructions from Old Armenian to Modern Armenian, presenting a complex scenario and offering a detailed discussion of the relationship between the multiple change mechanisms that have contributed to the development of these constructions. Old Armenian did not have a morphological future tense, instead, future events were typically expressed by means of modal forms. While such are a common starting point for the grammaticalisation of future markers, the Armenian language did not take that pathway. The Middle Armenian future has been shown to have developed from a former present through hypoanalysis and univerbation. In the Eastern varieties of Modern Armenian, additional analytic future periphrases with specific aspectual meanings have arisen, leading to an increase in aspectual differentiation.

Svetlana Petrova & Yen-Chun Chen address a special domain in the field of future research, the ‘future in the past’ (FiP). They give an account of the means

that were used to express this posteriority of an event in a past tense setting in Old High German and Middle High German, prior to the establishment of the *würde*-construction. In their onomasiological corpus study, they find that overall, the past subjunctive correlates most strongly with this function, however they also observe an increase in modal constructions. Comparing the usage patterns of these with the modal constructions found in the canonical future, they find them to behave similarly in both contexts. The diachronic evidence they provide sheds light on controversially discussed issues such as the nature of FiP as a temporal category. In their paper, parallels are outlined between FiP and the canonical future.

The contributions by **Kultida Khammee** and **Seongha Rhee** investigate the grammaticalization of future markers in Thai and Korean, respectively. Khammee's contribution shows that there are various ways to mark future time reference in present-day Thai, with *cà* being the primary future marker; it developed out of *càk*, which also continues to be used as a future marker. The origins of *càk* are unclear but Khammee hypothesizes that it may go back to a borrowing from Middle Chinese meaning 'know', while also outlining a number of other possibilities. In her paper, she discusses possible pathways of grammaticalization of Thai future markers against the background of Lehmann's (2015) grammaticalization parameters. Rhee's paper shows that Korean also has a large number of options for future marking, but unlike many other languages, it has two formants that inherently involve futurity, viz. suffixes with the meaning '(at a) later time'. More specifically, he shows that contemporary Korean has six future markers with varying grammatical status ranging from adnominalizing suffix (*-l*), to a modal auxiliary (*-li-*, *-l.ke(s).i-*, *-keyss-*), and sentence-ender (*-li*, *-l.key*). Similar to Khammee, he discusses to what extent the current status of the future markers can be explained in the light of Lehmann's (2015) grammaticalization parameters.

The chapter by **Kristian Roncero** leads us to a completely different language family. In his study of future constructions in West Polesian (East Slavic), he shows that the de-obligative (*xofu robŕŕ* 'I will (need to) do') and de-volitive futures (*majusⁱ(a) robŕŕ* 'I am planning to do') have had a highly unusual course of development in that they have inverted their original semantic function: while the de-volitive has synchronically an obligational nuance, the de-obligative synchronically expresses an intention. While he cannot offer a final explanation of this phenomenon, one potential explanation he proposes is that the overabundance of morphological forms for expressing futurity may have forced some of the patterns to specialize over time; however, he also points to potential additional factors as well as the remaining high degree of variability between the patterns.

Although the contributions to the present volume cannot represent the full typological breadth and diversity of future marking, they present insightful snapshots that demonstrate that future constructions offer exciting insights into processes of language change and grammaticalization, and can potentially contribute to a better understanding of the cognitive and sociocultural underpinnings of these processes.

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Chapter 2

Expressing future reference in Middle Low German

Sarah Ihden^a & Lena Schnee^b

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The article deals with future constructions in Middle Low German and with the question of which external and internal linguistic factors have an influence on the choice of the formally different variants referring to future events in Middle Low German. The study is based on transcriptions and annotations of the Reference Corpus Middle Low German / Low Rhenish (1200–1650). In addition to a purely quantitative part, the article presents a deep quantitative-qualitative investigation of future constructions in Middle Low German Bible texts and reveals various factors that have an impact on the choice of a variant – above all there is an influence of the translator, as well as an influence of the semantic context.

1 Introduction

Future constructions are intensively researched in Germanic languages both in their recent and in their historical varieties. However, with regard to historical varieties of the German language, there are still some research gaps, concerning for example the diachronic development of competing constructions expressing future and the role of internal and external factors influencing the use of a variant. This is especially true for Middle Low German,¹ the development of which is

¹Middle Low German is a group of historical varieties of Low German. It was used from 1200 until about 1650 not only in the northern German-speaking areas but also in the Baltic area and elsewhere in centres of the Hanseatic League such as London or Nowgorod, since it was the lingua franca of this confederation.



less researched compared to the High German varieties.² In the old Middle Low German grammars by Lübben (1882), Lasch (1974 [1914]), and Sarauw (1924) (see Section 2), future constructions are only briefly referred to. While the interest in Middle Low German grammar has increased in the last few years and a wide range of grammatical issues including complex verbal constructions have been investigated, future constructions have not been in the centre of interest yet. A recent corpus-based study by Ihden (2023) on *wērdēn/wēsen* + present participle contains some useful observations concerning the futurate use of *wērdēn* + present participle. However, this study did not provide a comprehensive description of the variation between *wērdēn* + present participle and *wērdēn* + infinitive as well as the other competing future constructions as the focus of the analysis was not on future expressions. Hence, findings on the development of future expressions in Middle Low German and on influencing factors remain to be obtained.

This key research desideratum is addressed in our paper focusing on the diachronic development of the formally different variants expressing future in Middle Low German and on interactions between them, especially on the reasons for the use of a specific variant. In order to do so, we use transcriptions and annotations from the Reference Corpus Middle Low German / Low Rhenish (1200–1650) (short: ReN) (ReN-Team 2021). The ReN is currently the largest corpus of Middle Low German texts. It is enriched with various linguistic annotations and the text selection is structured by time, language area, and genre, which allows us to analyse phenomena that vary along each parameter.

We will conduct a two-step analysis on the variation of future expressions in Middle Low German, starting with a purely quantitative approach to provide a bigger picture of frequency (Section 4.1) which we will complement with a quantitative-qualitative approach on variation of future constructions in Bible texts and on factors influencing the use of a variant (Section 4.2).

Our paper is structured as follows: In Section 2 we give a brief overview of the different future constructions in Middle Low German and on the previous research in this area. Section 3 contains our research questions, explains our method and presents the corpus we used. The results of our analyses are given in Section 4, starting with an overview on the distribution of two formally similar

²The development of future constructions in High German is intensively researched with differing focus regarding content and methodological focus. A complete overview of research cannot be given here, so reference is made only to selected works. See for example Saltveit (1962) analysing sentences with both *werden* ('become') with the present participle and with the infinitive from Old High German until the 16th century and Oubouzar (1974) providing a quantitative and qualitative study of *werden* + present participle and infinitive in a set of seven texts from the 11th until the 17th century.

variants, namely *wērdēn* + present participle and *wērdēn* + infinitive (Section 4.1), followed by a quantitative-qualitative analysis of variation in future references in Middle Low German Bible texts (Section 4.2). In Section 5, we summarise the main results and give an outlook on potential following studies based on our work.

2 Middle Low German future constructions in the research literature

The Middle Low German reference grammars by Lübben (1882), Lasch (1974 [1914]), and Sarauw (1924) describe four different grammatical structures for referring to future events.

- Modal verb constructions:
 - a) *schōlen* ('shall') + infinitive: *hê schal kōmen*
 - b) *willen* ('will') + infinitive: *hê wil kōmen*
 - c) *mōten* ('must') + infinitive: *môt kōmen*
- Present tense:
 - d) Main verb in simple present: *hê kōmet*
- Complex verbal construction with *wērdēn* ('become') in present indicative and a present participle:
 - e) *wērdēn* + present participle: *hê wērt kōmende*
- Complex verbal construction with *wērdēn* ('become') in present indicative and an infinitive:
 - f) *wērdēn* + infinitive: *hê wērt kōmen*

According to Lasch (1974 [1914]: 222–223, § 412, note 6), the most common choices, especially in earlier texts and in texts from the western areas, are modal constructions with the verbs *schōlen* ('shall') and *willen* ('will') as in *hê schal kōmen* in a) or *hê wil kōmen* in b) meaning 'he will come'. Sometimes the modal verb *mōten* ('must') is used in that sense as well, as in *hê môt kōmen* in c).³ Lübben (1882: 91, § 65) states that no differences in the use of these three variants could

³The forms of the lexemes given here are based on the Middle Low German dictionary (Lasch et al. 1956–) which differentiates between old length (e.g. *ē*) and new length (e.g. *ē̄*).

be detected – as an illustration, he cites a Bible translation by a chronicler from Bremen containing both *môten* + infinitive and *willen* + infinitive within the same sentence (*ik mot voruolgen myne vyende vnde ik wil nicht vmmekeren* ‘I will follow my enemies and I will not turn back’). Sarauw (1924: 225) considers the present as in *hê kômet* in d) the primary and mainly used option for expressing future. Aside from the present tense and modal verbs + infinitive, there are also two complex verbal constructions with *wêrden* (‘become’) in present indicative conveying a future meaning in Middle Low German, namely *wêrden* + present participle as in *hê wêrt kômende* in e) and *wêrden* + infinitive as in *hê wêrt kômen* in f). According to Lübben (1882: 91, § 65), a form such as *kômen* in f) must not be interpreted as an infinitive but as a present participle which has lost its ending *-de*. Through time, this new form without the *-de* becomes common for expressing a future event. It is described by Lübben (1882: 92, § 65) as typical for High German varieties where it already spread from the 14th century on. The formal ambiguity between the present participle and the infinitive is also mentioned by Lasch (1974 [1914]: 222–223, § 412, note 6). She states that the infinitive becomes the dominant variant within the complex verbal constructions with *wêrden* expressing future events.

Information on the different variants expressing future in Middle Low German can also be found in a grammar book from the Middle Low German time itself, the so-called *Münster Grammatik* (1451). According to the text, which explains tense and mood of the verb using the example of the Latin verb *legere*, the Latin form *legam* (‘I will read’) is future indicative and corresponds to Middle Low German “ick wil edder ick schal lesen, edder also de averlender seggen: ik werde lesen” (‘I will or I shall read, or as the foreigners (from the upper German regions, as opposed to the low lands) say: I become read’ (Wilken 1877: 44).⁴ The text thus contrasts the variants *willen* + infinitive and *schölen* + infinitive used in Middle Low German with the variant *wêrden* + infinitive used in High German areas. In addition, the author explains the subtle differences in the use of *willen* + infinitive and *schölen* + infinitive as future expressions. While *willen* + infinitive is used when someone has the will to carry out the mentioned action, *schölen* + infinitive is preferred when one is obliged to carry out this action (Wilken 1877: 44). Thus, a semantic difference between the two constructions is established here. Furthermore, the *Münster Grammatik* deals with Latin verb forms in the future imperative as in *legitō* (‘read!’) which means that someone is obliged to read in the future: “dat he in einer tokamender tyd lesen schal” ‘that he shall read in a future time’ (Wilken 1877: 45). Since such an obligation is connected with the variant *schölen* + infinitive as described above, it is most likely to serve

⁴See also the new edition of the *Münster Grammatik* by Caparrini (2018: 61–63).

as an equivalent for the Latin future tense in the imperative. Besides diachronic, diatopic and semantic differences between the variants, there are even more factors with an impact on the use of a variant for expressing future. An influence of the author, the scribe or the printer of a text is shown by Ihden (2023: 440), who provides a corpus-based analysis of *wērdēn* and *wēsen* + present participle with data from the ReN (see above). As Ihden (2023: 440) shows, *wērdēn* + present participle is rather frequent in Middle Low German Bible translations, where future expressions in the Latin Vulgate are translated very closely. Moreover, a cursory look at sentences with future expressions in the Bible texts from the ReN revealed that the same sentence from the Bible can be translated using formally different constructions for referring to future events (Ihden 2023: 436). Hence, the use of a variant seems to depend strongly on the person producing the text. In the case of the Middle Low German Bible texts, this is the translator.

As we have outlined, there are no comprehensive studies on the development of the different constructions expressing future in Middle Low German. Much research on future constructions has been done focusing on the older High German varieties, however. Regarding the future construction *werden* + infinitive in the German language, there has been a controversial discourse on where the construction comes from for decades, and several scenarios on the origin and the development of *werden* + infinitive have been worked out (also see Schnee and Hartmann, this Volume). One theory states that *werden* + infinitive derived from *werden* + present participle by a phonological reduction of the participle, namely the loss of the final *-e* and the assimilation of the consonant cluster *-nd* (Middle High German *-ende* > *-en*) (Weinhold 1967: 397–398, § 373).⁵ A second theory, the so-called analogy theory, explains the development of *werden* + infinitive by an analogy to Middle High German *beginnen* or modal verbs + infinitive (e.g. Wilmanns 1899: 177).⁶ See Westvik (2000) for an overview of the different theories.

Two studies, however, must be mentioned here that do include Middle Low German data. That is, for once, Kurrelmeyer's (1904) extensive study, which Westvik (2000) counts as a contribution to the analogy theory, and Schmid (2000), which according to Westvik proposes what he calls a theory of contamination,

⁵Weinhold (1967: 397, § 373) notes with reference to Bech (1901) that this loss of the *-e* was mainly spread in the Middle German varieties but not in Southern German varieties. In line with this, Kleiner (1925: 91) shows that in Alemannic texts from the 13th century there is no loss of the ending *-de* in present participle forms. Hence, the attestations of the reduced and assimilated forms are not in line with the areal development of the *werden* future. Because it extends from the Southern German speaking areas, one would have expected attestations for the loss of the participle ending if the theory was correct (Jäger 2024: 3).

⁶In contrast to these mono-factorial explanations, Diewald & Habermann (2005) propose a multi-factorial approach and explain the development of the *werden* future as an interaction between grammaticalisation, language contact and socio-cultural factors.

as he suggests that the contamination of formally different constructions with a similar meaning leads to the emergence of *werden* + infinitive.

Kurrelmeyer (1904) looks at a broad range of constructions potentially used for future reference (*wollen*+ infinitive, *sollen*+ infinitive, present, *werden* + infinitive, *werden* + present participle as well as *werden* + past participle, *werden* + adjective and *werden* + noun) and lists the proportions each of these have in the total number of future references for a total of 209 texts, 16 of which are Middle Low German. In the latter he identifies 497 future references. Nearly 50% of these are expressed with *wollen*⁷ + infinitive, followed by *sollen* + infinitive with about 20%. 13% of the future references use the present tense. *werden* + infinitive makes up 11%, however, this construction is only found in three of the texts. It has a substantial part only in one text, namely *Reynke de vos* (Lübeck 1498). Less than 3% of the instances have *werden* + present participle. The rest of the future references are expressed with constructions not typically counted as future markers but do have futurate semantics to them, namely *werden* with a past participle, an adjective or a noun. Overall, Kurrelmeyer (1904: 89) concludes for the Low German texts that “the first person is represented chiefly by *wollen* + infinitive, the second is not sharply defined, now using one, now the other form, the third uses all three constructions, that is, *present tense*, *sollen* or *wollen* + infinitive” (emphasis in original). He sees a tendency of the formerly prevailing form *sollen* + infinitive (p. 32) to be replaced by *wollen* + infinitive.

Schmid (2000) looks at 100 different versions of the *Fifteen Signs Before Doomsday* from the 12th to the 17th century. Due to the nature of its content (prophecies) and its consistency, this corpus allows for an onomasiological study, looking at formal variation. As the different versions come from different areas and centuries, it can be used for diachronic and diatopic studies.⁸ He reports about the five Middle Low German texts that in two of these, which date from the 15th century, the present construction is the more frequent variant, while in two texts from the 16th century, *sollen* + infinitive dominates.⁹

⁷Kurrelmeyer (1904) subsumes the Low German verbs under their High German counterparts. We adopt the author’s wording in this paragraph.

⁸The corpus is not balanced within area and time, however. It shows a preponderance of Upper German texts and other dialects areas are not represented either in the earlier or the later periods, compare Westvik (2000: 242).

⁹The numbers presented here are unfortunately somewhat elusive: In the overview of the texts in his corpus, Schmid reports four Middle Low German texts from the 15th, and only one from the 16th Century. While the overall difference could be explained by texts that have no clear leading variant, the discrepancy in the number of texts from the 16th century is not accounted for.

Overall, he notes that the *sollen* construction is particularly well established in Middle German, Middle Low German and Middle Dutch, while the Upper German area is characterised by the spread of the *werden* construction.¹⁰

As large corpora for historical German varieties have been published in recent decades, the development of future constructions is increasingly researched using corpus-linguistic methods. Regarding older varieties of German, the largest amount of data comes from the Reference Corpora on German language history (Ihden et al. 2023), namely the Reference Corpora of Old German (short ReA), Middle High German (short ReM), Early New High German (short ReF) and Middle Low German / Low Rhenish (short ReN) as well as German Inscriptions (short ReDI).

In a recent study based on Old High German and Old Saxon data from the ReA, Jäger (2024) provides an alternative scenario concerning the development of *werden* + infinitive in German. According to her, the origin of the construction lies in copula constructions with *werden* and a predicative past participle. Jäger (2024: 189) lists a number of conditions which have to be fulfilled in order to have a bridging context where there is an ambiguity between the past participle and the infinitive. In the data from the ReA she finds nine clear occurrences of such bridging contexts; one for Old High German and eight for Old Saxon. Jäger (2024: 183) points out that her scenario accounts for both the diachronic and the diatopic extension of *werden* + infinitive in High German. As the prefix *ge-* becomes an obligatory marker of the past participle in the 15th and 16th century, there is a higher chance that a speaker considers an ambiguous form without the prefix to be an infinitive (Jäger 2024: 229). This supports the reanalysis of *werden* + past participle as *werden* + infinitive. With regard to the areal distribution, Jäger (2024: 230) notes that the development differs between High German and Low German. In High German, the past participle is usually formed with the prefix *ge-* and the infinitive without it, while no such formal difference exists in Low German, as the past participle can still be formed without this prefix. Therefore, High German speakers tended to reanalyse a past participle without the *ge-* as an infinitive, unlike speakers of Low German, who were accustomed to the formal similarity of the past participle and the infinitive.

Hartmann (2021) provides an analysis of competing constructions expressing future in Middle High German and Early New High German using data from

¹⁰Westvik (2000) questions the validity of this assessment regarding the origin of the *werden*-construction, based on the already mentioned imbalance of the data set. This critique does not apply to the statements about Low German, for which, however, very little data was collected.

the ReM and the Bonn Early New High German Corpus (short FnhdC).¹¹ With regard to the frequency of the different constructions, Hartmann (2021: 376–378) shows that both the use of *werden* + present participle and of *werden* + infinitive increases during the 12th century. But, from the 13th century on the frequency of *werden* + present participle declines while the frequency of *werden* + infinitive increases even more. In contrast to the constructions with *werden*, the use of the modal verbs + infinitive remains at a relatively stable level through the Middle High German and the Early New High German period.

Besides frequency, Hartmann (2021) also analyses which semantic verb types (according to the aktionsart categorization of Vendler 1957) are used in the main verb slot of the different constructions. He finds a parallel development of both *werden* + present participle and *werden* + infinitive. First, in the 12th century, almost only activities are used as main verbs within the two constructions. From the 13th century on, Hartmann (2021: 378–379) observes a context expansion since not only activities but also achievements and accomplishments occur in the main verb slot. Hartmann (2021: 379) shows that in both constructions the frequency of telic verbs increases. In the modal verb constructions, which are used as a contrast here, the frequency of the different semantic verb types stays nearly the same through time (Hartmann 2021: 380–381). From these results Hartmann (2021: 382–383) draws the conclusion that both *werden* + present participle and *werden* + infinitive have gone through a similar development of grammaticalisation, since they could be used in the same contexts and therefore both were competing constructions for a specific period of time. This phase of *attraction* is followed by a phase of *differentiation*¹² concerning the semantics of the constructions. While nearly all of the few occurrences for *werden* + present participle in the FnhdC carry an inchoative semantic (Hartmann 2021: 390), with *werden* + infinitive, the future reading becomes stronger.

So far, no corpus-based studies have been conducted to assess the domain of future constructions in Middle Low German. With this paper we address this

¹¹See also Luther (2013) who uses a sample of texts from the ReM in order to investigate which different formal expressions are used for referring to future events in Middle High German. Unfortunately, her procedure for identifying the attestations for future reference is not completely transparent, which is reflected in the fact that she presents attestations that are not unambiguous future references.

¹²Hartmann (2021: 389–390) uses the terms *attraction* and *differentiation* according to de Smet et al. (2018), who explain that competing forms can change their functions by undergoing two phases: In the first phase, the attraction, two constructions with specific functional similarities become even more similar (de Smet et al. 2018: 203–204). In the second phase, the differentiation, “the functional domain competed over ends up being divided, with each expression filling a unique functional niche” (de Smet et al. 2018: 198).

deficit and contribute to a more complete picture of the development of future constructions in historical varieties of German.

3 Research questions, methods and corpus used

We work with a broad definition of (future) tense: Formally, this means that we include any grammatical form (i.e. morphological markers or periphrastic auxiliary constructions) that is used to express future (i.e. situate an event¹³ posterior to a reference point, i.e. the utterance time, compare Reichenbach 1947: 287–288.) We further acknowledge that referencing to the future is rarely the sole function of a construction (Comrie 1989: 52). The most commonly found atemporal functions of future markers are modal categories (such as imperative, obligation, necessity or polite request). Besides these, future markers can express e.g. aspectual categories (such as gnomic/general truth and habitual event), as well as some (non spatial) goal-oriented categories (Ultan 1978: 102–106). Markers of modality, particularly those that express obligation, volition, uncertainty or irreality, are also the most common source from which future markers develop, followed by aspectual markers or (spatial) goal-oriented categories (Ultan 1978: 118). The fact that modality and future tense are so closely connected is often attributed to “the relative uncertainty inherent in both future event and most of the categories subsumed under the general heading of modals” (Ultan 1978: 118).

It is noteworthy that this is but one of the many similarities and semantic overlaps between these categories that are ‘atemporal’ in a stricter sense but strongly connected to the category of future tense. Morris (1990: 73, 82) considers them subcategories of *future* (*future of intent*, *future of obligation* and *future of ingression*), which he suggests is to be seen as a broader, complex cognitive category. We do not (yet) make any claims about whether or not expressing future tense can be said to be the primary function of any of the forms looked at. The centre of interest of this paper is to identify the factors that influence the use of the different variants used in Middle Low German in order to refer to a future event (Section 4.2).

First, Section 4.1 gives an overview over the diachronic distribution of two different competing future expressions in Middle Low German. We will take a look at the frequencies of *wēden* in the present indicative with a present participle and with an infinitive in the ReN (ReN-Team 2021) and compare them with those found by Hartmann (2021) in the High German corpora. The ReN contains

¹³We use the term *event* in the broad sense to describe any type of activity, process or state (Ultan 1978: 85).

235 texts from 1200 until about 1650 with a total amount of 2.3 million tokens, of which 1.5 million tokens are grammatically annotated with respect to part of speech (pos), inflectional morphology, and lemma. The remaining 800,000 tokens are transcribed without annotations. The ReN is structured by the parameters of language area, time, and genre, which allows an examination of the potential influence of these factors on grammatical variation.

Due to their multi-functionality, the present forms as well as the modal constructions¹⁴ are not included in this first analysis. In order to collect all occurrences of *wērdēn* with either an infinitive or a present participle, we searched the corpus for sentences containing the lexeme *wērdēn* (annotated with the lemma *wērdēn*¹ in the ReN) as well as an infinitive or a present participle and then manually sorted out the cases where the latter is not a complement to the instance of *wērdēn*. We verified that each instance also carries future semantics. Instances of preterital *wērdēn* were excluded for this study, as future meaning is only expressed when the auxiliary verb is in the present tense. We also excluded instances of *wērdēn* that are unambiguous subjunctive forms¹⁵, leaving us with a data set of 614 instances in sum (326 instances of *wērdēn* + present participle and 288 of *wērdēn* + infinitive).¹⁶

The second part of our study (Section 4.2) aims to cover all the formally different constructions used for future reference (including the present tense and the modal constructions not considered in the purely quantitative approach). This onomasiological perspective entails the difficulty of obtaining an adequate data set, as the entirety of instances is not formally identifiable (due to the multiple functions most of the variants can have). For the present study, we draw on the fact that future tense is morphologically represented in Latin. We look at three Middle Low German Bible prints that are translations of the Latin Vulgate. More specifically, we examine the way each of the texts realises those sentences that have a future form in the Latin text. Notably, the formal identification of Latin future instances will not only yield such attestations with true futurate mean-

¹⁴Hartmann (2021) uses sets of modal constructions as a background to compare the semantic features analysed against, in order to control that the development he observes in the *werden* constructions can be considered distinctive for the latter and not merely a general development shared by all verbal phrases.

¹⁵The paradigm regularly shows syncretism of the indicative and the subjunctive in the first and second person singular as well as in the plural. Due to graphematic variation, further syncretism can arise. As the proportion of instances tagged as ambiguous is rather high, we decided to include both distinctly indicative forms and those that are not explicitly subjunctive (i.e. those tagged as ambiguous for mode).

¹⁶The relevant data and scripts for both sub-studies are available in an OSF repository, which can be accessed via <https://osf.io/4c6yv>.

ing, but also such instances where the Latin future marker serves an ‘atemporal’ function (see above). In addition to the true future meanings future and future progressive (Hale & Buck 1966: 243), the Latin future indicative can also have the following meanings: habitual action (Hale & Buck 1966: 252), the modal notion of command or prohibition (Hale & Buck 1966: 304), or the “discovery of a state of affairs now existing” (Hale & Buck 1966: 253). We include all instances of morphologically marked future indicative, regardless of their more specific function. We look at the different functions separately in Section 4.2.4.

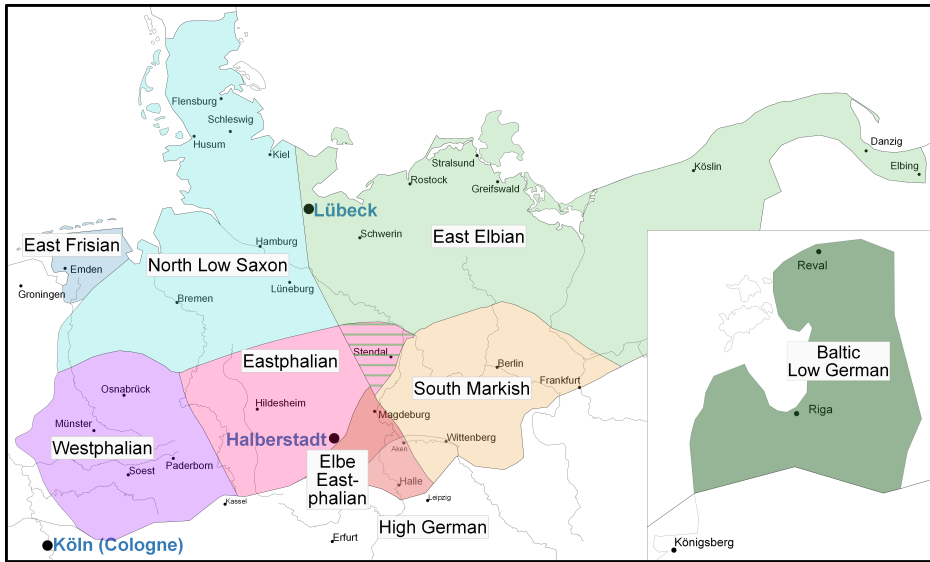


Figure 1: The language areas of Middle Low German

Figure 1¹⁷ shows the language areas of Middle Low German and the cities where the three Bible texts listed below come from.

1. The Bible of Köln (Westphalian) from 1478–1479;
2. the Bible of Lübeck (East Elbian) from 1494;
3. the Bible of Halberstadt (Eastphalian) from 1522.

When comparing the content and the language of these Bibles, one has to be aware of their complex text tradition. The Bible of Köln is directly translated

¹⁷This map was created by Sarah Ihden using <https://regionalsprache.de>. It is based on the map “Schreibsprachenlandschaften im Spätmittelalter” by Konrad Kunze and Robert Peters with additions by Libuse Spáčilová, drawn by Brigitte Pfeil.

from the Vulgate, and is itself the main source of the Bible of Lübeck. The Bible of Halberstadt is based on the Latin Vulgate and on both the other Middle Low German Bibles of Köln and Lübeck (Eichenberger & Wendland 1977: 66, 120, 144). For this study we used the transcriptions of the Bible texts provided by the ReN. Since only extracts of the Bibles are contained in the corpus, this means that we only look at a subset of Bible passages, namely those that we have from all three of the Bibles. Table 1 shows the number of sentences with a future verb in Latin for each of the Bible chapters covered by our data set.

Table 1: Number of Latin future forms in the Bible extracts included in our data set

ID	extract	n Latin futures
01–02	Gen. 2: The Creation of Man and Woman	2
03–10	Gen. 3: The Fall	8
11–18	Gen. 4: Cain and Abel	8
19	Gen. 11: The Tower of Babel	1
20–23	Gen. 22: The Sacrifice of Isaac	4
24–44	Exod. 20: The Ten Commandments	21
45–46	Luke 2: The Birth of Jesus Christ	2
47–63	Mt. 5: The Sermon on the Mount	17
64–83	Mt. 6: The Sermon on the Mount	20
84–95	Mt. 7: The Sermon on the Mount	12

Each sentence has been annotated for the following:

- construction used to translate the Latin futurate verb;
- lemma of the main verb;
- tense, mode, voice, person and number of the finite verb.

All annotations have been checked by both authors. The data set contains 35 future passive forms, which we look at separately in Section 4.2. They will be excluded for all remaining analyses, where we focus on the 95 future active instances. The latter have been further annotated for the overall semantics of the surrounding sentence. The categories that have been distinguished partially reflect the ‘atemporal’ meanings the Latin future indicative can express:

- **prognosis:**
 - a (**singular**) event is predicted to happen in the future:
[...] you **will find** a baby wrapped in swaddling cloths and lying in a manger. (Luk. 2:12)¹⁸
 - an event is predicted to **habitually** occur (in the future):
Because you have done this, cursed are you above all livestock and above all beasts of the field; on your belly you shall go, and dust you **shall eat** all the days of your life. (Gen. 3:14)
- **commandment:** an obligation or prohibition is expressed; it is valid from the moment of expression on and reaches into the future:
You **shall not steal**. (Ex. 20:15)
- **condition:** protasis of a conditional:
Or if he **asks** for a fish [...]
- **consequence:** the apodosis of a conditional:
[...] **will give** him a serpent (Mt. 7:10)
- **generic statement:** a statement not about a specific moment in the future but of general validity:
whoever murders **will be liable** to judgment (Mt. 5:21)

Besides the ‘future’ meaning in the narrower sense, which we label *prognosis* and which we subdivided into prognoses about singularly occurring events and such about habitually occurring events, we find the modal meaning of obligation or prohibition, which we labelled *commandment* here. Much less frequent are sentences that are part of a condition-consequence structure, especially the conditions are rare. As these are naturally non-assertive, they are considered unspecific regarding temporal reference (Dahl 2000: 314). Typologically only fully grammaticalised future markers (i.e. those that are obligatory) occur in these contexts (Dahl 2000: 314).¹⁹ Expressions of gnomic, general truth are (semantically)

¹⁸The annotations were made on the basis of the Latin references. For ease of comprehension, the examples given here for the individual categories are English translations based on the English Standard Version of the Bible provided by the *Deutsche Bibelgesellschaft* at <https://www.bibelwissenschaft.de/online-bibeln/english-standard-version/bibeltext/> (last accessed on October 1 2023). The English translations of Bible verses given in the remainder of this paper are taken from the same source.

¹⁹In modern German the usage of *werden* + infinitive is highly restricted in either part of conditional clauses (Vater 1975).

considered tenseless (Ultan 1978: 87). They tend to also formally be unmarked for tense (i.e. languages often use the present tense for these) (Ultan 1978: 87). However, some languages (e.g. Greek and English) also express it with future markers (Ultan 1978: 87). In the Latin Vulgate, we do find sentences with future markers that express a gnomic truth, which could hint at this being possible in Latin, too. However, it could also be due to our operationalisation of ascribing gnomic truth.

4 Analyses

4.1 Distribution of *wērdēn* + present participle and *wērdēn* + infinitive

Before providing a deeper analysis on factors influencing the use of the different future constructions in Middle Low German, this section gives a first, rather rough glimpse of the frequencies of two of the constructions. The research literature suggests a connection between the development of *wērdēn* + infinitive and *wērdēn* + present participle. As our quantitative analysis in this section treats these constructions only, we can only draw conclusions about these patterns and not about future marking in general. The entire range of Middle Low German future constructions will be investigated in the next sections.

Figure 2 shows the frequencies of *wērdēn* + present participle (red) and *wērdēn* + infinitive (blue) in the ReN.²⁰ The figure is based on the frequencies normalised to corpus size (n of finite verbs) with the raw frequencies given on top of each column. Before the 15th century, attestations for both constructions are marginal, ranging from 0 to 4 and remaining far below one token per 1000 finite verbs in the corpus. Starting in the second half of the 15th century, we find a sudden increase in attestations of both constructions, with *wērdēn* + present participle being more frequent at first. In the following half century, where most of the attestations are found, the two constructions' frequencies are roughly the same, with *wērdēn* + infinitive prevailing slightly. After that, we find, again, far fewer instances. For *wērdēn* + present participle we find only one attestation in the last 150 years covered by the corpus. *Wērdēn* + infinitive is also much less attested in the later period compared to the preceding one. However, as the corpus contains only a very small number of tokens from only three texts from this period, the relative frequency is still high.

²⁰ All figures are created with ggplot (Wickham 2016) in RStudio (R Core Team 2021).

2 Expressing future reference in Middle Low German

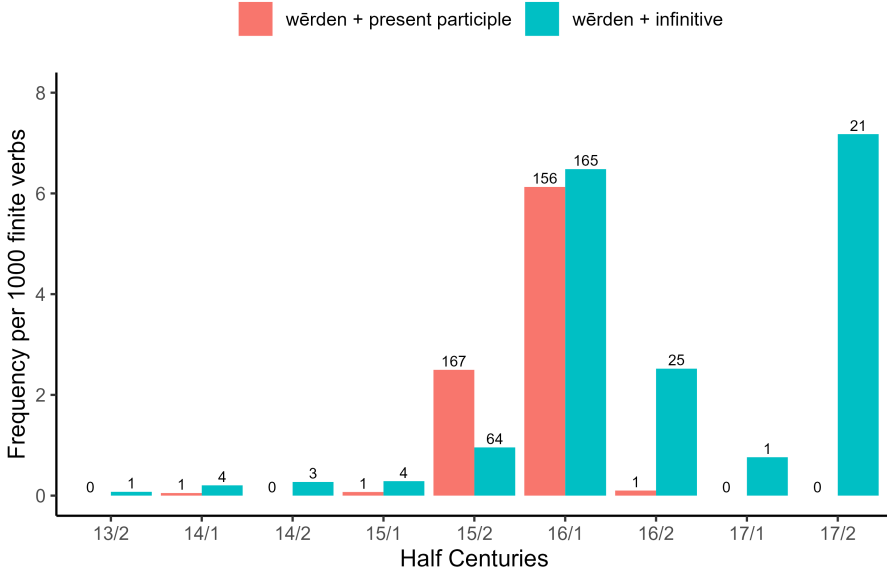


Figure 2: Diachronic distribution of *wēden* + infinitive and *wēden* + present participle in the ReN (normalised to corpus size, total counts given in labels)

While the frequency development is consistent with the observations in the traditional Middle Low German grammars,²¹ these results must be treated with caution. One caveat concerns the sparse data for the later periods of the corpus (the cessation of *wēden* + present participle could be due to the small range of texts covered). Another one concerns the uneven distribution of future references in texts, as future references are rather rare and the occurrence of future markers of course depends on whether future events play a role in each specific text.

The frequency peaks in Figure 2 can be attributed to some texts containing a large amount of future-related propositions: In the first half of the 16th century, many of the instances (for both constructions) come from *Der schapherders Kalender* (1523), a text containing a collection of prophecies for the different zodiac signs, which by nature of their content contain an exceptionally high number of future references. The second half of the 15th and the whole 16th century prominently feature Middle Low German Bible versions as well as other religious texts, which generally contain more references to the future than literary texts,

²¹For example, Lasch (1974 [1914]: 222–223, § 412, note 6) states that the construction with the infinitive predominates over the one with present participle over time.

for example. The majority of instances of *wērdēn* + infinitive in the second half of the 17th century can be attributed to the *Veer Schertz=Gedichte*, a compilation of humorous poems by Lauremberg (1652). Since in the later times of Middle Low German the influence of High German increases, the high frequency of *wērdēn* + infinitive in this text might be due to contact with High German where *wērdēn* + infinitive is already the established construction at that time.

The varying likelihood of future references between individual texts but also between genres/text types limits the reliability of the frequencies found, given that the corpus is not balanced overall regarding the distribution of genre over time. For the 13th and 14th century, the corpus predominantly contains legal texts and deeds, as well as some texts of knowledge transfer. For the 15th and 16th century, however, religious texts and literature predominate and the 17th century only features literary texts.²² As mentioned above, the texts in which we find most instances of the *wērdēn* constructions mainly belong to the fields of religious texts, literature and knowledge transfer (in the rather special case of the horoscopolical *Der schapherders Kalender*). Since these genres prevail in the later periods of the corpus, the results could possibly reflect this structure rather than a diachronic development in the usage of the constructions. We focus on one of the highly futurate text types found in the corpus, namely Bible translations, in the next section, while expanding the scope of our analysis with respect to the variants.

4.2 Variation in future references in Bible Texts

4.2.1 Frequency of future constructions in different texts

Before analysing the constructions used for translating future in the Latin Vulgate into Middle Low German in more detail, we will first look broadly at their frequencies in our three Middle Low German texts. As Figure 3 shows, the most frequent constructions are the ones described in the research literature, namely the present tense (purple) as in (1a), *wērdēn* + infinitive (blue) as in (2a), *wērdēn* + present participle (green) as in (1b), and modal verb constructions (orange). With regard to these modal verb constructions in our data, only *schōlen* + infinitive as in (1c) is used more frequently, while constructions with *willen* as in (2b) appear marginally and *môten* + infinitive is not used for future translations in the three Bible texts from the ReN. The frequencies in these texts do not reflect the trend

²²The distribution of tokens along the parameters of time, language area, and genre is visualised on the corpus website <https://www.slm.uni-hamburg.de/ren/korpus/korpusdesign.html> (last accessed on October 1 2023).

observed by Kurrelmeyer (1904), since *willen* + infinitive is not among the predominant constructions at all.

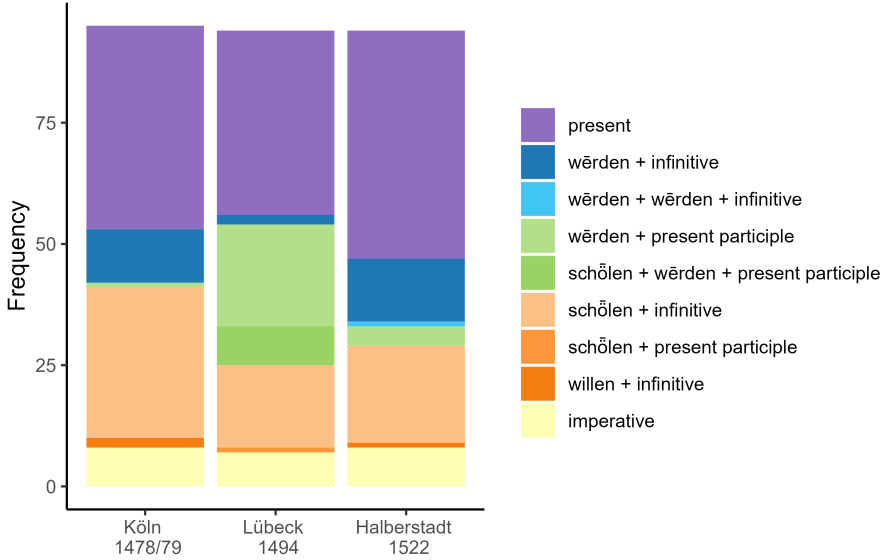


Figure 3: Verbal constructions used for translation of Latin future (active) in the three Middle Low German Bibles.

- (1) [...] nec enim *habebit*²³ insontem Dominus eum qui adumpserit nomen Domini Dei sui frustra (Exodus 20:7)
 [...] for the Lord will not hold him guiltless who takes his name in vain
 - a. [...] Wente de here *heft* on nicht entschuldich de synes godes namen ydelyken tho sik nympt . (Halberstadt, 39va, 19–22)²⁴
 - b. [...] wente de here *werd* ene nicht vnschuldich *hebbende* de to sik nemende werd den namen sines heren sines gades vorgheues . (Lübeck, 40va, 29–31)
 - c. [...] Vnde de here en *schal* en nicht *entschuldighen* . de sinen namen ydelik nomet . (Köln, 40ra, 22)

²³The Latin futurate verbs as well as the verbs used for their translation in the Middle Low German Bibles are italicised in all the examples.

²⁴The Middle Low German examples are given in the modernised transcription of the ReN; for better reading, hyphens and the alinea symbol are deleted. The different transcription views of the ReN texts in ANNIS (Krause & Zeldes 2016) are explained by Barteld et al. (2017) and Barteld et al. (2021: Section 6.3).

- (2) *benedicam* tibi et multiplicabo semen tuum sicut stellas caeli et velut harenam quae est in litore maris [...] (Genesis 22:17)
- I will surely bless you, and I will surely multiply your offspring as the stars of heaven and as the sand that is on the seashore [...]
- a. Jck *werde* dy *benedigen* vnde werde meren dyn saet alse de sternen des hymmels / vnd also dat sant dat dar ys in dem ouer des meres [...] (Halberstadt, 13ra, 24–28)
- b. so *wyl* ik dy *gebedien* vnde wil din geslechte mannichuoldighen . also de sternen des hemmels vnde also dat sand dat daer licht in deme ouersten des meres [...] (Köln, 13ra, 19–24)

Besides these constructions, the data also contain a few imperatives. Their use is evidence of the complex interplay between future and modal semantics. In Section 4.2.4, we show in detail the strong connection between commandments expressed in a sentence and the modal verb construction with *schōlen* + infinitive as in (3b). A semantic equivalent that also expresses obligation is the imperative, hence an imperative form combined with one or more negation elements can also be used in order to prohibit an action as in (3a).

- (3) *audistis quia dictum est antiquis non moechaberis* (Mt. 5:27)
- You have heard that it was said, You shall not commit adultery.
- a. Gij hebbet ghehort dat ghesecht is den olden nicht *breke* de ee . (Köln, 167rb, 26–27)
- b. Gy hebbet gehort dat gesecht ys den olden / du *scalt* neyne vnküscheyt *don* . (Halberstadt, 168vb, 02–04)
- c. Gi hebbet gehort dat ghesecht is den olden . du *scalt* nene vnkuscheit *don* . (Lübeck, 402va, 51–53)

In order to gain a more comprehensive picture of the future translations in Middle Low German, we not only look at the Latin active forms, but also at the passive forms shown in Figure 4. When analysing Latin passive forms one has to be aware that in Latin there are so called deponentia (i.e. verbs with passive forms in their paradigms carrying an active semantics, see Kühner 1912; Hale & Buck 1966: 74). Hence, the passive forms of these verbs are to be translated with an active form. For example the verb *moechāberis* in (4) is a future passive form of ‘commit adultery’ and in the sentence *non moechaberis* (Exodus 20:14) it is translated as ‘You shall not commit adultery’ in the English Standard Version of the Bible.

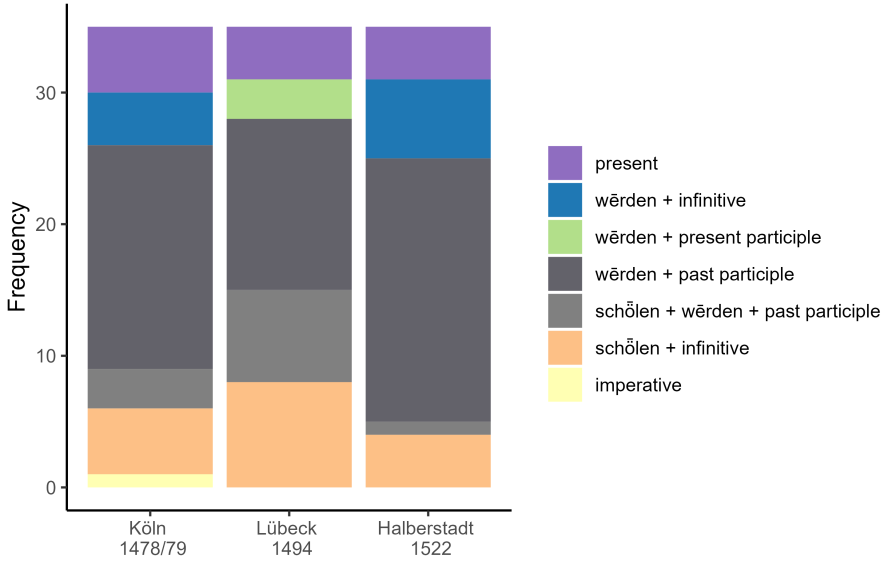


Figure 4: Verbal constructions used for translation of Latin future (passive) in the three Middle Low German Bibles.

Our data set contains 35 Latin future passive forms, 13 of which are deponentia. In the three Middle Low German Bibles, these deponentia are translated using different constructions, namely: the present tense (purple), *schōlen* + infinitive (orange), *wēden* + infinitive (blue), *wēden* + present participle (green) and the imperative (yellow) (in Figure 4), see for example the translation of the above-mentioned sentence *non moechaberis* as *Du schalt nene vnkuscheit dōn* in the Bible of Lübeck in (4). For the translation of the vast majority of the 22 verbs that are no deponentia, the construction *wēden* + past participle (grey) as in (5a) respectively *schōlen* + *wēden* + past participle as in (5b) is used – the formal equivalents of the Latin passive. Besides these, there are only a few occurrences of the present tense. In (6), the Latin passive form *adsimilābitur* ‘will be like’ is translated as *wert lik* in Middle Low German (i.e. the combination of the finite verb *wēden* and a predicative adjective). This is possible because of the strong syntactic and semantic relationship between the past participle form of a verb and a predicative adjective. Hence, in New High German, constructions like *die Fenster sind gewaschen* ‘the windows are cleaned’ are ambiguous and it is discussed in the research literature whether we have a passive construction here, the so called *Zustandspassiv* ‘state passive’, or a copula verb with a predicative

adjective (see, for example, Maienborn 2007). The translation in (6) is evidence for this aforementioned relationship between the past participle form of a verb and a predicative adjective.

- (4) non *moechaberis* (Exodus 20:14)
You shall not commit adultery.
a. Du *schalt* nene vnkuscheit *dôn* (Lübeck, 40va, 49)
- (5) et *benedicentur* in semine tuo omnes gentes terrae (Genesis 22:18)
and in your offspring shall all the nations of the earth be blessed
a. Vnd yn dynem sade *werden gebenediget* alle volk der erden (Halberstadt, 13ra, 28–30)
b. vnde alle lude der erden *scholen ghebenediet werden* in dime gheslechte (Köln, 13ra, 24–27)
- (6) omnis ergo qui audit verba mea haec et facit ea *adsimilabitur* viro sapienti qui aedificavit domum suam supra petram (Mt. 7:24)
Everyone then who hears these words of mine and does them will be like a wise man who built his house on the rock.
a. Darumme een iewelik de dar hort desse mine word vnde dar na deit .
de *wert lik* dem wisen manne de syn hus buwet vp den steen .
(Lübeck, 403rb, 60–62)

4.2.2 The influence of the text and the translator

Having gained an overview of the various options for translating Latin future expressions into Middle Low German, we will now take a closer look at the choice of the various constructions in the three biblical texts. Examples (1) and (2) in the preceding section already indicate that the three Middle Low German translations can use different constructions for the same Latin future expression. The Latin *habebit* ‘I will have’ in Exodus 20:7 is translated by a) a present tense form (*heft*) in the Bible of Halberstadt, b) *wēden* + present participle (*werd hebbende*) in the Bible of Lübeck, and c) *schōlen* + infinitive (*schal entschuldighen*) in the Bible of Köln (see 1a–1c). In Genesis 22:17, the Latin future *benedicam* ‘I will bless’ is translated by a) *wēden* + infinitive (*werde benedigen*) in the Bible of Halberstadt, whereas the Bible of Köln uses b) *willen* + infinitive (*wyl gebenedien*) (see 2a–2b).

The observation that the choice of a future construction strongly depends on the text also shows in the frequencies presented in Figure 3 in the preceding section. The most crucial difference concerns the two constructions *wērdēn* + infinitive and *wērdēn* + present participle. While with 12% in the Bible of Köln and 14% in the Bible of Halberstadt *wērdēn* + infinitive is an important option for translating Latin future, with 2% in the Bible of Lübeck it is only marginally used. Instead, the Bible of Lübeck contains a relatively high amount of *wērdēn* + present participle (22%), which is used infrequently in the other two texts. As mentioned in Section 3, the Bible of Lübeck is based on the Bible of Köln, and the Bible of Halberstadt is based on both of the others. Interestingly, the translators of the Bible of Lübeck and the Bible of Halberstadt make significant changes concerning the future expressions. They do not simply adapt the form from the base text – otherwise we would have similar frequencies for the different constructions in all three texts. In fact, the frequency of *wērdēn* + infinitive and *wērdēn* + present participle strikingly differs between the two Bibles from Köln and Halberstadt on the one hand and the Bible of Lübeck on the other hand.

As explained for the older varieties of High German in Section 2 and as is suggested for Middle Low German in Section 4.1, *wērdēn* + infinitive prevails over *wērdēn* + present participle diachronically. However, such a development is not reflected in the frequencies of these two constructions in the three Bibles – there is no rise in the use of *wērdēn* + infinitive from Köln via Lübeck to Halberstadt. Instead, the Bible of Köln and the Bible of Halberstadt show similar frequencies of the two constructions and the Bible of Lübeck stands out from the other two with a much higher proportion of *wērdēn* + present participle and a much lower frequency of *wērdēn* + infinitive. The frequent use of *wērdēn* + present participle in the Bible of Lübeck is a choice of its translator and therefore a characteristic feature of this text. Whether the differences are solely due to the translator or whether an influence of the language area manifests itself in the translator's language cannot be clearly answered on the basis of the texts we have examined. The observation that Lübeck distinguishes itself from Köln and Halberstadt suggests a possible influence of the language area. From around 1370 to 1530, the language of Lübeck functioned as a supra-regional lingua franca (Peters 2000a: 1483); it was regarded as the de facto Middle Low German standard. A higher proportion of *wērdēn* + infinitive in Köln and Halberstadt could be explained by a stronger influence of High German, because Köln has a special status due to its location in the High German language area and Halberstadt is on the south-eastern edge of the Eastphalian area on the border to the Elbe Eastphalian area (see Figure 1 in Section 3). The south of the Elbe Eastphalian area already switched to

Early New High German between 1350 and 1450 (Peters 2000b: 1417); a certain influence of High German can also be assumed in the neighbouring areas.²⁵

Besides these differences concerning *wērdēn* + present participle and *wērdēn* + infinitive, the three texts share many similarities. In all three texts the present tense is the dominating structure for the translation of a Latin future expression, the frequencies between the Bibles only differ minimally. Moreover, all three texts make use of the modal verb construction with *schōlen* rather frequently.

In order to gain deeper insights into the different choices of translating a Latin future form into Middle Low German, in the following we take a closer look at corresponding sentences with different future constructions in the three texts. Our focus lies on the sentences with *wērdēn* + present participle in the Bible of Lübeck, which are much more frequent in this text than in the other two. Tables 2 and 3 show which constructions are used for translating a Latin future form in the Bible of Köln and in the Bible of Halberstadt where the Bible of Lübeck has *wērdēn* + present participle.

Table 2: *Wērdēn* + present participle in Lübeck: Correspondences in Köln

KLN 1478/79	n
<i>schōlen</i> + infinitive	12
<i>wērdēn</i> + infinitive	7
present	5
<i>willen</i> + infinitive	3
<i>wērdēn</i> + present participle	1

In the Bible of Köln, the majority (12/28) of the corresponding constructions for *wērdēn* + present participle in the Bible of Lübeck have *schōlen* + infinitive, whereas in the Bible of Halberstadt we find *wērdēn* + infinitive in most of the corresponding sentences (12/28). One example illustrating this use of different constructions for the translation of the same Latin sentence is given in (7). The

²⁵Such an influence is confirmed by Czajkowski (2021), who examines Low German and High German forms on the phonological and graphematic level in the transitional area between Low German and East Central German. The locations of her corpus range from Sittichenbach in the south to Halberstadt and Aken in the north. The period examined extends from 1265 to 1490 (Czajkowski 2021: 31–36). Czajkowski (2021: 293–294) shows that the proportion of High German forms already increases from the second half of the 14th century until the end of the 15th century. It can be assumed that this language change manifested itself not only on the phonological and graphematic level but also in the syntax.

2 Expressing future reference in Middle Low German

Table 3: *Wērdēn* + present participle in Lübeck: Correspondences in Halberstadt

HBS 1522	n
<i>wērdēn</i> + infinitive	12
present	6
<i>wērdēn</i> + present participle	3
<i>schōlen</i> + infinitive	3
<i>willen</i> + infinitive	3
<i>wērdēn</i> + past participle	1

Latin future *comedes* ‘you will eat’ is translated as *scholt eten* in the Bible of Köln (7a), as *werst etende* in the Bible of Lübeck (7b) and as *werst eten* in the Bible of Halberstadt (7c).

- (7) ad [...] maledicta terra in opere tuo in laboribus *comedes* eam cunctis diebus vitae tuae (Genesis 3:17)
- [...] cursed is the ground because of you; in pain you shall eat of it all the days of your life.
- [...] de erde sij voruloket vnde dine wercke . du *scholt eten* vth er . in arbeit alle de dage dines leuendes . (Köln, 5va, 11–16)
 - [...] Vormalediet sy de erde . vmme di vnde diner sunde willen vnvruuchtbar to wesende in dyneme werke // Jn deme arbeyde so *werst* u van er *etende* alle de dage dines leuendes . (Lübeck, 8va, 64–8vb,04)
 - [...] vnde vormaledyget sy de erde Jn dynem arbeide *werst* u *eten* van ör / alle dyne daghe dynes leuendes . (Halberstadt, 4va, 04–10)

Besides the frequently used constructions *schōlen* + infinitive in the Bible of Köln and *wērdēn* + infinitive in the Bible of Halberstadt, both texts also contain sentences that use the present tense where the Bible of Lübeck has *wērdēn* + present participle. As shown in Figure 3, in the Bible of Halberstadt the present tense occurs more frequently than in the other two texts for translating future expressions from the Vulgate. An example of the use of the present tense in the Bibles of Köln and Halberstadt instead of *wērdēn* + present participle in the Bible of Lübeck is presented in (8). While the Bible of Lübeck translates the Latin *occidet* ‘he will kill’ as *werd dodende* (8b), the other two texts use a present tense verb: the Bible of Köln has *sleyt dot* (8a) and the Bible of Halberstadt *dódet* (8c).

- (8) [...] et ero vagus et profugus in terra omnis igitur qui invenerit me *occidet* me (Genesis 4:14)

[...] I shall be a fugitive and a wanderer on the earth, and whoever finds me will kill me.

- a. [...] vnde ich werde vluchtich vnde ellendich vppe der erden . daer vmme een iewelick de my werd vinden de *sleyt* my *dot* . (Köln, 6ra, 14–16)
- b. [...] vnde werde vnstedich vnde vorvluchtich vp der erden . Alle de my vindende werd . de *werd* my *dodende* . (Lübeck, 9ra, 23–27)
- c. [...] vnd werde eyn loder vnd eyn vorvlokeder in der erden / hijr vmme we my vindet de *dôdet* my / (Halberstadt, 5ra, 06–10)

Since the formally different options for expressing future in Middle Low German share the same basic semantics or the same grammatical function, they can easily be exchanged for each other within the same context as is illustrated by Tables 2 and 3 and by the examples analysed above. Hence, the use of one variant is clearly a choice of the translator. However, the reason why one translator prefers a variant against another one can hardly be discerned. One hypothesis might be that the translator uses one variant consciously in order to express a different semantic nuance of the sentence. Modal verb constructions could, for example, be preferred in order to convey the additional semantic nuance of modality. In order to verify our hypothesis, the following two sections take a closer look at the filling of the main verb slot of the future construction as well as at the semantic and syntactic context.

4.2.3 Filling of the main verb slot: the case of *wērdēn*

There is only one main verb occurring in the constructions analysed that shows a clear pattern regarding its effect on the choice of construction and that is *wērdēn*. In our data set, it is the most frequent verb in the main verb slot of the future constructions (n=45). Strikingly, it is nearly exclusively used in the present tense variant in all three texts (41/45, see Table 4) as in (9a–9b).

There are only two exceptions (see Table 5), one where the Bible of Köln and the Bible of Halberstadt use the main verb *wērdēn* in present tense where the Bible of Lübeck contains *schōlen* + *wērdēn* (see 10a–10c) and one where all three Bibles use *wērdēn* in the imperative. Consequently, there is a very close connection between the main verb *wērdēn* and the present tense variant.

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Table 4: The most frequent verbs occurring with the different constructions

Construction	nTypes	verb1	verb2	verb3
present	28	<i>wēdden</i> (41)	<i>gēven</i> (9)	<i>lōnen</i> (9)
<i>schōlen</i> + inf.	27	<i>māken</i> (8)	<i>begēren</i> (5)	<i>dōn</i> (5)
<i>wēdden</i> + pres. part.	19	<i>sēn</i> (4)	<i>ēten</i> (3)	<i>ingân</i> (2)
<i>wēdden</i> + inf.	14	<i>besitten</i> (3)	<i>ēten</i> (3)	<i>gēven</i> (3)

Table 5: Constructions occurring with the main verb *wēdden*

Construction	Bible	n
present	Köln	14
present	Lübeck	15
present	Halberstadt	15
<i>schōlen</i> + infinitive	Lübeck	1
imperative	Köln	1
imperative	Halberstadt	1
imperative	Lübeck	1

- (9) [...] *sin autem male statim in foribus peccatum aderit* [...] (Genesis 4:7)
 [...] And if you do not do well, sin is crouching at the door [...]
- [...] *Efft u euer ouel deist tohantes werd dyn sunde entegenwerdich* in den doeren [...] (Köln, 5vb, 03–11)
 - [...] *yodoch werst u quad donde. dinen broder to hatende // dine sunde werd drade ieghenwardich* in dem doren [...] (Lübeck, 8vb, 59 – 9ra, 04)
 - [...] *Jss et auer dat du ôuel werkest altohant ys dijn sunde in den dôren* [...] (Halberstadt, 4va, 49–53)
- (10) [...] *vagus et profugus eris super terram* (Genesis 4:12)
 [...] You shall be a fugitive and a wanderer on the earth.
- [...] *du werst vluchtich vnde ellendich vp der erden* . (Köln, 6ra, 07–09)
 - [...] *Vnstedich vnde vorvluchtich schalt u werden vp der erden alle de daghe dines leuendes* [...] (Lübeck, 9ra, 18–22)
 - [...] *Eyn loder vnd eyn vorvlokeder werdest du vp der erden* . (Halberstadt, 5ra, 01–04)

Most of the instances of *wērdēn* in present tense translate a Latin future form of the verb *esse* ‘be’. Instead of translating this future form into Middle Low German with the help of complex verbal constructions with the main verb *wēsen* or *sîn*²⁶ – the equivalent of Latin *esse* –, for example *schal sîn* or *wērt sîn*, the three texts prefer the present tense of *wērdēn*. This choice of construction is only possible because of the semantic connection between *wērdēn* in present tense and *wēsen* in future tense, as in *De sunde wert iegenwardich* ‘Sin becomes present’ and *De sunde wert sîn iegenwardich* ‘Sin will be present’. If *X* (here: the sin) becomes *Y* (here: present) today it means that *X* will be *Y* tomorrow.

This analysis showed a clear impact of verb semantics on the choice of future construction in the case of *wērdēn*. This effect can be accounted for by the verb’s semantics, as it denotes a change of state and thus already brings about a future reading when used in the present. An additional marking of futurity is unnecessary in such cases. The occurrence of the *wērdēn* + infinitive construction with *wērdēn* is considered to indicate an advanced level of grammaticalisation of the construction and is found from the 16th century on in High German (Schmid 2000: 17; Concu 2022: 28). It indicates an expansion of the contexts of occurrence and it shows the gradual bleaching of meaning in the auxiliary, which leads to a differentiation from its main verb usage and thus an ease of potential blocking effects to use both words in the same predicate. It is in line with previous research as well as our findings from Section 4.1 that we do not find such signs of grammaticalisation in the Middle Low German Bibles.²⁷

The role of sentence semantics, which has already been indicated in the preceding analysis, is examined more closely in the following section in order to identify potential correlations.

4.2.4 Syntactic and semantic context

The influence of syntactic context and sentence meaning is analysed with the help of a comparative view of the constructions used in the three texts. Figure 5 shows the corresponding Middle Low German passages in chronological order of appearance in the Bible. The first two lower bars represent the passages from *The creation of men and women* (2 instances), followed by *The expulsion from paradise* (8 instances) etc. (see also Table 1). At the sides, the abbreviated names of the Bible chapters are given. This type of figure illustrates the importance

²⁶Middle Low German has two different verbs meaning ‘to be’, namely, *wēsen* and *sîn*.

²⁷We find but 12 instances of *wērdēn* + *wērdēn* in the ReN. Most of them are instantiations of an embedded passive construction (*wērdēn* + *wērdēn* + past participle). Nine of them are from the first half of the 16th century, the rest from later periods.

of the semantic context for the choice of future construction. From the parallel layout of the three texts with the construction used encoded by different colours, we can easily assess where the texts agree and where they disagree in the future construction used. Concurrently we see clustering of certain constructions in certain passages of the Bible texts.

The lion's share of *schōlen* + infinitive appears in the Ten Commandments (Exod. 20). Our data set contains 21 sentences from this chapter (see also Table 1). The Bible of Köln nearly exclusively uses *schōlen* + infinitive here, with only one exception where it has *willen* + infinitive. In the Bible of Lübeck we find variation between *schōlen* + infinitive (n=8), *schōlen* + *wēden* + present participle (n=7), *wēden* + present participle (n=5), and *schōlen* + present participle (n=1). Halberstadt is similar to Köln in that it predominantly uses *schōlen* + infinitive (n=16). Apart from that, it uses the present, *wēden* + infinitive, *willen* + infinitive and the imperative in one sentence each. Outside of the Ten Commandments, Figure 5 illustrates the corresponding use of *schōlen* + infinitive (4) as well as the imperative (7) in the three chapters of the Sermon on the Mount (Mt. 5–7).

What further stands out from Figure 5 is the accumulations of corresponding use of the present tense variant (30) in the three chapters of the Sermon on the Mount (Mt. 5–7). More corresponding instances with the present (n=3) can be found in the chapters from Genesis. Overall, we observe more variation between the Bibles in the passages from Genesis and in the earlier passages of the Sermon on the Mount.

Before looking directly at the annotations of sentence semantics, let us examine the sentences that are consistently translated with one specific construction in all three Bibles, starting with the imperative which in Figure 5 forms three neat yellow stripes in the area of the three chapters of the Sermon on the Mount. In our data, there are for example two sentences from the chapter “Sermon on the Mount”, where the Latin future forms *peierabis* ‘you will lie’, *reddes* ‘you will give back’, *diliges* ‘you will love’ and *habebis* ‘you will have’ are in all of the three texts translated by the imperative forms *swere*, *gyff/ghif/gyf*, *hebbe* and *hate* (see 11a–11f). The content of the sentences indicates that not only an action taking place in the future is referred to but the sentences contain a demand note. This is supported by person and number of the verb, namely the 2nd person singular (in accordance with this, the other three occurrences with corresponding imperatives have verbs in the 2nd person plural). The translators of the Middle Low German Bibles transfer this semantic note of a request to do or to neglect an action by using the imperative which inherently expresses a demand.

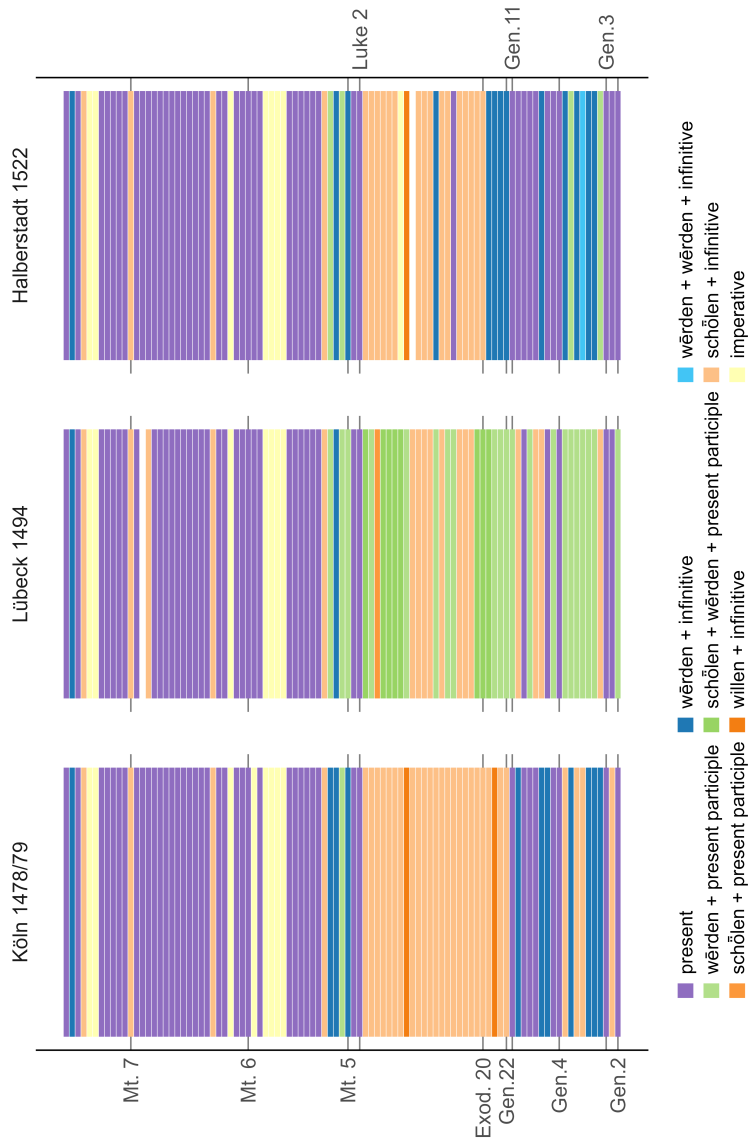


Figure 5: Comparative view of constructions used in the three texts

- (11) iterum audistis quia dictum est antiquis non *peierabis reddes* autem Domino iuramenta tua (Mt. 5:33)
 audistis quia dictum est *diliges* proximum tuum et odio *habebis* inimicum tuum (Mt. 5:43)
 Again you have heard that it was said to those of old, You shall not swear falsely, but shall perform to the Lord what you have sworn.
 You have heard that it was said, You shall love your neighbor and hate your enemy.
- a. Euer hebbe gij ghehort dat ghesecht is den olden nicht en *swere* meynede . sunder *gyff* de ede deme heren (Köln, 167rb, 43–46)
 - b. Gij hebben ghehort dat gesecht is *hebbe leff* dynen naber vnde *hate* dynen viant . (Köln, 167va, 08–09)
 - c. Ouer hebbe gi hort dat ghesecht is den olden . nicht en *swere* meenede . sunder *ghif* de ede dem heren . (Lübeck, 402vb, 03–04)
 - d. Gy hebben ghehort dat ghesecht is . *hebbe leff* dynen neghesten vnde *hate* dynen vint : (Lübeck, 402vb, 22–23)
 - e. Auer hebbe gy gehort dat gesecht ys den olden nicht *swere* meynede / sunder *gyf* de ede dem heren . (Halberstadt, 168vb, 19–21)
 - f. Gy hebben gehort dat gesecht ys / *hebbe leiff* dynen negesten vnd *hate* dynen vyent / (Halberstadt, 168vb, 37–39)

A similar observation can be made for the corresponding use of *schōlen* + infinitive in the three texts. Seven out of the eleven occurrences for such correspondences concentrate in the chapter the Ten Commandments. As examples (12a–12c) show, the Latin Vulgate uses a verb in the 2nd person singular, namely *adsumes* ‘you will take’ and, similar to the example in (11), the sentence contains a command or a prohibition. In the Middle Low German Bibles, this prohibition is expressed by the construction *schōlen* + infinitive, namely by *schalt (to)nemen*. Due to the special feature of the modal verb construction with *schōlen* expressing not only future reference but also modality, it is perfectly suitable for transferring the additional semantic nuance of demand.

These investigations of corresponding translations with the imperative and with *schōlen* + infinitive in the three Bible texts demonstrate that the meaning of the whole sentence and its context have a strong influence on the use of a future construction. That is, the imperative and modal constructions with *schōlen* are clearly and unsurprisingly connected to the domain of obligation. This connection is also reflected in the constructions’ usage with grammatical person and

number (see Figure 6). In all three texts, *schōlen* + infinitive predominantly occurs in the second person singular, as this is used in the directive contexts. The present tense is often used with the third person singular, but, since it occurs in all persons and numbers, no such clear preference as with *schōlen* + infinitive is observed here.

(12) non *adsumes* nomen Domini Dei tui in vanum [...] (Exodus 20:7)

You shall not take the name of the Lord your God in vain

- Du en *schalt* den namen dynes ghades nicht ydelick in dynen mont *nemen* [...] (Köln, 40ra, 20–24)
- Du *schalt* nicht *tonemen*. to sprekende // den namen dynes heren dines gades vorgheues [...] (Lübeck, 40va, 27–31)
- Du en *schalt* dynes heren godes name nicht vergheues yn dynen munt *nemen* [...] (Halberstadt, 39va, 19–22)

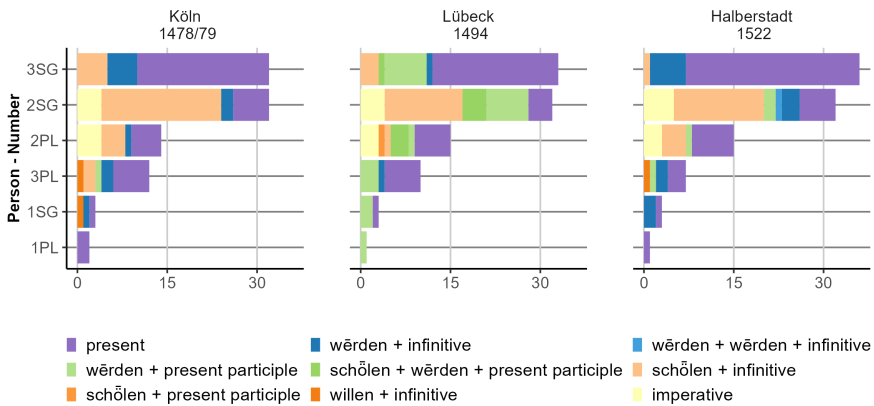


Figure 6: Constructions used by person and number in the three Bibles

The distribution of the future constructions between the chapters of the Bible indicated in Figure 5 is clearly influenced by different types of sentence semantics that predominate in the individual chapters. This connection between the sentence semantics and the choice of construction is illustrated in Figure 7,²⁸

²⁸Figure 7 is based on the 93 of the 95 sentences in our data set. Two were excluded because of ambiguous annotations for sentence semantics.

2 Expressing future reference in Middle Low German

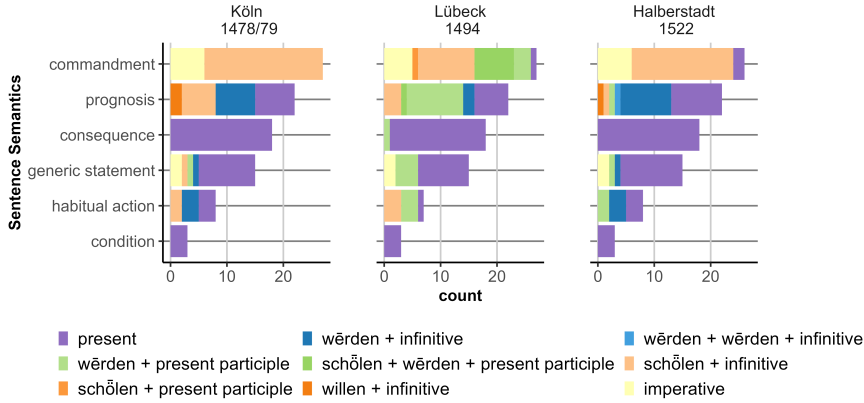


Figure 7: Correlation between sentence semantics and construction in the three Bibles (n=93).

which shows the distribution of the constructions for the different types of sentence semantics that we annotated for. While the most frequent correlation (that is, between commandments and the variant *schölen* + infinitive and the imperative) has already been discussed, this perspective offers new insights especially for the usage of the present: It is the only variant used (with only a marginal deviation in the Bible of Lübeck) in those sentences, that contain a) the consequence to a condition and b) a condition. Example (13) shows a case where both parts of the conditional clause have a future form in the Vulgate. Future encoding appears frequently with the consequent part of the conditional clause (n=18). The conditions themselves are rarely expressed in the future (we only find this three times in our data), which is to be expected given the typological evidence referred to in Section 3.

(13) aut si piscem *petet* numquid serpentem *porriget* ei (Mt. 7:10)

Or if he asks for a fish, will give him a serpent?

a. Edder effte he on *byddet* vmme eyen vysch *reket* he ohm nicht dan eine slangen (Halberstadt 169va, 21–22)

The apodosis examples all come from the second chapter of the Sermon on the Mount (Mt. 6) and each sentence is predicting that, if the preceding condition is met, God will reward the addressee, as in (14). The use of present tense (or also: the absence of overt future marking) in such (temporally unspecified) contexts

is typologically seen fairly common and not surprising for Middle Low German either, as it does not have one fully obligatory future marker (see Section 3).

- (14) [...] et Pater tuus qui videt in abscondito *reddet* tibi (Mt. 6:4)
[...] And your Father who sees in secret will reward you.
a. [...] vnde dyn vader de dar suet in vorborghenheit de *lonet* dij . (Köln 167va, 35–37)
b. [...] vnde din vader de dar suet in vorborghenheit de *lonet* di . (Lübeck 402vb, 47–49)
c. [...] vnd dyn vader de dat suet yn vorborghenheyt de *lonet* dy . (Halberstadt 169ra, 16–18)

The next category of sentence semantics that we differentiated in our data is *generic statements*, that is assertions that can be considered to be true generally, not merely in the future. This type of sentence shows a clear preference for the present, as illustrated in Figure 7. Our data contain a total of 15 verbal constructions in such general contexts that come from nine different Bible passages. More than half of these instances are convergently translated with the present constructions by all three Bibles, as for example in the passage in (15), which contains four constructions in such context.

- (15) nemo potest duobus dominis servire aut enim unum odio *habebit* et alterum *diliget* aut unum *sustinebit* et alterum *contemnet* [...] (Mt. 6:24)
No one can serve two masters, for either he will hate the one and love the other, or he will be devoted to the one and despise the other [...]
a. Nemant mach denen twen heren edder den enen *hatet* he . den anderen *hefft* he leff . edder he *beholt* den enen vnde *vorsmat* den anderen [...] (Köln 167vb, 29–33)
b. Numment mach denen twen heren. dar de ene deme anderen enteghen is // edder den enen *hatet* he . vnde den anderen *heft* he lef . edder he *behold* den enen . vnde *vorsmad* den anderen [...] (Lübeck 403ra, 28–32)
c. Nemant mach deynen twen heren edder den einen *hatet* he / vnd den anderen *hefft* he leyff edder he *beholt* den eynen vnd *vorsmadet* den anderen [...] (Halberstadt 169rb, 10–13)

The other types of sentence semantics we distinguished in our data show more variation. 22 sentences in our data set are classified as prognoses, making a statement over a singular event to take place in the future (for example 16). With this

type of sentence semantic, we find eight sentences that converge in the three Bibles using the present (n=5), *wēden* + infinitive (n=2) and *schōlen* + infinitive (n=1). In 14 sentences there is variation. While there are different combinations, there is a pattern within these sentences, namely the preference of the different texts for certain constructions, as was discussed in Section 4.2.1: We see a prevalence of the *schōlen* construction in Köln, *wēden* + present participle in Lübeck and *wēden* + infinitive in Halberstadt, which is illustrated in (16a–16c).

- (16) dixit Abraham Deus *providebit* sibi victimam holocausti fili mi pergebant ergo pariter (Gen. 22:8)

Abraham said, God will provide for himself the lamb for a burnt offering, my son.

- a. do sprak abraham myn sone god de *schal* *vorseen* vmme sinen offer vnde offerhande . (Köln 12vb, 28–29)
- b. doch sede abraam Sone myn . god *werd* sik *voresynde* dat dar schal werden ghedodet in een bernende offer (Lübeck 16va, 10–12)
- c. God de *wert* sick *voerseem* vmme dat gebrande opper . (Halberstadt 12vb, 14–15)

We differentiated the rather singular semantics of our category *prognosis* from clearly *habitual* statements that predict an event to occur habitually in the future. We have assigned this label to seven sentences in our data set (one including multiple future constructions, see (17)²⁹). This group does not differ strikingly from the sentences with the singular prognosis meaning. Only one of the sentences has the same construction (present) in all texts. For the other sentences, there is variation that is in accord with the observation made with the prognoses, complementing the pattern with an apparent preference for *schōlen* + infinitive in Köln and a preference for constructions with the present participle in Lübeck.

- (17) nolite ergo solliciti esse dicentes quid *manducabimus* aut quid *bibemus* aut quo operiemur (Mt. 6:31)

Therefore do not be anxious, saying, What shall we eat? or What shall we drink? or What shall we wear?

- a. Dar vmme nicht en wilt sin sorchuoldich sprekende . wat *ete* wy effte wat *drincken* wy edder war mede werde wy ghecledeet . (Köln 167vb, 52–55)

²⁹The equivalent of ‘What shall we drink?’ is omitted in the Bible of Lübeck.

- b. dar umme segghe ik iuw . nicht en syt sorchuoldich iuwer sele .
iuwem leuende // wat gi *schoelen eten* . noch iuwen liue . wor mede gi
dat kleden (Lübeck 403ra, 33–34)
- c. Dar vmme nicht wylyet syn sorchuoldich spreken / wat *ete* gy efte
wat *drinke* wy edder wor mede werde wy gekledet (Halberstadt 169rb,
30–33)

Let us return briefly to the construction *schōlen* + infinitive. In example (16) we saw it used in a prognostic (i.e. ‘real future’) utterance in the Bible of Köln, which we have found to show a preference of this construction. In (17) it is the Bible of Lübeck that uses *schōlen* while the other two Bibles have present forms. (18) shows the only non-directive sentence where Halberstadt uses the construction (alongside both other texts) and it might be up to debate in this case whether the sentence does or does not bear a modal meaning. Overall, regarding the different usages of *schōlen* + infinitive, the situation is as follows: As was shown in Section 4.2.1, the construction appears most frequent in Köln and less frequent in Lübeck and Halberstadt (see Figure 3). The proportion of non-directive (prognosis, habitual and generic) usage, however, is comparable in Lübeck and Köln, with about a third of the instances in Lübeck and more than a fourth in Köln. In Halberstadt, however, there are no non-directive instances of *schōlen* + infinitive except the one shown in (18c). Looking back at the frequency distribution found for the constructions with *wēden*, we could tentatively suggest a decline of *schōlen* + infinitive as a future construction, given that, in the period where we see an increase in the *wēden* constructions, we find no more usage of *schōlen* + infinitive in sentences that are futurate in the narrow sense (i.e. excluding directive) in the Bible translation. This conclusion is of course tentative and is in urgent need of verification on the basis of larger amounts of data.

- (18) non omnis qui dicit mihi Domine Domine *intrabit* in regnum caelorum
[...] (Mt. 7:21)

Not everyone who says to me, ‘Lord, Lord,’ will enter the kingdom of heaven [...]

- a. nicht alle de dar spreket tho my here here *shal gan* in dat rike der
hemmele [...] (Köln 168ra,56–168rb,02)
- b. Nicht alle de dar spreket to my . here here *scal gan* in dat rike der
hemmele [...] (Lübeck 403rb,50–403rb,54)
- c. Nicht alle de dar spricket tho my / here here *shal gaen* yn dat ryke
der hymmele [...] (Halberstadt 169va,46–169va,50)

This analysis showed that different types of sentence semantics clearly have an effect on the choice of the construction for translating the Latin future. We find a strong preference for the modal construction with *schölen* and the imperative for directive statements like commandments and interdictions. Both parts of conditional clauses clearly prefer the present construction, as do generic statements, albeit with a little more variation. The sentence types that seem to be the least determined with regard to the construction are prognoses, both about singular and habitual events. The latter are the sentences where most of the *wērdēn* + infinitive are found. Overall, the results regarding the ‘extra’ categories considered here, on top of the ‘real future’ categories, are inline with what would be expected based on the literature. We want to underline, however, that these results do contribute to an understanding of future reference in Middle Low German and that using a broader definition of future, encompassing constructions that overlap semantically, showed to be fruitful for an initial and exploratory study like the present one.

5 Summary and Outlook

Our empirical investigation of future constructions and the translation of Latin future forms in Middle Low German has brought the following results to light: In the first part of the study, we conducted a purely quantitative examination of the occurrence of the two competing constructions *wērdēn* + present participle and *wērdēn* + infinitive. Our data, taken from the ReN, show an increase in the occurrence of both variants in the second half of the 15th century and the first half of the 16th century. It is not possible to draw conclusions about the development of the constructions prior to the 15th century on the basis of the structure of the corpus, as texts that contain more references to the future are only represented from the 15th century on. A number of differences can still be noted in relation to their development in High German. Firstly, the development differs strikingly from that observed in High German in that *wērdēn* + infinitive does not seem to replace *wērdēn* + present participle over time. Instead, we see a rather sudden increase in both constructions. Furthermore, the frequency growth of *wērdēn* + present participle in Middle Low German takes place at a time when it no longer plays a central role in High German. In our data, we cannot trace its demise. Its absence in the 17th century should not be overestimated because of the few texts contained in that time span of the corpus. Overall, it must be noted that the suitability of reference corpora for the investigation of phenomena that are dependent on the content of the texts, such as future reference, is not necessarily given and must be checked and evaluated.

In the second part of our study, we undertook a quantitative-qualitative investigation of the translation of Latin future forms in three Middle Low German Bibles. The benefit of this analysis is its onomasiological approach: Starting from the function, namely expressing the future of an event, which in Latin is morphologically marked on the verb, we examined all the formally different constructions sharing this function (i.e. also the present tense and the modal constructions). Due to their multi-functionality, in a corpus linguistic approach, for example, finding attestations for these is rather difficult or requires a great deal of effort. In our analysis, we identified a set of intertwined factors influencing the choice of variant. One of them could be the translator, more precisely their preference for specific structures or their individual reading of the Latin original. However, the language area could also be a decisive factor, since Köln and Halberstadt are more susceptible to the influence of High German than Lübeck. With regard to constructions expressing future, the three texts differ above all in that the proportion of *wērdēn* + present participle is considerably higher in the Bible of Lübeck than in the Bible of Köln and the Bible of Halberstadt.

Regarding the main verbs in the Middle Low German future construction our examination revealed a connection with the type of construction for the verb *wērdēn*, that is, it occurs predominantly in present tense. As the change of state semantics of the verb itself can already bring about a future reading, an additional marking of futurity is not necessary. The fact that *wērdēn* does not occur in the verb slot of *wērdēn* + infinitive shows that the construction has not reached the same level of grammaticalisation in Middle Low German that it has in Early New High German, where *werden* + *werden* occurs from the 16th century on.

The semantic context also has an influence on the choice of variant. Here, a strong connection between commandments and the use of *schōlen* + infinitive was found. This in turn is also reflected in the fact that *schōlen* + infinitive is predominantly used in the second person singular. For this reason, the Ten Commandments of the Bible contain a particularly large number of verbal complexes with *schōlen*. We were also able to show that the present tense variant occurs mainly in conditions and consequences.

In order to gain an even more comprehensive insight into the use of future expressions in Middle Low German, it would be helpful to look at other text types besides the Bible texts studied here, such as the calendar mentioned above. However, it should be borne in mind that manual annotation of all future tense expressions in such a text is very time-consuming and unambiguous annotation is not always possible, especially in the case of modal verb constructions. Despite extensive corpora, the study of the future in German remains a complex and challenging task.

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Chapter 3

Constructional template or competitor? Revisiting the relation between German *werden* + Infinitive and *werden* + Present Participle based on new datasets

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The German future construction *werden* + Infinitive has been subject to considerable research efforts and controversial debates. In particular, it is a matter of debate which other constructions can be considered potential predecessors (in the sense of constructional “templates”) and/or competitors. The recent release of reference corpora of Middle and Early New High German offers the possibility of assessing its development based on balanced, systematically compiled datasets. In this paper, we present the results of an in-depth corpus-based analysis that sheds new light on some long-standing debates on the development of *werden* + Infinitive as well as its potential predecessors or competitors. More specifically, we investigate the distribution of slot fillers in *werden* + Infinitive and *werden* + Present Participle constructions using collocation analysis as well as dispersion measures. The results provide further evidence for the hypothesis that rather than *werden* + Participle being a predecessor of *werden* + Infinitive, the two constructions show parallel developments for a while before they specialize into different niches, with *werden* + Infinitive becoming a future marker and *werden* + Participle an inchoative marker that only survives in some dialects.

1 Introduction

In present-day German, there are essentially two main ways of marking future reference: On the one hand, the periphrastic form *werden* ‘become’ + Infinitive



can be used, as in (1a). On the other hand, one can make use of the futurate present, i.e. a present-tense form that refers to a future event, as in (1b).

- (1) a. *Morgen werde ich das Kapitel zu Ende schreiben.*
Tomorrow FUT I the chapter to end write.
'Tomorrow I will finish writing the chapter.'
- b. *Morgen schreibe ich das Kapitel zu Ende.*
Tomorrow write I the chapter to end.
'Tomorrow I will finish writing the chapter.'

The historical genesis of *werden* + Infinitive as well as its status in present-day German have been subject to considerable debate. To discuss the latter aspect first, i.e. its grammatical status in present-day German, it is a matter of debate whether the construction can actually be seen as a future construction (e.g. Saltveit 1960, Vater 1975). For example, Vater (1975) argues that *werden* in combination with the infinitive does not express temporal relations different from those that can be expressed with the futurate present, and thus cannot be considered a future tense form. Indeed, it can be considered an open question whether or not the *werden* + Infinitive construction should be regarded as a future *tense* form. However, even if we decide not to treat it as a tense in its own right, i.e. not on a par with tenses like the simple present or the simple past, it can still be considered a future *construction* in the sense that it is used in a systematic way to refer to future events. Thus, we agree with Hilpert (2008: 154) that "the modalist-temporalist debate should therefore be abandoned in favor of a discussion of how the temporal and modal meanings of *werden* relate to each other."

Regarding the historical development of *werden* + Infinitive, key questions revolve around the question of which constructions can be considered potential predecessors and/or competitors of the pattern. According to the textbook scenario, the construction *werden* + Present Participle, as in (2), played a major role in the development of *werden* + Infinitive, which prevailed over a number of other candidates that were occasionally used with future reference in earlier stages of German, especially modal verbs such as Old High German *skulan* 'shall' or, less frequently, *wellen* 'want' or *mugan* 'can' (see Szczepaniak 2011: 144–148).

- (2) *inti nu uuirdist thu suigenti*
And now become.2SG you be-silent.PRS.PTCP.
'And now you become silent' (Tatian 2,9, quoted from Szczepaniak 2011: 144)

Several scenarios have been proposed to explain the rise of *werden + Infinitive* (see Section 2 below). However, many studies had to work with comparatively limited datasets (e.g. Schmid 2000). Over the last decade, reference corpora covering the historical stages of (Low and High) German have become available that allow us to revisit such long-standing research questions on the basis of a fairly large amount of data.

In this paper, we revisit the competition between *werden + Infinitive* and what is arguably its most relevant competitor, *werden + Present Participle* (henceforth just *werden + Participle*), in Middle High German (c. 1050–1350) and Early New High German (c. 1350–1650)¹ based on data from the reference corpora of these two stages of German. We will first give a brief overview of previous research (Section 2), before we introduce our data as well as our annotation guidelines (Section 3). We focus on three aspects: firstly, we take a closer look at the frequency and productivity of both constructions in the light of the evidence that the newly available reference corpora of historical stages of German offer. Secondly, we investigate the (distribution of) the slot fillers in the two constructions that are usually seen as the most relevant competitors, *werden + Infinitive* on the one hand and *werden + Present Participle* on the other. And thirdly, we zero in on the question of whether or not the constructions actually clearly encode future reference in the contexts in which they appear. After presenting our results in Section 4, we discuss to what extent the results can shed new light on the question of how *werden + Infinitive* developed (Section 5).

2 The emergence of *werden + Infinitive*

2.1 State of the art

werden + Infinitive is probably among the most well-investigated constructions in German. Hence, this section cannot offer an even remotely comprehensive overview of previous studies. Instead, we will focus on two aspects that are particularly relevant for our research: we first discuss grammaticalization scenarios that have been proposed for *werden + Infinitive* (Section 2.1.1), before we turn to an overview of relevant studies that have taken a quantitative corpus-based approach to the *werden + Infinitive* construction and its competitors, including very early studies that can be regarded as examples of quantitative corpus linguistics *avant la lettre* (Section 2.1.2).

¹In the present paper, we will focus on the development of *werden + Infinitive* and *werden + Participle* in High German; see Ilden & Schnee (this volume) for an in-depth discussion of Low German.

2.1.1 Scenarios of the diachronic development of *werden* + Infinitive

As already mentioned in the Introduction (Section 1), there are a number of scenarios that have been proposed for the emergence of *werden* + Infinitive. In his comprehensive review paper on this topic, Westvik (2000) singles out six major hypotheses, or theories, as he calls them. We briefly summarize these hypotheses below, adding a more recent one proposed by Jäger (2024); Table 1 gives an overview of the different scenarios.

Table 1: The six scenarios for the emergence of *werden* + Infinitive proposed by (Westvik 2000), plus Jäger's (2024) reanalysis hypothesis. The German names in brackets are Saltveit's original terms for the hypotheses.

Theory	Explanation: <i>werden</i> + Infinitive ...
Erosion theory (<i>Abschleifungstheorie</i>)	...emerged from <i>werden</i> + Present Participle via phonological erosion
Confusion theory (<i>Konfusionstheorie</i>)	...emerged from <i>werden</i> + Present Participle through a “confusion” between present participles and inflected infinitives
Autonomy theory (<i>Autonomietheorie</i>)	...existed as an autonomous construction before it “merged” with <i>werden</i> + Participle
Analogy theory (<i>Analogietheorie</i>)	...emerged in analogy to other constructions, especially constructions with <i>werden</i> , but also e.g. inchoative constructions
Contamination theory (<i>Kontaminationstheorie</i>)	... emerged in analogy to constructions with modal verbs and infinitives
Interference theory (<i>Interferenztheorie</i>)	... as a loan construction modeled on Czech <i>budu</i> + Infinitive
Reanalysis theory	<i>werden</i> + Infinitive as a reanalysis of <i>werden</i> + Past Participle constructions

The oldest among the hypotheses reviewed by Westvik (2000) is what he calls the “erosion theory” (*Abschleifungstheorie*), according to which *werden* + Infinitive emerged from *werden* + Present Participle. On this view, the participle, as in MHG *ir werdint lachinde*, lit. ‘you become laughing’, underwent phonological erosion,

which caused the present participle to become homophonous with the infinitive (*lachend* ‘laugh-PRT’ > *lachen* ‘laugh-INF’). However, Kleiner (1925) already showed that this hypothesis is only plausible in very few dialects of German. More specifically, this hypothesis is not supported by evidence regarding the spatial development of the two phenomena: *werden* + infinitive first expanding in the South of the German-speaking area and *-d*-deletion starting prominently in Low German, i.e. in the North. Instead, Kleiner (1925) argued for what Westvik (2000: 236) calls the “confusion theory” (*Konfusionstheorie*). This hypothesis also assumes that the present participle forms lost their endings at some point, but offers a more complex explanation instead of the previous, purely phonological account. According to this hypothesis, in MHG, a “contamination” took place between the present participle on the one hand, and the inflected infinitive on the other: In Middle High German, the regular genitive and dative forms of the infinitive in its nominalized use – the so-called gerund – were *-ens* and *-ene* (e.g. *arbeitene* ‘work-DAT.’; see Hartmann 2021: 367). According to Kleiner (1925: 58), the ending *-de(s)* occasionally starts to occur in the oblique cases of the inflected infinitive. Thus, they become identical to the present participle. Kleiner now assumes that the present participle forms lost their *-de* endings in analogy to the regular forms of the gerund. However, both Schmid (2000: 7) and Pfefferkorn (2005: 325) point out that this hypothesis would only be convincing if such a contamination could be observed for present participles in general, not only for those that co-occur with *werden*. This, however, is not the case.

While both of the aforementioned early theories posit one specific construction, i.e. *werden* + Present Participle, as predecessor of *werden* + Infinitive, more recent ones instead assume a polygenetic origin of the construction. According to Saltveit’s (1962) approach, dubbed “autonomy theory” (*Autonomietheorie*) by Westvik (2000: 236), *werden* + Infinitive and *werden* + Present Participle used to be two semantically and syntactically different structures, with *werden* + Infinitive originally having modal meaning. Hence, *werden* + Infinitive is not seen as a secondary development: instead, Saltveit assumes that *werden* + Infinitive already existed as an autonomous construction before it “merged” with *werden* + Present Participle (see Westvik 2000: 236).

A number of approaches assume a central role of analogy, which is why Westvik (2000: 237) summarizes them under the label of “analogy theory”. Wilmanns (1899) assumed that *werden* + Infinitive was formed in analogy to verbs like *beginnan* ‘begin’, while also assuming a major role of the phonological erosion of the participial ending leading to a loss of the distinction between participle and infinitive. Kurrelmeyer (1904: 9) sees *werden* + Infinitive as “arising from a number of influences”, including the German passive form (*werden* + past participle),

the use of *werden* with a predicative noun or adjective, and the construction *werden* + present participle. In a similar vein, Krämer (2005) assumes that *werden* + Infinitive emerged in analogy to *werden* + present participle and other inchoative constructions such as *beginnen* + infinitive. However, Fleischer & Schallert (2011: 140) point out that Krämer does not substantiate her hypothesis based on authentic data.

In a similar vein, Schmid's (2000) "contamination theory" assumes that constructions with modal verbs and infinitive has acted as analogical templates for *werden* + Infinitive. Diewald & Habermann (2005) also argue that analogical influence was key in the genesis of *werden* + Infinitive, but they assume that not modal verb constructions but rather verbs that are similar to *werden* in *meaning*, such as *beginnen* 'begin', have played the most crucial role in this process. In addition, they hypothesize that language contact with Latin contributed significantly to the prevalence of *werden* + Infinitive as future marker (Diewald & Habermann 2005: 241).

Work on *werden* + Infinitive (and its competitors) has often singled out the fact that the grammaticalization of 'become' as a future marker is typologically quite unusual (e.g. Heine 1995: 125) – even within the West Germanic languages, German, along with its close relatives Yiddish and Luxembourgish, stands out in its use of *werden* 'become' as a future marker (Abraham 1989: 364). Kuteva et al. (2019: 86) only mention Russian *budu* 'I will be' + Infinitive and Estonian *saama* + *ma*-infinitive as further examples, with the latter being a contact-induced variant imposed on Standard Estonian by German language reformers. However, *budu* + Infinitive also exists in Czech. While Rösler (1952) assumes that the Czech construction is a structural loan from German, Leiss (1985) hypothesizes that it might actually have been the other way around, and that *werden* + Infinitive emerged as a loan from Czech. This is called "interference theory" (*Interferenztheorie*) by Westvik (2000: 236), but as Westvik (2000) also mentions, this idea has met with significant skepticism; see e.g. Kotin (2003: 156–157) for a critical discussion of Leiss' line of reasoning.

An entirely new explanation for the emergence and expansion of *werden* + Infinitive has recently been proposed by Jäger (2024). She hypothesizes that the construction emerged through a reanalysis of forms in the copula construction *werden* + past participle that show a syncretism between infinitive and past participle, as in (3) (for a brief discussion of this scenario, also see Wiemer, this volume). In the original structure, *werden* serves as a copula and the past participle functions as an adjective-like predicate noun.

- (3) *huan-ēr uurði imu thiū tīd kuman, that ...*
when would him the time come that...
‘when for him the time would come that...’ (Heliand, quoted from Jäger 2024: 191)

She argues that this scenario offers an explanation for several of the questions that are still open, all of which arguably none of the aforementioned theories can account for: (i) Why did *werden* as a copula verb, normally combining with NPs and AdjPs, combine with infinitival forms? (ii) Why did its expansion happen in the 15th and 16th centuries? (iii) Why did the construction only emerge in High German?

Starting out with the first of these questions, she shows that there are ambiguous instances where the complement of *werden* can be interpreted either as a past participle or as an infinitive. These only occur with a fairly limited subset of verbs that fulfills certain formal (syncretism of infinitive and past participle) and semantic (the past participle in predicative or attributive usage must refer to the subject) requirements (see Jäger 2024 for a detailed elaboration of both requirements). Based on these contexts, (Jäger 2024: 195) proposes a grammaticalization scenario parallel to that of the perfect with *sein* ‘be’, developing from full verb to copula to auxiliary (with the additional step of a reanalysis Part. II > infinitive in the case of the *werden* future). As for the second question, Jäger (2024) argues that the possibility of aspect marking with the prefix *ge-* decreased in the 15th and 16th centuries. *ge-* is usually said to mark perfectivity/telicity (but see Fleissner 2023 for an alternative analysis). From Old High German onwards, *gi-/ge-* mostly occurred in past participles, but in Old and Middle High German, the prefix was usually omitted in the case of past participles of verbs that were already perfective, e.g. *finden* ‘found’ (past participle), *komen* ‘come’ (past participle). She argues that the reanalysis Past Participle > Infinitive should predominantly occur in the case of those verbs because there is no *gi-/ga-/ge-* prefix that would set the infinitive apart from the infinitive (although they are only fully homonymous in some cases, e.g. *komen*, but not in others, e.g. *finden* vs. *finden*). The results of her corpus study of Old High German and Old Saxon offer some support for her hypothesis. First of all, she argues that the extremely infrequent occurrence of *werden* + present participle indicates that it is unlikely for this structure to be the origin of, or have served as a model for, the future construction *werden* + Infinitive. (Jäger 2024: 14). The passive construction, *werden* + past participle of transitive verbs, is by far the most frequent construction with *werden*. While the construction is found to be ‘well established’ with verbs that meet the semantic criterion, and ‘quite remarkable’ with such that meet the formal criteria, the

combination of both features is found quite rarely: looking only at instances with present tense forms of *werden* (excluding past and infinitive forms), Jäger only finds nine instances. While this does not necessarily weaken her hypothesis, the validity of these findings may have to be taken with caution, as eight of the instances she finds come from the Old Saxon *Heliand*. Apart from the fact that the evidence is predominantly Old Saxon (only one is from Notker), this is a text that is difficult to classify in many respects (see e.g. Glauch 2013). To conclude this section, some of the suggested scenarios appear to be more plausible than others. Both the erosion theory (I) and the confusion theory (II) could not be confirmed by empirical data, as is the case with the interference theory (VI). Generally, the prevailing opinion is that the most plausible scenarios are those that assume a more complex interaction of multiple factors, i. e. the analogy theory (IV) and the contamination theory (V).

In sum, then, the question of how *werden* + Infinitive developed is still very much open, and while it might be impossible to give a definitive answer, we can at least check the plausibility of the scenarios by taking a closer (quantitative and qualitative) look at the existing data. As the present paper focuses on the competition between *werden* + Infinitive and *werden* + Participle, we cannot do justice to all hypotheses outlined above to the same degree, but we can critically examine some of the most influential hypotheses, which tend to assume that *werden* + Participle has played some role in the development of *werden* + Infinitive, either as its main template or as one constructional template among others.

2.1.2 Quantitative corpus-based studies

Recent work has shifted the focus to a more thoroughly empirical investigation of the development of *werden* + Infinitive. A main tenet of this line of research has been to test existing theories by assessing the verbs that the construction combines with. However, it should be stressed that even long before the advent of linguistic corpora in the modern sense, a number of researchers have conducted large-scale analyses on the basis of impressive amounts of data. For example, Kleiner's (1925) aforementioned work or Rick's (1905) study on *werden* + Present Participle present comprehensive qualitative and even quantitative analyses of fairly large datasets. The same goes for Kurrelmeyer (1904), who looked at the distribution of future constructions in more than 200 texts from the 12th to the 15th century. For the High German dialects, he analyzed 20,236 instances of future reference, counting instances of present tense, *sollen* + infinitive, *wollen* + infinitive, *werden* + infinitive and *werden* + present participle, as well as *werden* + adjective or noun and *werden* + perfect participle. He further differentiated his

data by grammatical person. In his data, present tense and modal verbs make up the majority of the futurate items. He observes an increase in *werden + Infinitive* and *werden + Participle* from the 14th century onwards, which is in line with the observations reported above. While Kurrelmeyer's study makes use of a remarkably comprehensive dataset, it has some major limitations, which is why Luther (2013: 16) points out that this study does not fully meet today's scientific standards (in particular, his database is not fully transparent).

More recent quantitative studies have typically used smaller datasets and/or focused on only one of the constructions taken into account in the present paper. Hilpert (2008: 147–153) uses diachronic distinctive collexeme analysis to assess the semantic development of the *werden + Infinitive* construction over time. Distinctive collexeme analysis is a method that is usually employed to compare two (or more) constructions with an open slot in terms of their slot fillers (Gries & Stefanowitsch 2004). Hilpert's adaptation of the method instead compares the same construction at different points in time (Hilpert 2006). Drawing on the Bonn Early New High German Corpus (FnhdC) and texts from Project Gutenberg, and slicing the data into four periods, he shows that the verbs attracted to the first period are invariably dynamic, while the later periods contain stative verbs like *fehlen* 'be missing'. In later periods, terms like *verstehen* 'understand' and *vergessen* 'forget' become more relevant, which he interprets as evidence "that speaker-oriented modal meanings are on the rise in this period of German." (Hilpert 2008: 152).

Hartmann (2021) and Hartmann & Schnee (2023) investigate the diachronic development of *werden + Infinitive* and *werden + Participle* in terms of *Aktionsart*. As a lexical-semantic category, *Aktionsart* refers, roughly speaking, to the way the action described by a verb unfolds (Kątny 2018). The importance of *Aktionsart* is twofold: On the one hand, it has been argued that present tense forms can be inherently future-oriented depending on the verb's *Aktionsart* (Saltveit 1962: 9), with telic verbs like *retten* 'to save' or continuative ones like *bleiben* 'to stay/remain' typically evoking a future-oriented reading and atelic ones like *haben* 'to have' or *sein* 'to be' a present-oriented one. This entails that over the course of grammaticalization, we would expect a future construction to seize the latter type of verbs first before spreading to verbs that tend to evoke a future reading when used in the present form. On the other hand, *Aktionsart* can also help characterize a construction's selectional preferences, which, from a usage-based Construction Grammar perspective, can give valuable clues to its constructional semantics (Hartmann & Schnee 2023: 265). Drawing on the well-known typology proposed by Vendler (1957), and on data covering the time from the 14th up to and including the 17th century, it can be shown that both *werden + Infinitive*

and *werden* + Participle show an increase in the proportion of telic verbs they combine with.

Another recent corpus study was conducted by Concu (2022) on the basis of the Reference Corpus of Middle High German (*Referenzkorpus Mittelhochdeutsch*, REM) and the Bonn Early New High German Corpus (*Bonner Frühneuhochdeutschkorpus*, FnhdC), both of which are also used in the present study. Like many other studies on *werden* + Infinitive (e.g. Kotin 2003), Concu (2022: 29) explicitly focuses on the development of the verb *werden*. The goal of her study was to refute the popular scenario of *werden* + Infinitive emerging from *werden* + Present Participle. Her analysis is based on subsets from both corpora that are roughly balanced regarding language region over time, and include different text types. Her data contain 110 instances of the two constructions (from 30 different texts) from the REM and 410 instances (from 10 different texts) from the FnhdC. She only finds the constructions in the later texts (from the 13th and the 14th century) of her REM subset, *werden* + Present Participle prevailing by far over *werden* + Infinitive. The FnhdC subset contains *werden* + Infinitive in all centuries, while *werden* + Present Participle is only attested marginally after the 14th century, which is in line with the frequencies found in the entire FnhdC (see Hartmann 2021; Hartmann & Schnee 2023, also see Section 4 below). Like Hartmann (2021), she observes a coexistence of the two constructions in Middle High German with a clear overlap in the types of verbs they combine with. She sees this as evidence that the relationship between the constructions is one of siblings – referencing Kotin’s (2003) notion of a ‘twin construction’ – rather than parent and child, which entails that *werden* + Infinitive did not originate from *werden* + Present Participle.

The FnhdC data leads her to another conclusion concerning the time when *werden* + Infinitive grammaticalized. While in the REM Data both constructions also occur with *werden* in the preterite tense, this usage is becoming more restricted in the course of the 14th century, by the end of which she does not find any more preterite occurrences of indicative *werden* + Infinitive. According to her, this can be described as a loss in autonomy of the verb *werden*, and as an indicator that the overall construction is being strengthened as a unit. In the 16th century she finds the first instance of the main verb *werden* being used in the *werden* + Infinitive construction (e.g. *Er wirt ein priester Gottes werden* ‘He will become a priest of God’, Samson, 16.2, REF), indicating the full grammaticalization of the construction around this time (see also Schmid 2000: 17).

Concu’s results show interesting differences to Hartmann (2021) and Hartmann & Schnee (2023). Most importantly, she only finds durative verbs in the *werden* + Present Participle construction. Also, she reports inchoative readings

only for *werden + Present Participle*. This, however, can probably be attributed to different operationalizations of the *Aktionsart* categories, which shows that when analyzing such notoriously difficult data, it is essential to make the annotation criteria as transparent as possible, and to ideally share the annotated data publicly. The diverging results also show that the jury is still out regarding the development of *werden + Infinitive* and its relation to *werden + Participle*, which leads us to a brief discussion of some questions that are currently still open.

2.2 Some open questions

Despite extensive research, a number of questions regarding German *werden + Infinitive* remain unresolved. There may not be any definitive answers for some of those open questions; however, to the extent that we can empirically address them, the newly available reference corpora of Middle High German and Early New High German offer promising ways of looking at those long-standing questions. In particular, the overarching question remains which of the various scenarios proposed in the literature is most compatible with the available data.

Of course, even the most exhaustive set of historical data will hardly be able to resolve the open issues once and for all. The different scenarios entail fairly similar predictions, which is why a detailed look at specific aspects of the diachronic development of the different patterns is necessary. For example, as argued in Hartmann (2021) and Hartmann & Schnee (2023), the aspectual semantics of the slot fillers can give valuable clues to the development of the different patterns. On a more general level, it can be informative to take an explorative look at the collocational preferences of the constructions in question: if, for example, different patterns show similar developments regarding their “collo-profiles” (in the sense of Herbst & Uhrig 2019, i.e. the behavior that a pattern shows with regard to its collocational behavior), we can reasonably expect that they influence each other; if, by contrast, their “collo-profiles” seem to drift apart over time, we can assume that they are undergoing a process of differentiation in the sense of de Smet et al. (2018). This is the main focus of the present paper.

Another question that has not been fully and satisfactorily answered yet pertains to the semantics of the two constructions, especially to the division of labor between inchoative and futurate readings and how this division of labor changes over time. For *werden + Participle*, it is a matter of debate to what extent it can encode futurity in the first place. Welke (2005: 385) argues that all instances of *werden + Participle* can be interpreted as progressive forms. Pfefferkorn (2005: 325–326) takes a less radical stance, arguing that *werden + Participle* can have

futurate meaning but that it can only express a change of state in the immediate future, and is thus very similar to *werden* + Adjective, keeping the ingressive/inchoative semantics of the verb *werden*. But even for *werden* + Infinitive, as already hinted at in Section 1, it is a matter of debate whether its semantics can be described in terms of ‘pure’ future marking, free from any modal meaning, as claimed by Dal (1962: 132), or, to the contrary, whether *werden* + Infinitive has to be regarded as a modal rather than a temporal construction (Vater 1975; also see Heine 1995: 134). There is a broad consensus now that the construction combines modal and temporal meaning elements (see Hilpert 2008: 154). Note that both *werden* + Infinitive and *werden* + Participle are attested with the auxiliary *werden* in the past tense in the Middle High German and Early New High German periods, which indicates that the constructions must still have been closely connected to functions other than future marking at that time. A closer analysis of the contexts in which instances of the constructions occur, as well as of their slot fillers, can potentially help determine the degree to which the constructions actually encode future meaning at different points in time.

While much previous research, with some notable exceptions like (Hilpert 2008), has mostly focused on the role of the auxiliary *werden* in the grammaticalization process in question, we argue that taking the dependent verbs into account is at least as important, and therefore focus on their distribution in our corpora as well as their semantic properties and on the question of how those aspects change over time.

3 Data and methods

3.1 Corpora: REM, REF, and FnhdC

The data we draw on come from the reference corpora of Middle High German (Referenzkorpus Mittelhochdeutsch, REM) and Early New High German (Referenzkorpus Frühneuhochdeutsch, REF). Both corpora have been compiled with the aim to give a representative sample of the text production of the respective periods, balancing the data for different text types and regional variants (see e.g. Dipper & Kwekkeboom 2018).

The Reference Corpus of Middle High German (1050–1350) (Klein et al. 2016) contains nearly 2.5 million tokens from a total of 395² different texts. The texts have been assembled with the aim of obtaining a balanced set regarding time,

²See our documentation in the supplementary material for an explanation of the difference in the number of texts compared to the official website of the REM.

region and text type/topic; according to one assessment, this goal was met at least regarding the parameters of time and region (see Kopf 2019: p.6). As for text types and topics, there is a skew towards religious texts, which clearly predominate in the corpus (62 %), followed with a considerable gap by literary (17 %) and law texts (12 %). Other text types are marginally represented, namely science-related texts (8.7 %) and textual witnesses of everyday life (0.04 %).³

The Reference Corpus of Early New High German (1350–1650) (Wegera et al. 2021) contains nearly 3.5 million tokens from a total of 190 different texts. The corpus is comparably balanced regarding the three structuring parameters. Overall, religious texts predominate, if only slightly, amounting to 32 % of the tokens. They are more closely followed by science related (21 %) and literary texts (20 %). Textual witnesses from everyday life make up 17 % in the REF, the least represented category is law texts (10 %). There is a slight imbalance regarding the distribution over time, as the texts from between 1450 and 1550 make up nearly half of the tokens in the entire corpus. All other time slots are represented with at least 10 %, however, except for the late 17th Century. The different language areas are well covered in the corpus, Westmitteldeutsch being featured a little more (37 % of tokens) than the other areas.

For reasons of comparability and clarity, we have adapted the metadata of the corpora in such a way that we can use the same set of metadata for both corpora.⁴ In particular, we used unified tagsets for time, language area, and topic; for the latter, we adapted the REF tagset to the one used in the REM corpus, following the assignment proposed by Kopf (2019: 7).

For some analyses that required complex semantic annotation, we used the much smaller Bonn Early New High German Corpus (FnhdC) instead of REF. Both corpora partly overlap: REF contains 24 of the 40 texts contained in the older FnhdC. Among the FnhdC texts not included in REF are those dating after 1650 as well as the *Oxforder Benediktinerregel* which is already part of the REM.

3.2 Data retrieval and annotation

For the present study, we used version 1.0 of both REM and REF⁵. The corpora were queried using the Corpus Workbench⁶.

³The text type grouping used here can be found in the metadata entries for “topic” in REM and “genre” in REF.

⁴See the supplementary material for a more detailed description.

⁵An updated version of REM, version 2.1, has recently become available, but could not be taken into account here.

⁶See <https://cwb.sourceforge.io/>; the scripts for converting the original corpus files to CWB’s VRT format are available on GitHub: <https://github.com/hartmast/rem2cwb> and

In the REM we simply searched for words lemmatized as *werden*, preceeded or followed by an infinitive or a present participle with a distance of up to 5 tokens. We then manually excluded those cases where the infinitive or participle are not a complement to the *werden*-instance.

Unfortunately, the same procedure proved unfit for the REF, as much of the the (mostly automatic) lemmatization turned out to be incorrect. In order to obtain our data we therefore searched for a total of 127 word forms of *werden*⁷, resulting in 36,467 matches that were all checked manually. A total of 14,479 true positives were identified, among which 2,863 have an infinitive and 72 a present participle complement.⁸ We manually added lemmatization for the complement relying on the *Frühneuhochdeutsches Wörterbuch* and, supplementary, the *Deutsches Wörterbuch von Jacob Grimm und Wilhelm Grimm*.

Crucially, when identifying the true positives, we relied on formal criteria, and we did not take into account the semantic criterion of whether or not a specific use of *werden* + Infinitive or *werden* + Participle can be interpreted as referring to a future event: as we are interested in the polyfunctionality of both constructions, we also wanted to take instances into account that are almost certainly not futurate but rather inchoative, like e.g. (4).

- (4) *Die kraft des füres dū gat in den stein .dc er reht glüende*
The power of-the fire REFL goes into the stone . that it right glowing
wirt
becomes
'The power of the fire goes into the stone so that it becomes properly
glowing' (NikP, 14.2, REM)⁹

Another reason for using formal rather than semantic criteria was that futurate and other, e.g. modal, readings often blur into one another, as discussed in Section 2, and are therefore almost impossible to tell apart. Similarly, in the case of *werden* + Participle, we are confronted with the problem that present

<https://github.com/hartmast/ref2cwb> (all links last checked 22/12/2024).

⁷We included the forms given by the *Mittelhochdeutsches Handwörterbuch* (<https://www.woerterbuchnetz.de/Lexer>, last checked 16/10/2023) as well as the variants found in the FnhdC. We did not limit the search with regard to tense and mode of the *werden* instance. The full list of variants can be found in the supplementary material.

⁸No less than 12.5 % of these instances would not have been found if we had relied on the automatic lemmatisation.

⁹When quoting examples from our dataset in this paper, we give a) a short version of the text's title, using the short sigle in the case of REM, b) the half-century from which the text stems, here: second half of the 14th century, and c) the corpus – REF, REM, or FnhdC.

participles are arguably categorically ambiguous between verbs and adjectives; they are usually used like adjectives (Eisenberg 2020: 209), and at last in present-day German, some are clearly lexicalized and detached from their verbal origins (e.g. *blendend* ‘marvelous, brilliant’ from *blenden* ‘glare’). In our sample, there are also some cases where the participles might be considered lexicalized, or at least strongly adjectival in nature, as in the use of *sehend* meaning ‘seeing’ as opposed to ‘blind’, as in (5).

- (5) *vff das die do nit sehen sehendt werden vnd die do sehent*
so-that those there not see seeing become and those there seeing
blind werden
blind become
‘so that those who do not see become seeing and those who see become blind’ (Martin Butzers Predig [sic], 16.1, REF)

Although, from a present-day perspective, the use in (5) may seem quite different from e.g. the one in (6), it is hard to tell to what degree the present participle *sehend* is already lexicalized in our data, which is why the true positives were selected on the basis of purely formal criteria.

- (6) *dz er siech wirt in rechter liebe vntz er dich sehende wirt*
that he sick becomes in true love until he you seeing becomes
‘that he will become sick in true love until he will see you’
(Passionstraktet [sic], 14.2, REF)

As mentioned in Section 3.1, we additionally used the Bonn Early New High German Corpus (FnhdC). As this is a manually annotated corpus, we could rely on its lemmatization as well as its morphological annotation for retrieving instances of *werden* followed or preceded by an infinitive or participle in a distance of up to five words.

The lemmatization as well as the morphological annotation of all true positives was manually corrected. In addition, the data from REM and FnhdC was annotated for markers of future reference in the immediate context, i.e. whether an overt future marker such as a temporal adverbial appears in the context. This can help clarify to what extent the constructions convey futurity on their own.

3.3 Method: Collostructional analysis

Apart from simple frequency counts, we use collostructional analysis (Stefanowitsch 2013) to explore the semantic profiles of the verbs that combine with *werden + Infinitive* and *werden + Participle*. Collostructional analysis is a family of

Table 2: Overview of the true positives across the three corpora, including hits with *werden* in the past tense (excluded from the further analysis, hence greyed out).

	REM		REF		FnhdC	
	present	past	present	past	present	past
<i>werden</i> + Infinitive	56	30	2863	260	515	206
<i>werden</i> + Participle	145	75	72	11	11	19

approaches that are used to assess the interaction between words and constructions using association measures. Specifically, we use two variants of collostructional analysis: simple collexeme analysis (Stefanowitsch & Gries 2003) on the one hand, and distinctive collexeme analysis (Stefanowitsch 2006) on the other. The logic of collostructional analysis is very simple: In the case of simple collexeme analysis, an association measure is computed over a 2x2 table containing a) the frequency of a given word in a given construction, b) the frequency of any other word in the given construction, c) the frequency of the given word in all other constructions, and d) the frequency of all other words in all other constructions (approximated in the present study using the total number of tokens in the corpus). For example, in REF, a) the verb *tun* ‘do’ occurs 59 times in *werden* + Infinitive; b) there are 2,312 instances of *werden* + Infinitive with other verbs than *tun*; c) there are 255 further instances of *tun* in REF outside of *werden* + Infinitive (total frequency in the corpus: 314); d) there are 3,418,972 tokens in the REF corpus, from which the values in a)-c) are subtracted, leaving 3,416,346 as a proxy for “all other words in all other constructions”. This yields a log-likelihood ratio of $G^2 = 556.71$. The log-likelihood ratios of the different slot fillers can be used to rank them in terms of their degree of “attraction” to or “repulsion” from the construction in question.

Distinctive collexeme analysis follows the same logic, but instead of comparing the frequencies of words within and outside of a specific construction, it compares the frequencies of slot fillers in two (usually near-synonymous or functionally equivalent) constructions. Importantly, this method only takes the frequencies from the constructions themselves into account, not their frequencies in the entire corpus. Again, the method takes four frequencies as input: a) the frequency of a given word Construction A, b) the frequency of the same word in Construction B, c) the frequency of all other words in construction A, d) the frequency of all other words in construction B.

In the present paper, we use the log-likelihood ratio G^2 as association measure. There are some debates about which association measure to use in what context (see e.g. Wiechmann 2008), and many researchers argue that ideally, different measures should be triangulated (e.g. Gries 2024). We will not delve into this discussion here, but instead settle for one widespread measure that we report in the present paper; the supplementary material additionally contains the unidirectional word-to-construction and construction-to-word delta p measures (a unidirectional measure of association) and can be used to compute further measures.

4 Results

In this section, we summarize the main results of our corpus study, starting with general observations on the frequency development of the constructions and their distribution across text types (Section 4.1), and then moving on to a more in-depth analysis of the verbs they combine with, using collostructional analysis 4.2.

In each of the following subsections, we report the results both for Middle High German and for Early New High German, drawing on REM and REF for all our analyses except for the data annotated for future time reference: Given that this kind of semantic annotation is very complex, we used the Bonn Early New High German corpus (FnhdC) instead of the REF as a much smaller, yet balanced dataset.

4.1 Frequencies and distribution

We will first take a closer look at the frequencies of the two constructions and at their distribution across time, text types, and language areas. A crucial advantage of the reference corpora is that they allow for assessing these distributional facts in a very simple way based on their metadata.

One important observation when taking a closer look at the distribution of the constructions in the REM is that both constructions – especially, however, *werden + Infinitive* – are quite rare. *werden + Participle* occurs in 49 out of 368 REM texts, *werden + Infinitive* even in just 32. Also, those texts come from a fairly limited set of text types. 90 % of all instances are from the domain of religious texts. The few we find in the other domains all date from 1250–1300 or later, except for four instances from the domain of scientific writing, *Wiener Physiologus* (*Jüngerer Physiologus*) and *Kaiserchronik A*. Only in ten texts in the REM do we find five or

more instances of *werden* with either an infinitive or a present participle. Also, we find more than one occurrence of *werden* + infinitive in only two texts, both of which are East Middle German gospel texts from the 14th century: The *Berliner Evangelistar* (mid-14th century) has 13 occurrences (as well as two occurrences of *werden* + Present Participle), the *Evangelienbuch des Matthias von Beheim* (early 14th century) has 6 occurrences and no *werden* + Present Participle.

We find a more balanced distribution in the REF data, where religious texts account for only about half of the items. Approximately 25% come from literary texts and 16% from scientific texts. Much fewer instances are found in law and everyday life texts. Also, the distribution across texts differs considerably from that in the REM data: More than 90% of the texts in REF contain instances of one of the two constructions, and we find more than ten items in more than half of the texts. Among the three texts that have more than a hundred instances are two bible texts, *Das Neue Testament* by Martin Luther and one bible translation by Johannes Eck, as well as one literary piece (drawing on biblical material) *Samson* by Mattheus Rotbletz. While the REM data contain more instances of *werden* + Participle than *werden* + Infinitive, the REF data show a clear skew towards the latter construction. 170 out of the 190 REF texts contain instances of *werden* + Infinitive, while *werden* + Participle is attested in only 31 texts.

To assess the distribution of *werden* + Infinitive and *werden* + Participle across texts quantitatively in more detail, we calculated the normalized Kullback-Leibler divergence as proposed by (Gries 2022) for each of the time slices covered by the corpora (for an extensive discussion of Kullback-Leibler divergence and its applications, see Gries 2024: 80–169).¹⁰ As pointed out by Sönning (forthcoming), the term *dispersion* can either refer to the pervasiveness of an item (in our case: a construction) in a corpus, or it can denote the evenness of a distribution, with modern dispersion measures such as the Kullback-Leibler divergence being mostly concerned with the latter aspect. The Kullback-Leibler divergence is a directional measure of how much a probability distribution, usually the observed one, diverges from another one, usually the expected one under the null hypothesis of equidispersion (Gries 2022: 5). D_{KLnorm} is a normalized version of the Kullback-Leibler divergence; its values fall between 0 and 1, with 1 representing a perfectly even distribution and 0 an uneven one. For calculating D_{KLnorm} based on our data, we used, on the one hand, the frequency of *werden* + Infinitive or *werden* + Participle in each text and, on the other hand, the overall number of tokens per text, which allowed us to assess the dispersion of the two constructions across texts. As expected, the normalized D_{KL} values of *werden* + Infinitive see

¹⁰For the calculation of the association measure, we used code from Sönning (2022).

a steady increase, indicating that we see a more even distribution across different corpus texts over time, while *werden + Participle*, after an increase in the REM data, becomes limited to only a few texts in the REF data, which is reflected in the declining D_{KL} values (Figure 1).

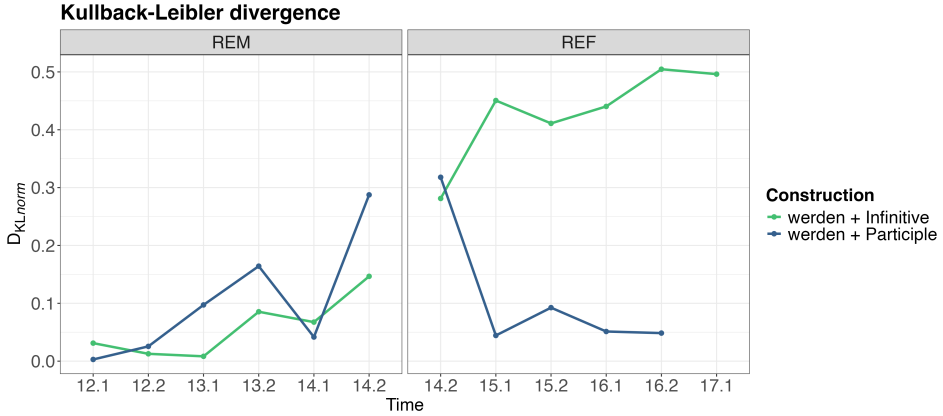


Figure 1: Dispersion of the instances of *werden + Infinitive* and *werden + Participle* as measured by the Kullback-Leibler divergence D_{KL}^{norm} .

Figure 2 shows the general frequency development of both constructions. While *werden + Participle* is quite frequent in the Middle High German period and even increases its frequency considerably over the time covered by the REM, it is far outnumbered by *werden + Infinitive* over the course of the Early New High German period.

However, the mere frequency of a construction does not tell us much about its productivity, i.e. the extent to which it extends to new cases. The type-token ratio of a construction can be informative in this regard, although it has to be taken with caution as it strongly depends on the sample size (see e.g. Baayen 2009). Figure 3 shows the type-token ratio of the two constructions in both corpora. As the sample size is very small for most periods, the datapoints are accompanied by information on the absolute frequencies of both constructions in each time slice. *werden + Infinitive* shows a fairly high type-token ratio from early on, indicating that it appears with a fair number of different verbs. *werden + Participle* has a comparatively low TTR in the 13th century, which indicates that there may be a few verbs that it preferentially occurs with, but this may be due to the composition of the corpus for these periods. The TTRs for the 14th century suggest that this construction, too, is quite promiscuous, while the data obtained for *werden + Participle* in REF from the 15th are not really revealing due to the extremely small number of attestations.

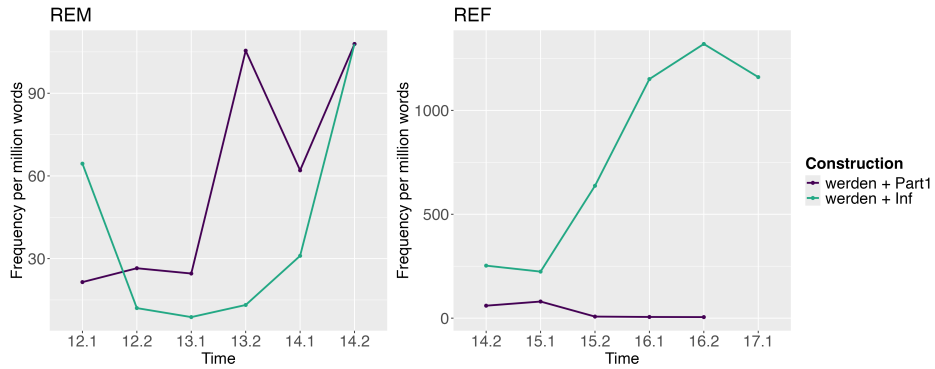


Figure 2: Frequency development of *werden* + Infinitive and *werden* + Present Participle in the Reference Corpus of Middle High German (REM, left) and the Reference Corpus of Early New High German (REF, right). Note the different scales of the y axis.

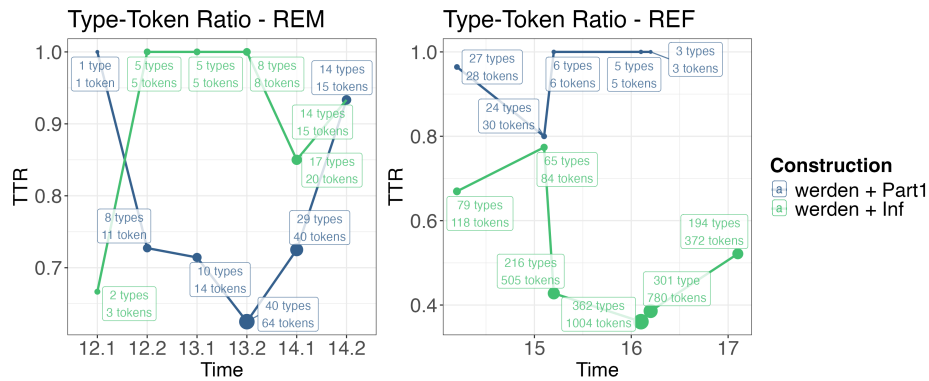


Figure 3: Type-token ratio in REM and REF

In the following section, we will take a closer look at the slot fillers of both constructions as well as their semantics.

4.2 Collostructional analysis

In this section, we summarize the results of the different collostructional analyses that we ran using the REM and REF data. For calculating the collostruction strengths, we used Flach’s (2021) package *collostructions* for R (R Core Team 2021).

Let us first take a look at the results of distinctive collexeme analysis. As discussed in Section 3.3, distinctive collexeme analysis is used to compare the frequencies of slot fillers across two constructions. In this case, the constructions to be compared are *werden + Infinitive* and *werden + Present Participle*. Table 3 shows the verbs that are significantly attracted to *werden + Infinitive* or *werden + Participle* in the REM data. The first column shows the verbs along with a rough translation, the second and third columns show the frequency of each verb in each of the two constructions. The fourth column indicates the direction of association, i.e. whether the verb is significantly attracted to *werden + Infinitive* or to *werden + Present Participle*. The fifth column shows the log-likelihood ratio G^2 , and the last column shows the significance (see the table caption for the p-value significance thresholds).

For all collostructional analyses, we used the lemma annotation available in REM, while in the case of REF, the data were lemmatized by ourselves. The lemmatization corresponds to the dictionary entries in the Middle and Early New High German dictionaries, respectively. Especially when it comes to variants that are ambiguous between a prefixed and a non-prefixed lemma, this annotation is necessarily speculative to some extent – for example, in the case of a participle form like *gehabt*, it is not necessarily clear whether the corresponding lemma is *haben* or *gehaben* (both mean ‘have’, with slight meaning nuances; as shown by e.g. Fleissner 2023, the exact function of the *ge-* prefix is subject to debate in the literature). Because of this, and even more so due to the small sample sizes especially in the case of the REM data, the results have to be taken with some caution.

Turning to the results in Table 3, only one verb emerges as a distinctive collexeme significantly attracted to *werden + Participle*, namely *ge-sēhen* ‘see’. As shown in (7), its use in *werden + Participle* usually refers to blind people regaining their eyesight (*sehend* ‘seeing’). Five verbs are significantly attracted to *werden + Infinitive*; given the overall low frequency of the *werden + Infinitive* construction in Middle High German, all of them are very infrequent, and

they mainly have in common that they do not occur in the *werden* + Participle construction (except for *entvâhen*, which occurs once in *werden* + Participle). The distinctive collexeme analysis on the basis of the REF data, the results of which are shown in Table 4, proves more informative thanks to the much larger sample size, despite the fact that the attestations *werden* + Participle are massively outnumbered by those of *werden* + Infinitive. Four items are identified as being significantly attracted to *werden* + Infinitive: *geben* ‘give’, *haben* ‘have’, *sein* ‘be’, and *werden* ‘become’. This fits in well with the results of the simple collexeme analysis discussed below, the results of which are reported in Table 7 and which shows that *werden* + Infinitive in Early New High German tends to combine with what, borrowing a term from first language acquisition, can be called general-purpose verbs (Clark 2009: 170; also see Goldberg 1995: 40), i.e. highly frequent and semantically “light” verbs. All remaining items in Table 4 are significantly attracted to *werden* + Participle – most of them, however, simply because they occur once in the construction and never or just once in *werden* + Infinitive. Among all the collexemes identified as significantly attracted to *werden* + Participle according to the distinctive collexeme analysis, only the verbum dicendum *sprechen* ‘speak’, which also emerges as a strongly attracted collexeme of *werden* + Participle in the simple collexeme analysis both for the REM and the REF data (Tables 6 and 8), occurs more than five times.

- (7) *sô wirdet ouch der denne dâ gesehente – der êr blint was* ‘so this there becomes seeing then – he who was blind’ (PrRoth R, 13.1, REM)

Table 3: Results of a distinctive collexeme analysis for *werden* + Participle and *werden* + Infinitive in the Reference Corpus of Middle High German. Only significantly attracted or repelled collexemes shown. ***** $p < .00001$, ****: $p < .0001$, *** $p < .001$, ** $p < .01$, * $p < .05$.

	Collexeme	Freq in werden+Inf	Freq in werden+Part	Direction	G ²	Significance
1	be-halten ‘keep’	4	0	Infinitive	10.44	**
2	ver-gêben ‘forgive’	3	0	Infinitive	7.79	**
3	ver-stôzen ‘outcast’	3	0	Infinitive	7.79	**
4	mēzzen ‘measure’	2	0	Infinitive	5.16	*
5	ent-vâhen ‘receive’	3	1	Infinitive	3.91	*
6	ge-sêhen ‘see’	0	10	Participle	6.73	**

Tables 5 and 6 show the results of simple collexeme analyses for *werden* + Infinitive and *werden* + Participle, respectively, in REM. Note that the frequency of

3 *werden* + Infinitive and *werden* + Present Participle

Table 4: Results of a distinctive collexeme analysis for *werden* + Participle and *werden* + Infinitive in the Reference Corpus of Early New High German. Only significantly attracted or repelled collexemes shown. *****: $p < .00001$, ****: $p < .0001$, *** $p < .001$, ** $p < .01$, * $p < .05$.

	Collexeme	Freq in <i>werden</i> +Inf	Freq in <i>werden</i> +Part1	Direction	G ²	Sign.
1	geben 'give'	111	0	werden + Inf	5.62	*
2	haben 'have'	101	0	werden + Inf	5.11	*
3	sein 'be'	274	2	werden + Inf	5.10	*
4	werden 'become'	100	0	werden + Inf	5.06	*
5	erquicken 'revive'	1	1	werden + Part1	4.71	*
6	gesehen 'see'	1	1	werden + Part1	4.71	*
7	irren 'err'	1	1	werden + Part1	4.71	*
8	liegen 'lie'	1	1	werden + Part1	4.71	*
9	scheinen 'shine'	1	1	werden + Part1	4.71	*
10	schwitzen 'sweat'	1	1	werden + Part1	4.71	*
11	schwören 'swear'	1	1	werden + Part1	4.71	*
12	spotten 'mock'	1	1	werden + Part1	4.71	*
13	wandeln 'stroll'	1	1	werden + Part1	4.71	*
14	zittern 'tremble'	1	1	werden + Part1	4.71	*
15	verdammen 'condemn'	5	2	werden + Part1	6.75	**
16	diessen 'sound, chime'	0	1	werden + Part1	7.43	**
17	einleuchten 'be(come) clear'	0	1	werden + Part1	7.43	**
18	fingerzeigen 'point-fingers'	0	1	werden + Part1	7.43	**
19	gagen 'rock back and forth'	0	1	werden + Part1	7.43	**
20	karen 'wheeze'	0	1	werden + Part1	7.43	**
21	kräftigen 'strengthen'	0	1	werden + Part1	7.43	**
22	nachgehen 'follow'	0	1	werden + Part1	7.43	**
23	rasen 'rage'	0	1	werden + Part1	7.43	**
24	tönen 'sound'	0	1	werden + Part1	7.43	**
25	verneuen 'rejuvenate'	0	1	werden + Part1	7.43	**
26	wallen, sieden 'simmer'	0	1	werden + Part1	7.43	**
27	zemen 'tame'	0	1	werden + Part1	7.43	**
28	sitzen 'sit'	4	2	werden + Part1	7.44	**
29	scheiden 'part'	2	3	werden + Part1	15.74	****
30	sprechen 'speak'	10	7	werden + Part1	30.01	*****

both constructions, and especially of *werden* + Infinitive, is very low in the Middle High German period, which is why the results have to be taken with caution and are arguably not particularly meaningful. What is striking, however, is the fairly equal distribution of the slot fillers: Most instances of *werden* + Participle and *werden* + Infinitive in our data are hapax legomena, i.e. attested only once. In

werden + Participle, *sēhen* and *gesēhen* (both ‘see’) are attested more than 10 times each, stemming from six different texts in the case of *sēhen* and from seven different texts in the case of *ge-sēhen*, but in most cases referring to regaining eyesight; notable exceptions include the attestation in (8).

- (8) *wā da w^sdēt fi mich fehende*
 because there become they me seeing
 ‘because they will see me there’ (PrSch, 13.2, REM)

Table 5: Results of a simple collexeme analysis of the REM data for
werden + Infinitive

	Collexeme	Cxn freq.	Corpus freq.	G ²	Signif.
1	ver-stôzen ‘outcast’	3	102	37.25	*****
2	be-halten ‘keep’	4	1022	33.55	*****
3	ver-gēben ‘forgive’	3	330	30.16	*****
4	mēzzen ‘measure’	2	75	24.40	*****
5	ge-dēnkenwider(e) ‘recall’ +	1	1	21.40	*****
6	ent-vāhen ‘receive’	3	1774	20.11	*****
7	ver-hol(e)n ‘acquire’	1	11	14.70	***
8	be-schatwen ‘shadow’	1	20	13.46	***
9	en-blanten	1	23	13.18	***
10	setzen ‘put/sit’	2	1286	13.04	***
11	tragen ‘carry’	2	1619	12.13	***
12	er-manen ‘urge’	1	55	11.41	***
13	hunger(e)n ‘hunger’	1	61	11.20	***
14	sunter(e)n ‘detach’	1	107	10.07	**
15	ge-līchen ‘equal’	1	144	9.48	**
16	lachen ‘laugh’	1	152	9.37	**
17	er-tēilen ‘judge’	1	156	9.32	**
18	ver-rāten ‘betray’	1	167	9.18	**
19	rüeren ‘move’	1	186	8.97	**
20	hazzen ‘hate’	1	205	8.77	**

The results of the simple collexeme analysis for *werden* + Infinitive in REF (Table 7) show that the construction tends to combine with general-purpose verbs such as *tun* ‘do’ or *sein* ‘be’. *geben* ‘give’, which occupies the second rank, can be used in an existential sense in German (*es wird Kuchen geben* ‘there will be

Table 6: Results of a simple collexeme analysis of the REM data for *werden + Participle*

	Collexeme	Construction freq.	Corpus freq.	G ²	Signif.
1	sēhen ‘see’	14	4058	87.89	*****
2	ge-sēhen ‘see’	10	1077	82.15	*****
3	nièzen ‘use’	4	179	39.74	*****
4	troumen ‘dream’	3	35	38.04	*****
5	wissagen ‘predict’	3	70	33.75	*****
6	bētel(e)n ‘beg’	2	9	29.46	*****
7	wider-gēben ‘return’	2	16	26.94	*****
8	hērschen ‘reign’	2	24	25.23	*****
9	vēhten ‘fight’	3	375	23.61	*****
10	wēinen ‘cry’	3	604	20.76	*****
11	rīches(en)en ‘reign’	2	76	20.50	*****
12	sēhen-ane ‘look at’	2	85	20.05	*****
13	stân-ûf ‘stand up/rise’	1	1	19.49	****
14	vâhen ‘catch’	3	853	18.72	****
15	lēben ‘live’	4	2939	17.59	****
16	vliēgen ‘fly’	2	175	17.15	****
17	toèten ‘kill’	2	195	16.72	****
18	sprēchen ‘speak’	6	11305	15.93	****
19	brinnen ‘burn’	2	239	15.90	****
20	be-schrēnken ‘compass’	1	3	15.67	****

cake’), but in the REF data, it is almost exclusively used in the literal sense, usually in religious contexts, e.g. *der heer wiert geben die seniftguettichait* ‘the Lord will give the gentleness’ (Rationale divinorum officiorum, 14.2, REF).

The fact that *geben* ‘give’ is among the top collexemes of *werden + Infinitive* coincides with Hilpert’s (2008: 142) results. However, in Hilpert’s data, most instances turn out to be cases of existential *geben* as in *Es wird keine Beatles-Tour mehr geben* ‘There will be no further tour of the Beatles’. In our data, by contrast, most attestations of *geben* are literal uses with a transfer meaning, as in (9). This is not surprising as the existential *es gibt* construction only emerged in the 16th century (Newman 1997, Lenz 2007). The fact that *geben* remains an attracted collexeme, albeit with a different predominant meaning, might be an

Table 7: Results of a simple collexeme analysis of the REF data for *werden* + Infinitive

	Collexeme	Cxn freq.	Corpus freq.	G ²	Signif.
1	tun ‘do’	59	314	556.71	*****
2	geben ‘give’	111	7504	474.06	*****
3	sehen ‘see’	82	4476	383.21	*****
4	sein ‘be’	274	117659	295.99	*****
5	hören ‘hear’	51	1995	270.99	*****
6	kommen ‘come’	79	8662	264.00	*****
7	finden ‘find’	46	2918	200.92	*****
8	werden ‘become’	100	23890	195.97	*****
9	sagen ‘say’	57	6545	185.14	*****
10	gehen ‘go’	53	5452	182.91	*****
11	bleiben ‘stay’	36	1971	167.37	*****
12	auferstehen ‘resurrect’	14	31	161.09	*****
13	vernemen ‘hear’	11	11	160.07	*****
14	haben ‘have’	101	38272	123.58	*****
15	vergeltten ‘retaliate’	10	36	103.01	*****
16	überantworten ‘sign over’	8	12	101.14	*****
17	erlangen ‘obtain’	14	257	95.38	*****
18	ergehen ‘fare’	13	211	91.82	*****
19	zergehen ‘dissolve’	6	6	87.30	*****
20	machen ‘make’	33	5774	81.72	*****

indication that *werden* + Infinitive has developed a preference for specific lexical items (as argued by Smirnova, this volume), regardless of their specific reading variants. Future work using data from the initial stages of the New High German period (1650 onwards) that is not covered in the present paper could address this hypothesis in more detail by investigating the distribution of different readings of *geben* within and outside of the *werden* + Infinitive construction.

- (9) Dye werden dir noch dreyssig phening geben ‘They will give you 30 more *pfennig* [currency]’ (Passion Sterzing, 15.2, REF)

It can also be instructive to take a closer look at the degree to which the slot filler types of both constructions overlap. A low degree of overlap could be a sign that the constructions are also functionally quite different, while a high degree of

Table 8: Results of a simple collexeme analysis of the REF data for *werden + Participle*

	Collexeme	Cxn freq.	Corpus freq.	G ²	Signif.
1	sprechen ‘speak’	7	8741	39.91	*****
2	scheiden ‘part’	3	542	28.41	*****
3	verdammen ‘condemn’	2	149	22.46	*****
4	rasen ‘rage’	1	1	21.82	*****
5	einleuchten ‘be(come) clear’	1	2	19.05	****
6	finden ‘find’	3	2918	18.38	****
7	fingerzeigen ‘hint’	1	3	18.00	****
8	kräftigen ‘strengthen’	1	8	15.79	****
9	sitzen ‘sit’	2	1211	14.09	***
10	verneuen ‘rejuvenate’	1	24	13.51	***
11	zittern ‘tremble’	1	30	13.05	***
12	erquickern ‘revive’	1	38	12.57	***
13	nachgehen ‘follow’	1	40	12.47	***
14	stinken ‘stink’	1	41	12.42	***
15	schwitzen ‘sweat’	1	49	12.06	***
16	spotten ‘mock’	1	67	11.43	***
17	irren ‘err’	1	150	9.81	**
18	reichen ‘hand’	1	165	9.62	**
19	wandeln ‘stroll’	1	178	9.47	**
20	aufheben ‘repeal/pick up’	1	214	9.10	**

overlap might indicate functional similarity (although this is not necessarily the case, as even functionally very different constructions might combine with the same slot fillers). As Figure 4 shows, both constructions overlap to some extent in the REM data, but there are also many slot filler types that seem to be unique to each of the two constructions. In REF, there are many verbs that only occur in the much more frequent *werden + Infinitive* construction, but only about 20% of all items attested in *werden + Participle* are unique to the latter construction. Table 9 shows the lemmas that are attested in both constructions in the REM along with their overall corpus frequencies.

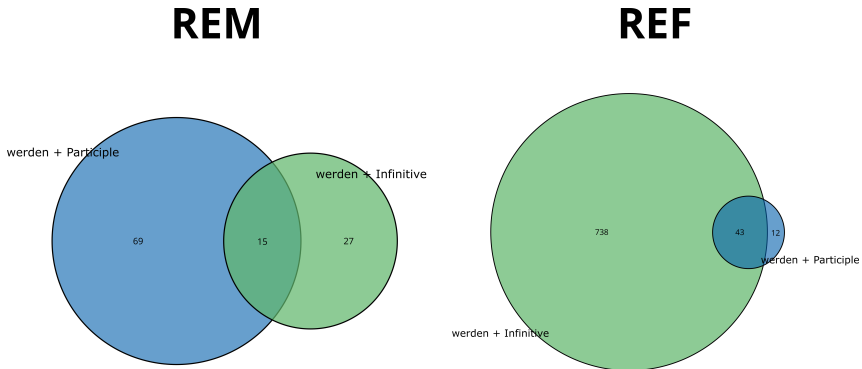


Figure 4: Overlap between *werden* + Infinitive and *werden* + Participle slot filler types in REM and REF.

Table 9: Lemmas attested in both constructions in REM

	Lemma	Corpus freq.	<i>werden</i> + Inf.	<i>werden</i> + Part1
1	sprèchen ‘speak/talk’	11305	1	6
2	gèben ‘give’	6432	2	3
3	sèhen ‘see’	4058	2	14
4	gân ‘go’	3808	1	1
5	var(e)n ‘go/drive’	2121	1	1
6	hoèren ‘hear/listen’	1850	1	1
7	ent-vâhen ‘receive/take’	1774	3	1
8	tragen ‘carry’	1619	2	1
9	sètzen ‘put/sit’	1286	2	1
10	suochen ‘search/seek’	895	1	1
11	be-kènnen ‘confess’	558	1	2
12	trinken ‘drink’	472	1	1
13	schinen ‘shine’	433	1	1
14	hazzen ‘hate’	205	1	1
15	lachen ‘laugh’	152	1	1

4.3 Future reference

As mentioned in Section 3, we annotated the data from REM and REF for the occurrence of overt markers of future reference in the immediate context of the instances of the *werden* + Infinitive and *werden* + Participle constructions in our

data. As Figure 5 shows, the vast majority of all instances of the two constructions come without an overt marker of future reference. Interestingly, however, we see a comparatively high proportion of overt markers of future reference with *werden* + Participle in the later Middle High German data, which might indicate that the construction, which is typically inchoative, is used in a futurate sense to an increasing extent, before *werden* + Infinitive takes over; in the entire FnhdC data, we find only one single attestation of *werden* + Participle with an overt marker of future reference. Of course, this does not mean that instances without an overt temporal marker cannot have a futurate reading as well, but most of the few attestations of *werden* + Participle in the FnhdC corpus are clearly inchoative, as (10c) and (10d) show; only very early attestations like (10a) and (10b) can be considered more futurate, although they also have a fairly clear inchoative meaning component.

- (10) a. we euch die ir nu lachet wanne ir wert her nach weinent ‘Woe to you who are laughing now because you will be crying afterwards’ (Altdeutsche Predigten I, 14th century, FnhdC)
- b. vntz der antichrift chvmt . Mit dem w^sde wir f^rreitend . [...] d^s wirt vns dāne töttend ‘... until the Antichrist comes. With him we will fight [...]. He will kill us then’ (KlostEvang, 14.1, REM)
- c. Ein stockblindes Kind wird vollkommenlich sehend ‘A completely blind child becomes fully sighted’ (Augsburgischer Arzt, Augsburg 1678)
- d. Ein junge Tochter so von Kindsweesen Stumm war wird redend ‘A young daughter who was mute from childhood becomes talking’ (Augsburgischer Arzt, 1678, FnhdC)

Overall, this is consistent with the idea that *werden* + Participle came to be used in futurate contexts more often and may have been on its way to becoming a future marker, although its function was never limited to marking future time reference. It also fits in well with the idea proposed in Hartmann & Schnee (2023: 274) that there may be a “co-evolution” of *werden* + Participle and *werden* + Infinitive in the sense of Zehentner (2019: 296), who argues that competing constructions can develop a “symbiotic” relationship, noting that in biology, the label *symbiosis* refers to any kind of interaction between organisms or species that can be antagonistic, parasitic, predatory or mutualistic. Our results provide tentative evidence that the two constructions undergo attraction and differentiation processes in the sense of de Smet et al. (2018), with the two patterns first becoming more similar and then coming to occupy different functional niches.

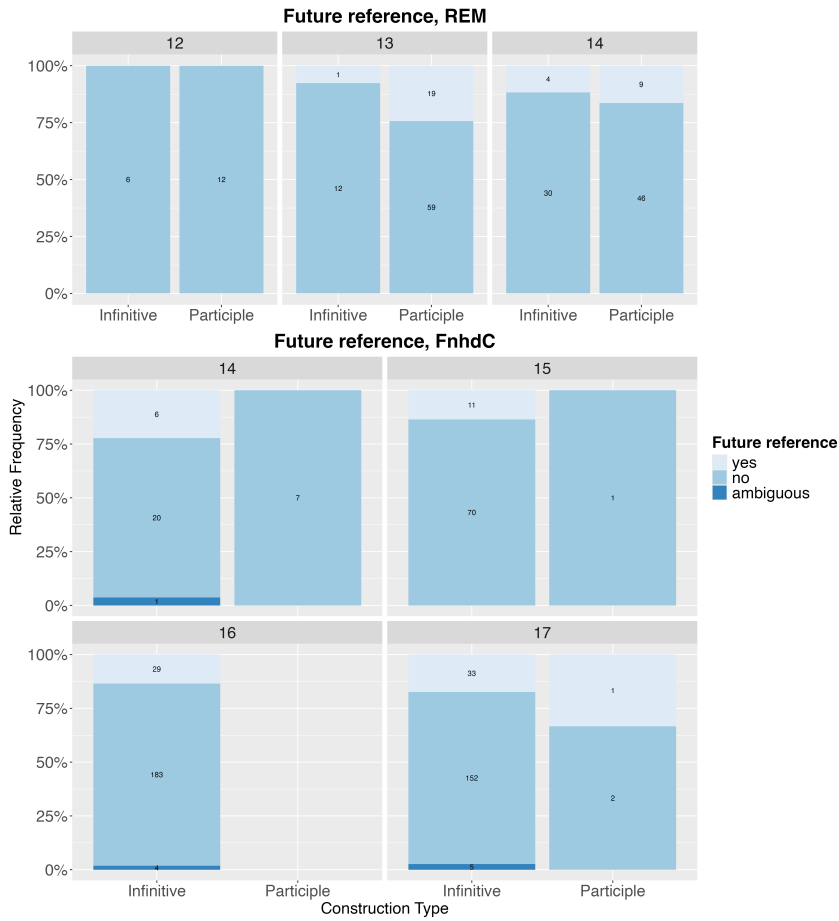


Figure 5: Overt markers of future reference in the immediate context of *werden* + Infinitive and *werden* + Participle, respectively.

5 Discussion and conclusion

A main goal of the present paper was to demonstrate that analyzing data from the newly available reference corpora of Middle High German and Early New High German can shed new light on the development of the German future construction *werden + Infinitive* in relation to its competitors. For the present study, we have focused on one particular competitor, namely *werden + Present Participle*. Following up on previous corpus-based studies that had focused on the role of aspectual patterns in the use of those two constructions, the present paper has taken a broader approach, targeting three aspects: firstly, the temporal and areal frequency distribution of the two constructions in the pertinent reference corpora of Middle and Early New High German; secondly, the collostructional preferences of the two constructions; and thirdly, the correlation of the two constructions with other markers of future time reference. As for the first aspect, the data from the reference corpora confirm the general tendencies observed in previous studies: *werden + Infinitives* experiences a slow but fairly steady growth (and spread) from the Middle High German to the Early New High German period. As for the second aspect, the results are quite mixed, especially due to the sparsity of the REM data for *werden + Infinitive* and the sparsity of the REF data for *werden + Participle*. Despite these limitations, the data are, overall, in line with the results of previous research, and additionally, they indicate that the two constructions share a fairly large number of collocates in the REM data, while this shared inventory becomes much smaller in the case of the REF data. To some extent, this is of course due to the fact that by that time, *werden + Participle* is vastly outnumbered by *werden + Infinitive*, but it is also in line with the hypothesis advanced in previous work (e.g. Hartmann 2021) that *werden + Infinitive* and *werden + Participle* first become more similar (“attraction” in the sense of de Smet et al. 2018) before starting to occupy different niches (what de Smet et al. 2018 call “differentiation”). This is also in line with our finding that *werden + Participle* is increasingly found in futurate contexts in the REM corpus.

Note that Smirnova (this volume), who investigates the development of *werden + Infinitive* in the time periods directly following the time span covered in the present paper, adds important observations that are directly relevant to those three aspects, namely that on its way to present-day German, *werden + Infinitive* seems to have undergone a clear specialization: *werden + Infinitive* has become less frequent in most future-related contexts compared to the futurate present, and the verb it combines most with, according to Smirnova’s data, is the copula *sein* ‘to be’.

As shown in Section 2.1, there are numerous hypotheses regarding the development of *werden* + Infinitive, and arguably the most convincing ones assume that a complex interaction of different factors gave rise to this construction. As such, our analysis of *werden* + Infinitive and one competing pattern, *werden* + Participle, can only address hypotheses related to a small part of the bigger picture of this development. Indeed, our results provide further evidence against a scenario in which *werden* + Participle is the main or even the only source for *werden* + Infinitive, regardless of whether one assumes a scenario of phonological erosion or an account that explains a purported extension from *werden* + Participle to *werden* + Infinitive via analogy. However, our results do not allow us to make a decision between the polygenetic accounts mentioned in Section 2.1.1 and listed in Table 1, as they are compatible with multiple conceivable scenarios. They do, however, provide tentative evidence that the two constructions may have undergone processes of attraction and differentiation. If this is the case, then the scenario would be compatible with different elements of different theories: The idea of an analogy between the two constructions is at the heart of both confusion and autonomy theory. However, while the former assumes that *werden* + Infinitive emerged from *werden* + Participle, the idea of the two constructions becoming more similar in the first stage of the process aligns more with autonomy theory, which assumes that the two constructions existed independently before they underwent partial assimilation.

Future research should take other competing constructions, e.g. modal constructions, but also the futurate present, into account more thoroughly. More specifically, future studies should also take an onomasiological perspective to explore the different possibilities that are taken in different contexts to encode future reference. Still, we hope to have shown with the present paper that data from the new reference corpora can help us better understand the competition between different ways of future encoding in historical German.

At the same time, our study also points to some limitations of working with data from those reference corpora. Among other things, despite the corpus compilers' efforts to make the corpora as balanced as possible, the reference corpora of Middle High German and Early New High German are not really balanced for factors like language areas or text types and, presumably most relevant for this kind of study, topics. As such, it is not always straightforward to compare the results between the two corpora, or even between different time slices within one corpus. Highly homogeneous corpora like Schmid's (2000) "Seven Signs corpus" exclusively comprised of a relatively small set of apocalyptic texts have a major advantage here. This does not mean, however, that the reference corpora are not well-suited for analyzing different ways of future reference in the history of

German; however, the aforementioned limitations should be kept in mind, and as virtually always, evidence from different sources, including smaller and more balanced corpora, should be triangulated to arrive at a more nuanced picture of the development of *werden* + Infinitive and its competitors.

Data availability

The data and scripts used for the analysis reported on in the present paper can be found on the Open Science Framework, <https://doi.org/10.17605/OSF.IO/4R9M7>.

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Chapter 4

German *werden*-future: A case of failed grammaticalization?

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In present-day German, the construction *werden* + infinitive is usually considered as an analytic future tense form, and the diachronic development of this construction is described as a typical case of grammaticalization. However, the synchronic status of this construction is still a matter of discussion. On the one hand, the function of this construction is not restricted to expressing futurity, but encompasses modal and evidential functions. On the other hand, when it comes to marking futurity, *werden* + infinitive is not obligatory. Although the relatively high grammaticalization status of the construction seems uncontested, especially when compared with the less grammaticalized copula verb *werden* from which it originated, this paper will argue that the grammaticalization process has not come to full completion and was halted at some point in time. What happened then is that the use of the futurate *werden* + infinitive was gradually restricted to a rather small range of particular contexts of use, i.e. a functional niche, where it unambiguously serves as a future tense marker.

1 Introduction

It is rather uncontroversial that present-day German does not have a default future construction in the sense that (i) there exists a verbal periphrasis with a dedicated future auxiliary and (ii) this periphrasis is highly grammaticalized and (iii) it is obligatory in all contexts with future time reference. Though most grammar books of present-day German present *werden* + infinitive as a future tense construction (see 1) along with other analytic tenses such as perfect or pluperfect, this construction in fact only marginally functions as an analytical future tense.



- (1) *Er wird im Urlaub sein.*
he werden.3SG.PRES on vacation be.INF
'He WIRD be on vacation.'
a. He will be on vacation. = future time reference
b. He might be on vacation. = epistemic assessment
c. He is bound to be on vacation. = evidential assessment

Without going into much detail on the longstanding debate regarding semantic and functional characteristics of *werden* + infinitive in present-day German, it is fair to say that it is not an easy task to determine the function of this construction. In the literature, there are three main accounts that may be briefly summarized as follows:

One branch of researchers, the so-called "temporalists" (see e.g. Matzel & Ulvestad 1982; Thieroff 1992) consider *werden* to be primarily a future tense auxiliary (see the paraphrase in 1a) and derive other readings from this future semantics.

The so-called "modalists" (see esp. Vater 1975) on the other hand, treat *werden* essentially as a modal verb, namely as an epistemic modal verb, see the interpretation given in (1b); and other readings are then derived from the modal semantics of this verb.

A third option is to see *werden* as primarily evidential in meaning (see esp. Diewald & Smirnova 2010) namely an inferential evidential verb, as in (1c).

The present paper does not attempt to answer the question of what counts as the central or core meaning of the construction *werden* + infinitive. Instead, it focuses on the temporal reading of this construction and argues that this reading is restricted to a very marginal set of contexts which constitute a functional niche of this construction. The goal of this paper is first, to show that the grammaticalization of *werden* + infinitive has not been proceeding further since late Early New High German (ENHG), and second, to explore the functional niche it occupies in present-day German.

The paper is structured as follows. In Section 2, I will briefly recall some known facts about the diachrony of several future tense candidates in the history of German and motivate my research questions. In Section 3, I will report the results of a diachronic corpus study that concentrates on some particular contexts of use, namely subordinate temporal and conditional clauses. It is in these contexts that the hampered grammaticalization of the future marking manifests itself prominently. In Section 4, I will focus on synchronic corpus data in zooming in on another context of use, namely the combination of *werden* and the infinitive verb

sein ‘to be’, in order to pinpoint the characteristics of the functional niche the construction occupies in present-day German. The paper will conclude with some thoughts on the reasons why *werden* has not reached an advanced degree of grammaticalization in German.

2 Future constructions in the diachrony of German

In present-day German, *werden* + infinitive is used to mark posteriority, that is, to locate the described event in the future relative to the speech time, as for example in (2).

- (2) *Bereits am 1. April wird das Strandbad Müggelsee seine*
 already on 1st April *werden*.3SG.PRES the Lido Müggelsee its
Pforten öffnen.
 gate.PL open.INF
 ‘The Müggelsee Lido will open as early as April 1st.’
 [Berliner Tageblatt 1931, DWDS-Kernkorpus]

From the diachronic perspective, it is well known that *werden* + infinitive was not the only construction on the way to becoming a future marker in the history of German. Prior and partly in parallel to *werden* + infinitive, especially in Middle High German (MHG), the combination of *werden* and present participle was used to mark future time reference (cf. 3; see Hartmann 2021; Ihden 2023).

- (3) *Salic fit ir die nu weinint. wan ir*
 blessed be.2.PL.PRES you who now weep.2PL.PRES. when you
werdint lachinde.
werden.2PL.PRES laugh.PRES.PTCPL.PL
 ‘Blessed are you who weep now, for you shall laugh.’
 [REM¹, 12th c., from Hartmann (2021: 367)]

As regards *werden* + present participle, it became obsolete in the course of the ENHG period, as has been clearly shown by Stefan Hartmann in his diachronic corpus study, cf.:

Während im REM *werden* + Partizip die deutlich frequentere Konstruktion ist und beide Konstruktionen vom 12. zum 13. Jahrhundert einen Anstieg

¹REM stands for *Referenzkorpus Mittelhochdeutsch*, a corpus of Middle High German data (see <https://www.linguistics.rub.de/rem/> for more details).

erfahren, sinkt die Gebrauchsfrequenz der *werden* + Partizip-Konstruktion vom 13. zum 14. Jahrhundert deutlich. [...] *werden* + Infinitiv erlebt einen geradezu explosionsartigen Anstieg, während *werden* + Partizip im eher niedrigen Frequenzbereich verharret. (Hartmann 2021: 377)²

But also the modal verbs such as *sollen* ‘shall’, *wollen* ‘want, will’, and *müssen* ‘must’ were all used with future time reference in the history of German (cf. 4; see Diewald & Habermann 2005 for further details).

- (4) *Ach Gott hilff* / *Ach Got hoere* / *Ach Got* / *wenn*
 oh God help.IMP.SG / oh God hear.IMP.SG / oh God / when
will *es ein end werden?*
 will.3SG.PRES it a end werden.INF
 ‘Oh God help / Oh God hear / Oh God / when will it end?’
 [Dietrich, 16th c., Diewald & Habermann (2005: 232)]

As for the modal verbs, they too did not develop future marking as their central semantics, and did not proceed further on the grammaticalization path towards future constructions in German (cf. Diewald & Habermann 2005: 232): “Die Entwicklung von *werden* zum Futurmarker hat ganz offensichtlich den Grammatikalisierungsprozess der Modalverben zu Futuren gebremst”.³

Werden + infinitive is hence a clear winner in this diachronic competition with modal verbs and with the construction *werden* + present participle, as only this construction successfully developed into a future tense marker. However, the story of *werden* + infinitive appears to be less successful as it might seem at first glance.

For present-day German, it has been repeatedly observed that this construction is generally very infrequent. To refer to future time, German speakers clearly prefer present tense forms (cf. Brons-Albert 1982; Hilpert 2008). For example, Brons-Albert (1982) found that only 30% of all futurate contexts contain *werden* + infinitive. The unmarked expression of future time today is not *werden* + infinitive, but the (furate) present tense.

²Translation: While *werden* + participle is the considerably more frequent construction, and both constructions experience an increase from the 12th to the 13th century, the usage frequency of the *werden* + participle construction decreases considerably from the 13th to the 14th century. [...] *werden* + Infinitive experiences an enormous increase, while *werden* + participle remains in a rather low frequency area.

³Translation: The development of *werden* to a future marker has quite obviously curbed the grammaticalization process of modal verbs to futures.

Thus, it seems that no significant semantic and formal changes have occurred since *werden* + infinitive prevailed over the other constructions in the late Middle High German period, and, most importantly, the frequency of use has remained very low or even decreased since then. In the remainder of this paper, I will argue that this construction has not progressed further on the way towards becoming a highly grammaticalized construction. Instead, it has even regressed, and is now limited to some specialized niches.

In the following, I will focus on two local contexts of use of this construction, which I will examine further in order to substantiate my claim:

1. From the diachronic perspective, I will look at conditional and temporal subordinate clauses, i.e. the contexts in which *werden* + infinitive can be unambiguously interpreted as conveying future time reference. Starting from the assumption that the appearance of *werden* + infinitive in conditional clauses represents an advanced step in the grammaticalization process towards a future tense marker, a detailed exploration into these contexts will give us valuable insights into the diachronic development of this construction.
2. From the synchronic perspective, the contexts with the infinitive verb *sein* 'be' will be dealt with in a more detailed quantitative and qualitative analysis of corpus data from present-day German. As previously shown in studies by Hilpert (2008) and Hartmann (2021), and as will be confirmed by my own diachronic analysis in Section 3, this verb is the most frequent infinitive verb that combines with *werden*. Given the fact that *sein* itself is a highly polyfunctional verb in present-day German, the mere co-occurrence of the two does not provide us with any substantial information about the functional and/or semantic range of *werden* + *sein*. The in-depth analysis of these contexts presented in Section 4 will identify the relevant characteristics of the semantic niche *werden* + infinitive occupies in present-day German.

3 *Werden* + infinitive in conditional and temporal clauses

In a comprehensive diachronic corpus study of *würde* + infinitive in MHG and ENHG (Smirnova 2006), I also looked at the indicative form of this construction, i.e. *werden* + infinitive. One of the central insights of this study was that, in its diachronic development, *werden* + infinitive passed through an intermediate stage

of semantic change where it clearly was used to mark a textually salient sequential relation. *Werden* + infinitive acquired this semantics in the MHG period via the reinterpretation of the original *aktionsart* ingressive meaning (see Smirnova 2006: 259–267 for further details).

In this respect, the development of *werden* + infinitive towards a future tense construction follows a rather unusual trajectory. Starting as a copula verb with aspectual change-of-state semantics (i.e. ingressive *aktionsart*), it developed into a marker of textual sequence relation before turning into a temporal marker of posteriority. These semantic shifts proceeded incrementally and gradually along a series of syntactic contexts, and they show a particular directionality: (i) from main clauses preceded by subordinate clauses to (ii) independent clauses and (iii) finally to subordinate clauses.

In main clauses preceded by subordinate clauses, i.e. in stage (i) of the development, the proposition introduced by *werden* + infinitive is marked as following from some important pieces of information given in the text, as shown for example in (5) and (6).

- (5) *Denn wenn solch stück der Königin Auskopmt zu den weiberen hin, So*
then when such piece the queen come.out.3.SG to the women to so
werden *sie verachten gar All ir menner ganz offenbar.*
werden.3.PL they despise.INF even all their men quite obviously
‘For if such a piece of the queen’s head is sent out to the women, they
will despise all of their men quite obviously.’

[ENHGCORPUS, Smirnova (2006: 262)]

- (6) *Wer nun Tugend / gute Sitten vnd Kuensten liebhat/ wird*
who now virtue / good morals and skills like.3.SG/ werden.3.SG
wissen was er tun sollte.
know.INF what he do.INF shall.3.SG
‘Now he who loves virtue / good morals and arts / will know what he
should do.’

[ENHGCORPUS, Smirnova (2006: 263)]

In (5), *werden* + infinitive occurs in the apodosis of a conditional sentence and introduces a conditional consequence. In (6), on the other hand, *werden* + infinitive introduces an effect of a condition that appears in the form of a free relative clause. Additionally, in both examples (5) and (6), this construction can be interpreted as referring to future time. What is important is that in these contexts, the predicates do not refer to a specific future event, but they refer generically to certain types of events.

In the course of ENHG, the temporal future reading of the construction *werden* + infinitive gets gradually emancipated from this kind of textual support and expands into new contexts of use. This is especially the case for uses of the construction in independent clauses, see (7). In these contexts, the marking of textual sequence relation is no longer the function of this construction.

- (7) *Von dem andern geschriben ist: Dann **wierst** du **nemmen***
of the other write.PAST.PART be.3SG: then werden2.SG you take.INF
das oppher der gerechtichait.
the sacrifice the justice
‘Of the other it is written: Then you will take the sacrifice of justice.’
[ENHGCORPUS DR, Smirnova (2006: 261)]

The next and most advanced step in the semantic development of future tense marking is reached when the construction unambiguously bears future time reference and does not bear any textual relation to the preceding text. This reading is most pronounced in the subordinate temporal and conditional clauses, as shown in (8)–(10).

- (8) *Dem Lande wird wiederum wohl seyn **wenn** Lycogenes die*
the country werden.3.SG again happy be.INF when Lycogenes the
*Krone tragen **wird***
crown wear.INF werden.3.SG
‘The country will again be happy when/if Lycogenes will wear the crown.’
[ENHGCORPUS 147, Smirnova (2006: 301)]

- (9) ***Wenn** er aber finden **wird**, daß Jhr Euer Hertz nicht zu ihm*
when he but find.INF werden.3.SG that you your heart not to him
*lencken wollet, **nachdem** Jhr von allem fogenannten Zwange*
lead.INF want.2.PL after you of all so-called constraint
*befreyet feyn **werdet**, so wird er aufhören, an*
free.PAST.PART be.INF werden.2.PL so werden.3.SG he stop about
Euch zu dencken.
you to.think.INF
‘But if he finds that you do not want to turn your heart to him, after you have been freed from all of the aforementioned constraints, he will stop thinking about you.’

[Richardson, 1748, DTA]

- (10) *fo will ich dasjenige, was der meinigen fehlt, sobald es*
 so want.1.SG I that what from mine miss.3.SG as soon as it
möglich feyn wird, hinzufügen
 possible be.INF werden.3.SG add.INF

‘I will add what is missing from mine as soon as it will be possible.’

[Gessner, 1745, DTA]

It should be noted that the temporal subordinators *nachdem* ‘after’ and *sobald* ‘as soon as’ are much younger than the temporal-conditional *wenn* ‘when’: *Nachdem* is first attested since ca. 1550, and *sobald* appears at around 1600, whereas *wenn* ‘when’ was present since the Old High German period. In the context of this study, only the temporal and realis-conditional uses of the temporal-conditional subordinator *wenn* are relevant, as only those are associated with the development of the construction *werden* + infinitive.⁴ These subordinators share a general conceptual schema that is relevant for the history of *werden* + infinitive that is investigated in this study.

Essentially, the general temporal schema of these contexts may be conceptualized as follows. The scene consists of two events: the event *e1* that takes place at time *t1* and the event *e2* at time *t2*. The event *e1* is expressed in the subordinate clause with the subordinators *wenn* ‘when’, *nachdem* ‘after’, and *sobald* ‘as soon as’, and the event *e2* is depicted in the main clause that usually follows the subordinate clause. The time *t1* precedes the time *t2*, what is often iconically represented in the syntagmatic ordering of the clauses; both events are usually located in the future relative to the speech time.

Importantly, in contexts like (8)–(10), *werden* + infinitive can only be read as future tense construction. That is, it cannot be read in one of its present readings mentioned in Section 1, i.e. neither as a modal nor as an evidential verb, as these markers are not compatible with subordinate clauses.

The expansion of *werden* + infinitive into temporal and conditional subordinate clauses along the trajectory presented above thus represents a very advanced step on a grammaticalization path towards future tense marker. From the usual way grammaticalization processes proceed, it is to be expected that in the further stages of this process, the construction *werden* + infinitive should get firmly established in these contexts and become obligatory.

⁴Whereas irrealis-conditional contexts with *wenn* are clearly marked by subjunctive verbal morphology, realis-conditional contexts with indicative verbal form are often ambiguous between temporal and conditional reading throughout the time period.

However, from what we know about present-day German, *werden* + infinitive is no longer used in these contexts.⁵ The following case study is based on data consisting of conditional and temporal clauses, in order to trace recent changes in the development of the construction *werden* + infinitive and see what happened in the time between ENHG and today.

3.1 Data

The present study is based on data from three recent diachronic slices of German: 1600–1649, 1700–1749, and 1950–1999. The data comes from two corpora of German: Deutsches Textarchiv (DTA) for the first half of the 17th and of the 18th centuries, and the DWDS-Kernkorpus for the second half of the 20th century, which in my data represents present-day use. Table 1 gives an overview of the total occurrences of temporal and conditional clauses in the data. The number of subordinators *wenn* ‘when/if’, *nachdem* ‘after’, and *sobald* ‘as soon as’, tagged in the corpus as KOUS ‘subordinating conjunction’, is taken as a proxy for the number of the clauses they introduce.

Table 1: Subordinate clauses with *wenn*, *nachdem*, and *sobald*

period	corpus	corpus size	<i>wenn</i> clauses	<i>nachdem</i> clauses	<i>sobald</i> clauses
1600–1649	DTA	7 203 918	9 462	347	5
1700–1749	DTA	23 500 609	64 483	7 767	437
1950–1999	DWDS	58 068 993	88 782	5 548	2 147

Figure 1 depicts the normalized frequencies per million words of the same types of subordinate clauses. As can be seen, *wenn* is the most frequent subordinator across the three periods, the other two being far less frequent.

The qualitative analysis is mainly based on the data presented in Table 2, where only subordinate clauses containing the construction *werden* + infinitive are shown. The data amounts to a total of 740 observations that are distributed over three periods of time and the three subordinators under investigation.

⁵The fact that *werden* + infinitive occurred in these contexts at all is particularly interesting from the cross-linguistic point of view. As is known, ‘real’ future tense constructions behave differently with respect to this type of subordinate clause. The English *will*-future or the synthetic future in French are banned from conditional clauses, whereas in Slavic languages, future markers appear naturally in these contexts.

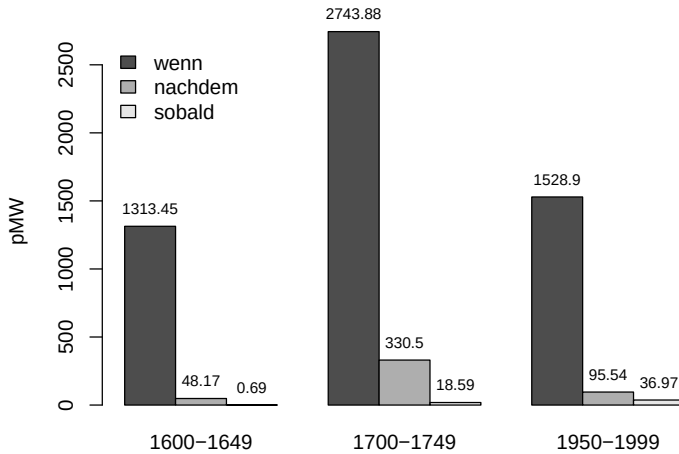


Figure 1: *wenn*, *nachdem*, and *sobald*, normalized frequency

Table 2: Subordinate clauses with *werden* + infinitive

period	corpus	<i>wenn</i> clauses	<i>nachdem</i> clauses	<i>sobald</i> clauses
1600-1649	DTA	213	2	0
1700-1749	DTA	312	23	4
1950-1999	DWDS	162	10	14
		687	35	18

3.2 Analysis

Figure 2 shows the relative frequency of the clauses containing the construction *werden* + infinitive in the three types of subordinate clauses.

The numbers in Figure 2 reveal that the proportions are extremely low: Most bars hardly reach the one percent mark. That means that subordinate clauses with *werden* + infinitive are extremely rare. In the first period, *wenn*-clauses have the highest proportion of *werden* + infinitive (2.25%). Interestingly, a decline in the use of *werden* + infinitive can be observed in all types of subordinate clauses; it is most pronounced in *wenn*-clauses (dark bars in Figure 2), but a similar progression can be observed for *nachdem*- and *sobald*-clauses. What we see here is a clear tendency for *werden* + infinitive to slowly disappear from the conditional

4 German *werden*-future: A case of failed grammaticalization?

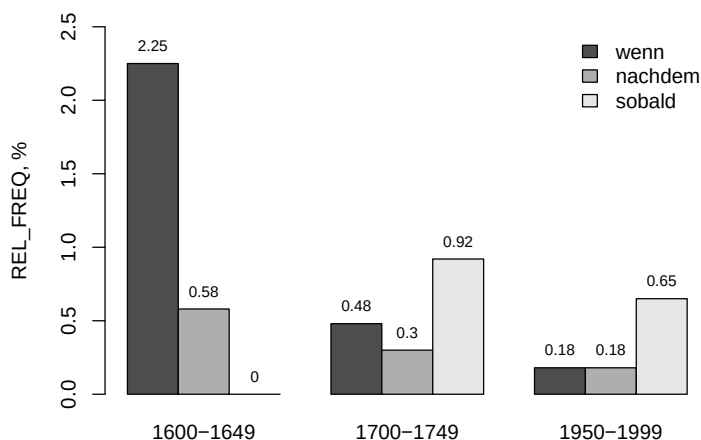


Figure 2: Relative frequency of *werden* + infinitive in subordinate clauses with *wenn*, *nachdem*, and *sobald*, in percent

and/or temporal clauses in the course of time. If we look at the younger subordinators *nachdem* and *sobald*, we can see that *werden* + infinitive initially finds its way into these structures, but fails to gain ground and declines in use in the subsequent periods.

From this, it can be concluded that *werden* + infinitive gradually loses its contexts of use. In Section 3, conditional and temporal subordinate clauses were described as representing an advanced stage of grammaticalization of *werden* + infinitive as a future tense marker. These contexts were the last into which the construction expanded, and the further grammaticalization process would normally imply further expansion and consolidation of use in these contexts. What we observe, however, is the opposite – *werden* + infinitive is gradually withdrawing from these contexts. The construction does not proceed further on the grammaticalization path.

Furthermore, a closer look into the data reveals that, in the third period, the subordinate clauses differ considerably from each other as concerns the form of the infinitive (see Table 3).

Whereas *wenn*-clauses still predominantly contain *werden* + infinitive (see e.g. 11), *nachdem*- and *sobald*-clauses show a clear preference for *werden* + infinitive perfect, usually called future perfect or *Futur II* in German (see e.g. 12).

Table 3: Form of the infinitive, 1950–1999

subordinator	<i>wenn</i>	<i>nachdem</i>	<i>sobald</i>
<i>werden</i> + infinitive	137	4	1
<i>werden</i> + infinitive perfect	25	10	9
% infinitive perfect	15%	71%	90%

- (11) *Wenn wir trotzdem an diesen ungesicherten Allgemeinbegriffen*
when we nonetheless to these unsecured generalities
festhalten werden, so liegt darin mehr als ein unentschuldbarer
cling.INF werden.3.PL so lie.3.SG there more than a inexcusable
methodologischer Leichtsinns.
methodological recklessness
‘If we will nevertheless cling to these unsecured generalities, there is
more than an inexcusable methodological recklessness in this.’
[Dahrendorf, 1957, DWDS-Kernkorpus]
- (12) *Sie hat versprochen sich zu entscheiden, sobald sie das Leichentuch*
she has promised REFL to decide as soon as she the shroud
für ihren Schwiegervater Laertes fertiggewebt haben wird
for her father-in-law Laertes finish.PAST.PART have.INF werden.3.SG
‘She has promised to decide as soon as she will have finished weaving the
shroud for her father-in-law Laertes.’
[Schwanitz, 1999, DWDS-Kernkorpus]

The form *werden* + perfect infinitive is usually employed to mark a completion of an event in the future and is particularly tied to contexts of temporal clauses such as those analyzed here.

Another important observation concerns the verb form in the main clauses. The relative proportion of different verb forms is depicted in Figure 3.

What we see in Figure 3 is a clear tendency towards the decline of the form *werden* + infinitive across the three periods: Whereas in the 17th century, *werden* + infinitive was the dominant form in both clauses of the sentence, in the 20th century, *werden* + infinitive in the main clause amounts only to 40% of all verb forms. On the other hand, the rise of the use of simple present forms is particularly pronounced: The use of the present tense increases from ca. 10% in the first period to 40% in the third period. That is, main clauses follow the same general

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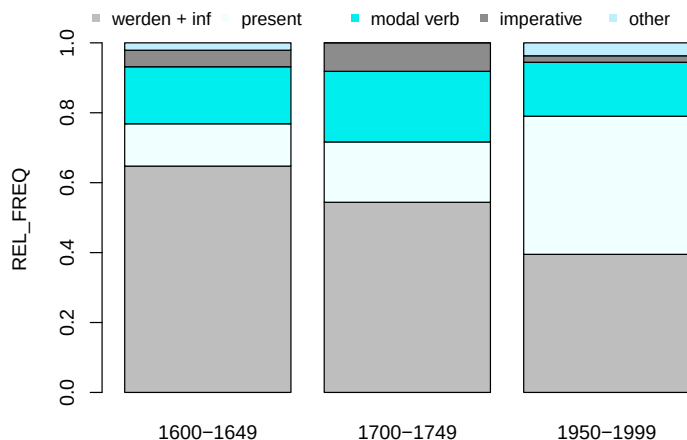


Figure 3: Relative frequency of different verb forms in main clauses, in percent

tendency we have observed above, and the construction *werden* + infinitive gets gradually replaced by the present tense forms, both in main and in subordinate clauses of temporal and conditional sentences.

The data presented so far shows that *werden* + infinitive gradually disappears from temporal and conditional clauses with *wenn*, *nachdem*, and *sobald* (see Figure 2). In present-day German, *werden* + infinitive is almost completely absent from the temporal clauses with *nachdem* and *sobald* (see Table 3); *werden* + perfect infinitive (= “future perfect”) is used there instead. As seen from Figure 2, *werden* appears first in these contexts, but fails to gain ground and disappears gradually from them. Moreover, this downward tendency holds for main clauses of these sentences as well (see Figure 3).

A last observation to be drawn from the analysis of this dataset concerns the infinitive verbs combining with *werden*.⁶ Table 4 gives an overview of the ten most frequent infinitive verbs in each period.

As can be seen from Table 4, some of the infinitive verbs, such as *kommen* ‘come’ or *haben* ‘have’, have combined with *werden* from the beginning; others have become more or less associated with the construction in the course of time. In the verb list of the last period, however, *sein* ‘be’ is not only the most frequent verb, but it is also almost the only verb in this combination. With 64 occurrences,

⁶For the infinitive verbs, only the forms with the present infinitive, and not those with the perfect infinitive, were analyzed.

Table 4: Infinitive verbs, top ten collocates

1600–1649			1700–1749			1950–1999		
verb	gloss	N	verb	gloss	N	verb	gloss	N
<i>kommen</i>	come	33	<i>kommen</i>	come	41	<i>sein</i>	be	64
<i>erscheinen</i>	appear	18	<i>sein</i>	be	28	<i>haben</i>	have	9
<i>sein</i>	be	16	<i>haben</i>	have	19	<i>kommen</i>	come	4
<i>anschauen</i>	look	5	<i>erscheinen</i>	appear	13	<i>gelingen</i>	succeed	3
<i>offenbaren</i>	reveal	5	<i>sehen</i>	see	9	<i>lesen</i>	read	3
<i>sagen</i>	say	5	<i>werden</i>	become	8	<i>werden</i>	become	3
<i>sehen</i>	see	5	<i>hören</i>	hear	7	<i>ausmachen</i>	turn out	2
<i>werden</i>	become	5	<i>posaunen</i>	broadcast	7	<i>führen</i>	lead	2
<i>angehen</i>	concern	4	<i>aufhören</i>	stop	5	<i>gehen</i>	go	2
<i>brechen</i>	break	4	<i>offenbaren</i>	reveal	5	<i>heiraten</i>	marry	2

it stands in sharp contrast to other infinitive verbs, which are attested in the data with very rare occurrences.

Hartmann (2021: 385) analyzed the usage of the construction *werden* + infinitive in the ENHG period and found that towards the end of the period, i.e. in the 17th century, the most frequent infinitive verb in this construction is *sein* (23 occurrences in his data), followed by *haben* ‘have’ (with 8 occurrences). In a study based on data from the 19th and 20th century, Hilpert (2008: 148) detected the same verbs, that is, *sein* ‘be’ (n=172), *kommen* ‘come’ (n=62), *haben* ‘have’ (n=40), and *werden* ‘become’ (n=40) among the most frequent infinitive verbs in this construction. In each study, it is thus the verb *sein* that turns out to be the most frequent infinitive verb that combines with *werden*. This suggests that the changes detected above and that concern the temporal and conditional clauses are not only characteristic of these particular contexts, but manifest a general tendency in the development of the construction *werden* + infinitive.

In sum, the results of the diachronic corpus study show that in the last four centuries, the construction *werden* + infinitive has not progressed further on its way towards becoming a grammaticalized future tense marker:

1. With regard to usage frequency, there is a decrease in use and a gradual replacement of *werden* + infinitive by the futurate present tense. As concerns its use in temporal and conditional subordinate clauses, after initial uses, it failed to expand further, and this tendency holds also for independent uses of this construction. That is, no further context-expansion of the construction in the sense of Himmelman (2004) can be observed since ENHG. Instead, the contexts of use become more and more restricted.

2. With regard to the infinitive verbs used in the construction, the range of regularly used verb types decreased, most of them being used rarely in the construction. Importantly, *werden* + infinitive established a particular preference for the infinitive verb *sein* 'be'. This limitation to one dominant usage pattern with *sein* clearly goes against the tendency of obligatorification (Lehmann 1995). During the advanced stages of a grammaticalization process, obligatorification would manifest itself in the increasing token and type frequencies of the infinitive verbs. However, this is clearly not the case for *werden* + infinitive, as shown above.

To conclude this section, in the light of the evidence presented above, I would like to argue that the development of the construction *werden* + infinitive in the last four centuries constitutes a case of failed grammaticalization. What has happened instead is that *werden* + infinitive has limited itself to some non-frequent and specialized uses. This empirical evidence points to the establishment of a possible functional niche of this construction in modern German. In Section 4, I will deal in more detail with the combination of *werden* + *sein* to see what kind of functional niche this particular pattern occupies in present-day German.

4 The functional niche of *werden* + infinitive in present-day German

Let me start this section by giving some illustrative examples of the combination of *werden* and *sein* (cf. 13–15).

- (13) *Du wirst mir eines Tages dankbar sein.*
 you werden.2.SG me one day thankful be.INF
 'You will be grateful to me one day.'

[Noll, 1996]

- (14) *Was wird inzwischen zu Hause alles geschehen sein?*
 what werden.3.SG meanwhile at home all happen.PAST.PTCP be.INF
 'What will have happened at home in the meantime?'

[Grzimek, 1954]

- (15) *Ob und wie dieser Ratschlag funktionierte, darüber wird noch zu sprechen sein.*
 whether and how that advice work.3.SG.PAST about it werden.3.SG
 yet to speak.INF be.INF
 'Whether and how this advice worked, will have to be discussed yet.'

[P. M., 1993]

The examples (13)–(15) show that the combination of *werden* + *sein* occurs in different syntagmatic configurations and displays an array of possible functions. In (13), *sein* is a copula verb that combines with an adjective (*dankbar* ‘grateful’); the combination with *werden* renders future time reference. In (14), on the other hand, *sein* is a perfect tense auxiliary modifying the full verb *geschehen* ‘happen’, and the combination *werden* + *past participle* + *inf* constitutes a case of Future II or future perfect. This combination may refer to future time, but mostly, it is interpreted as a modal construction, i.e. expressing the epistemic uncertainty of the speaker. In (15), *sein* appears as a part of another construction, i.e. *sein* + *zu* ‘to’-infinitive with the modal meaning of obligation. The combination with *werden* can be interpreted as referring to an obligation that will hold at a later point in time, or as a textual lookahead, where the author announces that something will (have to) be treated later in the text.

Given this variety of functions the syntagmatic combination of *werden* and *sein* may carry in individual contexts, it is worthwhile to inspect the individual contexts in more detail in order to see whether there are patterns in the data that point to the existence of a particular functional niche of the pattern *werden* + *sein*. This is the aim of the following synchronic case study.

4.1 Data

The present analysis is based on a random sample of 1000 observations that stem from the most recent slice of the DWDS-Kernkorpus20 (Geyken 2007), i.e. from the years 1950–1999.⁷ The data were manually annotated with respect to the type of the element the infinitive verb *sein* combines with. The following categories were used for annotation:

- ADJECTIVE (see 13 above)
- NOUN PHRASE (see 16)
- PREPOSITIONAL PHRASE (see 17)
- ADVERB (see 18)
- ZU-INFINITIVE (see 15 above)
- PAST PARTICIPLE (see 14 above)
- (ADJECTIVAL) PAST PARTICIPLE (see 19)

⁷The search string used to extract the examples of the combinations of *werden* and *sein* from the corpus was the following: /werde|wirst|wird|werden|werdet/ && “sein with \$p=/V.*INF/” #asc_date[1950-00-00, 1999-99-99]; i.e. I searched for all present indicative forms of the verb *werden* in combination with the infinitive form of *sein* within a sentence. The obtained hits were then randomized, manually inspected for false positives and annotated according to the variables described in this section.

- (16) *Das ausführende Organ der Bank wird der Vorstand*
 the executive body of the bank werden.3.SG the board of directors
sein.
 be.INF
 ‘The executive body of the bank will be the board of directors.’
 [AdG 40, 1970]
- (17) *Ich bin zu Hause, wie gesagt, und solange es hell ist,*
 I be.1.SG at home as said and as long as it light be.3.SG
werde ich im Garten sein.
 werden.1.SG I in the garden be.INF
 ‘I am at home, as I said, and as long as it is light outside, I will be in the garden.’
 [Born, 1979]
- (18) *Peter wird natürlich auch da sein.*
 Peter werden.3.SG of course too there be.INF
 ‘Peter will of course be there too.’
 [Horster, 1950]
- (19) *Dann, und nur dann, wird ein erster Schritt auf dem langen Wege*
 then and only then werden.3.SG a first step on the long way
zur nuklearen Abrüstung getan sein.
 to nuclear disarmament do.PAST.PTCPPL be.INF
 ‘Then, and only then, will a first step on the long road to nuclear disarmament be taken.’
 [AdG 28, 1958]

In combinations with adjectives (13), noun phrases (16), prepositional phrases (17), and adverbs (18), *sein* qualifies as a copula verb. For this reason, these combinations will be treated further below as one sub-group of *sein*-usages. As noted above, the combination of *sein* with *zu* ‘to’-infinitive qualifies as a construction on its own, and therefore it will be treated as such in the following. In the contexts like (14), *sein* will be considered a perfect auxiliary.

In combinations like (19), *sein* is usually classified as passive auxiliary, and the construction *sein* + past participle is called stative passive or *Zustandspassiv* (cf. e.g. Helbig 1983; Duden 2016: 567–568). However, the grammatical classification of the construction as stative passive is not uncontroversial. Without going into much detail at this point, I will follow here the account of Maienborn (2007) who

has convincingly shown that the past participle in this construction behaves like an ordinary adjective, and *sein* in these combinations can be considered a “normal” copula verb. For this reason, I will treat such combinations as pertaining to the sub-category of copula verb *sein* mentioned above. Table 5 gives the frequency of these elements in the sample.

Table 5: Complements of *sein*-infinitive

complement	N
adjective	388
noun phrase	264
<i>zu</i> -infinitive	85
adjectival participle	81
prepositional phrase	81
adverb	54
past participle	47
TOTAL	1000

Figure 4 summarizes the combinations of *werden* and the infinitive of *sein* according to the three categories of the verb *sein* introduced above:

- copula verb in combination with adjectives, adjectival past participles, noun phrases, prepositional phrases, and adverbs
- modal construction *sein* + *zu* ‘to’-infinitive and
- perfect auxiliary in combination with past participles of full verbs.

As can be already seen from Table 5 and Figure 4, the distribution of different variants of the infinitive verb *sein* is extremely skewed.⁸ In the next section, I will look more closely into the functional range of these combinations.

4.2 Analysis

In the preceding section, three categories of the verb *sein* were distinguished according to the elements the verb combines with, and to the semantics the whole syntagmatic configuration with *werden* conveys. Let me start by discussing the two categories that are only sparsely presented in the data and make up 13.2% of the analyzed sample.

⁸Even if I had considered the 81 cases of so-called *Zustandspassiv* as a separate category, this would not have considerably changed the distribution.

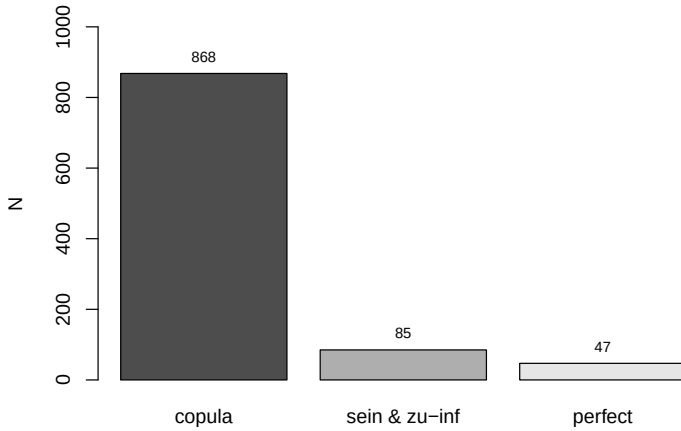


Figure 4: Frequency of different *sein*-variants in the data

4.2.1 The future perfect form *werden* + past participle + *sein*

The future perfect form is attested 47 times in the sample. As mentioned in Table 5 and Figure 4, in this context *werden* functions as a modal-evidential verb, and it cannot be considered exclusively a future tense auxiliary. In this reading, *werden* behaves similarly to other epistemically used modal verbs like *müssen* ‘must’ or *dürfte* ‘may’ (see 20a–20b and 21a–21b):

- (20) a. *Es **wird** wohl ’ne größere Taube gewesen sein.*
 it werden.3.SG probably a larger pigeon be.PAST.PTCPL be.INF
 ‘It will probably have been a larger pigeon.’
 [Bieler, 1983]
- b. *Es **muss/dürfte** wohl ’ne größere Taube*
 it must.3.SG/may.3.SG.SUBJ probably a larger pigeon
gewesen sein
 be.PAST.PTCPL be.INF
 ‘It must have been a larger pigeon.’
- (21) a. *Für die “Verräter” **wird** dieser Rausschmiß nicht unerwartet*
 for the traitors werden.3.SG this expulsion not unexpectedly
gekommen sein.
 come.PAST.PTCPL be.INF
 ‘For the “traitors” this expulsion will not have come unexpectedly.’
 [ZEIT, 1954]

- b. Für die “Verräter” **muss/dürfte** dieser Rausschmiß nicht unerwartet
 for the traitors werden.3.SG this expulsion not unexpectedly
 gekommen sein
 come.PAST.PTCPL be.INF
 ‘For the “traitors” this expulsion should not have come unexpectedly.’

These combinations are very low in frequency (see Figure 4 in Section 4.1). Moreover, they are functionally similar to other means of expressing epistemic and/or evidential assessment in present-day German. Furthermore, future time reference, i.e. future in the past, constitutes only one possible interpretation of these combinations, whereas the modal nuances are much more pronounced.

This suggests that this usage of *werden* + *sein* does not represent a special functional niche of the more abstract construction *werden* + infinitive. Rather, the future perfect forms attested in the analyzed sample should be seen in a broader context of the usage of perfect infinitive forms in present-day German, including perfect infinitive forms with the perfect auxiliary *haben*, which in many contexts of use display very similar formal and functional characteristics.

4.2.2 *Werden* and the modal construction *sein* + *zu* ‘to’-infinitive

The combination of *werden* with the modal construction *sein* + *zu* ‘to’-infinitive amounts to 85 observations in the analyzed sample (= 8.5%). What makes these contexts particularly interesting is the fact that the combination with *werden* unambiguously renders a future tense reading. Compare for example (22a) and (23a) with (22b) and (23b) respectively, where *werden* has been removed, and only the present tense form of *sein* + *zu* ‘to’-infinitive is present.

- (22) a. Es **wird** nie wieder gutzumachen sein.
 it werden.3.SG never ever to.make.good be.INF
 ‘It will never be possible to make up for it.’

[Bachmann, 1971]

- b. Es **ist** nie wieder gutzumachen.
 it be.3.SG never ever to.make.good.INF
 ‘It can never be made up for.’

- (23) a. *Ihre Aufenthaltserlaubnis wird jedes Jahr zu erneuern*
 their residence permit werden.3.SG every year to renew.INF
sein, und ihre Arbeitserlaubnis wird jedes Mal zu
 be.INF, and their work permit werden.3.SG every time to
erneuern sein, wenn ...
 renew.INF be.INF when ...
 ‘Their residence permit will have to be renewed every year, and their
 work permit will have to be renewed every time when ...’
 [AdG 39, 1969]
- b. *Ihre Aufenthaltserlaubnis ist jedes Jahr zu erneuern, und ihre*
 their residence permit be.3.SG every year to renew.INF, and their
Arbeitserlaubnis ist jedes Mal zu erneuern, wenn ...
 work permit be.3.SG every time to renew.INF when ...
 ‘Their residence permit must be renewed every year, and their work
 permit must be renewed every time when ...’

In (22b) and (23b), the future tense reading gets lost, and only the modal semantics of the construction *sein* + *zu* ‘to’-infinitive remains, which in turn may be read as a general, tenseless predication. In opposition to (22a) and (23a), however, it is clear that the added value of *werden* consists of conveying future tense semantics. That is, the modal semantics of the construction *sein* + *zu* ‘to’-infinitive is compositionally combined with the future tense semantics of *werden*. Thus, in these contexts, we find *werden* in clear opposition to the simple present tense, and in these contexts, the simple present is not able to produce clear futurate readings.

To sum up so far, the combination of *werden* and a conventionalized construction *sein* + *zu* ‘to’-infinitive with its own modal passive meaning serves to unambiguously mark the future time reference of the proposition. In these contexts, *werden* + infinitive is not functionally equivalent to the present tense, as the present tense form of the construction does not allow for a future tense interpretation. I will return to this point in Section 4.3.

4.2.3 *Werden* and the copula verb *sein*

The great majority of observations of *werden* + *sein* contains the infinitive of the copula verb *sein*, i.e. 868 observations (= 86.8%). Only in a small fraction of these observations, namely in 12 sentences, does *werden* not primarily evoke a future tense interpretation, but may be attributed a modal-evidential reading (see 24–25). Note that in these contexts, modal adverbials such as *wohl* ‘surely/probably’

are present in the text and contribute as well to the modal meaning of the proposition.

- (24) *Was bei Mikrowellen, Joysticks und Zahnbürsten erlaubt ist, **wird** doch wohl bei der Soziologie nicht verboten **sein**.*
what for microwaves joysticks and toothbrushes allow.PAST.PTCPL
be.3.SG werden.3.SG MP⁹ surely for the sociology not forbidden be.INF
'What is allowed for microwaves, joysticks and toothbrushes will surely not be forbidden for sociology.'
[ZEIT, 1996]

- (25) *Das **wird** doch nicht etwa schon die Polizei **sein**?*
that werden.3.SG MP not MP already the police be.INF
'Surely that won't be the police already?'
[Grass, 1959]

In the vast majority of cases, however, the future temporal interpretation is clearly the dominant reading present in the context. Sometimes (in 196 out of 856 or in ca. 23%), temporal adverbials or other explicit contextual information with future time interpretation are present, as in (17)–(19) and (26), but this certainly does not need to be the case (see 16 and 27).

- (26) *Im Jahr 2005 **wird** Sprache im Umgang mit dem Computer in in year 2005 werden.3.SG language in handling with the computer in jeder Hinsicht verfügbar **sein**.*
every aspect available be.INF
'In 2005, language will be available in every aspect of computer use.'
[BILD, 1998]

- (27) *Viele potentielle Produzenten **werden** nicht in der Lage **sein**, das many potential producers werden.3.PL not able be.INF the erforderliche Kapital zu erhalten.*
necessary capital to obtain.INF
'Many potential producers will not be able to obtain the necessary capital.'
[AdG 46, 1976]

⁹German modal particles (MP) are notoriously difficult to translate. For this reason, I resort to the label MP in the glosses.

It may be concluded that for the combinations of *werden* with the copula verb *sein*, the function of *werden* unambiguously and almost exclusively consists of marking future time reference. It is in these contexts that the future meaning of the construction *werden* + infinitive is highly distinctive from the other contexts of use.

The frequency-based evidence presented so far is already very telling: (i) *Sein* is by far the most frequent infinitive verb that combines with *werden* in present-day German (see Section 3.2); (ii) it is the copula verb *sein* that mostly combines with *werden* (see Section 4.1). This suggests that the combination of *werden* and the copula verb *sein* constitutes a very significant use of *werden* + infinitive and may be qualified as a niche. The qualitative evidence presented in this section certifies that in these contexts, *werden* functions as a future tense marker. In other words, the future tense reading of *werden* + infinitive is largely restricted to the combinations of *werden* and *sein*, whereas in all other contexts, it is the (futurate) present tense that is employed to mark future tense reference. In Section 4.3, I will present a tentative account of why combinations with the copula *sein* represent a suitable context for future uses of *werden* + infinitive in present-day German.

4.3 Discussion

It is generally acknowledged that German has three copula verbs: *sein*, *werden*, and *bleiben*. In present-day German, the copula verb *sein* is a stative copula used to ascribe qualities to all kinds of referents, and it stands in semantic opposition to two further copula verbs, i.e. to the inchoative *werden* on the one hand, and to the durative stative *bleiben* on the other, as in (28).

- (28) *Lena ist/wird/bleibt* *böse_{Adj}/meine Freundin_{NP}*.
 Lena be./werden./remain.3.SG angry/my friend
 ‘Lena is/becomes/remains angry/my friend.’

In the following, only the pair *sein* vs. *werden* will be relevant for our purposes. The semantic difference between the two is one of aktionsart: Whereas *sein* is stative and atelic, *werden* is inchoative and telic. These copula verbs are usually seen to behave very similarly with respect to the predicative elements they combine with – that is, they may combine with adjectives, with noun phrases, with prepositional phrases, and with adverbs (see Section 4.1).

However, there is an important difference in syntactic behaviour between the two. The range of the usage contexts of the copula *werden* is more restrictive than those of *sein*. Whereas *sein* may be combined with all types of predicative

elements, irrespective of their semantics, the use of *werden* seems to be restricted to complements that express incremental changes. That is, the use of *werden* is ruled out in combination with locative expressions such as *im Garten* ‘in the garden’ or with non-gradable adjectives such as *tot* ‘dead’. Table 6 summarizes these differences.

Table 6: Complements of *sein*- vs. *werden*-copula

complement	<i>sein</i>	<i>werden</i>	gloss
gradable property	A ist böse.	A wird böse.	‘angry’
non-gradable property	A ist tot.	*A wird tot.	‘dead’
location	A ist im Garten.	*A wird im Garten.	‘in the garden’
location	A ist dort.	*A wird dort.	‘there’

These semantic differences manifest themselves in an interesting way when it comes to future time reference. Combinations with the copula *sein* unambiguously render stative interpretations, and usually locate the ascribed state or property in the present time. This applies to all kinds of expressions, irrespective of their gradable/non-gradable nature, as in (29)–(30).

- (29) *Er ist (immer) böse, wenn er das hört.*
 he be.3.SG (always) angry when he it hear.3.SG
 ‘He is always angry when he hears that.’

- (30) *Sie ist Astronautin.*
 she be.3.SG astronaut
 ‘She is an astronaut.’

Combinations with the copula *werden* with complements that refer to gradable properties or imply incremental changes, on the other hand, are essentially ambiguous between the inchoative reading with present time reference (= ongoing change of state) and the future reading (= result state in the future). It is only by means of further contextual elements such as temporal adverbials that a non-ambiguous present or future reading is achieved, as in (31a–31b).

- (31) a. *Er wird langsam böse.*
 he werden.3.SG slowly angry
 ‘He is getting angry.’

- b. *Er wird böse, wenn er das hört.*
 he werden.3.SG angry when he it hear.3.SG
 ‘He will be/get angry when he hears that.’

Combinations of the auxiliary *werden* with the copula *sein*, on the other hand, clearly result in a non-ambiguous future tense reading, as in (32):

- (32) *Er wird böse sein.*
 he werden.3.SG angry be.INF
 ‘He will be angry.’

For the expressions that refer to non-gradable properties or non-incremental changes, the situation is rather different. They may be combined with the copula *sein* to refer to a present state,¹⁰ as in (33). When combined with the auxiliary *werden* and the infinitive of the copula *sein*, they unambiguously refer to a future state, as in (34). Note that the option with the copula verb *werden* alone is not available (see Table 6).

- (33) a. *Sie ist bereit.*
 she be.3.SG ready
 ‘She is ready.’
 b. *Sie wird bereit sein.*
 she werden.3.SG ready be.INF
 ‘She will be ready.’
- (34) a. *Ich bin im Garten.*
 I be.1.SG in.the garden
 ‘I am in the garden.’
 b. *Ich werde im Garten sein.*
 I werden.1.SG in.the garden be.INF
 ‘I will be in the garden.’

From what has been observed in previous sections, and from the semantic analysis just presented, it is reasonable to assume that for present-day German copula verbs, there exists a firmly established paradigmatic pair *sein* with present

¹⁰The copula verb *sein* in its present tense form is in principle suited for futurate contexts and can be used to refer to future states, especially if some adverbial modification is present in the text that unambiguously refers to future, cf. *Morgen ab 16 Uhr bin ich im Garten* ‘Tomorrow from 4 p.m. I will be in the garden.’ However, temporal adverbials are obligatory in such contexts.

time reference and *werden* + *sein* with future time reference. Though *werden* is essentially an inchoative copula verb and can be used as such in contexts with gradable expressions, its use in the construction *werden* + *sein* has been conventionalized as strong expression of future tense, in opposition to the present tense of the copula *sein*.

This paradigmatic relation is strongly pronounced in case of more or less fixed multi-word expressions containing the copula verb *sein*. These include, for example, conventionalized and idiomatic expressions such as *an der Reihe sein* ‘be next, lit: be in the turn’, *in der Lage sein* ‘be able to, lit: be in the position’, *zu Ende sein* ‘be finished’, and the like. The only way to mark future time reference is to employ the future auxiliary *werden*, as in (35)–(36).

- (35) *Unsere Kultur* **wird** *in der Lage sein, sich unabhängig von*
our culture werden.3.SG able be.INF REFL independently of
der menschlichen Biologie und ihren Gesetzen zu entwickeln.
the human biology and its laws to develop.INF
‘Our culture will be able to develop independently of human biology and its laws.’

[Rechenberg, Informatik, 1991]

- (36) *Und auch hier* **wird** *es von entscheidender Bedeutung sein, wie*
and too here werden.3.SG it of critical significance be.INF how
sich die DDR-Führung verhält.
REFL the GDR leadership behave.3.SG
‘And here, too, it will be crucial how the GDR leadership behaves.’

[DEDok., 1990]

To sum up, I would like to argue that the future construction *werden* + infinitive have “survived” in this particular niche – that is, in combination with the infinitive of copula verb *sein*. In combination with other infinitive verbs, *werden* + infinitive did not develop further, and the futurate present tense was more successful in expressing future time reference.

In this diachronic situation, the contexts with the copula verb *sein* offered a welcome niche for the future construction *werden* + infinitive. And this is, as I believe, at least partly due to the asymmetries of usage contexts of the copula verb *werden* described above. As discussed above, *werden* as a copula cannot be used in present tense contexts with a significant number of complements due to their non-incremental and non-gradable semantics. The future auxiliary in combination with the copula *sein* is, however, fine. It could be hypothesized that

the combination of *werden* and *sein* serves as a sort of compensating strategy for this gap. Once established in this particular context, it could survive there and be preserved till the present-day German. Of course, more research is needed to substantiate this idea.

5 Conclusions

In this paper, I have argued that the development of the future construction *werden* + infinitive constitutes a case of failed grammaticalization. Although the relatively high grammaticalization status of the construction seems uncontested in present-day German, especially when semantic and formal characteristics of this construction are concerned, a closer look at the frequency of use and the distribution over different infinitive verbs offers a more differentiated picture. Namely, these factors suggest that the grammaticalization process has not come to its full completion, leaving *werden* + infinitive “just” an alternative means of expressing futurity.

Not being successful and leaving the grammaticalization path does not need to mean, however, that the construction dies out and disappears. As this study has shown, *werden* + infinitive has survived in a particular context of use, i.e. in combination with the infinitive of the copula verb *sein*. In these contexts, the auxiliary *werden* has become nearly obligatory, and it serves here to mark future time reference. The synchronic motivations of this situation as well as the diachronic details of this process are complex and concern not only the construction *werden* + infinitive itself but also the related copula verb *werden*. In Section 4.3, I offered a first tentative account of this functional niche of *werden* in present-day German. But of course, this idea needs to be followed up in further research and substantiated with more data, synchronically and diachronically.

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4 *German werden-future: A case of failed grammaticalization?*

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Chapter 5

The future that did not happen: On the history of the verb *varda* in Swedish

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This paper considers the constructions with the verb *varda* ‘to become’ in the history of the Swedish language, in particular such uses of the verb (with infinitives, present participles or in the present tense) which could constitute future reference. We follow the distribution and usage of each construction across a time span between 1300 and 1750 and complete with a qualitative analysis of the extant examples. We also consider the reasons for the demise of the futurate construction with the infinitive, in particular its low frequencies, limited distribution and the role played by the rise of a passive construction.

1 Introduction

Typologically, future constructions develop from a relatively small set of lexical sources and follow restricted developmental paths. Cross-linguistically, the main lexical sources of future constructions are motion verbs such as ‘come’ and ‘go’ on the one hand and verbs of ability, obligation, and desire on the other hand (Heine 1995). The grammaticalization into future tense markers involves a convergence of these different sources and a development along similar paths, most notably including the notion of intention. As argued by Bybee et al. (1994: 254) ‘all futures go through a stage of functioning to express the intention, first of the speaker, and later of the agent of the main verb.’ Heine (1995) argues that even those future tense constructions which are derived from other sources than the ones mentioned above, e.g., the German *werden* ‘become’, at one stage become markers of intention before proceeding along the grammaticalization cline



(Heine 1995: 127). However, it has also been demonstrated that the development of de-venitive future constructions need not involve the notion of intentionality (Dahl 2000: 322), but rather the development proceeds through ingressive, inchoative or resultative meanings (Traugott 1978: 378).

Apart from the lexically based futures (future constructions which grammaticalize from lexical sources), also aspectual forms can acquire future time reference, so that present tense forms can be used to refer to future events (so-called aspectual futures, see Bybee et al. 1994: 275, or futurate present, see Hilpert 2008: 157–180). However, in contrast to the lexically based futures, with the aspectual futures, future time reference is not part of their semantics, but is only achieved as a contextual effect.

In Modern Swedish (MSw), there are a number of grammaticalized means of expressing futurity, which include: de-venitive *komma att V*, lit. ‘come to V’, modal verbs, in particular *ska* ‘shall’ and the use of the present tense, typically with temporal adverbials indicating its future reading. The constructions are exemplified in (1–3).

- (1) *Jag kommer att resa imorgon.*
I come.PRS to travel.INF tomorrow.
‘I am going to travel tomorrow.’
- (2) *Jag ska resa imorgon.*
I shall.PRS travel.INF tomorrow.
‘I will travel tomorrow.’
- (3) *Jag reser imorgon.*
I travel.PRS tomorrow.
‘I travel tomorrow.’

There is in MSw no construction corresponding to the German *werden* construction, although a cognate of the verb, *varda* ‘become’, is present in some Swedish dialects (in particular in the northern regions, Eklund (2010), Loenheim (2013)). The verb is also well-attested in the oldest extant text from the Swedish territory, the runic inscriptions, and in the oldest Swedish prose written with Latin script, from early 1200s (Skrzypek 2020). There are no studies devoted exclusively to *varda* in the history of Swedish; however, for the cognate *verða* in closely related Icelandic we find suggestions that it was used as a futurity marker in the Old and Middle Icelandic period, i.e., 1050–1350 and 1350–1550 respectively (Halbe 1963: 39–46). For the Swedish material, Svenska Akademiens

Ordbok (SAOB, The Dictionary of the Swedish Academy) quotes several examples of the use of the verb as a future tense marker, with either infinitive (4) or present participle (5) complements, some as recent as early 1900s, both constructions are marked as obsolete in the dictionary.

- (4) (SAOB, example dated 1595)

At gud alzmectig warder at senda sit harda straff öffuer
that God almighty varda.PRS to send his.REFL severe punishment over
oss alla.
us all

‘That God almighty will send his severe punishment over us all.’

- (5) (SAOB, example dated 1928)

En hel del annat som du snart varder skådande.
a whole bit other which you soon become.PRS seeing

‘Other things, which you will soon be seeing.’

The verb *varda* is taken to signify inchoative or ingressive meaning. Some authors, notably Toyota (2009) and Propst (2001), narrow it down to a mutative meaning – not a general beginning of a new state, but rather a change of state from one to another (see also Denison 1993:426) or coming into being. Some of the early examples of *varda* illustrate this narrower meaning, as shown in (6).

- (6) (Pent, 1300)

Swa lot han oc ewam wardha.
so let he also Eve become

‘So let he [also] Eve come into being.’

Otherwise, *varda* has been studied mainly as a passive auxiliary (Markey 1969, Elmevik 1970, Skrzypek 2020, Skrzypek 2024), a function in which it also appears in the oldest extant sources. A lot of research has centered around the question of the loanword *bliva* ‘remain’, which was borrowed into Swedish at some time in the late 1200s (the oldest examples are found in the early 1300s). The verb gradually developed a new meaning ‘become’ alongside the original ‘remain’ and replaced *varda* in all constructions, including the passive one (see 7 and 8).

- (7) (Bur, ca. 1330)

þa varþ han takin mz myclom hepar i efeso staþ
then became he taken with much glory in Ephesus town

‘He was then taken with much glory to the town of Ephesus.’

(8) (Brahe, 1585)

Anders Person bleff strax greppen på sin gård Ranchyttan
Anders Person became immediately captured on his farm Ranchyttan
'Anders Person was immediately captured on his farm Ranchyttan.'

As a result, the usage of *varda* declined, and the verb lives on only in dialects, but is not part of the standard idiom.

The aim of the present paper is to discuss the history of the verb *varda* in the Swedish prose from 1300 to 1750, in particular whether it could be considered to be a marker of future reference, in combination with non-finite forms of other verbs. The time frame of the study coincides with the major developments of the passive construction and with the documented presence of *varda* in all prosaic genres. We aim to establish whether *varda* could be found in futurate constructions and we will examine the circumstances in which a potential grammaticalization into a future tense marker was halted. The term construction is used in the Construction Grammar (CxG) sense, i.e., we take constructions to be emergent clusters of memory traces, aligned within our conceptual space on the basis of shared form, function, and contextual dimensions (Goldberg 2019: 7).

The paper is structured as follows: in Section 2 we give a brief account of diachronic studies of future constructions in Germanic languages, with a focus on the German *werden*. Section 3 includes information about sources, methods and tools which were employed to study the development of *varda*. Section 4 gives the general results in terms of frequencies and construction types connected with *varda* in the history of Swedish. In Section 5 we discuss the material in more detail and attempt a qualitative analysis of the examples. Finally, Section 6 concludes the study with a discussion of the reasons for why the Swedish *varda* construction never grammaticalized into an expression of future tense.

2 Future tense in a diachronic perspective

Most attempts to define future tense refer to the classical study by Reichenbach (1947: 287–298) and posit that any definition of temporal relations must take into account three locations in time that are positioned relative to each other on a straight line. They are: the moment of speech (S), the moment of an event (E), and a point of reference (R). The latter signifies a vantage point from which an event is viewed, which need not coincide with S. R can be mentioned explicitly, e.g., 'When we arrived at the station, the train had already left', or it can be understood from the context. It is possible to define each tense uniquely with

reference to these three locations in time, e.g., the present tense is the coincidence of E, R, and S on a single point of the time line.

Although very elegant from a logical and formal point of view, Reichenbach's model has some shortcomings which become apparent in historical linguistic studies. Most importantly, it fails to capture constructional polysemy, i.e., the cross-linguistic generalizations that future constructions commonly express other meanings or are derived from other constructions, which are still present in language. Furthermore, it does not account for the fact that many languages have more than one future construction, with different distributional restrictions placed on them (see Hilpert 2008: 17–18 for a discussion and Comrie 1981 for critique). This makes it difficult to apply the model in a diachronic study.

The rise of future tense markers has mainly been approached from a grammaticalization perspective and, in recent years, within the framework of Diachronic Construction Grammar. It is impossible to refer to the entirety of literature on the topic, but to name just a few relatively recent approaches to English modal verb constructions it includes Traugott (1989), Gotti (2003), Aijmer (1985), Visser (1963), Ziegeler (2006). A number of Germanic future tense markers are investigated diachronically in Hilpert (2008), including the Swedish de-venitive *komma att* and the modal *ska*. The development of *varda* is understandably not included, as the grammaticalization, if at all initiated, did not progress to a fully-fledged marker of future tense.

There are relatively fewer studies which consider other sources of future tense markers. However, for the Swedish *varda* there is a parallel development in German, with the cognate verb *werden* 'become', which has been studied extensively (Diewald 1997, Kotin 2003, Hilpert 2008, Hartmann 2021, Krämer 2005, Thieroff 1992, Vater 1997 to name just a few). As there are unequivocal similarities between the two verbs, we will briefly recapitulate the developments in German, which will be of relevance for the present study.

German future tense construction with *werden* has developed from an inchoative or ingressive aspect marker (Diewald 1997: 121). In contrast to the aspectual futures discussed in Bybee et al. (1994), the German *werden*-construction is periphrastic rather than morphological (or zero) and it is a fully grammaticalized future tense marker. The verb itself can also be a copula, which conveys the meaning of an incipient change of state and takes nominal and adjectival complements, e.g. *Peter wird Lehrer* 'Peter is training to be a teacher.' or *Peter wird unruhig*. 'Peter is getting nervous.' (German examples after Hilpert 2008: 132). It can be also used as a passive auxiliary, with a past participle, e.g. *Peter wird beobachtet*. 'Peter is being observed.' (Hilpert 2008: 132). We find parallel examples in the Swedish material, here given in the adapted version, *Han vard kung*

‘He became a king’, *Han vard vred* ‘He turned furious.’ and *Han vard gripen* ‘He was captured.’ The similarities between German *werden* and Swedish *varda* (and its later replacement *bliva*) are in this respect striking, as *varda* is found in exactly the same array of functions and has also grammaticalized into a passive auxiliary when accompanied by a passive past participle.

There is, however, one more construction in Modern German, in which *werden* is found, this time with an infinitive complement, e.g., *Peter wird singen*. ‘Peter will sing’. There is some controversy in the interpretation of the last sentence, which is ambiguous between the temporal interpretation ‘Peter will sing (at some given point in the future)’ and the epistemic interpretation ‘Peter probably sings (right now or generally)’¹. The ambiguity of the construction is the source of much controversy in the account of its development into the future tense construction, dating back at least to Saltveit (1962). Saltveit (1962: 175) argues that perfective verbs such as *kommen* ‘come’, tend to receive a temporal interpretation, because of the implied reference to the end point of an event. Stative verbs such as *sein* ‘be’ on the other hand give the whole construction an epistemic flavour. In this account, Saltveit is followed by a number of authors, although some, notably Leiss (1992: 196), posit the inverse relation². In his corpus analysis of the rise of the future construction with *werden*, Hilpert (2008: 133–156) tests whether perfective and continuative verbs are similarly attracted to the construction throughout different periods in time, in order to establish which of the uses, the modal or the temporal one, is the primary one, from which the other is derived. Hilpert (2008) shows through a diachronic distinctive collexeme analysis that there is evidence against the account that future meaning developed out of the meaning of intentionality, as suggested in Heine (1995). Further, there is evidence for the primacy of temporal meaning over the epistemic modal one (Hilpert 2008: 155).

In the diachronic accounts of the rise of *werden*-future it has been further observed that there are in fact two constructions which are in competition, *werden* with an infinitive complement and *werden* with a present participle complement. Some accounts give one primacy over the other, but in the most recent proposal

¹Note that the English future construction with *will* may also be used in this way: *We keep telling him to stop but he will do it* can be interpreted as future but also as epistemic, that is just how he is, he does and will keep doing this whatever we say.

²It is impossible to give even a most succinct overview of the vast body of research devoted to the grammaticalization of *werden* as future tense marker in the present paper. The reader is directed to Hilpert (2008) and Osterkamp (2013) for an exhaustive account of current research as well as novel proposals and to Hartmann (2021) for the most recent advances into the formation of the future tense construction.

by Hartmann (2021) the two constructions are shown to first mutually attract each other and finally merge rather than develop from each other, a process facilitated by the collapse of the infinitive and the present participle into one form. The development entails an expansion in terms of verbs which can occupy the infinitive or present participle slot in the constructions, so that the group of verbs which can be used becomes more heterogeneous in terms of *aktionsart*, beginning with verbs denoting states and activities and gradually encompassing verbs expressing accomplishments and achievements (Hartmann 2021).

As exemplified in (4) and (5), for convenience repeated as (9) and (10), *varda* in Swedish could take both infinitives and present participles as complements; there are, however, no corresponding processes of reduction and the two forms are distinct throughout the whole documented history of the Swedish language.

- (9) (SAOB *varda*, example dated 1595)

At gud alzmectig warder at senda sit harda straff öffuer
that God almighty *varda*.PRS to send his.REFL severe punishment over
oss alla.
us all

‘That God almighty will send his severe punishment over us all.’

- (10) (SAOB *varda*, example dated 1928)

En hel del annat som du snart varder skådande.
a whole bit other which you soon become seeing
‘Other things, which you will soon be seeing.’

No similar convergence could therefore have taken place in Swedish. However, we can study the development in terms of both general frequency of the constructions with the infinitive and the present participle, as well as in terms of the verbs occupying the open slots in the construction (infinitives and present participles).

The account of previous studies of the diachrony of *werden* is relevant to the subject of the present paper, as there are similarities, both in etymology and use, between the two verbs, which may be reflected in similar developmental paths.

3 Methods, sources and tools

For the present study we excerpted a total of fifteen Swedish texts written down between 1300 and 1750. We chose prose texts, with both religious and secular prose included. For the sake of the present analysis, the texts were divided

into five sub-periods: 1300–1350, 1350–1450, 1450–1550, 1550–1650, and 1650–1750, with three texts in each period. The division of the corpus roughly corresponds to the conventional periodisation of the Swedish language, which is illustrated in Table 1 below. The acronyms used in referring to the texts are explained in the Sources section.

Table 1: The corpus and periodisation of Swedish

	Conventional nomenclature	Conventional periodisation	Periods in the present study	Texts included in the cor- pus
Old Swedish	Early	1225–1350	1300–1350	Bur, Pent, KS
	Late	1350–1526	1350–1450	KM, ST, SVM
			1450–1550	Linc, Luc, Troj
Modern Swedish	Early	1526–1732	1550–1650	Brahe, Gyll, Swart
	Late	1732–	1650–1750	Hiarne, Högström, Spegel

Details of each can be found in the Sources section at the end of this paper. Digitalised versions of the Swedish texts are freely available from *Fornsvenska textbanken* (<https://project2.sol.lu.se/fornsvenska>). In some cases the paper editions of the texts were double-checked to confirm that the digitalised versions do not contain errors. For each of the hundred-year periods, approximately 1800 VPs were annotated, evenly distributed across three texts (per period). In some cases there were slightly more, so that complete portions of text could be analysed. Table 2 shows the size of the corpus and the number of annotated VPs. Over 9000 VPs were tagged with information about (morphological) tense, mood, diathesis, subject animacy, position in sentence and type of clause (main vs subordinate)³. In the present paper, this data is used mainly to illustrate the relative frequency of *varda* and all verbs, as well as the relative frequency of *varda* and *bliva*. As *varda* is not high in frequency in proportion to all verbs, we have gleaned only 180 examples from the entire annotated material. This gives us some idea of the verb's constructional potential (i.e., what constructions it can be found in), but is not enough to follow its development. Therefore this study was completed with manual excerption of the 15 texts in their unabridged form. They were searched for all occurrences of *varda* in their entirety⁴. The search returned 1422 examples, which will be analysed qualitatively.

³The research presented here is part of a larger project centering around the rise of the periphrastic passive in Swedish and Danish, thus the paper does not utilize all annotated information. The tool used is DiaPass, created specifically for the project.

⁴All spelling variants and tense forms of *varda* were included in the search; it is however possible that a handful of examples were missed in the search, which should not affect the overall results.

Table 2: The material

Text	Entire text no of words	No of instances of <i>varda</i>	<i>varda</i> per 1000 words	Date (assumed)	Period
Pent	140760	356	2.53	1300	I (1300–1350)
Bur	191808	33	0.17	1330	I (1300–1350)
KS	21600	74	3.43	1330	I (1300–1350)
KM	10930	29	2.65	1380	II (1350–1450)
SVM	16240	40	2.46	1420	II (1350–1450)
ST	145479	497	3.42	1420	II (1350–1450)
Lucidarius	28175	151	5.36	1487	III (1450–1550)
Linc	9824	37	3.77	1520	III (1450–1550)
Troj	44380	8	0.18	1529	III (1450–1550)
Swart	51940	61	1.17	1560	IV (1550–1650)
Brahe	26380	4	0.15	1585	IV (1550–1650)
Gyllenhielm	53010	113	2.13	1640	IV (1550–1650)
Hiarne	11030	7	0.63	1660	V (1650–1750)
Spegel	32720	12	0.37	1680	V (1650–1750)
Hogstrom	17990	0	0.00	1747	V (1650–1750)

The number of hits for *varda* in each text varies not just according to the text's length (which is trivial, as we expect fewer hits in shorter texts), and not just according to the chronology, although we see a tendency for *varda* to gradually disappear throughout the periods studied. It seems also that, especially after 1450, the use of *varda* was very much dependent on the scribe's preferences (or rather dialect). Thus Brahe (1585) and Hiarne (1660) show very low values, while Gyllenhielm (1640) uses the verb often. We return to these differences and their likely origins in Section 4.

It is notoriously difficult to establish the meanings of constructions in historical texts. There is the risk of imposing modern reading on the material, as well as a lack of clear diagnostics in the context, which would clarify the meaning of a given structure. This problem becomes paramount when incipient structures are considered, as we expect them to be ambiguous and constitute bridging contexts (in the sense of Heine 2002), and an unambiguous analysis of such examples may not be possible.

There seems to exist only one unequivocal diagnostic of future reference, which is the presence of temporal adverbials indicating a future location in time. Disregarding the construction type (be it present tense, modal verb, verb of motion, etc.), if such an adverbial is present in the clause, the entire clause can be taken to have futurate reference.

A problem which nevertheless presents itself is the distinction between the immediate and the remote future. While a temporal adverbial indicating remote

future, e.g. *on the day of judgement*, may give an unambiguous future reading (see 11), it is less clear whether adverbials indicating immediate future, e.g. *immediately*, result in the same reading (compare with 12).

- (11) (Pent, ca. 1300)

Nw sighir war hærra innan læstinne at wi vardhom like
now says our lord within script.DEF.DAT that we become.PL like
ænglom æptir domadagh.
angel.PL.DAT after doomsday

‘Now our Lord says in the script that we will be like angels after the Day of Judgement.’

- (12) (Brahe, 1585)

efter theras fader war slagin i krich, som strax omtaladt
after their father was killed in war, which immediately discussed
warder.
becomes

‘After their father was killed in war, which will immediately be discussed.’

It may also be said that the futurate interpretation of (12) arises from the present tense and is mainly aspectual.

In this section we have presented the material used for the present study and discussed some aspects of approaching the data. The following sections are devoted to presenting an overview of the quantitative results (Section 4) and a more detailed and mainly qualitative analysis (Section 5).

4 *varda* in Swedish – an overview

In this section we present the use of *varda* in Swedish between 1300 and 1750. As shown in Table 3, *varda* is not a common verb in the OSw material. Table 3 illustrates the frequencies of both *varda* and its competition, the loanword *bliva*, in each of the periods studied. As we can see, the frequency of *varda* is relatively stable between 1300 and 1550 and falls significantly after 1650, while *bliva* gains significantly in frequency after 1550.

The verb *varda* could be used in a wide array of constructions, taking both verbal (infinitive INF, past participle PTCP or present participle PPart) and nominal (NP, AdjP) complements. Representative examples are quoted below (13–17).

Table 3: Frequencies of *varda* and *bliva* in the corpus

Period	Entire text no of words	<i>varda</i>	<i>varda</i> per 1000 words	<i>bliva</i>	<i>bliva</i> per 1000 words
1300–1350	354168	463	1.31	7	0.02
1350–1450	172649	566	3.28	211	1.22
1450–1550	82379	196	2.38	166	2.02
1550–1650	131330	178	1.36	926	7.05
1650–1750	61740	19	0.31	328	5.31
Total	802266	1422		1638	

- (13) [SBJ *varda* INF] (KM, 1400)
Nw wardhom wi tala om them som fore waro stadde.
 now become.1PL we talk.INF about them who before were met
 ‘We shall now tell about those who were met before.’
- (14) [SBJ *varda* PTCP] (Bur, ca. 1330)
þa varþ han takin mz myclom hepar i efeso stap.
 then became he taken with much glory in Ephesus town
 ‘He was then taken with much glory to the town of Ephesus.’
- (15) [SBJ *varda* PPart] (SVM, 1420)
Aff hwilko ther riddare-n wardh syrghiande.
 of which there knight-DEF became grieving
 ‘What made the knight grieve.’
- (16) [SBJ *varda* NP] (Bur, 1330)
tel han varþ røuare ok røuar-a mæstare.
 until he became robber and robber-PL.GEN master
 ‘Until he turned into a robber and master of the robbers.’
- (17) [SBJ *varda* AdjP] (SVM, 1420)
Tha wardh swin-ith spakare.
 then became pig-DEF clever.COMP
 ‘Then the pig became more clever.’

The loanword *bliva* is found in the same constructions, although it takes about 200 years since its introduction into Swedish for it to be used with all types of

complements (Skrzypek 2024). Throughout the entire period studied, *bliva* can also be used with locative adverbials. This use is clearly connected with the verb's original meaning 'remain', see (18).

- (18) [SBJ *varda* PTCP] (Hiarne, 1665)
Och alltså bleef Celadon uthi Epheso.
and thus stayed Celadon in Ephesus
'And so Celadon stayed in Ephesus.'

No corresponding examples of *varda* with locatives as complements were found in the material.

There are also less common uses of the verb *varda*, which are mainly found in the oldest material. They include an 'absolute' use, without any complements, when the verb can be taken to mean 'come into existence', see (6) above and (19) below.

- (19) [SBJ *varda* INF] (ST, 1420)
tha wardh swa mykyt thrang a gatumen / at the ey
then became so much throng on street.DAT.PL.DEF that they not
kundo fram koma.
could forward come
'Then (suddenly) there was such a throng on the streets that they could not come out.'

This construction is found mainly in Pent, the oldest text in the sample; some instances were identified in SVM and Luc as well, but none after 1550 in the corpus.

In Table 4 we show the distribution of *varda* across the constructions which were most common in the corpus, divided by periods. We give the absolute numbers of occurrences, as the percentage values are too low.

The quantitative overview shows that *varda* can appear in a number of constructions throughout the period studied, including constructions which are strong candidates for expressions of future tense, especially with regard to similar developments taking place in the closely related German and with the cognate *werden* at the centre (e.g. Hartmann 2021). We may note, however, that by 1450 *varda* is mainly found with past participle complements and the entire construction is passive (much like in German). Development beyond 1550 involves mainly the suppression of *varda* in favour of *bliva* (Skrzypek 2024).

In the following section we will present a qualitative analysis of the potential future constructions with *varda*.

Table 4: Occurrences of *varda* in the major constructions

	[SBJ <i>varda</i> NP]	[SBJ <i>varda</i> AdjP]	[SBJ <i>varda</i> PTCP]	[SBJ <i>varda</i> PPart]	[SBJ <i>varda</i> INF]	Total
1300–1350	78	145	94	9	22	348
1350–1450	66	199	195	17	11	488
1450–1550	16	44	95	6	0	161
1550–1650	9	17	122	15	3	166
1650–1750	1	2	12	3	0	18
Total	170	407	518	50	36	1181

5 *varda* in Swedish – a detailed analysis

As we have discussed in Section 2, it is challenging to discern future reference in historical material and it is only seldom that the context provides enough cues to be able to do so with any confidence. Linguistic meaning is notoriously difficult to quantify and measure objectively, but in this section we will attempt to discuss in more detail the material collected through manual excerption, sorted by construction type, including the three constructions that make the best candidates for a future tense construction, i.e., one with the infinitive, one with the present participle and the aspectual future. In Table 5 we show the frequencies of [SBJ *varda* INF] and [SBJ *varda* PPart] across the texts in the sample.

The most likely candidate for the future expression is the construction with infinitive as a complement of *varda*, [SBJ *varda* INF]. It is exemplified in (20) below, see also (13) in Section 4.

- (20) (Pent, 1300)
Nw varda the göra som han sagde.
 now become they do as he said
 ‘Now they will do as he said.’

There are only 36 similar examples in the entire corpus (see Table 5), the majority in the oldest text in the sample, Pent. This, however, is not the only text where we find *varda* with the infinitive. There are some examples in KM (1400), see (21), ST, see (22) and, somewhat surprisingly, as late as in 1640, in Gyll, see

Table 5: Occurrences of *varda* with the infinitive and with the present participle

Text	Entire text no of words	No of instances of <i>varda</i>	[SBJ <i>varda</i> INF]	[SBJ <i>varda</i> PPart]
Pent	140760	356	22	4
Bur	191808	33	0	1
KS	21600	74	0	2
KM	10930	29	2	0
SVM	16240	40	0	1
ST	145479	497	9	19
Lucidarius	28175	151	0	2
Linc	9824	37	0	1
Troj	44380	8	0	1
Swart	51940	61	0	13
Brahe	26380	4	0	2
Gyllenhielm	53010	113	3	1
Hiarne	11030	7	0	3
Spegel	32720	12	0	0
Hogstrom	17990	0	0	0

(23). In this respect Gyll is completely isolated, and most likely the use of *varda* there is strongly influenced by German.⁵

(21) (KM, 1400)

Nw wardhom wi tala om them som fore waro stadde.
 now become we speak about them who before were set
 ‘We will now speak of those who were living before.’

(22) (ST, 1420)

swa længe gudh ær j hymerike tha wardher iak her swa brinna.
 so long god is in heaven then become I here so burn
 ‘As long as God is in heaven, I will burn here so.’

⁵In his study of word order in Swedish, Magnusson Petzell (2014) shows some examples from Gyll which strongly suggest German influence on the syntactic level visible in the text. It is not unlikely that such influence could also include constructional interference with the verb *varda*, due to its similarity to *werden*.

- (23) (Gyll, 1640)
som historien uthvijsa varder.
 as history show will
 ‘As history will show.’

In KM, the construction is only found sporadically and seems to be restricted lexically, so that only infinitives of *verba dicendi* can appear there. It seems therefore that the construction was present in the oldest material⁶, but rather uncommon, and its later occurrences are a result of the influence of German.

We have remarked before that the borrowed *bliva*, originally found only with locative adverbials (see 18), gradually developed the polysemous meaning ‘remain/become’ and started appearing with the same complements as *varda*, i.e., with NPs, AdjPs and participles. However, we have not found a single example of *bliva* with the infinitive, which suggests that by the time the polysemy arose, the construction with the infinitive was no longer found in the material and the new verb could not be used with infinitives.

Even though rarely found, the construction with the infinitive comes closest to what might be regarded an (incipient) expression of futurity. Discussing the grammaticalization of the future tense, Bybee et al. (1994) state the following:

We hypothesize that all futures go through a stage of functioning to express the intention, first of the speaker, and later of the agent of the main verb (Bybee and Pagliuca 1987, Bybee 1988). The meanings that can feed the future path must be meanings that appropriately function in statements that imply an intention on the part of the speaker. (Bybee et al. 1994: 254)

It is not entirely clear whether there is any intentionality on the part of the subjects of the sentences we find where *varda* is complemented by an infinitive, with the exception of sentences with *verba dicendi* and first person subjects (as in 13). These contexts, however, are all of the subject being identical with the speaker. The structure can be seen as a bridging context, allowing for a broader interpretation of volition and intention as not restricted to the speaker but rather as characterizing the subject. In use with other verbs, the subject may not be identical with the speaker, yet retain intentionality (see Heine (1995) for a similar explanation of the development of the German *werden*, based on synchronic considerations and Hilpert (2008) for a diachronic corpus account).

⁶SAOB quotes more recent examples of *varda* with infinitive, the youngest are from the 1500s (see also (4)).

In sentences where the subject and the speaker are not identical (see 20), what we find may equally well be the original, lexical meaning of *varda*, i.e., ‘to begin’. Translated into English, *varda* in these examples can mean ‘will’, but it can also mean ‘begin’. It seems that to consider them futurate, we would impose an anachronistic reading. However, in examples similar to (22), ‘as long as God is in heaven, I will burn here’, the wording strongly implies that the burning has already started (and is not about to begin) and the speaker expects it to continue for a time yet, rendering the construction futurate. Such examples are confined to the earliest texts in the corpus. However, they seem to have been productive (in the sense of Barðdal (2008), based on the concept of extensibility, i.e., being used with new verbs) and not merely calqued into translated texts from the original ones, as we find examples of futurate *varda* with infinitive also in the provincial legal texts from late 1200s.

(23) on the other hand comes from a relatively late text, from 1640. It is therefore striking. The only explanation of its existence (and two more similar examples can be found in the same text) that we can offer is that it is the author’s proficiency in German (see also Magnusson Petzell 2014) that allows him to use this construction. This is strengthened by the fact that the examples in Gyll are isolated, and similar constructions were not found in other contemporary texts.

Summing up, there are too few examples of [SBJ *varda* INF] to categorically state that what we see is incipient grammaticalization, but enough to demonstrate that the construction was present in the language use, albeit under some constraints, at least in the early 1300s, and that it was acceptable enough to be used in the translation of ST in the early 1400s. Its occurrence in Gyll in 1640 is very difficult to explain unless we assume German influence. The most diverse examples (in terms of verbal complements) are found in Pent, the oldest text in the sample. Many authors who have studied its linguistic form have noted its archaic nature, in particular Hesselman (1927)⁷, see also Thorell (1959). Furthermore, Pent is generally considered free from foreign influence (Wollin 2001). This suggests that the use of *varda* in this material represents an authentic construction.

Another potential candidate for a future tense construction is one with the present participle in the complement position, [SBJ *varda* PPart]. It is exemplified in (24) below.

⁷Då man läser något parti av MBI, särskilt i de berättande styckena, t.ex. delar av Genesis, får man rätt snart ett högst tilltalande intryck av en mycket gammaldags och kärnfull svenska, en språklig äkthet och på samma gång rikedom, som man eljest icke lätt finner motstycken till i vår gamla litteratur. (Hesselman 1927: 21)

When one reads an excerpt from MBI, especially in the narrative passages, e.g. parts of Genesis, one gets a highly pleasing impression of a very old-fashioned and pithy Swedish, a linguistic authenticity and at the same richness which one otherwise does not easily find parallels to in our old literature.

(24) (Petri, 1530)

*Allom them som thenna Cröneke läsande eller hörande warda, önskar
all.DAT them who this chronicle reading or hearing varda wish
jach Olauus Petri, predikere i Stocholm, saligheetennes kundtskap.*

I Olaus Petri preacher in Stockholm bliss.GEN wisdom

‘All those who are/will be reading or hearing this chronicle, I, Olaus Petri, preacher in Stockholm, wish the wisdom of bliss.’

This construction is reminiscent of the German construction with *werden* and the present participle, which under a long period of time competed with the construction with the infinitive, for the status of a futurity marker (Hartmann (2021) and references therein). Most likely due to a blurring of present participle and infinitive forms (*lachend* vs. *lachen* ‘laughing’ vs. ‘(to) laugh’, see Hartmann 2021: 367) the construction was attracted to the one with the infinitive and the two converged, resulting in [SBJ *werden* INF] as the only marker of futurity in Modern German (Hartmann 2021).

In [SBJ *varda* PPart] the mutative, inchoative or ingressive meaning is apparent. In each of the examples (25–27) below, we see an event presented by means of [SBJ *varda* PPart] construction, which is shown to be the result of an earlier event (suggested in the context by *aff hwilko* ‘by which’, *nar han saat* ‘when/after he sat down’ and by *after thett att* ‘after that’).

(25) (SVM, 1420)

Aff hwilko ther riddaren wardh syrghiande.

of which there knight-DEF became grieving

‘What made the knight grieve.’

(26) (Linc, 1520)

tha böriadhe hālga iomffrun til at mykyt sokkas oc gratandhe wardha.

then began saint virgin.DEF to to much sigh and crying become

‘And the saint virgin started to sigh and cry.’

(27) (Brahe, 1585)

Theras fader war slagin i krich, som strax omtaladt warder.

their father was killed in war which immediately discussed become

‘Their father was killed in the war, which will now be discussed.’

We can see that the [SBJ *varda* PPart] construction has the potential to signal futurity, or at least, to show one event as taking place as a result of (and therefore also later than) another, earlier event. However, the [SBJ *varda* PPart] construction does not show any signs of grammaticalizing beyond the resultative meaning (as witnessed by 27 from late 1500s) and gradually disappears as *varda* gives way to *bliva*.

Finally, we consider the constructions in which *varda* is found in the present tense, [SBJ *varda*_{pres}], irrespective of the complement, which means we include also instances of the periphrastic passive construction [SBJ *varda* PTCP], as exemplified in (28).

(28) (SVM, 1430)

Waktin fore thy idher thz i wardhin ey swik in aff gire.
 guard for it.DAT you that you *varda* not disappointed of greed
 ‘Be on your guard so that you will not be let down by greed.’

A first observation is that there is a significant difference between time before 1450 and after this date. We have noted before that after 1450, *varda* appears mainly with past participles and these constructions are instances of passive voice. Interestingly enough, this change coincides with a change in the distribution of *varda* amongst different tenses. The numbers, based on the annotated material of ca. 9000 VPs, are given in Table 6. Examples with modal verbs were excluded from the analysis, as in these it is the modal verb that is the locus of temporal information.

Table 6: Distribution of *varda* across tenses

Period	Present	Past	Perfect	Past perfect
1300–1350	38.10%	52.38%	0%	0%
1350–1450	31.11%	66.67%	0%	0%
1450–1550	6.52%	76.09%	0%	15.22%
1550–1650	7.14%	85.71%	0%	7.14%
1650–1750	0%	93.75%	6.25%	0%

The sharp fall in numbers after 1450 is partly to be seen as a result of there generally being very few clauses in the present tense in the material in the period 1550–1650. However, in 1450–1550 and 1650–1750, the percentage of clauses in the present tense is much higher than the numbers for *varda* would suggest. For the sake of clarity, in Table 7 we show the results for the present tense first for all annotated VPs (including any finite verb, with the exception of modal verbs), then those with *varda* (repeated from Table 6).

Table 7: The present tense VPs in the corpus

Present tense	All VPs	<i>varda</i>
1300–1350	47.06%	38.10%
1350–1450	25.76%	31.11%
1450–1550	31.79%	6.52%
1550–1650	3.87%	7.14%
1650–1750	18.79%	0%

Tables 6 and 7 show that the frequency of *varda* in the present tense is on a comparable level to the frequency of all verbs in the present tense until 1450. Then a sharp drop occurs, which cannot be explained by the lower values for the present tense alone (although period 1550–1650 seems to be rather extreme and *varda* reaches higher values than all verbs). What mainly transpires from these data is that *varda* is by no means strongly affiliated with the present tense, apart from the first two periods studied.

How is the distribution of *varda* in different tenses, in particular between the present and the past tense, relevant to the present study? We would like to argue that the potential futurate meaning of constructions with *varda* is aspectual. Since *varda* denotes a change of state (mutative meaning), or (in many contexts) can be read as ‘begin, come into existence’, in combination with the present tense it will render the whole construction futurate, as the change is usually not instant, but takes some time, shorter or longer. Therefore the speakers are aware of the result of the action (the end-point of the change) being located further from the moment of speech, not being simultaneous with it. The idea of future reference is thus inherent in the meaning of the verb and the future, relative to S (speech time), and becomes fully actualized in the present tense.

We are thus left with a last diagnostic of possible future reference, the presence of temporal adverbials.

(29) (Pent, 1300)

thy at sanctus gregorius sighir aff oss at hælge mæn æptir
 it.DAT that sanctus Gregorius says of us that holy men after
domadagh vardha swa lyusir oc gønom skærir at (...)
 doomsday become so light and through transparent that (...)
 ‘Thus Saint Gregory tells us that holy men (will) become so light and
 transparent after the Day of Judgement that (...)’

We note that the verb *varda* is given in the present tense in these examples. It is very difficult, if not outright impossible, to tease apart the different cues through which the future reference seems to arise. Apart from the verb *varda* itself, and the present tense form, we find adverbials such as *strax* ‘immediately’. Also the adverbial *nw* ‘now’ could be taken to bring to the sentence the reading of immediate future, see (21).

In this section we have discussed examples of potentially futurate constructions with *varda*, i.e., constructions with the infinitive or the present participle or the aspectual uses of the verb. The common denominator of the constructions is that the verb’s lexicality is present, i.e. it means ‘become’ or ‘come into being’ or even ‘begin’ and the whole construction is inchoative or ingressive.

In the following section we bring all the results together to discuss to what extent *varda* could be considered a future tense marker in OSw and why there was no further grammaticalization.

6 Discussion

The aim of the present paper was to give a brief history of the verb *varda* in Swedish and, given that some of its uses in the oldest extant texts is reminiscent of nascent future tense constructions, examine the circumstances in which this potential grammaticalization into a future tense marker was halted.

We have seen examples of *varda* being used as a marker of future tense (with infinitival complements), reminiscent of the development that has taken place in German (of a cognate verb *werden*). In other words, we have some arguments beyond the cross-linguistic comparison and the developments in a closely related language to support the hypothesis that *varda* can be treated as an incipient futurity marker, whose grammaticalization into a fully-fledged exponent of future reference is for some reason halted. The examples are sporadic, and concentrated in the oldest extant texts, with the exception of ST from ca. 1420, which is a translation of a MLG text, heavily influenced by the MLG original and Gyll from 1640, where many other syntactic influences of German can be found. These examples are all of the construction with the infinitival complement, illustrated in (22) and (23).

It is not possible to discuss *varda* without making a reference to *bliva*, a verb borrowed from MLG in the late 1200s, which gradually takes over all of *varda*’s original domains, with one significant exception. As we have noted, there were no examples found with *bliva* occupying the finite verb slot in the construction in the entire material. It seems that this is the only construction where *varda* could

be found in which *bliva* does not suppress it, which suggests that by the time *bliva* developed towards the polysemous meaning ‘remain/become’ and started appearing in constructions with adjectival and nominal complements, the construction with the infinitive was no longer found in the material and the new verb could not be used with infinitives. Interestingly enough, there is contemporary evidence for the rise of a futurate construction with *bliva* and an infinitival complement in Norwegian (see Mikkelsen, this volume). To my knowledge, similar uses of *bliva* are not found in MSw.

Another candidate for a future tense construction is one with the present participle in the complement position. This construction survives as long as the verb does in the standard language, and its counterpart in MSw is [SBJ *bliva* PPart]. The construction is slightly more common than the one with the infinitive, but its overall usage remains low throughout the entire period studied. In contrast to German, there is no attraction between the constructions, and the present participle and the infinitive are formally distinct in MSw. As a consequence, any future reference which might be inferred from the construction is a result of the verb *varda* used in the present tense rather than the use of the present participle.

On the other hand, we see *varda* become increasingly associated with the slot in a participial construction (with past rather than present participles), at the expense of its earlier association with other verbal, adjectival or nominal complements (see Table 4). The mutative, or ingressive, meaning is present throughout the period studied, but since the verb is more and more often found with passive participles (which presuppose an agent for the action), the original non-agentive meaning of the verb (‘it came to pass’, ‘it happened’ rather than ‘it was done’) is backgrounded and *varda* can more easily enter into constructions with participles derived from highly agentive verbs, shifting the entire construction towards passive rather than inchoative, ingressive or resultative ones. We may see that there is a parallel here with the grammaticalization of the future tense marking, where there is an increase in intentionality, which is closely intertwined with agentivity. In fact, both can be seen as two sides of the same coin: while agentivity is mainly focused on the past or its present results, intentionality concerns the future, yet in both cases we find that the Agent is in control. However, the orientation of the verb is different: while in the passive voice construction, it is oriented towards the subject that is not the Agent but rather the Patient (and thus does not entail volition or intentionality⁸), in the future tense construction,

⁸As this paper concerns the (potential) future tense constructions, we cannot go into too much detail concerning the passive voice construction here. However, the periphrastic passive construction [SBJ *bliva* PTCP] in MSw does entail some intentionality on the part of the subject

it is oriented towards the subject that is the Agent and capable of intentional activity.

It is naturally not a given that it was this process alone that halted a potential development towards future marking, as the German *werden* does grammaticalize in both functions. Rather, it seems that the lack of a similar development in Swedish is a result of several factors. Firstly, the nascent future constructions (with the infinitive or the present participle) have a very low usage throughout the periods studied. Secondly, as soon as *bliva* develops the polysemous meaning ‘to remain / to become’, all the developments in *varda* are closely linked to what is happening with *bliva*, and its further development into a passive auxiliary. From being an antonym to *varda* (in the sense of Löbner (1990: 93)), *bliva* gradually converges with it, entering into the same array of constructions in which *varda* was previously found. Significantly, however, not a single example of *bliva* with the infinitive is attested, which could suggest that the construction itself was of a highly limited use.

Most importantly, however, at no stage does the use of *varda* involve the notion of intentionality, apart from the few examples with the infinitive. Admittedly, some authors have pointed out that intentionality need not be part of the development, e.g. Traugott (1978) and Dahl (2000), which may instead proceed through inchoative, ingressive or resultative meanings. These meanings are present in all uses of *varda*. Yet it seems that only when coupled with some measure of intentionality of the subject they can lead to the development of the future reference.

7 Conclusions

We have discussed the developments and ultimate demise of a construction [SBJ *varda* INF], which was present in the oldest extant Swedish texts, and could be argued to have a futurate reference, but which quickly disappears from the documented language use. At the same time, we see that where *varda* is retained, it can still appear with infinitival complements as late as in the early 1900s (according to SAOB). Even though the futural use never becomes part of the standard idiom, it is present in the Swedish language for several centuries. One of the reasons for the incomplete grammaticalization must be sought in the replacement of *varda* by *bliva*. Had it been retained, its original constructional potential, in particular

(= the patient), in contrast to a morphological passive construction [SBJ V_{pass}], which does not. The reader is directed to Engdahl (1999) and Laanemets (2012) for a more detailed account on the synchronic aspects and to Skrzypek (2024) for diachronic ones.

the possibility to take an infinitive complement, could have been strengthened by the renewed contact with German (Middle Low German, as illustrated by the ST and High German, as seen in the 1600s), with its already grammaticalized future tense *werden*-construction. It is of course ‘the land of might-have-been’. Nevertheless, the study of change that did not happen, despite there being a potential for a certain change, is an interesting exercise in language history and the forces which influence language change. Another candidate for such a study would be the Swedish *vilja* ‘will’, which appears with future-like uses in OSw, but then develops into the marker of volition alone (unlike the Danish and Norwegian *vil*).

We know that *varda* does not grammaticalize into a future tense marker, but our intention was to test the possible differences in context and use which may have led to this divergence of closely related languages, Swedish and German. Naturally, language change which takes place in one language need not translate into identical or even similar developments in another. Yet there are developmental patterns we repeatedly find cross-linguistically, which synchronically take the form of implicational universals and diachronically of paths or clines. Much research is devoted to judging why a change takes place, but it is also relevant to ask why it does not, when apparently there is room for it.

The conclusion is that to understand language change we need to consider not only the particular construction, which is the locus of change, but also the constructional potential of its components. What happens to a given item outside a given construction is of consequence for language change. Either the divergent development results in a split in the lexeme (separate lexical entries for a given item, originally one word) or the course of the change may be stayed or altered.

Abbreviations

1	first person	INF	infinitive
COMP	complementizer	PL	plural
DAT	dative	PRS	present
DEF	determiner	PASS	passive
GEN	genitive	REFL	reflexive

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Chapter 6

The future that may still be: The spread of *blir å INF* in contemporary Norwegian

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This article investigates the semantics of the incipient future construction *blir å INF* in contemporary Norwegian. Traditionally considered a feature of northern dialects, the construction is now used in writing by speakers throughout the country. We take data from a corpus consisting of blogs and manually annotate uses for various semantic features which we then investigate quantitatively. We show that the construction is largely non-distinct with respect to the other future constructions. However, there does appear to be evidence that the construction is used significantly more frequently as indicative of low-certainty predictions. Yet, in this respect, the construction overlaps with uses typical of the *vil INF* construction. The study considers these results in terms of the diachronic spread and synchronic polysemy. First, we discuss whether *blir å INF* is encroaching on the semantic territory of *vil INF* specifically or is being used as an unmarked marker of futurity relative to the other constructions. Second, we consider the two main hypotheses concerning the diachrony of *blir å INF*: a development of an intentional construction, or a loan from Sámi. Our findings seem to support the second hypothesis. However, the synchronic nature of the data and their geographic distribution constitute two caveats that impede strong conclusions.

1 Introduction

The verb *bli/bliva* ('become') was borrowed by mainland Scandinavian from Middle Low German and/or Middle Dutch in the late Middle Ages (Markey 1969). Despite the cross-linguistic observation that verbs signifying 'be' or 'become' often



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develop into future markers (Bybee et al. 1994), this did not happen in any standard variety of Scandinavian (Skrzypek 2026 [this volume]). There is, however, evidence that one such instance is generalising to standard Norwegian. Example (1) is taken from Norwegian national media and shows the periphrastic *blir å INF* ('become to'), indicating futurity, has made its way into mainstream discourse.

- (1) *Hvis jeg ikke får jobb i Tromsø, BLIR jeg Å SØKE i Canada.*
if I .NEG get job in Tromsø FUT I to seek in Canada
'If I don't get a job in Tromsø I will apply in Canada.'¹

Traditionally, the *blir å INF* construction is considered a distinct feature of Northern Norwegian dialects and there have even been discussions regarding its appropriateness in writing in the language magazine of the official Norwegian Language Council (Jahr 2009). Over the past 10 years, however, the construction has received attention on social media (*Språkspalta* on Facebook) and radio (*Språkteigen* on NRK, 3. October 2014), and according to some informants in these channels, it has been adopted by speakers in other parts of the country.

In this paper, we investigate the possible spread of this construction and its place alongside other verbal expressions of futurity. First, we need to establish the extent to which examples of *blir å INF* can be found in written contemporary Norwegian. Given that it is considered a dialectal feature, we have chosen to work with a corpus consisting of blogs, in which speakers are less likely to be influenced by normative pressure on their language use than in the data typical of the national reference corpora. Second, having established that there likely is some spreading of this construction, we compare the semantics of *blir å INF* with those of alternative future referring constructions, such as *skal INF* ('shall'), *vil INF* ('will') and *kommer til å INF* ('come to'). In this, we seek to answer the question: is *blir å INF* encroaching on the semantics traditionally associated with one of the other three constructions, or is it an entirely new aspectual encoding of the future in Norwegian?

To answer this question, we use a methodology that combines corpus data, qualitative analysis and multivariate statistics (Section 3), and compare *blir å INF* with the other future constructions in Norwegian along four dimensions of meaning:

- Future function: whether the construction is used to express an intention, a plan or a prediction.

¹Source of example NRK Troms og Finnmark 13th October, 2023

- Conditionality: whether the conditions for carrying out the future event are given at the time of speech, dependent on a future contingency or unknown.
- Temporal proximity: how near or far the future event is located on a temporal scale relative to the moment of speech.
- Speaker certainty: how certain the speaker seems to be that the future event will actually be carried out.

The results of the analysis (Section 4) show that *blir å INF* is encroaching on a part of the semantic space of *vil INF*, namely as an expression of low-certainty predictions. We suggest two possible (independent, but not mutually exclusive) motivations for this: the first has to do with the social meaning traditionally associated with *vil INF*, the second is the absence of an “unmarked” or “neutral” future marker, i. e., one that is not associated with the specialised contexts of *skal* (high certainty intentions and plans), *vil* (low certainty predictions, contingent and temporally indefinite future events) and *kommer til å INF* (high certainty predictions and atemporal future events).

However, due to limitations in the availability of data (Section 3), the results should be seen as preliminary. Furthermore, the data consists of utterances produced by what may turn out to be two different groups of speakers: the “natives”, for whom the construction has been part of their dialect for generations and the “adopters”, for whom the construction is likely to be an innovation. For this reason, we cannot exclude the possibility that the data represent two separate grammatical systems: one in which the construction has developed from childhood on a par with the other constructions; another in which it is a new addition to an already established system of future constructions. When we seek to explain a phenomenon such as the spread of *blir å INF*, it is important to take such a divide into account and ask whether we are witnessing a gradual spread of a linguistic form from one variant to another, or a more sudden change where the new form is adopted, first by a reduced number of speakers (the “early adopters”), before it rapidly generalises across the broader community.

Finally, we discuss our findings from the perspective of grammaticalisation. Some authors have argued that all futures go through a stage of intention (Bybee et al. 1994, Heine 1995), while others defend alternative paths of grammaticalisation, such as ingressive, inchoative, or resultative meanings (Traugott 1978, Dahl 2000). While intentional readings are part of the meaning of *blir å INF*, our data suggest that it is not a defining feature of the construction today. This result must be interpreted in light of what was said in the preceding paragraph: if what

we are seeing is representative of the “native” grammatical system, then the intentional readings could be remnants of a previous, intentional stage; if, on the other hand, we interpret our data as representative of the grammatical system of the “adopters”, then the results would suggest that the construction is establishing itself as a prediction-based future marker from the outset. The results will be interpreted in light of the two dominant hypotheses concerning the origin of *blir å INF*: a development of the intentional-obligational *det blir å INF* construction or an analogy on the Sámi verb for ‘become’ (Section 2).

The rest of the paper is organised as follows: Section 2 presents a short literature review of the *blir å INF* construction in particular and Norwegian future constructions in general. Corpus data and methods are introduced in Section 3, Section 4 presents the results, and Section 5 discusses the results both synchronically and diachronically.

2 Norwegian future markers and *blir å INF*

The origin of *blir å INF* as a future marker in Norwegian is unclear, but there are three main hypotheses. Iversen (1954) proposes, first, that *blir å INF* is a development of the expletive *det blir å INF* (‘it becomes to’) construction (2), which is found across most varieties of Norwegian (including the standard):

- (2) *Skal du studere, så BLIR DET Å LEVE på lån.*
 shall you study then FUT EXPL to live on loan
 ‘If you are going to study, you will have to live off loans.’ (Iversen 1954)

His second hypothesis is that *blir å INF* is a loan construction from the Sámi verb *šâd’dât* (‘become’) (3):

- (3) *Don ŠÂDDÂK oaid’net.*
 you FUT see
 ‘You will see.’ (Nielsen 1926)

This hypothesis is questioned by Vannebo (1979) because of the low social status of the Sámi languages relative to Norwegian, and he suggests, as a third hypothesis, that *blir å INF* is formed by analogy to the Norse verb *verða* ‘become’, which developed into a future marker in some Norwegian dialects (also in earlier Icelandic and Swedish cf. Skrzypek 2026 [this volume]).

While it is not the goal of this paper to investigate the origin of *bli å INF*, the original geographical extension of the construction certainly lends credibility to

the Sámi hypothesis, as it corresponds precisely to those areas of Norway (blue dots, Figure 1) where there has been widespread contact between Norwegian and Sámi. Over time, the construction has spread from Finnmark and Troms to the entire Nordland region (red dots, Figure 1).

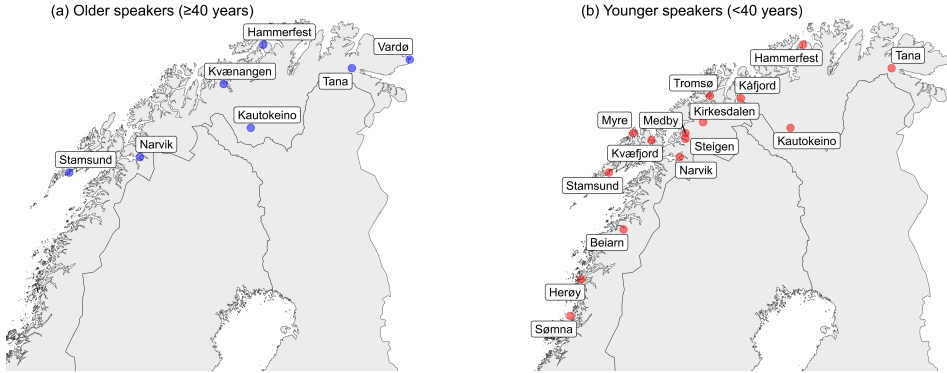


Figure 1: The use of *blir å INF* in the Nordic Dialect Corpus (Johannessen et al. 2009). Image created with the R packages *maps* (Becker et al. 2025) and *giscoR* (Hernangómez 2026) using geodata from GISCO.

In our data, consisting of written online personal blogs (see Section 3), we find that *blir å INF* is used not only by speakers from the North, but by speakers from all the other major dialectal areas of Norway: the East, the Southwest and the Centre (Trøndelag). While still infrequent in writing, we assume that the construction is in the incipient stage of spreading well outside of its traditional area, and that a comparison with the other expressions of futurity in Norwegian (*skal INF*, *vil INF* and *kommer til å INF*) is warranted.

It should be noted that the literature on futurity in Norwegian is somewhat limited. Perhaps due to the fact that Norwegian does not have a single morpheme to express futurity (i.e., “future tense” in the traditional sense) and that the future constructions are used to express many other (mostly modal) meanings, the Norwegian linguistic community has preferred to focus on modality more broadly (Eide 2020). Nonetheless, there seems to be a distinction in the usage of the “intention-based” *skal INF* on the one hand, and the “prediction-based” *vil INF* and *kommer til å INF* on the other (Dahl 2000). That is, the former is mainly used to express what the speaker intends to carry out, while the latter two are preferred when talking about what the speaker thinks will happen. Earlier authors have focused on distance in time and degree of certainty: *skal INF* is used to talk about events that are happening in the near future and which the speaker

feels more confident about, relative to the more distal and uncertain *vil INF* (Mac Donald 1982). Lie (2005) points out that *skal INF* is dispreferred with certain conditional clauses, and this observation is corroborated by Mikkelsen & Hartmann (2022) using data from the NoTa and BigBrother corpora. These authors also observe that *skal INF* prefers syntactically more complex environments (i.e., context of negation, interrogation, and subordination) than the other two constructions. As for the difference between *vil INF* and *kommer til å INF*, Mac Donald (1982) and Vannebo (1985) claim that it is mainly one of register (*vil INF* being the more formal of the two), while Hasselgård (2015) finds that *kommer til å INF* favours confident predictions on the basis of present knowledge and events that come about as a matter of course (i.e. predestination).

To the authors' knowledge, no study to date has attempted to give a detailed description of the semantics of *blir å INF* in relation to the other future constructions in Norwegian. However, the different hypotheses concerning its origin may give us some clues: should the construction be a development of the expletive *det blir å INF* construction, we might expect it to retain more intentional readings; if the construction is formed by analogy on the Sámi *šad'dât*, which according to Nielsen (1926) translates to *kommer til å INF* in Norwegian, then we might expect it to behave more like the predictive-based futures. In any case, the differences in meaning between the different constructions are subtle, and there are many contexts in which speakers judge the constructions as equally grammatical (4), and where the change in meaning between them is minimal, if at all noticeable:

- (4) *Jeg (SKAL/VIL/BLIR/KOMMERTILÅ) SØKE i Canada hvis jeg ikke får*
 I FUT seek in Canada if I NEG get
jobb i Tromsø.
 job in Tromso
 'I will apply in Canada if I don't get a job in Tromso.'

This does not imply, however, that these constructions are simply different means of saying the same thing. In certain specialised contexts (possibly remnants of different stages of grammaticalisation diachronically / different uses synchronically) (5–9), the constructions are non-interchangeable and may or may not have future time reference:

- (5) *Du SKAL ikke STJELE. (future time reference, deontic modality)*
 you FUT NEG steal
 'Thou shalt not steal.'

- (6) *Han SKAL VÆRE dyktig.* (present time reference, epistemic modality)
 He FUT be capable
 ‘He’s supposed to be good.’
- (7) *Jeg venter på at han SKAL BLI ferdig.* (future time ref., inceptive)
 I wait on that he FUT be finished
 ‘I’m waiting for him to finish.’
- (8) *VIL du GIFTE deg med meg?* (present time reference, deontic modality)
 FUT you marry you with me
 ‘Will you marry me?’
- (9) *Jeg KOMMER TIL Å TENKE på noe.* (present time ref., inceptive)
 I FUT think on something
 ‘I just remembered something.’

Examples such as these illustrate that the future-referring constructions have many other distinct functions, and this is before we take into account highly frequent chunks related to particular subject-construction-verb combinations, such as *det vil si* (elucidation construction), *jeg vil si* (hedging construction) or *det skal sies* (concessive construction), to name but a few. All this complexity is part of speakers’ knowledge about these constructions and may contribute to biases in the choice of constructions, even in contexts where there is no apparent difference between them.

3 Data and methods

As noted, *blir å INF* is considered a dialectal feature that is rare in formal writing. For this reason, we looked at informal written language drawn from the Norwegian Blog Corpus (Mikkelsen 2021). This corpus was compiled using the Python script *Trafilatura* (Barbaresi 2021) and consists of a total of 841 blogs (approximately 18 million words), with entries spanning the years 2010–2020. During the scraping process, partial speaker metadata (age, sex, and region) were retained where possible (extracted from the “about me” pages or similar). All instances of the verbs *skal*, *vil*, *kommer* and *blir* in the present tense were drawn using the *CasualConc* concordancer version 2.1.6 (Imao 2021) with a context of 500 characters left and right of the verb. Using different sorting techniques, instances without an infinitival complement were discarded. This process resulted in a pool of roughly

100,000 items, from which subsamples of 200 occurrences per construction were drawn.

The subsamples were manually checked and both non-interchangeable instances (5–9) and interchangeable instances with no future time reference (habituals and generics) were removed. Negated and interrogated instances with future time reference, which both correlate with *skal INF* and are often seen as indices of processing factors (Mikkelsen & Hartmann 2022), were also excluded from study. The *blir å INF* construction (1) was separated from the expletive *det blir å* construction (2) and borderline cases with no overt subject (10). Since the expletive construction is considered a different construction altogether and instances like (10) are ambiguous with respect to the interpretation (the unexpressed subject could be interpreted both as an expletive *det* ‘it’ or as a first person *jeg* ‘I’), these types of examples were removed from further analysis. Furthermore, the latter null-subject construction might be perceived as a bridging-context in the sense of Heine (2002), facilitating the acceptability (and thus spread) of *blir å INF*.

- (10) *BLIR nok Å STRIKKE fremover, uansett.*
 FUT enough knit forwards anyway
 ‘I’ll keep knitting anyways.’ / ‘There’ll be plenty of knitting anyways.’
 (strikkehege)

The pseudonym of the blogger will be given in parentheses. Each subsample of *skal INF*, *vil INF* and *kommer til å INF* was controlled for potential intralinguistic variation by extracting only one item per speaker per construction. In the case of *blir å INF*, this was not possible, as the total number of instances after manual disambiguation was reduced to 76 items (34 speakers). This imbalance in the data in disfavour of *blir å INF* is a disadvantage that is not easily resolved. While it would be possible to pool data from other sources, this will not be easy as the construction is still infrequent in writing, and it will also raise issues of comparability. The distribution of speakers by region following the division traditionally used in Norwegian dialectology (East, Southwest, Centre and North) are given in Table 1.

The total dataset of 676 items was subject to feature analysis, the first step of behavioural profile approach (Dirven et al. 1982, Geeraerts et al. 1994, Gries 2003, Divjak & Gries 2006), which uses manual analysis of each sample to identify patterns of usage interpretable as functional criteria for a quantified description of grammar. The patterns are typically identified with the use of multivariate statistics, and in this study, we employ both multiple correspondence analysis (MCA) and loglinear analysis. MCA is a multivariate exploratory space reduction technique for categorical data analysis that reveals frequency-based associations in

Table 1: Constructions and speakers by region

Construction	East	Southwest	Centre	North	Unknown
<i>Blir</i>	10	18	15	31	2
<i>Kommer</i>	98	52	13	12	25
<i>Skal</i>	96	55	12	14	23
<i>Vil</i>	91	44	14	17	34

the form of biplots. These biplots seek to represent the degree of likelihood of combinatorial possibilities of the different features. As such, they produce maps of the different uses, understood as clusters of those combinatorial possibilities (Glynn 2014). Loglinear analysis allows us to examine some of the more frequent combinatorial possibilities to see if the frequency or infrequency of their occurrence is statistically significant (Glynn 2014).

After sample extraction, the data were submitted to blind annotation, i.e., the actual construction was hidden from the annotator to minimise the effects of annotator bias. Annotation was performed by a single native speaker (the first author of this study). From the literature review in Section 2, four semantic dimensions were distinguished. First, the basic distinction between intention-based and prediction-based was operationalised as Future Type: ‘intentions’, ‘plans’ and ‘predictions’. These were distinguished by virtue of an entity called the “director”, defined as the entity willing and able to carry out the future event. When the speaker is director, we are dealing with an intention (11):

- (11) *På Facebook BLIR jeg Å DELE coachteknikker.*
 On Facebook FUT I to share coachingtechniques
 ‘I will share coaching techniques on Facebook.’ (slottsfruen)

Note that the director can be backgrounded, i.e., not overtly expressed, so that examples like (12), in which the speaker is clearly both willing and able to carry out the future event (talking about bullying) even though the syntactic subject (“today’s topic”) is not the director, were annotated as intentions:

- (12) *Dagens tema BLIR Å OMHANDLE mobbing.*
 Todays topic FUT to concern bullying
 ‘Today’s topic will be bullying.’ (=In this blog entry I will talk about bullying.)’ (renatev)

When the speaker is not a director, but merely a reporter of a future event that is to be carried out by the director (typically backgrounded or expressed syntactically as the subject of the construction), we are dealing with a plan (13):

- (13) *Boken BLIR Å INNEHOLDE (...) inspirerende ideer.*
 bookDEF FUT to contain (...) inspiring ideas
 ‘The book will be filled with inspiring ideas.’ (livetdetgode)

Predictions are statements in which there is no director, neither explicitly mentioned nor backgrounded (14), and where the speaker takes on the role of an evaluator, assessing the likelihood of the future event:

- (14) *Ikke hjelper det å fortelle at taxi hjem til meg BLIR Å KOSTE en formue.*
 Not helps it to tell that taxi home to me FUT to cost a fortune
 fortune
 ‘There’s no use in telling [them] that taking a taxi home will cost a fortune.’ (majbee)

Both Temporal Proximity and Speaker Certainty were annotated using a nine-point Likert scale, which were then collapsed to binary distinctions. In this study, for Temporal Proximity, ‘proximal’ (15) future uses refer to events that will take place within the same day relative to the moment of speech, while events that are located further away (days, weeks, months, years...) are ‘distal’ (16). For Speaker Certainty, expressions that contain an overt statement of certainty (17) and also unmarked future statements (14) are annotated as ‘high certainty’. On the contrary, statements in which there is some degree of doubt or reservation (18) are annotated as ‘low certainty’.

- (15) *Snart kommer X hit, regner med at vi BLIR Å CHILLE masse.*
 Soon come X here, calculate with that we FUT to chill a-lot
 ‘Soon X will be here, I’m sure we will have a good time.’ (sophieelise)
- (16) *Får håpe at dei BLIR Å SKAPE mykje glede i åra som kjem!*
 get hope that they FUT to create much happiness in years to come
 ‘Let’s hope they will be the source of much happiness in the coming years!’ (ladyingunn)
- (17) *Har innstilt meg på at jeg BLIR Å STRYKE uansett.*
 Have adjusted me on that I FUT to fail anyway
 ‘I have come to terms with the fact that I will fail anyway.’ (sandraeliene)

- (18) *Langt fra overbevist om at dette er en metode jeg BLIR Å BRUKE*
 far from convinced about that this is a method I FUT to use
flere ganger. (strikkepiken)
 more times
 'I'm far from convinced that I will use this method again.'

Certain examples cannot, however, be located on a temporal or epistemic scale. Cases in which the speaker is not addressing a particular reader, but anyone who happens to read the blog, can only receive a temporal interpretation relative to the time it is read (which can happen at many different points in time). Such cases are referred to as temporally 'indefinite'. Expressions with universal quantifiers such as *alltid* ('always') or *aldri* ('never') are in this study marked 'atemporal' when they have future time reference (if not, they are generic or habituals). Finally, certain expressions, notably indirect questions and protasis (the clause introducing the condition of a conditional clause) do not express a degree of certainty. These are referred to as epistemically 'non-committing'.

Finally, Conditionality is an operationalisation of the "inevitable" uses associated with *kommer til å INF*, defined as future culmination of present cause, or 'given' conditionality (19); and the more hypothetical uses often associated with *vil INF*, defined as dependent on a future condition (semantic if-clause) or 'contingent' (20). These two uses are seen as two extremes on a scale of conditionality, with uses that are neither given nor contingent in the middle. We will refer to the middle cases as 'stand-alone'.

- (19) *Tromsø ble klare til opprykk. Så til neste år blir de å spille i*
 Tromsø was ready for promotion. So to next year FUT they to play in
eliteserien. (vegardsverden)
 topdivisionDEF
 'Tromsø was promoted, so next year they'll be playing in the Premier.'
- (20) *Snart senga, men må vente litt til ellers så blir jeg å våkne*
 Soon bed but must wait abit more other wise FUT I to wake
kl.3. (majbee)
 3o'clock
 'Bed soon, but have to wait a bit, otherwise I'll wake up at 3 am.'

Table 2 provides an overview of the distribution of the semantic variables and their features by construction. It is important to note that all of the semantic features are possible for all the constructions. These data serve as input for both the MCA and the loglinear models.

Table 2: Distribution of semantic features wrt. future constructions

Feature	<i>Blir</i>	<i>Kommer</i>	<i>Skal</i>	<i>Vil</i>
Intention	29	70	142	36
Plan	3	5	23	17
Prediction	44	125	35	147
Proximal	8	14	58	9
Distal	52	150	136	145
Indefinite	7	8	4	29
Atemporal	9	28	2	17
High Certainty	44	149	159	102
Low Certainty	29	38	37	86
Non-Committing	3	13	4	12
Given	10	30	21	34
Stand-Alone	61	151	172	139
Contingent	5	19	7	43

4 Results

Given that there was no clear hypothesis presented by the previous literature concerning the precise semantics of *blir å INF* relative to the other constructions (Section 3), we started out by exploring the distribution of all the constructions in relation to the semantic variables, described in the preceding section, using MCA. This method is a useful technique for visually identifying potential structure in complex data. Figure 2 presents the results of a multiple correspondence analysis that includes all the features annotated for all four constructions.

In Figure 2, degree of association is represented by relative proximity between the usage-features and the constructions. The plot depicts a 2-dimensional representation of these associations across 17 dimensions. The inertias indicate the relative contribution of each dimension, where dimension one constitutes the x -axis and dimension two the y -axis. The visualisation was generated in R with the *explor* package (Barnier 2023) and uses the Burt method to generate its representation. The scree-plot was produced in R with *ca* (Nenadic & Greenacre 2007) using the so-called adjusted method, since it is argued to more realistically represent complexity in multiple correspondence analysis (Greenacre 2010). The size of the data-points represents contribution.

Beginning on the left in the bottom quadrant, we see *skal INF* dominating the visualised structure, being associated with proximal temporal reference and both

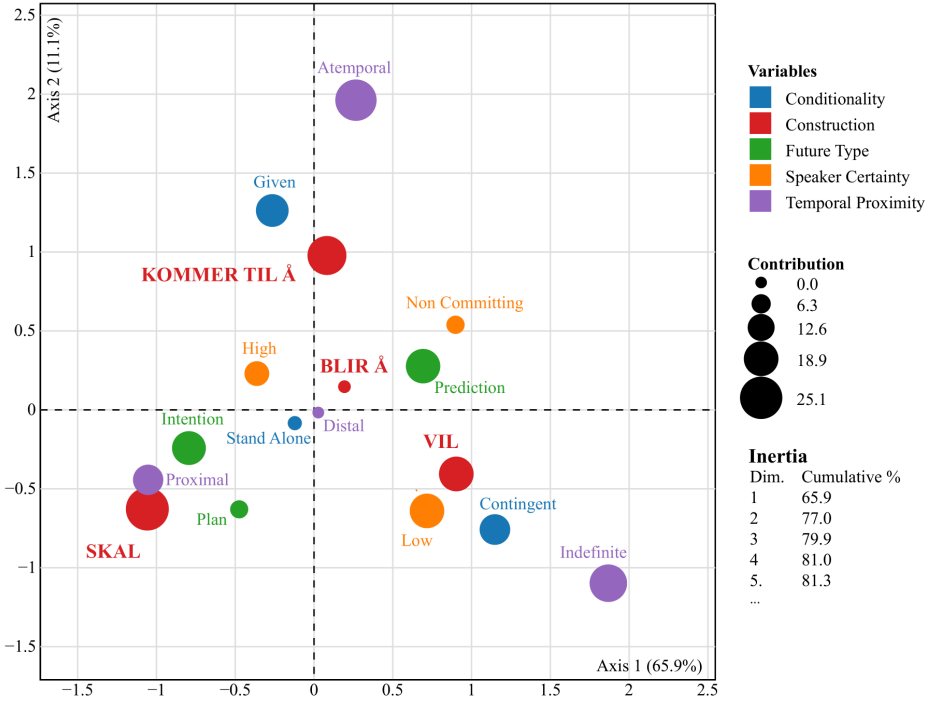


Figure 2: Correspondences in the distribution of future constructions and semantic variables

intention and plan future types. However, the future type associations are not distinctive, even if their association with the construction is strong. The structure visualised in bottom-right quadrant is equally clear. The construction *vil INF* is both highly and distinctly associated with indefinite temporal type, contingent conditionality and low speaker certainty. The top two quadrants are entirely structured by a combination of atemporal temporal proximity and given conditionality, two features distinctly associated with the *kommer til å INF* construction. The rest of the features and the *blir å INF* construction group around the centre, at the intersection of the two axes, indicating that their use is relatively non-distinctive.

In summary, what we see largely echoes previous introspective accounts (Section 3): *skal INF* is the intention-based marker of near-future, while *vil INF* is the future of uncertain and contingent events. Both *vil* and *kommer til å INF* are prediction-based, but the latter is associated with 'high certainty' (just like *skal INF*) and 'given conditionality' (future culmination of present cause). In addition to this, we see that *vil INF* is associated with 'indefinite temporality' (habitual-like

statements directed at more than one reader), and *kommer til å INF* with 'atemporal' readings (often strong assertions or promises). The division of labour seems quite clear between three of the constructions: *skal INF* would mostly be used to express intentions or plans that take place in the near future (the same day) and with a high degree of speaker certainty; *vil INF* could be characterised as the more hypothetical future, used to express predictions with some degree of uncertainty; *kommer til å INF* seems to specialise in future culmination of present cause ('given conditionality') and high certainty predictions.

Where does *blir å INF* fit into the picture? Its defining characteristic seems to be that it is nondistinctive, that is, a future construction that is unmarked with regards to the four semantic variables in question when we analyse all of the different usage events (the data points) collectively. There are several possible (not mutually exclusive) explanations:

- The construction is highly polysemous, and thus the plot is a poor visualisation,
- The construction has different meanings for different speakers, which again would lead to a visualisation that does not reflect the complexity of the data.
- The construction is used as a neutral way of referring to the future (predictions about non-near and non-contingent future events), and the plot is a good representation of the linguistic reality.

It is practically impossible to distinguish the first two alternatives using these data (we would need substantially more data from different speakers), but we may get an idea about how probable the third alternative is by examining the actual data points that are distributed "behind" the labels presented in Figure 2. The plot in Figure 3 presents such a plot of the data points. We see that while *skal INF* (blue dots), *vil INF* (green dots) and to a lesser degree *kommer til å INF* (red dots) are neatly separated, *blir å INF* (yellow dots) is overlapping with all of them. Can this be taken to support the first two alternatives mentioned above? Is the defining feature of *blir å INF* that it can be used in most contexts, unlike the other, more specialised constructions?

In order to try to tease apart some of the interactions visualised above, we first conducted a series of univariate analyses (chi-squared tests) with each of the constructions for each of the four variables. In contrast to the other constructions, the chi-squared tests revealed no significant interactions between *blir å INF* and any of the variables. This lack of distinctiveness with respect to the semantic

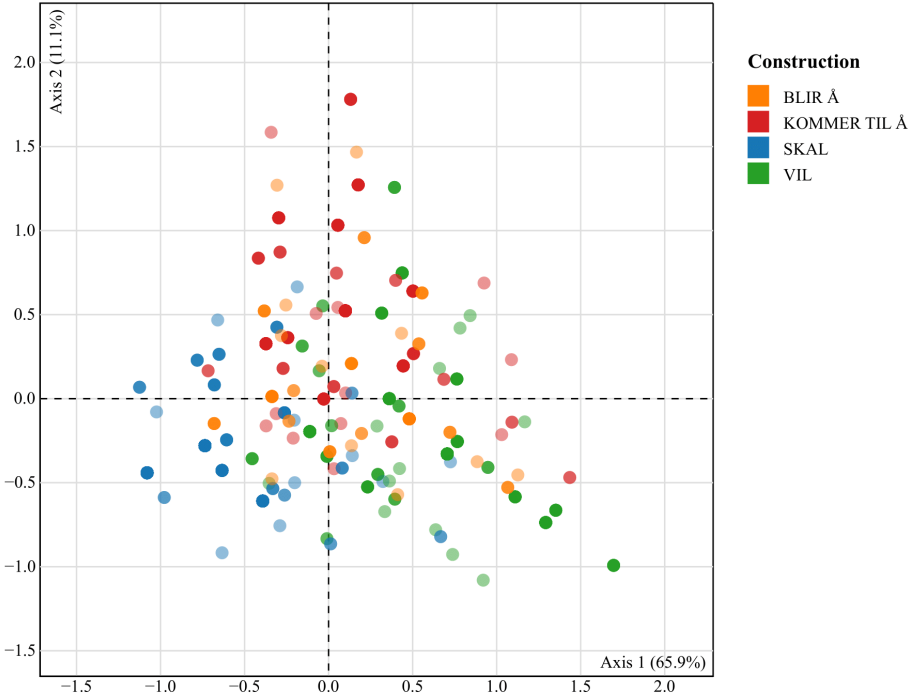


Figure 3: Distribution of data points for the four constructions viz. semantic variables

variables is in line with what we observed in the correspondence analysis. However, it is perfectly reasonable that an interaction of certain semantic variables is significantly associated or dissociated with the construction *blir å INF* even if independently they are not. For these reasons, we submitted a simplified set of data to a loglinear analysis. This multivariate approach models the combinatorial possibilities of all the semantic variables, testing for significant association and disassociation.

It was necessary to simplify the data because a complex multivariate analysis requires a larger sample than we have for this construction. In order to fit the model, ‘plans and intentions’ were collapsed into a single category labelled ‘intention - plan’, and ‘given’ and ‘stand-alone’ conditionality were both treated as ‘non-contingent’. The instances that were annotated as ‘non-committing’ with regards to speaker certainty were excluded. Finally, the variable Temporal Proximity was not included.

The mosaic plot in Figure 4 visualises a model combining the remaining simplified variables of future type, conditionality and speaker certainty, with respect

to the constructions. In this plot, there are four columns representing each of the four constructions (*blir å INF* on the far left, *vil INF* on the far right and the other two in the middle). Each column is split into two, accounting for the variable conditionality (contingent and non-contingent). The variable future type represents the two main rows (prediction is bottom, and the combination of intentions and plans is top). These two rows are, in turn, subdivided according to the variable speaker certainty (high and low). The size of the boxes indicates frequency, and the colours indicate significance: blue boxes indicate a significant association, red boxes a significant dissociation, while grey boxes are an indication of no significance.

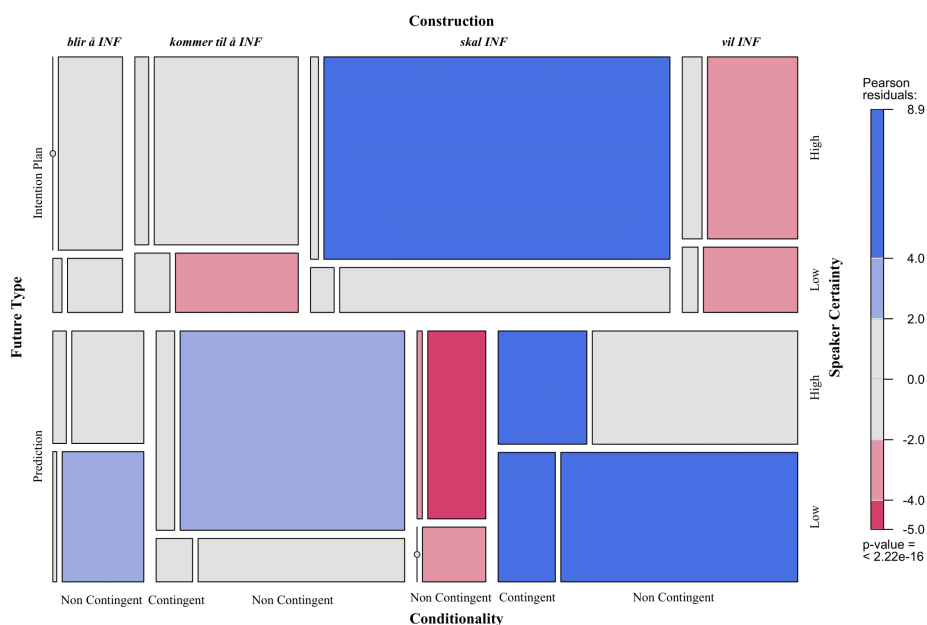


Figure 4: Distribution of Constructions, Future Type, Conditionality, and Speaker Certainty

The plot shows that *blir å INF* is not significantly associated or dissociated with respect to any uses except with respect to the combination of prediction, low-certainty, and non-contingent future events. However, in this, *blir å INF* is like *skal INF*. In contrast to *blir å INF*, we see that other constructions do have clear and quite distinctive profiles of use. For example, *skal INF* is significantly associated with high-certainty intentions and plans, and that *vil* and *kommer til å INF* divide up the “predictive” semantic space between them: the former is used to

talk about predictions that are dependent on a future contingency or predictions that are low in certainty; the latter to talk about predictions with a higher degree of confidence.

In summary, our results confirm previous introspective research concerning the semantics of *skal* and *vil INF*: the former is typically used to express plans and predictions with a high degree of certainty, the latter is used to express predictions that are seen as more uncertain. We can also provide a more accurate description of the difference between *vil* and *kommer til å INF* than what has been done in earlier research: these two constructions are semantically distinctive when we take into consideration speaker certainty and contingency, *vil INF* being associated with low certainty predictions, or predictions that are contingent and *kommer til å INF* with high certainty predictions. With regards to *blir å INF*, the result of the loglinear analysis shows that the construction is largely non-distinct with respect to the other future constructions. While it is used to express low certainty predictions significantly more frequent than would be expected by chance, so is *vil INF*.

As mentioned above, the variable temporal proximity was not included in the final loglinear analysis. A more sophisticated statistical technique would be necessary to account for the added complexity of indefinite and atemporal future time reference, but also to determine whether the observed correlation between *skal INF* and proximal future time is simply due to intentions occurring in near future. Furthermore, the variables speaker certainty and temporal proximity are here treated as nominal variables, even though they are inherently scalar. This was done to avoid having to exclude the items that do not fit on the scales (atemporal and indefinite temporal reference, non-committing speaker certainty). In future studies, speaker certainty and temporal proximity should be treated as ordinal variables rather than binary distinctions as in the present study.

5 Discussion

We have seen that *blir å INF* is not semantically distinct from the other three constructions of futurity in Norwegian. While it is true that the construction is significantly associated with expressions of low certainty predictions, this feature does not constitute a distinct way of referring to the future since it is also associated with *vil INF*. This could mean that *blir å INF* is encroaching on the semantic territory of *vil INF*. Provided that this is the case, what would be the motivation? As mentioned in Section 2, *vil INF* is seen as a more formal way of talking about the future, and having a more colloquial variant at one's disposal to describe predictions that are less certain may be desirable for many speakers.

Another possibility would be to say that *blir å INF* is not replacing *vil INF* specifically, but that it constitutes an unmarked way of referring to the future relative to the other constructions. In Section 2, we saw that the choice of future construction in Norwegian is complex: there are contexts in which it is nearly impossible to make a distinction between the different forms, and at the same time each of the constructions has restrictions related to specific contexts or lexical items. Could it be that a new construction not only does not increase, but actually reduces complexity and makes speaker choice easier when it is less burdened with linguistic excess baggage in the form of contextual specificities and lexical biases? Fleischman (1982) likens the development of future markers to turning wheels: as the older markers adopts more non-future (modal, irrealis) uses, new lexical material is recruited, first with aspectual meanings of imminence or prospection, before they consolidate as new future markers. In the case of Norwegian, the modals have both retained their original meaning of obligation (*skal*) and volition (*vil*) and developed new epistemic uses. Such a panorama would favour the apparition of a new future marker, following Fleischman's analysis.

Here an important caveat concerning the geographical distribution of the data must be added. In Section 3, we saw that the examples were divided between speakers of Northern dialects (31 items) and other varieties (43 items) / unknown origin (2 items). As was mentioned in Section 1, this could mean that the data reflect two subsystems: that of the "natives" who learned *blir å INF* from childhood and that of the "adopters" who incorporated the construction into their adult grammars. In other words, it is possible that *blir å INF* only is a "neutral" or "unmarked" future in appearance when we look at the population as a whole and that it takes on more specific meanings when we look at individual users. If we subdivide the sample of *blir å INF* into two separate populations ("North" and "Other"), we find no significant difference between the two in terms of the semantic variables investigated. Because of the small data set, however, this should not be interpreted as solid evidence against the parallel grammar hypothesis. At present, we do not have enough data to investigate the matter further. Furthermore, it would be useful to ascertain the motivations for why speakers from other regions have started adopting the construction, whether it is due to prolonged contact with Northern varieties, perceived language "fashion" or a mixture of the two factors.

A secondary question is what our findings on the synchrony of *blir å INF* can tell us about its diachrony. In Section 2, we noted that there are two main hypotheses concerning the development of *blir* as a future marker in Northern dialects: a development of the expletive *det blir å* construction or an analogy of the Sámi verb *šâd'dât*. A preliminary investigation of the expletive construction

in our corpus (199 items) suggest that, despite not having an animate subject, most of the examples are used to express near-future intentions (21).

- (21) *Ferdig på jobb 22:00, da blir det å dra rett på Kaliber etterpå.*
 Finished at job 10pm, then FUT EXPL to go straight at Kaliber after
 ‘I finish work at 10 pm and will be heading straight to Kaliber afterwards.’
 (idadir)

If (pending more detailed analysis) this use is prototypical of the expletive *det blir å* construction today, we might infer that the construction went through a stage of intention at some point during its history. Therefore, if *blir å INF* has developed from the expletive construction, we would expect it to behave like an intention-based future. The Sámi *šâd’dât*, on the other hand, is described as a prediction-based future in previous literature (Section 2), and if *blir å INF* has developed from *šâd’dât*, we would expect it to behave like a prediction-based future. Given our findings that *blir å INF* is associated with low-certainty predictions, this would lend some support to the second (Sámi) hypothesis concerning the diachrony of the construction.

We must be cautious not to make any strong claims, however, since *blir å INF* is not significantly associated with predictions when this feature is examined independently of speaker certainty (it is only when these variables combine that we see a significant association). In other words, the construction is used to express both intentions and predictions. Another caveat is that we know too little about how *blir å INF* was used when people first started adopting the expression in the North of Norway. Vannebo (1979), looking at the spread through the Northern regions, distinguishes three phases: intention (22), plan (23) and prediction (24). According to his informants, the construction was only used to express intentions and plans in the southernmost areas. In the “core” area, meanwhile, intentions were more common than plans, which in turn were more common than predictions. This suggests that the construction was in fact originally intention-based.

- (22) *Jeg blir å reise i morgen.*
 I FUT to travel to tomorrow
 ‘I will travel tomorrow.’ (Vannebo 1979)
- (23) *Båten blir å gå i morgen.*
 The boat FUT to go to tomorrow
 ‘The boat will depart tomorrow.’ (Vannebo 1979)

(24) *Jeg tror det blir å regne til natta.*

I think it FUT to rain to night

‘I think it will rain tonight.’ (Vannebo 1979)

When we interpret our findings in light of these caveats, new and interesting questions emerge. The first of these questions is whether the “natives” are still using *blir å INF* in its original, mostly intention-based uses, while the “adopters” are responsible for the new prediction-based uses. This could mean that the new construction fills a gap in the existing system of speakers from outside the Northern areas. It is here important to point out that such a gap may not only refer to a lack of specialised functions or contexts, but also to a lack of a more general or less specific function (in this case, a “neutral” future). A parallelism within the grammatical system of Norwegian would be that of the epistemic stance verbs. Over the past decades, a new use of *tenke* ‘think’ has established itself in detriment of *tro* ‘believe’, *synes* ‘appear’ and *mene* ‘mean’, and many speakers find this new use “too generic” (*Språkrådet*, 16.02.2016).

Another possibility could be that when *blir å INF* enters a new variety of Norwegian, it is compared to the formally similar expletive *det blir å* construction. In Section 3, we noted that there are contexts (with a null subject) in which both the obligational-intentional and the future meanings are possible, depending on the interpretation. Could it be that the formal and semantic similarity between these two constructions functions as an underlying motivation for speakers to use *blir å INF* to express predictions, in order to separate the two?

It is, of course, possible that the situation is reversed entirely: the adopters prefer the original intentional meanings, while the natives innovate. This would follow the development outlined in Vannebo (1979) and would also be in accordance with the view that all futures go through a stage of intention (Bybee et al. 1994, Heine 1995). It should not, however, be interpreted as an absolute law, as other authors defend alternative paths of grammaticalisation, such as ingressive, inchoative, or resultative meanings (Traugott 1978, Dahl 2000). In the case of the Swedish future constructions, Hilpert (2008) did not find evidence for an intentional phase in the collocation patterns of *komma att INF*. If the Norwegian *kommer til å INF* followed a similar path as its Swedish cognate, i.e., from ingressive aspect directly to prediction, without having passed through intention, then it is possible that *blir å INF* is doing the same now.

Future researchers wanting to disentangle these questions need to take several routes to reach their objective. First, more synchronic data is needed to test the possible difference between native users and adopters of *blir å INF*. This includes uses of the expletive *det blir å* construction as well as the null subject bridging

contexts. Second, diachronic data is needed to establish the construction's path of grammaticalisation: prediction via intention, or prediction directly through the inchoative. Finally, it will be necessary to look at both social variables such as prestige (in our data we find famous bloggers like Sophie Elise) and individual variation, as new constructions often are more polysemous and more prone to idiolectal effects not necessarily visible in frequency-based data.

6 Online repository

The annotated data, R-history and R plots can be found at the following address: <https://osf.io/mh4n6/>

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Chapter 7

The expression of future in the past in Old and Middle High German

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We adopt an onomasiological approach to the description of future in the past in Old and Middle High German. In these stages of German, the present-day main correlate of future in the past, the *würde* + infinitive construction, is not established yet. We search the reference corpora ReA and ReM and compile a database consisting of sentential complements of past-tense matrix predicates embedding a posterior event. We describe the forms and constructions occurring as correlates of future in the past in these clauses and investigate their formal and semantic properties. The past subjunctive is the most relevant correlate of future in the past in these periods, but it alternates with various types of periphrastic constructions which are known to express posteriority in the respective periods of German. Modal verb periphrases are the most frequent ones among these. We compare their frequency and distribution in contexts pertaining to future in the present and future in the past. We consider factors such as the choice of the modal verb, its relation to person, and the actional properties of the embedded verb. We find that modal verb periphrases act similarly in both contexts.

1 Introduction

The category of the future and the development of forms and constructions expressing reference to future events are well-established topics in the diachronic description of German. There is a long tradition of research investigating the origins and the grammaticalization of the construction *werden* + infinitive (*werden-INF*) (see the overview and the references in Luther 2013: 19–45), which is considered a periphrastic future tense in present-day German (PDG) and which



emerges relatively late, towards the end of the Middle High German period (MHG, c. 1050–1350) (Behaghel 1924: 261, Paul 2007: 296, Luther 2012: 11). There is also corpus-based work describing the forms and constructions used to express future time reference prior to the establishment of the *werden*-INF or during the grammaticalization of this construction (Schmid 2000, Luther 2012 and 2013, Hartmann 2021), which takes place during the Early New High German period (ENHG, c. 1350–1650).

A property unifying these approaches is that they address the future as a deictic category, i.e. as a category expressing posteriority relative to the actual utterance time. Yet there is another type of future time reference which has attracted less attention from a comparable corpus-based perspective, namely the one describing events or actions subsequent to some relevant point in the past. This kind of future time reference, also known as *future in the past* (FiP) (see Comrie 1985: 75 and 128, Dahl 2000: 797), is illustrated in (1) taken from Kamp & Reyle (1993: 595):

(1) English

Mary got to the station at 9:00. Her train **would arrive** at 10:00.

In the small discourse in (1), the event in the second conjunct, namely the arrival of the train, is subsequent to the event in the first conjunct, Mary's arrival at the train station, which, for its part, precedes the time of utterance.

In PDG, the expression of FiP is associated with the *würde*-infinitive construction (*würde*-INF) as its most salient formal correlate (see Herdin 1903 and 1905, Petkov 1991: 35–36, Thieroff 1992: 140–159, Szatzker 2002: 60–67, among others), next to some additional forms, most of all the past indicative and subjunctive, which are also attested in such contexts (see Fabricius-Hansen 2000: 94, Szatzker 2002: 177–184 or Welke 2005: 307). In the German translation of the small discourse in (1), the relations pertaining to FiP are most naturally expressed by means of *würde*-INF, see (2):

(2) PDG

Maria erreichte den Bahnhof um 9 Uhr. Ihr Zug **würde** um 10 Uhr **kommen**.

The nature of *würde*-INF in PDG has raised considerable debate, due to the exceptional status and the functional ambiguity of this construction (see Section 2.1). Its development has also received some attention (see Section 2.2). Szatzker (2002) provides a detailed account of the functions of *würde*-INF and the expression of FiP in corpus data from the 15th century to 2000. She finds that

würde-INF is robustly attested as a correlate of FiP in her material, but that it alternates with other expressions, such as the modal verb periphrases and the past indicative. However, there is no comparable, corpus-based investigation of the expression of FiP in earlier periods, in which this construction is still missing. The present paper aims at filling this gap. It adopts an onomasiological approach, proceeding from meaning to form. It identifies specific contexts in which temporal relations pertaining to FiP occur, and searches the digital reference corpora of the respective periods to determine the inventory of forms and constructions appearing in these contexts.

The paper is structured as follows. Section 2 describes the notion of FiP from a synchronic perspective (Section 2.1) and summarizes the state of the art regarding the diachronic research on FiP in German (Section 2.2). Section 3 describes the corpora and the query methods, including the caveats related to the identification of the relevant data. Section 4 describes the inventory and distribution of the correlates of FiP found in the database. Section 5 summarizes the results of the analysis.

2 Previous research

2.1 The notion of FiP

The complex interplay of temporal and modal features, which is constitutive of the category of the future as a whole, finds its proper manifestation in the expression of FiP as its representative as well. The problematic status of this type of future results from the observation that in a series of languages, the formal correlates of FiP display temporal and modal functions alike (Comrie 1985: 75). This also applies to the construction *würde-INF*, which is considered the most salient correlate of FiP in PDG (see Section 1). As a member of the paradigm of the future construction *werden-INF*, *würde-INF* is exceptional both with respect to form and meaning (Thieroff 1992, Fabricius-Hansen 1998 and 2000, Smirnova 2006). In terms of form, it involves the past subjunctive (*Konjunktiv II*) of the future auxiliary *werden* ‘become’, and lacks a morphologically indicative counterpart of the type **wurde-INF* (but see Section 2.2 below). In terms of meaning, it is polyfunctional in that, next to its use as a correlate of FiP, it also occurs in prototypical subjunctive contexts, such as conditional and counterfactual utterances or instances of indirect speech, where it is partially interchangeable with the synthetic past subjunctive. For these reasons, the classification of *würde-INF* as part of the system of moods and tenses in PDG has been particularly difficult (see the overview in Smirnova 2006: 11–40). Thieroff (1992: 141–159) argues

that *würde*-INF is syncretic, being part of the subjunctive paradigm when used in counterfactual and reportive contexts, and of the indicative tense system when used to express FiP.

The following observations are taken to suggest that despite its subjunctive morphology, *würde*-INF as a correlate of FiP is part of the indicative tense system (see Petkov 1991: 136, Thieroff 1992: 145–147). First, it can be observed that *würde*-INF alternates with the indicative of the future construction *werden*-INF when the temporal orientation of the context is changed from past to present, as shown in (3a) and (3b). Note that the respective variant of the subjunctive derived from the present stem of the future auxiliary *werden*, the so-called *Konjunktiv I Futur* given in (3c), is ungrammatical in this case. Second, it is observed that in the domain of FiP, *würde*-INF alternates with the indicative of the simple past, i.e. with the so-called *Präteritum Indikativ* (see also Welke 2005: 306–307), see (3d):¹

(3) PDG

- a. *Er wusste, was geschehen würde.*
he know.3SG.IND.PST what happen will.SBJV.PST
'He knew what would happen.'
- b. *Er weiß, was geschehen wird.*
he know.3SG.IND.PRS what happen will.IND.FUTI
'He knows what will happen.'
- c. *Er weiß, was geschehen *werde.*
he know.3SG.IND.PRS what happen will.IND.FUTI

¹Fabricius-Hansen (2000: 94) provides examples in which the past subjunctive (*Konjunktiv Präteritum*) also expresses relations pertaining to FiP, see (i). According to her, the past subjunctive is interchangeable with *würde*-INF in this context, as shown in (ii). This alternation, in turn, is parallel to the one between the present indicative and the future construction *werden*-INF, see (iii)-(iv):

- (i) *Ich ging schon beinahe in die fünfte Klasse des Gymnasiums [...] Nicht mehr lange, und ich bekäme einen Schülerausweis mit dem begehrten roten Stempel „über 16“.* 'I almost went to the fifth class of high school already (...) It wouldn't take much longer, and I would get a student ID card with the coveted red stamp 'over 16' [...]
- (ii) *Nicht mehr lange, und ich würde einen Schülerausweis [...] bekommen.* 'It wouldn't take much longer, and I would get a student ID.'
- (iii) *Nicht mehr lange, und ich bekomme einen Schülerausweis [...]* 'It wouldn't take much longer, and I [would] get a student ID.'
- (iv) *Nicht mehr lange, und ich werde einen Schülerausweis [...] bekommen* 'It wouldn't take much longer, and I will get a student ID.'

- d. *Er wusste, was gleich geschah.*
 he know.3SG.IND.PRS what next happen.IND.PST
 ‘He knew what happened thereafter.’

Thieroff (1992) explains this variation along the lines of his assumption that the indicative of the simple past represents a kind of a secondary (or shifted) present within a system of tenses directed towards a point preceding the time of utterance. As such, the simple past may have a futurate meaning and alternate with the respective future construction, namely *würde*-INF, in the same way in which the canonical present tense is able to express future events, alternating with the *werden*-INF in the system of tenses oriented towards the actual utterance time.

2.2 State of the art of the diachronic description

The development of *werden*-INF has given rise to much debate in the literature. Despite the controversial treatment of the origins and grammaticalization of this construction (see Jäger 2024 on a novel account), there is agreement that *werden*-INF emerges towards the end of the MHG period (Behaghel 1924: 261, Paul 2007: 296)² and advances as a periphrastic future construction only during the subsequent ENHG period (Luther 2012: 5).

The rise of *würde*-INF has also attracted much attention, see Szatzker (2002, 2003), Kotin (2003), Smirnova (2006), Durrell & Whitt (2016), among others. Research has commonly assumed that this expression is the subjunctive variant of the inchoative construction *ward*-INF which was temporarily present in the history of German (Paul 2007: 307). The loss of *ward*-INF in the 16th century is taken to indicate the crucial point in the grammaticalization of *werden*-INF as a future construction, and in the consolidation of *würde*-INF as a special representative of this paradigm (see Oubouzar 1974: 87, Guchmann & Semenzuk 1981: 221, Ebert et al. 1993: 392, Szatzker 2002: 58–61, Kotin 2003: 175–189, Welke 2005: 469, Smirnova 2006: 41–54 and 283–295, Paul 2007: 307 and Durrell & Whitt 2016: 335). At the same time, it has been observed that *würde*-INF is used to express FiP already at the beginning of the attestation of *werden*-INF. The earliest conclusive example is the one given in (4), dating to the first half of the 14th century (c. 1320):

²Note that Luther (2013: 52) characterizes the earliest attestations of the *werden*-INF in MHG as ingressive and inchoative, expressing the beginning of a new state or action. This is identical with the position held by Behaghel (1924: 261, fn. 1). Putative instances of the *werden*-INF in earlier periods have been rejected as inconclusive (Saltveit 1962: 185–194 and Jäger 2024: 216–226).

- (4) Ottokar v. Steiermark, *Österr. Reimchron.* vv. 4707–4713 (cit. in DWB vol. 14, 1/2, col. 255; online vol. 29, col. 273.),
 context: und swer ir îlen het gesehen, / der müeste des fürwâr jehen, / daz
 nie von kuniges kinde / wart gesehen alsô swinde / über velt gegangen
 = ‘and whoever has seen her haste would have to admit that no king’s
 child has been seen going so fast over fields’
dâ si wurd enphangen / von Arrogûn diu kunigin
 where her.ACC will.SBJV.PST receive.INF / of Aragon the queen
 ‘[to the place] where the Queen of Aragon would receive her’

The example in (4) is cited in the entry of the verb *werden* ‘become’ in Grimm’s *German Dictionary* (*Deutsches Wörterbuch*, DWB) and is interpreted as expressing “zukunft von einem standpunkt der vergangenheit aus”, i.e., future from the perspective of the past (DWB vol. 14, 1/2, col. 255; online vol. 29, col. 273.). This opinion is shared by the subsequent literature, see Wolf (1981: 206), Kotin (2003: 180), Smirnova (2006: 51), and Durrell & Whitt (2016: 335).

The role of *würde*-INF as a correlate of FiP is consolidated in the subsequent periods. Szatzker (2002 and 2003) examines the forms and constructions used to express FiP in a corpus of texts from the 15th century to 2000. She finds that *würde*-INF is the most prominent correlate of FiP in her material (Satzker 2002: 180–181 and 2003: 199–203), and that it co-varies with other forms and constructions whose frequency changes during the investigated period. She shows that at the beginning, the most prominent competitors of *würde*-INF are the past subjunctive and modal verb periphrases involving *wollen* ‘will/want to’ and *sollen* ‘shall’, but their frequency continuously declines, while especially after 1850, the past indicative becomes the most frequent alternate of *würde*-INF (Satzker 2002: 181).

The results obtained by Szatzker (2002) are consistent with the observation that *würde*-INF alternates with the past indicative in the domain of FiP in PDG, but also with insights regarding the expression of FiP in the earlier periods. The reference grammar of MHG (Paul 2007) describes a special function of the past subjunctive expressing “präteritales Futur” (Paul 2007: 291), i.e. past futurity, for example in subordinate clauses expressing posteriority relative to a matrix clause in the past (Paul 2007: 291, 401–402, 434 and 444–445). Consider the examples in (5) and (6):

- (5) Iwein 1532, cit. in Paul (2007: 291)
er weste wol daz Keii in niemer gelieze vrî von
 he know.IND.PST well that Keie him.ACC never let.SBJV.PST free from
spotte
 mockery
 ‘He knew very well that Keie would never spare him from mockery.’
- (6) Nibelungenlied 1131, 1–2, cit. in Paul (2007: 445)
ih swuor [...], daz ih ir getaete nimmer mêre leit
 I swear.IND.PST [...] that I her.DAT do.SBJV.PST never more harm
 ‘I swore [...] that I would never (again) do her any harm.’

The same use of the past subjunctive is also observable in data from Old High German (OHG, c. 750–1050). Petrova (2008) provides the example in (7), in which the past subjunctive occurs in the same context as in the MHG instances in (5) and (6), namely in a complement clause conveying an event that is subsequent to a past-tense matrix clause. At the same time, she observes that the modal verb construction involving OHG *sculan* ‘shall’ is also found in such contexts. Consider the example in (8), in which the modal verb construction and the past subjunctive alternate in two subsequent clauses expressing posteriority relative to the matrix clause, which itself is situated in the past:

- (7) O I 10, 13–14, cit. in Petrova (2008: 13), her ex. (10)
gihiaz [...] / thaz ér uns sin gisiuni in lichamen
 promise.IND.PST [...] / that he us.DAT his likeness in human_body
gábi
 give.SBJV.PST
 ‘[God] had promised [...] to give us his likeness in the shape of a human body’
- (8) O I 3, 37–38, cit. in Petrova (2008: 175), her ex. (245)
Iro dāgo ward giwāgo fon alten wíragon, tház
 her.GEN.PL day.GEN.PL become.IND.PST known from old prophets COMP
si uns bēran scolti thér unsih gihéilti
 she us.DAT bear shall.SBJV REL us.ACC save.SBJV.PST
 ‘Of her days [of the Holy Virgin Mary], it was prophesied by the ancient prophets that for our sake, she would give birth to the one who would save us’

Petrova (2008) provides the examples above to exemplify the temporal underspecification of the past subjunctive, which conveys all kinds of temporal relations between main and embedded clauses, including those of posteriority, sharing the inherently prospective semantics of the deontic modal verb OHG *sculan* ‘shall’. Schönherr (2011) confirms these observations but claims that in the contexts at issue, the past subjunctive and the modal verb construction leave their original modal domain and obtain a purely temporal interpretation, sharing the polyfunctionality observed for *würde*-INF as a member of the paradigm of the past subjunctive in PDG.³

Two important conclusions may be drawn from these observations. First, they provide diachronic evidence suggesting that members of the paradigm of the past subjunctive display the polyfunctionality described for *würde*-INF as a correlate of FiP in PDG (see Section 2.1) as their historically constant property. Second, they show that during the periods in which *werden*/*würde*-INF are still missing, modal verb constructions act as periphrastic expressions of the future not only in its canonical domain, namely the one oriented towards the utterance time (Behaghel 1924: 258–259, Luther 2012: 4–10, Luther 2013: 90–100), but also in the domain of FiP. Szatzker (2003: 190) also expresses this assumption, suggesting that the modal verb constructions act as forerunners of *werden*-INF and *würde*-INF alike. However, it is still unexplained if the properties governing the distribution of modal verb constructions within the domain of the canonical future and described in the literature (Behaghel 1924: 258–259, Luther 2012 and 2013) also apply to their function in the domain of FiP in the same periods.⁴ These ques-

³In addition, Schönherr (2011) elaborates on differences in the interpretation of the past subjunctive and the indicative of *sollen*-INF that are related to effects of perspective taking. These effects are well-known from the semantic literature on FiP. For instance, Eckardt (2015, 2017) explains that the English counterpart of *würde*-INF, namely *would*-INF, conveys two different perspectives on events subsequent to a reference point in the past, namely the protagonist’s perspective on events held possible in the future of a past point of reference, as in *free indirect discourse*, and the narrator’s preview into events to come, also called *objective interpretation* or *future of fate*. Thieroff (1992: 145–152) also discusses similar effects related to the interpretations of *würde*-INF in PDG, see also Zeman (2018 and 2020) on parallel suggestions on *sollte* + infinitive in PDG. With respect to the expression of FiP in OHG, Schönherr (2011) suggests that the construction involving OHG *sculan* correlates with the objective interpretation of FiP, i.e., with the expression of future of fate.

⁴Luther (2013: 104) observes that past tense forms, including those of the modal verbs MHG *mugan* ‘may’, *sol(e)n* ‘shall’ and *wellen* ‘want to/will’, also occur in her database (a total of 37 cases). She remarks that the verbal mood is indistinguishable in the past paradigm of *wellen* and *sol(e)n*, but in the remaining cases, i.e. in the active or passive paradigms of main verbs as well as of *mugan*, in which the subjunctive can be distinguished from the indicative, the former is found in 16 out of 25 instances. Luther does not elaborate on the question of whether the respective instances express FiP. She rather interprets the subjunctive forms as hypothetical and therefore basically modal in nature.

tions regarding the formal properties and the distribution of correlates of FiP in the history of German will be addressed on the basis of corpus evidence in the following sections.

3 Methods of data collection and properties of the database

The present investigation pursues an onomasiological approach, seeking to identify the inventory of linguistic means used to express FiP in OHG and MHG. The database is retrieved from the reference corpora of the respective periods, namely *Referenzkorpus Altdeutsch* (ReA, Version 1.2, Zeige et al. 2024) and *Referenzkorpus Mittelhochdeutsch* (ReM, Version 1.0, Klein et al. 2016). The corpora are lemmatized and annotated for morpho-syntax, searchable via ANNIS, an open-source web browser-based search tool facilitating queries in complex, multilayer linguistic corpora (Krause & Zeldes 2016).

Onomasiological approaches essentially depend on the identification of well-defined contexts in which the phenomenon at issue is present. The underlying semantic property of FiP is that it conveys events or actions which take place after some relevant point in the past. But at the same time, there are no formal diagnostics occurring in contexts related to FiP, such as temporal adverbs or related lexical expressions. Also, the syntactic environments in which FiP can occur are quite heterogeneous, ranging from declarative and non-declarative main clauses to various types of dependent clauses, including relative and complement ones. This is demonstrated by Thieroff (1992: 149–152) for PDG, on the basis of data from original prose. The same kind of heterogeneity is present in the historical data. Consider the examples in (9) and (10), retrieved by way of manual search. Relations pertaining to FiP are conveyed in a relative-like main declarative clause given in (9) and in two different types of subordinate clauses in (10), namely in a complement clause of MHG *gehèizen* ‘promise’ used in the past, and a relative clause modifying the head noun MHG *kint* ‘child’ in the preceding complement clause:

- (9) DDD-AD-Tatian_1.2 > T_Tat82, 12
Quad uuarlichō Iudam Simonem Scariothen, dér uuas
 mean.IND.PST in-fact Judas son_of_Simon Iscariot DEM be.IND.PST
selanti inan
 betray.PTCP.PRS him.ACC
 ‘he meant Judas Iscariot, the son of Simon, for he would betray him’ / Lat.
Dicebat autem Iudam Simonis Scariothis; hic enim erat traditurus eum (Jo
 6,71)

- (10) 12_2-bair-V_2-X > M087-N1 (tok_dipl 3875–3890)
got gehiz abrahame. daz ein kint dannen chome.
 God promise.IND.PST Abraham COMP a child thereof come.SBJV.PST
daz di werlt alle. fuorte uon der helle
 REL the world all lead.SBJV.PST from the hell
 ‘God promised Abraham that a child would thereof [from Abraham’s
 people] come which would lead all world from hell’

The lack of specific diagnostics associated with the presence of relations of FiP, and the heterogeneity of the syntactic contexts in which FiP may occur, make it difficult to formulate queries for an automatic corpus search.

In a questionnaire prepared for linguistic fieldwork and designed to determine the linguistic devices used to express tense and aspect cross-linguistically, Dahl (2000: 797) provides the following five contexts in which FiP is present, each accompanied by questionnaire items selected to retrieve the respective linguistic correlates:⁵

- a. sentential complements of past-tense verbs of saying and believing, e.g. *My brother said yesterday that he would come here {today/next week}* or *My brother said yesterday that it would {be cold/rain} today* (see questionnaire items 70, 72–74 in Dahl 2000: 792)
- b. sentential complements of past-tense verbs of wishing and hoping, e.g. *My brother hoped yesterday that {you would come here/it would be cold} today* (see questionnaire items 71 and 75 in Dahl 2000: 792)
- c. purpose clauses specifying a past-tense matrix clause, e.g. *I wrote a letter to my brother in order that he knew that I would come to see him* (see questionnaire item 96 in Dahl 2000: 793)⁶
- d. *before*-clauses, e.g. *Yesterday evening, I went to bed before my brother came home* (see questionnaire item 22 in Dahl 2000: 790)

⁵In the original questionnaire, the individual items are supplemented by notes specifying the context in which the sentences are uttered, and the respective verbs are provided in their non-inflected lemma form, put in small caps. For illustration, we use the full forms of the sentence items in English and provide inflected verb forms.

⁶In fact, this test item involves not only one, but two different cases related to FiP, namely the purpose clause itself and another context, namely the complement of the verb *to know* describing a situation subsequent to the past-tense matrix clause.

- e. main clauses of the type *He would return at seven o'clock* (uttered at eight o'clock, while the person has left at six) (see questionnaire item 109 in Dahl 2000: 794).

However, applying the questionnaire by Dahl (2000) to OHG and MHG faces some serious methodological problems. First, it is well-known that in OHG and MHG, similarly to PDG, complements of verbs like *say* and *believe*, but also independent clauses uttering indirect speech or thought, tend to display the subjunctive as a marker of reported speech. Note also that in OHG and MHG, the past subjunctive may reveal all kinds of temporal relations between a matrix clause and its sentential complement, namely those of simultaneity, anteriority, and posteriority alike (see Petrova 2008 for discussions and examples).⁷

Second, it has been observed that adverbial clauses like purpose clauses and *before*-clauses, but also clausal complements of verbs like *wish* systematically display the subjunctive in OHG (see Coniglio et al. 2018). In the typology proposed by Giannakidou (1998 and later), purpose clauses, *before*-clauses and complements of predicates of wish and desire display conditions pertaining to non-veridicality. The respective clauses share some formal cross-linguistic properties, most importantly the use of the subjunctive in Romance and of the class of so-called subjunctive complementizers in Greek and the Balkan languages (see Farkas 1985 and 1992, Giannakidou 1998 and later, Roussou 2009, inter alia). In OHG and MHG, but also in other early Germanic languages, the verbal mood in these clauses is prototypically the subjunctive (see Coniglio et al. 2018).

Taken together, the above-mentioned observations suggest that in the contexts described in Dahl (2000), we expect to find a high proportion of subjunctive forms in OHG and MHG. But despite the fact that the respective types of clauses are intrinsically prospective in nature, and despite the observation that in some of its functions, the subjunctive bears future implications in general (Ullmann 1978: 94–95), its presence in the respective clauses may be the consequence

⁷As in PDG, the system of OHG and MHG provides two paradigms of the subjunctive, the present subjunctive (*Konjunktiv I*) and the past subjunctive (*Konjunktiv II*), the latter being derived from the plural past stem of the verb. Despite of the implied terminology, these two subjunctive paradigms do not differ in terms of temporality in PDG. E.g., in indirect speech, both the present and the past subjunctive may refer to an event or situation which is simultaneous to the time at which the report is uttered. This holds for historical German in the same way, while the distribution of the different subjunctive paradigms may be governed by the rules of *consecutio temporum*. But in addition to that, and in contrast to PDG, the past subjunctive in OHG is able to express posteriority and anteriority as well, see Petrova (2008: 13 and 162–190) for examples.

of the principles of mood selection, rather than a strategy of expressing FiP in the respective periods of German.⁸

As the contexts provided by Dahl (2000: 797) cannot be used to retrieve conclusive examples, we decided to apply alternative criteria to discerning the relevant contexts. We restricted the database to sentential complements of matrix predicates which select the indicative in OHG and MHG, unless they appear in the scope of a negative or non-assertive modal operator (see Schrodtt 1983, 2004: 185). These are the OHG and MHG equivalents of the verbs of mental, perceptual, and emotional states like *know*, *realize*, *feel*, *see*, *be clear*, *fear*, *be afraid*, *be anxious* etc. as well as other verbs implying the posteriority of their complement, such as *indicate*, *predict*, *foresee*, *promise*, *pledge* etc. In the queries, we used the lemma information provided in the annotation, and specified that the matrix predicate should be used in the past tense. We extracted those cases in which the respective verbs select complements which are finite and posterior, relative to the event in the matrix clause.

Table 1 provides a list of the matrix verbs which yielded conclusive examples, including the respective lemmata in OHG and MHG.

On the basis of the queries described above, we compiled a database consisting of a total of 130 instances, 26 OHG ones found in ReA, and 104 MHG ones found in ReM. The inventory and distribution of the variants attested as correlates of FiP in the database is discussed in Section 4.

4 Results of the corpus study

4.1 Inventory

Four types of constructions functioning as correlates of FiP can be identified in the OHG part of the sample. First, as expected from the descriptions found in previous research, the past subjunctive of the lexical verb is found in contexts expressing FiP in OHG. Representative examples are provided in (11a) and (11b):

- (11) a. DDD-AD-Z-Notker-Psalmen_1.2 > N_Ps_030, 91, 4–5
So hábest dû getân also dû gehiêzze. daz du mit
 so have you done as you promised.IND.PST COMP you with
mînemo bluôte mînen liût irlôstist
 my blood my people saved.SBJV.PST
 ‘So you have done as you promised, that you would save my people
 with my blood’

⁸Following similar considerations, Luther (2013: 63) rejects the status of the present subjunctive as a potential marker of futurity in MHG, because its use very probably results from the selectional properties of the matrix predicate or the semantic type of the adverbial clause.

Table 1: Matrix verbs yielding conclusive examples of FiP in OHG and MHG

Matrix verb	OHG	MHG
<i>know, see, recognize</i>	<i>inkennen, sehan, wizzan</i>	<i>wizzen, erkennen</i>
<i>foreknow, predict, indicate</i>	<i>forasehan, wizagon, bouhhanen, bizeinen/bizeinon/ gizeihanon</i>	<i>wîsen, zèigen</i>
<i>hope</i>	<i>wanen, warten</i>	<i>gedingen, hoffen, wænen, verwænen, sich versëhen</i>
<i>be afraid</i>	<i>forhten</i>	<i>vürhten</i>
<i>promise, pledge</i>	<i>giheizan, sih biheizan, gilobon</i>	<i>gehèizen, geloben</i>

- b. DDD-AD-Otfrid_1.2 > O_Otfr.Ev.4.19, 45

Bizéinta thaz sin wîrði zi niwîhti scioro
 demonstrate.IND.PST COMP his priesthood in tatters soon
wûrði
 became.SBJV.PST

‘[This] indicated that his priesthood would soon be ruined’

Second, as also mentioned in the literature, we find modal verb constructions occurring as correlates of FiP in OHG. The modal verbs attested are OHG *sculan* ‘shall’ and *wellen* ‘want’. Their distribution will be subject of analysis below (Section 4.2). Note that these modal verbs are attested both in the indicative and the subjunctive, as shown in (12a) and (12b):

- (12) a. DDD-AD-Z-Notker-Psalmen_1.2 > N_Ps_015, 23–24

Aber in carne uuésendo. fôre sah ih. daz ih ze gôte
 but in flesh being fore- see.IND.PST I COMP I to God
fölletânero uërte iruuînden sôlta
 accomplish.DAT earth_life.DAT return shall.IND.PST

‘But as human I foresaw that I would return to God after my earthly existence is fulfilled’

- b. DDD-AD-Otfrid_1.2 > O_Otfr.Ev.5.9, 31–32
Wir wântun thes giwísso [...] er únsih scolti
 we hope.IND.PST DEF.GEN.SG truly [...] he us.ACC shall.SBJV.PST
irlären thes mánagfalten wéwen
 exempt DEF.GEN.SG manifold.GEN.SG pain.GEN.SG
 ‘we truly hoped [...] that he would exempt us from this manifold pain’

The third correlate of FiP found in the OHG part of the database is the construction involving the OHG equivalents of the copula verb *be*, namely *sîn* and *wesan*, in combination with the present participle of the lexical verb. As shown for the modal verb constructions above, this construction may also involve both the past indicative and subjunctive of the finite verb, as demonstrated in (13a) and (13b):

- (13) a. DDD-AD-Tatian_1.2 > T_Tat135, 30
uuizagota thaz ther heilant sterbenti uuas furi
 predict.IND.PST COMP the Savior die.PTCP.PRS be.IND.PST because-of
thiota
 people
 ‘predicted that the Saviour would die for the sake of this people’ / Lat. *prophetavit quia Ihesus moriturus erat pro gente*
 b. DDD-AD-Tatian_1.2 > T_Tat225, 3
Uuir uuantumes thaz her uuari arlosenti Israhel
 we expect.IND.PST COMP he be.SBJV.PST save.PTCP.PRS Israel
 ‘We expected that he would save Israel’ / Lat. *Nos autem sperabamus quia ipse esset redempturus Israhel*

Finally, we find a lexical periphrasis involving the predicative adjective *zuowart* ‘future, to come’ as a complement of the copula verbs OHG *sîn/wesan* ‘be’ or OHG *werdan* ‘become’. Representative examples are provided in (14a) and (14b). This construction is only attested in the past indicative in the database:

- (14) a. DDD-AD-Tatian_1.2 > T_Tat80, 8
soso her thaz inkanta thaz sie zuouuerte
 as_soon_as he DEM.ACC.SG realize.IND.PST COMP they to_come
uuarun
 be.IND.PST
 ‘as soon as he realized that they would return’ / Lat. *cum cognovisset quia venturi essent*

b. DDD-AD-Isidor_1.2 > I_DeFide_6, 1

Uuexsal dhes nemin huuazs bauhnida? Nibu
 change DEF.GEN.SG name.GEN.SG what signified.IND.PST if-not
dhazs after moysise dodemu [...] uns zuouuert leididh
 COMP after Moses' death [...] us.DAT to_come leader
uuardh unser druhtin ieselus christus
 become.IND.PST our lord Jesus Christ
 'What signified the changing of Christ's name? It meant that after
 Moses' death, [...] our lord Jesus Christ would become our leader' /
 Lat. *Mutatio nominis quid significabat, nisi quia defuncto Moyse [...] dux nobis Dominus Jesus Christus erat futurus*

The total of examples found in OHG is very small (n=26), but nevertheless, we want to analyze the quantitative distribution of the four types of correlates of FiP included in the OHG part of the database. The absolute numbers and the proportions of the OHG correlates of FiP are provided in Table 2. With 11 out of 26 cases (42.3%), the past subjunctive of the lexical verb provides the most frequent correlate of FiP in OHG. The modal verb periphrases and the construction OHG *sîn/wesan* + present participle share the same proportions in the database, occurring in 6 out of 26 instances (23.1%) each. The most infrequent variant is the lexical construction *zuowart sîn/wesan/werdan* which is attested only three times (11.5%) in the database.

Table 2: Types, absolute numbers, and percentages of OHG formal correlates of FiP in the database

Formal correlate of FiP in OHG	n (%)
past subjunctive	11 (42.3%)
modal verb construction	6 (23.1%)
<i>sîn/wesan</i> 'be' + present participle	6 (23.1%)
<i>zuowart sîn/wesan/werdan</i> 'be to come'	3 (11.5%)
Total	26 (100%)

Let us look into the distribution of the indicative and the subjunctive in the domain of FiP in OHG. The relevant figures are provided in Table 3. With simple forms of the lexical verb, the past subjunctive occurs in all of the instances involved in the database. In contrast, the lexical periphrasis *zuowart*

sin/wesan/werdan is only attested in the past indicative. In the remaining domains, there is alternation of the indicative and the subjunctive. The former is present in 66.7% of the modal verb constructions and in 50.0% of the construction OHG *sin/wesan* + present participle. In total, the subjunctive is present in 69.2% of the contexts involving relations pertaining to FiP in the OHG part of the database.

Table 3: Verbal mood of formal correlates of FiP in the OHG part of the database. IND = Indicative, SBJV = Subjunctive.

Formal correlate of FiP in OHG	IND	SBJV (%)	Total
past subjunctive	0	11 (100%)	11
modal verb construction	2	4 (66.7%)	6
<i>sin/wesan</i> 'be' + present participle	3	3 (50.0%)	6
<i>zuowart sin/wesan/ werdan</i> 'be to come'	3	0 (0.0%)	3
Total	8	18 (69.2%)	26

Let us compare this picture with the one obtained for MHG. As expected from the previous literature, the past subjunctive is found as a correlate of FiP in this period as well. A representative example is provided in (15a). But in MHG, the verbal mood of lexical verbs used in the past cannot be determined in each individual case, because the formal distinction between the indicative and the subjunctive is blurred, due to phonological reductions. The formal syncretism of the indicative and the subjunctive is particularly strong in the paradigm of the weak verbs, as in (15b), where the indicative and the subjunctive completely overlap in the past. But there are also syncretic forms in the past paradigm of some strong verbs, as in (15c).⁹

⁹In the domain of the strong verbs, distinctions between the indicative and the subjunctive are partially provided by the property of *i*-mutation of stem velar vowels (/a/, /o/, /u/ and respective diphthongs), caused by the subjunctive suffix *-i* present in OHG. But note that except for the case of short /a/ (*Primärumlaut*, see Sonderegger 1979: 312), the effect of *i*-mutation is not systematically marked in manuscripts of the OHG and MHG period. For example, the form *wären* in (i) can be both indicative and subjunctive, provided that the diacritics can signify both the length of the vowel, or *i*-mutation:

- (i) 12_2-bair-PV-G > M157-G1 (tok_dipl 6456 - 6476)
den gehiez er daz si gotes chint wâren
 DEM.DAT.PL promised he that they God.GEN.SG children were.IND/SBJV.PST
 'to them, he promised that they were God's children'

At least in one case, given in (15d), the attested form is indisputably one of the indicative. Given this, it cannot be excluded that the indicative was already used in those cases in which FiP is expressed by way of a simple past form of the lexical verb in the MHG period. In some few cases, we also find forms of the present subjunctive in the database, as shown in (15e).

- (15) a. 12_2-bair-V_1-X > M088-N1 (tok_dipl 7609 – 7622)
er ime do gehiez. daz er niemmer mêre die wærlt
 he him.DAT then promise.IND.PST COMP he never again the world
flure mit wazzere
 flood.SBJV.PST with water
 ‘he promised him that he never again would flood the world’
- b. 12_2-obd-PV-X > M199B-N1 (tok_dipl 3946 - 3956)
Vnder in sie tho gelobeten thaz sie iz
 among them.DAT they then swear.IND.PST COMP they it.ACC.SG
ire nicht ne=sageten
 her.DAT not NEG=say.PST
 ‘Among themselves they swore that they would tell absolutely nothing of it to her’
- c. 12_2-bair-V_Kchr2-X > M121y2-N (tok_dipl 44714 – 44724)
di=baier im alle gehiezen. daz si
 the=Bavarians him.DAT all.NOM.PL promise.IND.PST COMP they
in niemer uerliezen
 him.ACC never leave.SBJV.PST
 ‘The Bavarians all promised him that they would never desert him’
- d. 13_1-obd-V-G > M312-G1 (tok_dipl 9538 – 9549)
da gelobte si wider in. daz si sit
 then promise.IND.PST she towards him.ACC COMP she thereafter
allez war liez
 everything real make.PST
 ‘then she promised him that she would make thereafter everything true’
- e. 12_2-bair-V_1-X > M088-N1 (tok_dipl 23212 - 23217)
er vorhte, daz ez ime zerrinne
 he fear.IND.PST COMP it him.DAT run_out_of.SBJV.PST
 ‘he was afraid that he would run out of it’

Modal verb constructions are found as correlates of FiP in MHG as well. Next to the etymological equivalents of the modal verbs found in OHG, namely MHG *sol(e)n* and MHG *wellen* exemplified in (16a) and (16b), the modal verbs MHG *mügen* ‘may’ and MHG *müezen* ‘must’ also occur, see (16c) and (16d). Except for *mügen*, all past forms of the modal verbs occurring in the database are syncretic between the indicative and the subjunctive:

- (16) a. 13_2-bair-V-X > M012-N0 (tok_dipl 15882 - 15890)
Do gehiez er in baeiden [...] Als si
 then promise.IND.PST he them.DAT both.DAT [...] as_soon_as they
des obzes enbizzen. daz si fvr war solden
 DEF.GEN.SG fruit.GEN.SG eat COMP they for true shall.PST
wizzen
 know.INF
 ‘Then he promised both of them that they would truly become wise
 as soon as they eat the fruit’
- b. 13_1-alem-PV-X > M113y-N1 (tok_dipl 8308 – 8319)
er=giheiz noe er ne=wolte uns nicht mere
 he=promise.IND.PST Noah he NEG=would us.ACC no more
ertrenchen inder sintu-luote
 drown.INF in_the deluge
 ‘he promised Noah that he would never again drown us in the flood’
- c. 14_1-omd-V-G > M311-G1 (tok_dipl 2752 - 2759)
vnd waz er recht erkande daz in
 and whatever he justly discern.IND.PST COMP.REL him.ACC
gewirden mochte
 honour.INF can.PST
 ‘and whatever he justly discerned that would increase his dignity’
- d. 12_2-wmd-PV-G > M199A-G1 (tok_dipl 109 - 123)
se gelouede êr that se mit in uore.
 she promise.IND.PST rather COMP she with them.DAT go.SBJV.PST
that thes genesen moste ere herre.
 COMP DEF.GEN.SG recover must.PST their lord
 ‘she rather promised immediately that she would go with them, and
 that their lord would recover from it [i.e. the sickness]’

Finally, we find the construction MHG *künftic sîn/wesen* ‘be to come’, which corresponds to the lexical periphrasis OHG *zuowart sîn/wesan*. There is a single

example of this type in the database, provided in (17). As in OHG, the copula verb in this example is in the past indicative.

- (17) 12-2-bair-V_Kchr1-X > M121y1-N (tok_dipl 45193 - 45205)
aines nahtes er an dem fternernen fach.
 one.GEN.SG night.GEN.SG he at the.DAT.PL stars see.IND.PST
waz im des tages kunftich was
 what.COMP him.DAT the.GEN.SG day.GEN.SG future be.IND.PST
 ‘one night, he saw in the stars what would happen to him the next day’

The construction involving the present participle of the lexical verb does not occur in the MHG part of the database.

The quantitative distribution of the forms and constructions acting as correlates of FiP in MHG are provided in Table 4. The simple forms of the lexical verb, i.e. the past indicative and the subjunctive as well as the present subjunctive, are taken together. They provide the largest group, present in 65.4% of all data. In this group, distinctive forms of the past subjunctive represent the most significant part, appearing in 44.2% of all data. This shows that despite the process of loss of mood distinctions, the proportion of forms of the past subjunctive remains the same as in OHG, where it makes up 42.3% (see Table 3). At the same time, the proportion of the modal verb construction increases from 23.1% in OHG to 33.6% in MHG. The frequency of the lexical periphrasis involving the adjective OHG *zuowart*, namely MHG *künftic*, declines from 11.5% in OHG to 1.0% in MHG.

Table 4: Types, absolute numbers, and percentages of MHG formal correlates of FiP in the database

Formal Correlate of FiP in MHG		n (%)
simple forms of the lexical verb 68 (65.4%)	past subjunctive	46 (44.2%)
	past indicative = past subjunctive	15 (14.4%)
	present subjunctive	6 (5.8%)
	past indicative	1 (1.0%)
modal verb constructions		35 (33.6%)
<i>künftig wesen</i> ‘be to come’		1 (1.0%)
Total		104 (100%)

The proportions of the indicative and the subjunctive in the MHG part of the database are given in Table 5. Simple forms of the lexical verb are taken together.

Of those, 76.5% are clearly in the subjunctive. Modal verb constructions are overwhelmingly syncretic regarding verbal mood, i.e., the formal overlapping of the indicative and the subjunctive holds for 94.3% of the occurrences. As explained above, the lexical periphrasis *künftig wesen* only attested in the indicative, but it occurs only once in the sample.

Table 5: Formal correlates of FiP in the MHG part of the database

Formal Correlate of FiP in MHG	IND n	SBJV n (%)	IND = SBJV	Total
simple forms of the lexical verb	1	52 (76.5%)	15	68 (100%)
modal verb construction	2	0 (0.0%)	23 (94.3%)	25 (100%)
<i>künftig wesen</i> 'be to come'	1	0 (0.0%)	0	1 (100%)
Total	4	52 (50.0%)	48	104 (100%)

Let us summarize the observations regarding the inventory of forms and constructions expressing FiP in the OHG and MHG part of the database.

Note that first, the corpus search reveals that the past subjunctive is the most significant correlate of FiP in OHG and MHG. But at the same time, there is variation between the indicative and the subjunctive in the domain of FiP. This is reminiscent of the situation observed for FiP in PDG, in which the morphologically subjunctive form *würde*-INF co-varies with the forms of the past indicative (next to the past subjunctive, see Section 2.1 above).

Second, the database confirms previous observations according to which there is parallelism in the expression of the future as a deictic category and FiP in OHG and MHG, in that modal verb constructions act as correlates of both, prior to the emergence of the *werden/würde*-INF. This parallelism is even expanded, as the results of our study suggest. As an example, the corpus search revealed that the construction *be* + present participle acts as a correlate of FiP in OHG. This construction is documented in descriptions of historical German where it is generally related to the expression of duration (Wilmanns 1906: 171–172, Paul 1968: 72). But according to Rick (1905: 20), *be* + present participle in OHG is also used to express reference to the future, similarly to its more prominent counterpart involving the later future auxiliary *werden* 'become' and taken as the potential source construction of *werden*-INF (Behaghel 1924: 262). Note that *be* + present participle is documented as an equivalent of the Latin future in vernacular translations of the OHG period (Hinsdale 1897: 11–12). The same applies to the construction OHG *zuowart wesan/werdan*, as Hinsdale (1897: 11) demonstrates. In the database of the present study, both constructions are used to translate the periphrastic future of the type Lat. *venturus esse*, see (13a)–(13b) and (14a). At the same time, the

putative source construction of *werden*-INF, the construction *werden* + present participle, is missing in our sample.

In the following subsection, the distributional properties of the constructions acting as competitors of the single forms of the lexical verbs in the domain of FiP will be investigated. We will compare the distribution of correlates of FiP with that of forms expressing the canonical future in the earlier periods of German (Behaghel 1924, Luther 2012 and 2013, Hartmann 2021). The focus will be on the variation between the modal verb periphrases and the respective single verb forms of the lexical verb. The construction *be* + present participle or *be* + OHG *zuowart* or MHG *künftig* will be excluded because they are not considered in the relevant work on the distribution of future constructions in the history of German.

4.2 Distributional properties of the correlates of FiP

Three aspects regarding the distribution of the simple forms of the lexical verb versus modal verb constructions will be considered, namely i. the development in the proportion of these variants, ii. the choice of modal verbs, incl. the presence of restrictions related to special positions in the verbal paradigm, and iii. the semantic properties of the lexical verb in terms of *aktionsart*.

4.2.1 General observations

Behaghel (1924: 253–266) provides an overview of forms and constructions used to express relations of the future in OHG and MHG, including their estimated frequencies. He observes that as a rule, the future is expressed by forms of the paradigm of the present indicative (Behaghel 1924: 253), and that gradually, different periphrastic expressions emerge, which compete with the futurate present. One of these is the combination of the OHG and MHG equivalents of the PDG modal verbs *sollen*, *wollen* and *müssen*, which convey the meanings of necessity, volition, and ability in the historical periods (see also Lühr 1997). According to Behaghel, *sollen* is the preferred modal verb until the end of the MHG period (Behaghel 1924: 259), while the remaining two evolve later and display low frequencies (Behaghel 1924: 259–260). As for the construction *werden* + present participle, Behaghel (1924: 261) mentions that it is still rare in OHG, having a strong inchoative meaning.

Behaghel's observations for OHG are consistent with the results of Hinsdale (1897), who examines the translations of the Latin future in vernacular texts of the OHG period. She also finds that in the predominant part of the data, the correlate of the Latin future in OHG is the present indicative (Hinsdale 1897: 37). The

modal verb periphrases are generally infrequent, and only *sollen* can be found in a genuine future function in the earliest translations (Hinsdale 1897: 9), while *wollen* and *müssen* start to appear in translations of the late-OHG period (Hinsdale 1897: 37). In addition, the construction *werden* + present participle is only marginally attested in translations of Latin futures and probably the result of a syntactic loan, because in the respective cases, the present participle is contained in the Latin original as well.¹⁰

This picture is still in place in MHG, as the corpus study by Luther (2012 and 2013) reveals. The present indicative is the main correlate of the future (Luther 2012: 4 and 6; Luther 2013: 90), co-varying with modal verb periphrases using *sollen*, *wollen* and *müssen*. The construction *werden* + present participle is still rare and strongly inchoative (see also Wilmanns 1906: 173 and Paul 1968: 72), and so is the later future construction *werden*-INF. Both *werden*-periphrases advance as forms conveying future meaning in late-MHG and ENHG (Luther 2012: 5, Hartmann 2021).

4.2.2 Development in the proportion of variants

Luther investigates the development in the frequency of the futurate present and of modal verb periphrases during the MHG period, calculating the percentages of these forms in total, as well as in main and dependent clauses individually. To account for changes within the MHG period, she identifies 5 time spans, each covering 50 years of duration, with the exception of the first sub-period, which comprises the time span from the beginning of MHG in the second half of the 11th century until the middle of the 12th century. In brief, Luther (2012 and 2013) discovers that there is change in the overall distribution of the futurate present and of modal verb periphrases within the MHG period, in that the proportion of the former continuously declines, while the one of the latter gradually increases. She also discovers some differences in the change of these proportions in main versus subordinate clauses, namely that the decline of the futurate present is stronger in main clauses than in dependent ones, while the opposite holds for the modal verb constructions. In other words, over time, the modal verb constructions become more strongly associated with the expression of the future in main clauses, while the futurate present is more consistently maintained in dependent clauses.

Let us look if the same kind of development also applies to the proportions of simple forms of the lexical verb versus the modal verb constructions in the

¹⁰As in OHG *nû uuirdist thû suîgenti* ‘and now, you will become silent’ in T 6, 9 for Lat. *erit* [sic!] *tacens*, see Hinsdale (1897: 10 and 38).

domain of FiP. Recall that only complement clauses are included in the database of the present analysis. Consequently, the comparison will only consider the figures applying for dependent clauses in Luther's study (2013).

Table 6 provides the percentages of the futurate present (listed as “ PRS_{FUT} ”) and of modal verb constructions (listed as “ MV ”) in dependent clauses in MHG, taken from Luther (2013: 92, Table 2 and 93, Table 3). The time spans distinguished by Luther (2013) are provided in the titles of the respective columns, e.g. “12_2” stands for the second half of the 12th century, etc. The first sub-period is represented as “-12_1”. Table 7 provides the absolute numbers and the percentages of simple forms of the lexical verb (listed as “ v ”) and of modal verb constructions expressing FiP in the OHG and MHG sample. The figures for OHG are taken together; those for MHG are classified according to the time spans identified by Luther (2013). In addition, separate figures are provided for distinctive forms of the past subjunctive (listed as “ SBJV_{PST} ”) and for the total of simple forms of the lexical verb found in the database.

Table 6: Proportion of futurate presents and modal verb constructions in dependent clauses in MHG (Luther 2013: 92, Table 2 and 93, Table 3)

	-12_1	12_2	13_1	13_2	14_1
PRS_{FUT}	64.6%	59.7%	59.2%	59.0%	55.9%
MV	14.4%	24.6%	24.6%	21.1%	22.2%

The proportions of the modal verb constructions as correlates for FiP in OHG and MHG must be treated with caution, because of the low number of occurrences obtained for the individual sub-periods. But let us nevertheless analyze the tendencies revealed by the figures in Table 7, compared to those described by Luther (2013) and revealed in Table 6. The data for OHG as well as for the first period of MHG must be neglected, because of the low frequency of conclusive data. For the remaining sub-periods, we can observe that there is a decline in the proportion of simple forms of the lexical verb and an increase in the proportion of modal verb periphrases over time. The modal verb constructions even reach higher percentages in the domain of FiP than in the domain of the canonical future. But this might be an effect of the database and must be corroborated on the basis of a bigger sample of data.

Table 7: Proportions of simple forms of the lexical verbs and of modal verb constructions as expressions of FiP in the OHG and MHG part of the database

	OHG	12_1	12_2	13_1	13_2	14_1
Total	n=26	n=1	n=51	n=24	n=14	n=14
SBJV _{PST}	11 (42.3%)	0 (0.0%)	25 (49.0%)	11 (45.8%)	5 (35.7%)	5 (35.7%)
v	11 (42.3%)	1 (100%)	42 (82.3%)	15 (62.5%)	5 (35.7%)	5 (35.7%)
MV	6 (23.1%)	-	8 (15.7%)	9 (37.5%)	9 (64.3%)	9 (64.3%)
Other	9	-	1	-	-	-
Periphrases	(34.6%)	-	(2.0%)	-	-	-

4.2.3 The choice of modal verbs

Recall that according to Behaghel (1924) and Hinsdale (1897), *sollen* has a future function already in the earliest OHG texts, while *wollen* and *müssen* appear only later in future constructions. These observations are only partially confirmed by the data examined by Luther (2013) for MHG. She observes that in total, *wollen* is the more frequent modal in the beginning of the MHG period, appearing in 6.8% of the cases referring to the future in the first period (-12_1) (see Luther 2013: 95, Table 5). At the same time, *sollen* is present in 4.6% of the data in the same period (see Luther 2013: 97, Table 7). The modal verbs *müssen* and *mögen* constantly display low percentages (see Luther 2013: 98, Table 9 and 100, Table 11).

The proportion of the modal verbs *sollen*, *wollen*, *müssen* and *mögen* in future constructions in MHG dependent clauses are provided in Table 8, following Luther (2013). The proportions of the same modal verbs found in constructions expressing FiP in OHG and MHG are given in Table 9. No data is available for the first period of MHG (-12_1).

The numbers in Table 9 show that *sollen* is more frequent than *wollen* in OHG, as accounted for in the literature. In MHG, however, *wollen* becomes the more frequent modal, as also observed by Luther (2013). Nonetheless, the relation of *sollen* and *wollen* is dynamic among the sub-periods of MHG and the individual

Table 8: Proportions of modal verbs as expressions of the canonical future in dependent clauses in MHG (Luther 2013: 95, Table 5, 97, Table 7, 98, Table 9 and 100, Table 11)

	12_1	12_2	13_1	13_2	14_1
<i>wellen</i>	6.0%	5.7%	10.4%	11.7%	8.0%
<i>soln</i>	5.6%	4.8%	8.3%	7.0%	12.5%
<i>müezen</i>	2.8%	10.9%	5.7%	2.5%	1.5%
<i>mügen</i>	3.2%	3.2%	5.5%	6.0%	0.9%

Table 9: Distribution of modal verbs in constructions expressing FiP in OHG and MHG in the database

	OHG	-12_1	12_2	13_1	13_2	14_1
Total	n=26	n=1	n=50	n=24	n=14	n=14
MV	n=6	n=0	n=8	n=9	n=9	n=9
<i>wollen</i>	1	-	5	3	5	5
<i>sollen</i>	5	-	2	6	3	3
<i>müssen</i>	-	-	1	-	-	-
<i>mögen</i>	-	-	-	-	1	1

records, as Luther (2013: 94–97) explains. She refers to one phase in which *sollen* is more frequent than *wollen* in her database, namely the final one (14_1), see Table 8. This dynamic behavior in the ratio of *sollen* and *wollen* is reflected in the database on FiP, but here, the phase in which *sollen* exceeds *wollen*, is the first half of the 13th century (13_1).

Similar to the findings of the previous literature, the frequency of *müssen* is lower than that of *wollen* and *sollen* in the database on FiP. Interestingly, the single example involving *müssen* in FiP is found in the second half of the 12th century, for which Luther (2013: 98) documents an exceptional rise in the proportion of this modal verb in dependent clauses in her data as well. As for *mögen*, which according to Luther (2013: 100, Table 11) reaches its highest frequency in the two phases of 13th century, only a partial correspondence with the FiP can be observed, namely that this verb is attested in the second half of the 13th and the first half of the 14th century. But as a whole, the figures for *müssen* and *mögen* must be treated with caution, because the quantity of data found in the present database is extremely low.

Let us turn to the validity of well-known paradigmatic restrictions governing the choice of modal verbs in future periphrases in OHG and MHG. Hinsdale (1897: 37) observes that *wollen* displays a preference for the 1st person singular in OHG, which she explains along the lines of the genuine modal semantics of this verb, expressing intention and volition on the part of the speaker. In MHG, the preference of *wollen* in the 1st person singular is maintained. Luther (2013: 94–95) observes that on average, half of the constructions involving *wollen* are in the 1st person singular in MHG.

The present database does not display a preference for *wollen* in the 1st person singular. There are 3 examples in the 1st person, two in the singular and one in the plural. All of them display the modal verb *sollen*, as exemplified in (18):

- (18) DDD-AD-Notker-Psalmen_1.2 > N_Ps_015, 23–24
Aber in carne uuésendo. fôre sah ih. daz ih ze gôte
 but in flesh being fore- saw.IND.PST I COMP I to God
fólletânero uérte iruuínden sólta
 accomplished life return.INF should.IND.PST
 ‘But as human I foresaw that I would return to God after [my] earthy
 existence is fulfilled’

At the same time, *wollen* is used in 19 of the examples contained in the database. None of these is used in the 1st person singular. But in all of them, the original semantics of *wollen* is present, in that the event in the dependent clause is interpretable as expressing the intention or volition of the matrix subject, see (19). The same interpretation is sometimes available in clauses involving *sollen*, see (20), which is obviously less limited to its original semantics of obligation or determination. In (21), for example, the construction involving *sollen* is plainly prognostic rather than related to necessity or obligation:

- (19) 13_1-bair-PV-G > M329-G1 (tok_dipl 11261 - 11272)
vnd gehiez in daz er in ze=trôste
 and promised.IND.PST them.DAT COMP he them.DAT for=consolation.DAT
wólte sênten den heiligen geist
 would.PST send.INF the Holy Spirit
 ‘and promised them that he would send them the Holy Spirit to console
 them’

- (20) 13_2-omd-PV-G > M408-G1 (tok_dipl 8912 - 8926)
vnd ime gelobite he daz he in na drin tagin
 and him.DAT swore.IND.PST he COMP he them.DAT around three days
volgin scholde an der martire
 follow.INF should.PST in the martyrdom
 ‘and he swore to him that he would follow them in around three days in
 the martyrdom’
- (21) 13_1-wmd-PV-X > M505-N1 (tok_dipl 510 - 518)
Alfos wisd=ons di iunfrouwe wa wir
 so indicated.IND.PST=US.DAT the damsel where.COMP we
uh finden fulden
 you.ACC.PL find.INF should.PST
 ‘So the damsel pointed us to where we would find you’

In conclusion, it can be observed that there are parallels in the inventory and distribution of modal verbs in constructions expressing the canonical future and FiP in OHG and MHG. The same modal verbs are used, with roughly the same development of frequencies over time. A difference is present regarding the observed restriction of use of *wollen* in the 1st person singular in the expression of the canonical future in late-OHG and MHG. This preference was not confirmed for the use of *wollen* in contexts pertaining to FiP. However, more data is needed to establish a reliable generalization regarding this issue.

4.2.4 Semantic properties of the lexical verb

In this sub-section, we will investigate if there is a preference for a particular type of expression of FiP, depending on the semantic properties of the lexical verb in terms of *aktionsart*.

It is well-known that verbs with different aspectual properties behave differently with respect to the way in which they express reference to future events. A prominent example is provided by Russian, in which the present tense of perfective verbs was reanalyzed as a future form, while imperfective verbs have retained their original present-tense reference and demand a special, periphrastic construction when referring to future events (see the discussion in Leiss 1992: 191–198). Similar effects have been discussed for German since the seminal work by Streitberg (1891, see Leiss 1992). Perfective verbs have been considered as inherently prospective in meaning, which makes them suitable to refer to the future even if they are used in the present, while imperfective verbs lack this prop-

erty and require special markers of futurity (see the discussion and the references in Hartmann 2021: 373 and 375).

Hartmann (2021) investigates the actional properties of the embedded verbs in *werden*-INF and its putative source construction *werden* + present participle during the MHG and ENHG period, comparing them with modal verb constructions as already well-established means of referring to the future in the same periods. He seeks to identify effects related to context expansion as an indicator of the ongoing grammaticalization process of novel constructions and tries to determine if *werden*-INF can be considered more restricted in terms of the actionality of the embedded verbs than its putative forerunner *werden* + present participle. The result is negative. Hartmann (2021) applies the seminal classification of verbs proposed by Vendler (1957), according to which predicates can be divided into telic and atelic, depending on whether verbal semantics involve an inherent terminal point (telic) or not (atelic). Telic verbs are further subdivided into so-called achievements and accomplishments, depending on whether the terminal point is reached within a single, definite moment (achievements), as in *discover*, *reach a top* etc., or involves a continuous process (accomplishments), as in *draw a circle* or *run a mile*. Atelic verbs are also subdivided into two classes, namely activities and states, the former referring to processes which involve homogeneous sub-intervals, e.g. *run*, *draw*, or *work*, while the latter denote unitary non-dynamic situations, such as *know*, *feel*, *see* or *have*. The result is that in the beginning of the 12th century, both constructions involving *werden* are limited to verbs representing the class of atelic verbs (Hartmann 2021: 378), while representatives of the class of telic verbs start to appear only later in these constructions, namely during the 13th century for *werden* + present participle (ibid.) and even later for *werden*-INF. By contrast, in the domain of the modal verb periphrases, telic and atelic verbs display an even distribution over the entire period of investigation (Hartmann 2021: 381).

Let us look at the distribution of *aktionsart* classes in the database obtained for FiP in OHG and MHG. Table 10 reveals the absolute number of verbs representing the various classes of *aktionsart* distinguished by Vendler (1957). It shows that all classes of *aktionsart* are attested in the database, although the class of telic verbs is predominant in both parts of the sample.

Let us check if there is a correlation between the *aktionsart* of the lexical verb and type of expression of FiP included in the database. Table 11 shows the absolute numbers of telic and atelic verbs found in the database, classified according to the type of expression of FiP. Again, all simple forms of the lexical verbs are taken as a total and listed as “v”, while all kinds of periphrastic constructions are taken together and listed as ‘marked construction’. Table 12 provides the same numbers

Table 10: Distribution of *aktionsart* categories of the lexical verb in the database

	Telic		Atelic	
	Achievement	Accomplishment	Activity	State
OHG	18	4	2	2
MHG	62	17	8	17
n=130				

for telic and atelic verbs occurring in any simple form of the lexical verb (“v”) and in modal verb periphrases (“mv”) in the OHG and MHG part of the database.

Table 11: Aspectual properties of verbs in FiP-constructions – single verb and verbs in all kinds of marked constructions

	Telic		Atelic	
	v	Marked construction	v	Marked construction
OHG	9	13	2	2
MHG	52	27	16	9
n=130				
OHG: Fisher’s exact value = 1, not significant				
MHG: $\chi^2 = 0.03$, p-value = .87, not significant				

The figures in Table 11 and Table 12 reveal that there is no preference for any of the periphrastic constructions included in the database, depending on the *aktionsart* of the lexical verb. We calculated the p-values for the individual periods using standard correlation tests. The results revealed that there is no statistically significant correlation between the *aktionsart* of the verb and the type of expression of FiP found in the database.

Table 13 provides the development in the distribution of telic and atelic verbs embedded in modal verbs in constructions expressing FiP in the database. We give global numbers for the whole OHG period; the MHG data is provided according to the individual sub-periods distinguished in the sections above.

Table 12: Aspectual properties of verbs in FiP-constructions – single verb and verbs in modal verb constructions

	Telic		Atelic	
	v	MV	v	MV
OHG	9	5	2	1
MHG	52	26	16	9
MHG: $\chi^2 = 0.06$, p-value = 0.81, not significant at $p < .01$				

Table 13: Aspectual properties of verbs in FiP-constructions – single verb and verbs in modal verb constructions

	OHG	-12_1	12_2	13_1	13_2	14_1
Telic	5	-	6	6	7	7
Atelic	1	-	2	3	2	2

The figures in Table 13 show that in OHG and over the entire MHG period, both telic and atelic verbs fall within the scope of modal verbs in constructions expressing FiP. Throughout the entire database, telic verbs are continuously more frequent in each of the periods distinguished.

These results confirm the picture obtained by Hartmann (2021) for MHG and ENHG, who shows that in modal verb periphrases used as future constructions, representatives of the various *aktionsart* classes are evenly distributed over the entire investigation period.

5 Summary and conclusion

In this paper, we discussed the results of a corpus study which was designed to determine the inventory of forms and constructions expressing FiP in OHG and MHG. We found that the past subjunctive is the most relevant correlate of FiP in these periods, and that it alternates with various types of periphrastic constructions, which are known to express futurity in the history of German. These are modal verb constructions, but also two additional types of periphrases, namely *be* + present participle and the lexical periphrases *be/become* in combination with an adjective meaning ‘future, to come’. We also found that the verbal mood in

these constructions is not consistently the subjunctive but rather, variation between the indicative and the subjunctive is present in the domain of FiP from the beginning of its attestation.

We observed that, while *be* + present participle and *be/become* + adjective decline over time, modal verb periphrases advance as the major competitors of the past subjunctive (or any simple form of the lexical verb) during the MHG period. Considering the observation that modal verb constructions represent a kind of a periphrastic future expression and a forerunner of the later construction *werden/würde*-INF, we investigated whether the distribution of these constructions is parallel in contexts of the canonical future, and in FiP.

Three factors were tested, namely the development in the percentages of variants over time, the choice of modal verbs, including the observed preference for *wollen* in the 1st person singular, and the presence of a correlation between the aspectual properties of the lexical verb and any type of expression of FiP.

The results revealed that the development of the proportions of modal verb constructions as competitors of simple forms of the lexical verb follows the same tendencies in both domains, in that the relative frequency of the former gradually increases, while the relative frequency of the latter declines over the investigation period. There is a difference, however, in that modal verb constructions yield higher proportions in the domain of FiP than in the domain of the canonical future. This, however, needs to be corroborated on the basis of a bigger sample of data from the respective periods.

With respect to the choice of modal verbs, the situation is similar, in that the inventory and relative frequency of modal verbs in FiP is comparable to the one in the domain of the canonical future. However, the preference for *wollen* in the 1st person singular observed for the canonical future cannot be confirmed by the data expressing FiP.

Finally, the choice of the type of expression and the semantic properties of the lexical verb in terms of *aktionsart* were examined. We found that verbs displaying all types of *aktionsart* may occur both as futurate simple forms of lexical verbs as well as in a modal verb construction expressing FiP. It was shown that the lack of restrictions regarding the *aktionsart* of verbs embedded in modal periphrases in the domain of FiP is a common property shared with modal periphrases as markers of the canonical future in MHG and ENHG.

The results of the present investigation add diachronic evidence to some controversial issues regarding the nature of FiP as a temporal category, supporting the formal compositionality and functional ambiguity of the correlates of FiP observed in modern languages, including PDG, and highlighting the parallels

between FiP and the future as a deictic category. However, future research investigating a larger sample of data is needed to obtain a more reliable picture of the parallels and differences regarding the formal and distributional properties of forms and constructions expressing the two types of the future in the history of German, including the later period in which *werden/würde*-INF raises and consolidates as a special future construction.

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Chapter 8

The entanglement of grammaticalization and hypoanalysis in the rise of futures in Slavic languages

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This article pursues two aims. The first one is to show the interrelation of grammaticalization and hypoanalysis in the evolution of future tenses in Slavic languages. For this purpose, hypoanalysis is introduced as a major mechanism of language change, distinct from grammaticalization, and exemplified by the shift of the present tense of perfective stems to mark perfective future by default, which is characteristic of entire North Slavic. This type of hypoanalysis is set off from futurate presents. The second aim consists in specifying the place of this type of hypoanalysis among the means of future marking which have been attested in Slavic languages over the past 1,000+ years. Against this backdrop, the article details the chronological relationship and the dynamics of the areal distribution of the main sources of future tenses in Slavic languages. The author asks about the reasons for North-South splits between Slavic languages in the domain of futures tenses and forwards proposals in order to understand how, in this grammatical domain, grammaticalization and hypoanalysis might have interfered to yield the contemporary distribution of future tenses in Slavic languages.

1 Introduction

The main purpose of this article is to show and, to some extent, explain the interrelation of grammaticalization and hypoanalysis in the evolution of future tenses in Slavic languages. In the first place, this presupposes that grammaticalization and hypoanalysis are distinguished as two separate mechanisms of change, that



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is, they must be disentangled conceptually, before we can demonstrate how they get entangled in observable or reconstructible diachronic processes. I will argue that the comparative study of future tenses in Slavic languages, both in their evolution and their contemporary stage, supplies excellent ground for a demonstration of grammaticalization and hypoanalysis. In connection with this, the present contribution makes an attempt at giving a comprehensive survey of the conceptual and empirical issues, based on the facts of Slavic languages, that have to be considered when trying to disentangle the often entangled relation between grammaticalization and hypoanalysis.

To achieve this aim, I first provide some general information on future tenses in Slavic languages against the backdrop of their inherited tense-aspect system. Concomitantly, I will point out the issues that have to be addressed in order to comply with the main purpose mentioned above and to disclose some diachronic and areal patterns characteristic of Slavic languages with regard to future tenses (Section 2). These issues will be specified at the end of Section 2.

A few words are due concerning the standard divisions among Slavic languages: ‘North Slavic’ is the conjunction of East and West Slavic, that is Slavic spoken north of the Danube and the Pannonian Plain. ‘South Slavic’ is the remainder, roughly south of these topographic points of orientation, with the exception of Serbian in the Vojvodina (spoken north and east of the Danube). South Slavic further divides into Balkan Slavic in the southeast (= Bulgarian, Macedonian, SE-Serbian dialects), Slovene in the northwest, and, in between, Central South Slavic¹ (CentSS), which corresponds to BCMS, i.e. SerBoCroatian incl. Montenegrin. CentSS subdivides into Štokavian in the east, which also forms the dialectal basis of Standard Serbian and Croatian, Čakavian on the Dalmatian coast, and Kajkavian in the northwest, which comprises entire Slovenian (as the basis of its standard variety) and a small region of adjacent Croatian dialects.

One further “technical” remark: extensive glossing will be supplied only if required for the particular point being made. Otherwise, I will just indicate aspect and tense with superscript indexes (^{IPFV}, ^{PFV}).

2 General background: setting the stage

Like predecessors of modern Germanic languages, Common Slavic (3rd–8th centuries AD)² lacked a dedicated future tense (henceforth ‘future’). That is, there

¹This term was suggested by a reviewer, which I hereby gratefully take over.

²This term relates to the late Proto-Slavic period (with an assumed still relatively homogeneous dialect continuum) and can be considered equivalent to proto-stages of other IE (sub)branches (cf. Andersen 1985, 2006b; Townsend & Janda 1996).

was no regular form, or construction, employed as a predicate whose core function was the denotation of a single ('episodic') situation located in time after a reference interval (either the speech interval, in deictic tense use, or some other anchor, e.g. in past-tense narratives).³ In Common Slavic (CS), apart from a system of perfects (based on a BE-copula and an active anteriority participle suffixed with *-l-*), there only existed a contrast between a past domain (with an aorist-imperfect distinction) and an undifferentiated non-past: only present tense stems were used to refer to events posterior to speech time (as for an inherited future perfect see Section 3.2.1). Various periphrastic ways of marking future were in an embryonic stage and could be best considered minor patterns. Possibly already in the CS period, i.e. in times prior to documentation, these patterns were distributed with different biases within the Slavic territory (cf. Andersen 2006b for an idea of what this distribution might have looked like), but one can hardly speak of any kind of grammaticalized stage (*pace* Andersen 2006b).

However, already by CS times a perfective-imperfective (pfv.-ipfv.) distinction (a.k.a. perfectivity opposition) had started developing. This opposition is based on stem derivation, consequently pfv. vs ipfv. aspect can be determined in every (finite or non-finite) verb form (see Section 3.1). Over time, this overarching aspect system started interacting with other categories relevant for verb morphology and clausal syntax (Wiemer & Seržant 2017, Wiemer 2020: 267–268; 2022: 70–86). In particular, the strengthening of the perfectivity opposition began to exert its impact on a differentiation of the non-past domain early since it considerably predated the rise of future constructions whose nuclei (auxiliaries) derive from specific lexical sources. These sources include verbs meaning BECOME or WANT, but also BEGIN (mainly in East Slavic), TAKE (> BEGIN, Ukrainian), STAND UP (Russ. *stan-*), and most recently the isolated case of GO in Slovak (see below); moreover HAVE, prominent at the earliest attested stages, continues playing an important role in Bulgarian and Macedonian (see Section 3.2.2).

Thus, on the one hand, futures that have emerged from grammaticalization are comparatively recent in Slavic languages, and this manifests itself in considerable variation of their various stages over the last 1,000+ years. On the other hand, even a superficial glance at the last centuries (approximately since 1500) reveals a basic North-South split concerning the predominating source domains of morphemes that have turned into nuclei of future constructions: while in the Northern area auxiliaries originating in an inchoative verb (CS **bǫd-*, BECOME)

³This more precise understanding of 'future' excludes transpositional use of present tense (as with futurate presents, see Section 3.1.3), but also reference to future events by some generalized non-past (as, e.g., in Finnish).

have come to prevail, the southern area is dominated by clitic auxiliaries deriving from the volition verb *xotěti/xătěti* (WANT). Slovene is an exception, as it marks the future with a BECOME-auxiliary (*bô-* < **bqd-*). Otherwise, however, Slovene “behaves” like any other South Slavic language (and markedly differs from North Slavic), namely: in all South Slavic languages, future auxiliaries can be combined with both pfv. and ipfv. stems; this lack of restrictions as for aspect entails a clear morphological contrast between present and future not only for ipfv., but also for pfv. stems. By contrast, in North Slavic languages future auxiliaries (BECOME or other, minor ones) do not combine with pfv. stems (Wiemer 2020: 266–270, 275–279). Exceptions are known for colloquial and dialectal varieties of Sorbian (cf. Schuster-Šewc 1996: 169, Scholze 2008: 253–254, Breu 2012: 251–252), and very few instances appear to be attested for historical stages of other North Slavic varieties. Another exception is the Slovak ‘prospective’ construction with a GO-verb, which probably appeared in the 18th century and at present shows signs of developing into a future. In this construction pfv. stems can also appear, and this can be explained by the semantics of this construction (cf. Giger 2008).

These exceptions do not change the general picture: North Slavic languages use the morphological present tense of pfv. stems to mark pfv. future, which acts as a default interpretation of their present tense inflection. There is no additional marker, and the process responsible for this change may be captured under “hypoanalysis”: peripheral and contextually conditioned functions of a construction, or a categorial opposition, turn into its inherent property due to a shift of prominence within an inherited function range. This shift in default meaning distinguishes hypoanalyzed future from futurate presents (see Section 3.1.3). Simultaneously, this shift cannot be subsumed under any of the current understandings of secondary grammaticalization:⁴ the present>future shift is conditioned by the interplay of an age-honoured present tense morphology and a more recent, though likewise well-established stem-derivational opposition of pfv. and ipfv. aspect (see Section 3.1.1), but the latter has by no means resulted from any usual grammaticalization process (Wiemer 2020: 269–270; 2022: 67–90).

At least against a European background, the interaction of futures based on the grammaticalization of lexical source expressions with the pfv. future resulting from hypoanalysis looks unique. This justifies differentiating between grammaticalization and hypoanalysis as two distinct types of grammatical change, and the diachrony of future marking in Slavic languages can be considered a paradigm example for the study of their interaction.

⁴For a critical overview of phenomena that have been called ‘secondary grammaticalization’ cf. Breban (2014).

Accordingly, this article pursues two aims. A more general goal consists in outlining the diachronic relation between futures arisen from grammaticalization and from hypanalysis in Slavic languages; this implies asking for the reasons for the aforementioned North-South split within Slavic. Among the now grammaticalized futures, I will focus on the predominant “players” based on WANT (South) and BECOME (North), while the others take a backseat.

The second topic pursued is more specific and can, as it were, be located inside this general picture. The basic North-South split for constructions marking future started gaining salience during the 13th–15th centuries. This follows from the available data and insights gained so far concerning the role of **bqd-* ‘become’ in North Slavic and of *xotěti* ‘want’ in South Slavic (with the NW-corner of the latter as an intermediate area); see Section 3. Upon closer inspection, these insights correlate with the behaviour of certain verbal prefixes in the languages at the western edges of North and South Slavic, which can be analyzed as future markers on the present tense of ipfv. stems of certain verb lexemes. Compare (1), where square brackets indicate that the prefix *po-* takes scope over the entire present tense form of the ipfv. stem (infinitive *sunout se* ‘move’) and thereby causes its future interpretation; by contrast, with the pfv. stem *objevit se* ‘appear’ it is the present tense itself which moves the reference to a future event:

(1) Czech

Objeví se mrak a celé tři hodiny

appear.PFV-(PRS>FUT.3SG) REFL cloud-(NOM.SG) and entire three hours

se po-sun-e obloh-ou.

REFL PFX>FUT-[move.IPFV-PRS.3SG] firmament-INS

‘A cloud **will** appear and **move** across the sky for three hours.’

(from Ševčíková & Panevová 2018: 185)

While this phenomenon has been known for quite some time and described reasonably well, primarily for contemporary Czech, a couple of unsolved questions have remained, which I will identify in Section 4. First of all, these questions should be considered together with differences in the gradual strengthening of the stem-derivational aspect opposition and the split between South and North Slavic concerning the interaction of non-past tense with auxiliaries as explicit markers of future. Most briefly, I claim that prefixation as a means to mark future of ipfv. present tense forms was caused by actional lability among certain semantic classes of stems which was more typical of earlier stages in the differentiation among Slavic languages. That is, in the 13th–15th centuries the relevant stems were not yet firmly assigned to pfv. or ipfv. aspect, and the opposition

between a class of pfv. and a class of ipfv. stems was less stable on the whole. In addition, future as a function of prefixes added to ipfv. present tense stems resulted from ousting alternative functions that are regularly associated with PFV.PRS. More or less at the same time, we observe the onset of a major split between North and South Slavic concerning the predominant source expressions of future auxiliaries (see above).

Correspondingly, the remainder of this article is structured as follows. In Section 3, I supply the necessary background concerning the main future marking devices, namely the default interpretation of PFV.PRS as (pfv.) future in North Slavic and its preconditions (Section 3.1), and the most prominent constructions based on auxiliaries, namely on *BECOME* and *WANT* (Section 3.2). In Section 4, I concentrate on future marked by prefixes on ipfv. present tense stems (mainly in Czech). In Section 5, both threads are brought together, before I provide a summary and tentative conclusions in Section 6.

3 Future marking in Slavic

The onset of the stem-derivational aspect system clearly predates historical records. We can confidently locate its emergent stages in Common Slavic (CS), which overlapped with the time of the Great Migrations (3rd–7th centuries AD). In Old Church Slavonic (since late 9th century AD) we already encounter the basic architecture of the contemporary system (Eckhoff & Janda 2014, Eckhoff 2020 [2018], Kamphuis 2020), although it was less consistent in terms of morphological patterns (combining prefixes and suffixes) and of the functional distribution between related stems. Moreover, morphological patterns and functional distribution had still less consistently spread over different actionality classes (cf. Wiemer & Seržant 2017: 260–272, with further references). The particular relevance of this last point will become evident in Section 4. Overt future markers, to the contrary, started emerging on the basis of an already well-established stem-derivational PFV:IPFV opposition. CS did not have any dedicated future, and the earliest attested stages of Slavic languages display a variety of periphrastic constructions used to mark future events, which were based on etyma with the meanings *BECOME* (> *BE*), *HAVE*, *WANT*, *BEGIN*, *TAKE* (> *BEGIN*); cf. Birnbaum (1958), Křížková (1960), Večerka (1993: 174–185), Grković-Mejdžor (2012: 91–93), also Borokovskij (1949: 146–157) and Ševeleva (2021) on East Slavic, as for *TAKE* cf. Danylenko (2012: 18–26) on Ukrainian and Andersen (2006b: 31) on North Russian.

I will begin with the premises that made hypoanalysis possible in North Slavic (Section 3.1). Subsequently, I will dwell upon the two most prominent construc-

tion types, BECOME for North Slavic, WANT for South Slavic, with some more background (Section 3.2).

3.1 From perfective present to perfective future

In order to understand the conditions for the pfv. present > pfv. future hypoanalysis in North Slavic we need to spell out a few basics about the architecture of the stem-derivational aspect opposition (Section 3.1.1) and its interaction with non-past tense (Section 3.1.2), before we turn to hypoanalysis (Section 3.1.3).

In general, I take pfv. aspect as a grammatical property by which a verb form indicates that the denoted situation is bounded (limited). Importantly, this need not imply the attainment of some inherent goal (as with telic verbs or clausal configurations); instead, boundedness may simply mark atelic situations as temporally limited (e.g., Russ. *Oni polčasa* [a] *posideli*^{PFV} *na skamejke, potom* [b] *vstali*^{PFV}, [c] *poguljali*^{PFV} i [d] *rozošlis*^{PFV} ‘They [a] sat on a bench for half an hour, then [b] got up, [c] walked (for a while) and [d] separated’).

3.1.1 From prolific stem derivation to grammatical complementarity

In Slavic, aspect is not indicated by unequivocal, monofunctional morphemes. Instead, the morphology of aspect builds on the functional reinterpretation of patterns of stem derivation; these patterns consist in the extension of simpler stems by the (usually consecutive) addition of prefixes and suffixes, or in a suffix change (Breu 2000, Wiemer 2008, Wiemer & Seržant 2017, among many others). Apart from minor, mostly lexically restricted or obsolete patterns, two patterns predominate and are productive across contemporary Slavic. See (2–3) with infinitives in Polish. The concrete shape of the suffixes is of minor concern (see fn. 5); stems are given in square brackets.

- (2) Predominant patterns of stem derivation and their grammatical reinterpretation
 - a. simplex $\Rightarrow$ PREFIX + simplex
 - b.

prefixed stem $\Rightarrow$ [prefixed stem] + SUFFIX
$\Downarrow$ reinterpretation: (a) (potentially) identical lexical concept
(b) different grammatical distribution

- (3) a. simplex^{IPFV} — prefixed stem^{PFV} (e.g., Pol. [gotowa]-ć ⇒ [**u**-gotowa]-ć ‘cook’)
 b. prefixed stem^{PFV} — [[prefixed stem] + SUFFIX]^{IPFV}
 e.g., Pol. [prze-kaz-a]-ć ⇒ [prze-kaz-**ywa**]-ć ‘convey’ (suffix addition⁵)
 [s-praw-i]-ć ⇒ [s-praw-i-**a**]-ć ‘cause’ (suffix change)

In a nutshell, the reinterpretation consists in the derivation of more complex stems from simpler stems which can (but need not) share an identical lexical meaning but in any case differ in their distribution over grammatically or pragmatically definable contexts and in their sensitivity to actionality (events, processes, states) and external pluractionality (regular or irregular iteration of distinct eventuality tokens). Grammatical contexts may be defined by the interaction of stems with other categories marked on the verb, the verb phrase or on clause level (such as tense, voice, or polarity); alternatively, there may be pragmatic contexts to which aspect choice is sensitive, such as illocutionary contrasts (e.g., warning vs prohibition) or presuppositions that the speaker assumes to be shared (or not) with the addressee. Importantly, morphemes marking tense, mood, agreement (person/number), but also non-finite forms, all occur outside of stems.⁶ Thus, a form like Pol. *pisala* in (4a) has the same structure of tense-agreement marking as has its pfv. counterpart *napisala* in (4b) derived with the prefix *na*-.

- (4) Polish, past tense
 a. [pis-a]^{IPFV}-ł-a
 write-THV-PST-SG.F
 ‘She wrote^{IPFV}’
 b. [**na**-pis-a]^{PFV}-ł-a (list)
 PFX-write-THV-PST-SG.F letter-(ACC.SG)
 ‘She wrote^{PFV} (a letter)’

The same applies to non-past stems, as in Pol. *roz-wiż-q* (5a) vs *roz-wiż-uj-q* (5b), or the infinitives *prze-kaza-ć* ⇒ *prze-kaz-ywa-ć* ‘convey’ in (3b), in which a suffix is used to derive an ipfv. stem:

⁵Most probably, the suffix -ywa- is an accretion of an earlier suffix -va- with a preceding vowel (associated with causatives). On a synchronic level, the relation between (prefixed) pfv. stem and its suffixed ipfv. derivative looks like a suffix change (here: with the thematic vowel -a-). For the consistency of the pattern itself this is of minor concern.

⁶Present, or non-past, tense can also be indicated by the choice of the stem (compare 4 vs 5). However, distinctions between present/non-past stem and infinitive (< aorist) stem are older than the prefixation-suffixation patterns in (2–3).

(5) Polish, present tense

- a. [roz-wiąż]^{PFV} -q (sznurk-i)
 PFX-tie- PRS.3PL lace-ACC.PL
 ‘They (will) untie (the shoelaces).’
- b. [roz-wiąz-uj]^{IPFV} -q (sznurk-i)
 PFX-tie-SFX -PRS.3PL lace-ACC.PL
 ‘They untie / are untying (the shoelaces).’

Present tense stems, as in (5a) and (5b), often show morphonological alternations at the end of roots; compare -wiąż- [v’õʒ] in (5a) vs -wiąz- [v’õz] in (5b). Likewise, suffixes may alternate between the infinitive and the present (or non-past) stem; compare -ywa- (see *przekaz-ywa-ć* in 3b) vs -uj- (see *rozwiąz-uj-q* in 5b). These processes affect inflectional morphemes only insofar as present tense stems imply a specific set of person/number endings.⁷ Otherwise stem derivation does not affect the explicit marking of the respective categories, nor is it itself affected by them. However, it often restricts combinability with these categories or their interpretation. This is what happens, in particular, to tense distinctions and the combinability of the infinitive with all kinds of auxiliary verbs. Thus, forms like *rozwiążq* ‘they (will) untie’ in (4a), which are based on the non-past stem, are by default interpreted as future.⁸

Suffixation techniques had been part and parcel of stem derivation since Proto-Indo-European (PIE) times, while more recent prefixation developed from morphologically independent verbal modifiers and became commonplace in practically all IE groups of Europe. However, only Slavic brought prefixation beyond the stage of merely “telicizing bounders” and applied it all over the verbal lexicon in combination with suffixes.⁹ Since both strategies started to combine (no later than in CS), they “stayed together” as prolific devices that continued to be productive through the entire history of Slavic languages, albeit to varying degrees in different subareas.

As a consequence of prolific stem derivation, patterns like those in (2–3) have been creating a tremendous number of morphologically related verb stems. In

⁷This principle is an IE heritage. One may thus say that present tense is indicated twice: by the stem and by its correlated endings.

⁸Strictly speaking, such forms might better be glossed as non-past (NPST) – first of all, because non-future interpretations have not been ousted (see Section 3.1.2).

⁹Practically all other IE languages of Europe lost suffixes as a technique to mark actionality distinctions independently of tense and valency. Lithuanian is an exception, possibly via contact with (East) Slavic (Arkadiev 2014; 2015, Wiemer & Seržant 2017: Section 5, Wiemer 2023: 205–212).

many cases, the derivational affixes did not change the lexical meaning, but yielded synonyms, which started to be distributed over different grammatical contexts, with an increasing tendency towards complementary distribution. In the Slavist tradition, pairs of morphologically related pfv. and ipfv. stems that are able to convey an identical lexical meaning but occur in complementary contexts¹⁰ are called aspect pairs. These can, in a sense, be considered the backbone of the system.

However, the aspect membership of a particular stem does not hinge on entering into a relationship of strict replacement conditions or on looser relations, e.g. between stems denoting atelic activities (e.g., Russ. ipfv. *plakat'* 'weep', *letet'* 'fly') and their ingressive modifications (pfv. *za-plakat'* 'start weeping' and *po-letet'* 'fly off', respectively), which can be considered 'aspect partnerships' (Lehmann 1988). Ultimately, membership in pfv. or ipfv. aspect depends on whether the respective stem adds a meaning of boundedness (limitation) to the context, or is compatible with this feature ($\Rightarrow$ pfv.), or whether it does not, or is indifferent in this respect ($\Rightarrow$ ipfv.). This can be tested in various ways, albeit to different degrees of reliability. For instance, a reliable indicator of ipfv. aspect is the ability of the finite verb to denote ongoing processes (as in *Right now, X is eating his porridge*); however, not all ipfv. stems are capable of denoting ongoing processes.

After all, the spread of derivational patterns like those in (2–3) has turned into a productive technique that opposes stems which are morphologically related but show complementary distribution over actional and pluractional contexts (and thence to more and more contrasts). This was a long-lasting process, which only stepwise involved different actionality groups with varying speed and consistency. There are some indications that aspect pairs and partnerships appeared first for stems that coded strictly goal-directed ('strongly telic') situations and for some verbs denoting speech acts; only later did pairs for punctual events and for atelic activities appear.¹¹ For some of the latter aspect partnerships proved unstable; in particular, this applies to verbs of (directed) motion, as we will see in Sections 4.3–4.4.

3.1.2 Irrealis meanings and the conventionalization of invited implicatures

Old Church Slavonic (OCS) presents us with the earliest Slavic text monuments (New Testament, gospels) that have been passed down to us from the 10th cen-

¹⁰The conditions and tests on aspectual pairedness are concisely reviewed in Wiemer (2017: 227–228), Wiemer et al. (2020: 236–240), and Łaziński (2020: 27–29).

¹¹Cf. Bermel (1997), Mende (1999), Eckhoff & Janda (2014), Eckhoff (2020 [2018]), Kamphuis (2020). On the significance of perfectivized atelic stems for the aspect system cf. Arkadiev (2014; 2015: 82–89) and Wiemer & Seržant (2017: 265–269).

tury onwards. They already show a fairly well-developed distinction between morphologically related stems with a tendency toward complementary division of labour of actionality distinctions, which can be discerned primarily in narrative past tense contexts. As for the non-past domain, matters might seem less clear inasmuch as there was no dedicated future marking. However, provided pfv. and ipfv. stems can be sufficiently distinguished on the basis of their stem-derivational relation, non-past forms of pfv. stems are also encountered in pluractional (i.e. habitual or gnomic) contexts. In such contexts they do not only signal ‘that the event occurs on more occasions, but rather that the event will occur when the right circumstances occur’ (Kamphuis 2020: 162; emphasis added); see similarly in Večerka (1993: 152). This amounts to dispositional or circumstantial, i.e. non-deontic modal, readings. Compare, for instance, (6), in which PFV.PRS translates Greek present tense forms, and (7), which contains a relative clause with a generic head noun (*vsěkb* ‘anybody’).¹² In either case, the non-past forms do not refer to any single, let alone ongoing situation, and the named events do not (or not necessarily) occur in some interval posterior to the time of utterance; instead, time reference is generalized, or unspecific:

- (6) *i g[lagol]ljo . semu idi^{IPFV.IMP} . i idet^{IPFV.PRS} i drugumu prid^{PFV.IMP} . i pridet^{PFV.PRS} . i rabu moemu s^tvorⁱ^{PFV.IMP} se i s^tvorit^{PFV.PRS}*
 ‘And I say to this one, “Go!” and he goes, and to another, “Come!” and he comes, and to my slave, “Do this!” and he **does** it.’

(Matthew 8:9; Kamphuis 2020: 162)

- (7) *az^o že g[lagol]ljo vam^o ěko vsěkb iže v^oz^orit^{PFV.PRS} na žen^o s^o poxotij^o. uže ljuby s^tvorⁱ^{PFV.AOR} s^o nej^o v^o sr^odcⁱ svoem^o [Z, M]*
 ‘But I say to you that everyone who **looks** at a woman with lust for her has already committed adultery with her in his heart.’

(Matthew 5:28; Kamphuis 2020: 162)

However, also well-attested are pfv. non-past forms with an unambiguous future reference, i.e. with reference to single events after (presumed) utterance time; compare, for instance, (8–9):

- (8) *t^ogda ěgla im^o is. vsi vy s^oblaznite^{PFV.PRS} s^o o mně v^o s^oj^o nošt^o*
 ‘Then Jesus told them: all of you (will) **become concerned** about me this night.’

(Matthew 26:31; Večerka 1993: 175)

¹² *v^ozr^oeti* ‘look’ (ipfv. *zr^oeti*) translates a Greek present active participle.

- (9) *az̃ poml̃čŋ^{PFV.PRS} sebe radi. da povēdēt̃^{PFV.PRS} čudesā °ba mojego*
'I will keep silent myself for them to tell the miracles of my God [or: so
that they may/will tell the miracles of my God].'
(Suprasl' chronicle; Ivanova-Mirčeva 1962: 35)

In fact, pfv. non-past was a frequent, if not the most frequent, means to express future events in OCS, also as an equivalent of the Greek future (cf. Birnbaum 1958: 17–18, Bunina 1959, Ivanova-Mirčeva 1962: 28–49, Večerka 1993: 175–176). Moreover, it was widespread after the irrealis connective *da*, particularly in contexts of volition (e.g., purpose clauses as in 9 and 10), and in the protasis of conditionals. The latter belong to cognition-based contexts with downgraded epistemic support, and they are associated with relative clauses with generic referents (as in (7)).

- (10) *s̃lěz̃ ot̃ °nbse °adl̃ sedm̃. da ot̃ženŋt̃^{PFV.PRS} vse z̃zlo*
'Seven angels came down from heaven to drive away [lit. that (they)
drive away] all evil.'
(from Večerka 1993: 92)

That is, pfv. non-past was encountered in all kinds of irrealis contexts, i.e. in 'the realm of the imagined or hypothetical' to denote a potential event, but 'not an observable fact of reality' (Elliott 2000: 66–67). Future events are closely associated to the realm of irrealis inasmuch as they cannot be checked at the moment of speaking.

For all contexts just mentioned, the [+bounded]-feature of pfv. aspect is crucial, in particular for pluractional contexts (and their non-deontic modal implications): in order to conceive of repetitions, situations need to be discernible at distinct (= limited) intervals, and each potential occurrence of the denoted situation type can easily be linked to conditions (= circumstances) that make it more likely to happen and allow it to represent the situation type as its imagined token. Likewise, if one knows about somebody else's inclination or about some lawful behaviour (i.e. a situation type), one may – via abduction – infer from circumstances observed, or holding, at the moment of speech that a specific (singular) manifestation of this behaviour (i.e. a situation token) will happen. There is thus a natural connection between habituality, the assignment of stable, and thus predictable, properties and predictions based on 'the right circumstances' (see quote above) that are observed in a particular setting. These connections have been recognized as a chain of conceptual bridges to future reference (cf. Tatevosov 2004, Shluinsky 2006, Sonnenhauser 2008). The last link of this chain is predictions

(i.e. epistemic judgments) concerning specific events. These events cannot be “checked” since they are thought to occur after the speech act (or another reference interval), nor can habitual or associated modal meanings be checked since they are not about specific intervals.

Clearly, the present tense, or rather: non-past, forms of pfv. stems in Slavic were unable to denote any ongoing situations right from the start whenever the perfectivity opposition established itself (during CS) on the basis of the past:non-past contrast in verb morphology. However, given the tight connection between the aforementioned usage types of the non-past forms of pfv. stems in Slavic – and since no further morphology is involved – one wonders how present and future can be told apart for these forms. After all, all sorts of readings as “inactual present” were no less inherent to these forms from the start than reference to the future, i.e. to single events after speech time. Therefore, if future (i.e. reference to a single situation token after speech time) has become predominant for pfv. non-past in North Slavic (see Section 3.1.4), this can only mean that this function has become more salient within this domain. It does not mean that non-future functions (related to irrealis) have vanished; instead, they have only been driven into the background as being less frequent.

3.1.3 Hypoanalysis

Before we continue browsing through diachrony, let us examine to what extent such a shift of saliency within the non-past can, and should, be subsumed under hypoanalysis. Croft (2000: 126–127) defines hypoanalysis as follows:

In *HYPOANALYSIS*, the listener reanalyzes a contextual semantic/functional property as an inherent property of the syntactic unit. In the reanalysis, the inherent property of the context (often the grammatical context (...)) is then attributed to the syntactic unit, and so the syntactic unit in question gains a new meaning or function.

In Croft’s wording, hypoanalysis applies to syntactic units and it results in a shift to a new meaning that previously was only implied by favourable types of context. Strictly speaking, the case of North Slavic PFV.PRS acquiring the meaning of (pfv.) future as its default function slightly differs in both respects. First, the new prominent function of PFV.PRS was not really new, but already implied as just one among others that an undifferentiated non-past form could acquire, before and after the perfectivity opposition came into play (see Section 3.1.2). Thus, the shift rather consisted in highlighting a subdomain within the

non-past, no matter what caused this conventionalized shift of prominence. Second, and more importantly, we are not dealing with a change applying to syntactic units, nor does the change hinge on any source expressions (etyma) in their syntagmatic settings; instead it pertains to the interpretation of functional oppositions marked, or induced, by patterns of stem derivation. It is, thus, paradigmatic contrasts between two classes of stems (called ‘pfv.’ and ‘ipfv.’) against the background of morphological patterns which favour the default shift into future. Therefore, in subsuming the conventionalized future shift of PFV.PRS under hypoanalysis we should be aware that this brings about a certain expansion of this notion beyond Croft’s original definition.

Moreover, paradigmatic contrasts lie at the core of Andersen’s (2006b) considerations about types of language change which he applies to the evolution of futures in Slavic languages. Andersen understands grammaticalization as a complex of changes, more or less in line with mainstream literature on grammaticalization, but he neither explicitly mentions hypoanalysis, nor does he consider the default shift of PFV.PRS from undifferentiated non-past to future.¹³ However, hypoanalysis comes close to Andersen’s (2006b: 39) ‘regrammation’, which amounts to a “change by which a grammatical expression through reanalysis is ascribed different grammatical content”.

Apart from these caveats, another point is crucial: the conventionalized default shift of PFV.PRS to pfv. future in North Slavic cannot be subsumed under futurate presents. The latter have been well described for Germanic languages, but they are certainly widespread beyond. Futurate presents do not change their morphology, which is also the case for the North Slavic “hypoanalyzed” pfv. future. However, futurate presents acquire future reference not by virtue of a conventionalized default; instead, they are just one, non-dominant reading among many. Futurate use is often difficult to predict, although plans and schedules may trigger it (Copley 2014); see (11). Moreover, the present tense form can be ambiguous between an ongoing present and a future reading (Hilpert 2008: 157, 159). This is excluded for North Slavic PFV.PRS; instead, in all (not only North) Slavic languages it is the present tense of ipfv. stems which is widely used for scheduled future, just like the English present progressive or the German present.

This comparison also shows that futurate presents need not build on a contrast of grammatical aspect nor of present vs future. In English, it is the “non-perfective” member of the aspect opposition which is used for scheduled events (although the Simple Present can be used as well; cf. Copley 2008; 2014: 72) and

¹³Simultaneously, Andersen (2006b: 31) erroneously describes pfv. aspect as referring to states. Of course, this misunderstanding precludes an adequate analysis of what happens to PFV.PRS.

there is a clear opposition between present and future morphology. None of these properties holds, e.g., for German (there is no grammatical aspect and the *werden*-future is much less consistently employed than the English *will*-future), in which futurate presents are quite commonplace.

We thus have to distinguish between common shifts of temporal reference of present tense forms that, after all, do not change the default function of these forms, on the one hand, and a conventionalized shift (or narrowing) leading to a new default function, on the other. Compare the following translational equivalents: the Russian ipfv. present (*vstrečāemsja* ‘we meet’) for scheduled events is equivalent to the present tense in German, English, and Bulgarian (see 11); by contrast, PFV.PRS (*vstretimsja* ‘we (will) meet’) is functionally equivalent to the future form in languages like English and Bulgarian, where the present-future contrast is more firmly established than in German (see 12). This corresponds to the default of future reference for PFV.PRS, which is not the case with the futurate interpretation of present tense in (11).

(11) **scheduled (planned) events**

Russ.	<i>My vstrečāemsja zavtra.</i>	ipfv. present
Bulg.	<i>Sreštame se utre.</i>	ipfv. present
Engl.	<i>We are meeting / meet tomorrow.</i>	present (progressive / simple)
Germ.	<i>Wir treffen uns morgen.</i>	present (no aspect distinctions)

(12) **future (predictive, but not necessarily planned)**

Russ.	<i>My vstretimsja skoro / kogda-nibud’.</i>	pfv. present (non-past) > pfv. future
Bulg.	<i>Šte^{FUT} se sreštnem skoro / njakoj den.</i>	pfv. future
Engl.	<i>We will^{FUT} meet soon / some other day.</i>	future
Germ.	<i>Wir treffen uns bald / ein andermal.</i>	present
	<i>Wir werden^{FUT} uns bald treffen.</i>	future (optional)

After these clarifications let us now return to the reconstruction of the “chronology of changes” that contributed to the differentiation of future marking among Slavic languages.

3.1.4 Beginning split between North and South Slavic

The first written documents of East Slavic date from the 11th century, the first West Slavic documents are from the late 13th century in Old Czech, and the first Old Polish texts are from the mid-14th century. The earliest documents in western varieties of South Slavic are from the 12th century in the Štokavian-Čakavian

area (Old Serbian-Croatian),¹⁴ whereas it is difficult to name a clear “cut-off point” between Old Church Slavonic and the emergent stages of Balkan Slavic (leading to modern Bulgarian and Macedonian), but Middle Bulgarian is considered to comprise the 12th–14th centuries. Regardless of these different “onsets” of historical attestation, already at these earliest accessible stages some differences can be observed among these groupings (East, West and western vs eastern South Slavic) when it comes to marking future events. As for the emergence of auxiliary-constructions, the employment of BEGIN-verbs (+ infinitive), apart from HAVE, was strongest in early East Slavic (up to the 16th–17th centuries), but unremarkable in the remainder. In turn, in the earliest West Slavic texts we already encounter a clear preference for BECOME (CS **bǫd-*), which proved a serious “rival” of WANT and HAVE in the early Štokavian-Čakavian area, before it was superseded by WANT (see Section 3.2).

Moreover, in South Slavic dedicated future markers have appeared that are not sensitive to aspect, as they also combine with pfv. stems (see Section 3.2.2). This stands in stark contrast to North Slavic and justifies the assumption that the conventionalization of future meaning as a default of PFV.PRS (= hypanalysis) was blocked by new constructions with BEGIN- and BECOME-auxiliaries.

What happened to PFV.PRS? The shortest answer is that in North Slavic the future implicature turned into a conventionalized default, but non-future functions of PFV.PRS have not been abandoned and are still well-attested even in Russian (see 13, with a habitual reading), although with heavier restrictions than in the western parts of North and of South Slavic.

(13) Russian

*I s ego točki zrenija, novyj den’ – èto novoe čudo. On ne speša
prosnetsja*^{PFV.PRS}, *plavno primet*^{PFV.PRS} *vannu, vdumčivo vyp’et*^{PFV.PRS}
kofe, (...). (RNC)

‘And from his point of view, a new day is a new miracle. He slowly **wakes up**, **takes** a gentle bath, thoughtfully **drinks** coffee (...)’

By comparison, in contemporary Bulgarian (and Macedonian) PFV.PRS has largely been restricted to co-occurrence with irrealis-marking connectives, mainly with the verbal proclitic *da*, but also with conditional or temporal conjunctions (e.g. Bulg. *ako* ‘if’, *kogato* ‘when’), to relative clauses (especially headless ones, see 14) and to biclausal proverbs with a sequential structure (15).

¹⁴The oldest texts with clearly Slovene features are the *Freisinger Denkmäler* (about 1000 AD), but they are very limited and do not contribute anything to the issue of (emergent) futures (see Section 3.2.1.1).

In all such cases, conditionality ('if..., then...') keeps these clause units together. In addition, they are deprived of specific temporal reference (see 14–15).¹⁵

(14) Bulgarian

Kakvoto reši^{PFV.PRS}, *go pravi*^{IPFV.PRS}.

'What(ever) s/he **decides**, s/he realizes (it).'

(adapted from Lindstedt 1985: 192)

(15) Bulgarian

Izlāžeš^{PFV.PRS} *li vednāž, ne ti vjarvat*^{IPFV.PRS} *vtori pāt*.

'If you **lie** once, people don't believe you a second time.'

(from Maslov 1956: 231)

In all these contexts, PFV.PRS continues to be freely used in Russian and all other North Slavic languages. Therefore, non-future meanings of PFV.PRS are not only original throughout Slavic, but they have stayed the main shared usage domain of all South and North Slavic languages until today. Even in the eastern part of Slavic (both North and South) non-future uses form a strong "residual" of PFV.PRS irrespective of whether dedicated future markers have arisen and how they interact with aspect. That is, neither the default shift of PFV.PRS to pfv. future (hypanalysis) in North Slavic nor the grammaticalization of future auxiliaries that are "insensitive" to aspect (in South Slavic) have caused any major changes in the irrealis usage domain of PFV.PRS (cf. Wiemer 2026 for a detailed discussion).

Against this backdrop, can the period when this default shift became salient somehow be specified? According to specialists, already by the 12th century North and South Slavic seem to have begun to divide up the preferred domains of usage of PFV.PRS. As for South Slavic, Grković-Mejdžor (2012: 87) remarks that already in the oldest Štokavian/Čakavian documents (12th–15th c.) PFV.PRS exclusively appears in all irrealis domains except the future (cf. also Belić 1999: 443); evidently, future reference had been taken over by other means (see Section 3.2.2). At first sight, this conclusion seems to contrast with analyses of the evolution of future marking in Balkan Slavic which suggest several centuries of overlap between PFV.PRS and the developing WANT-future. Thus, following Ivanova-Mirčeva (1962: 42–49, 67–68) or Asenova & Guentchéva (2022) one would conclude that, in Bulgarian, PFV.PRS continued to mark future events (also in main clauses) up to the 16th century and, in this usage, ultimately disappeared only

¹⁵For comprehensive accounts of PFV.PRS-uses in Balkan Slavic cf. Maslov (1956: 231–233), Lindstedt (1985), Tomić (2006; 2012).

by the 18th century. However, such a conclusion would mainly be based on texts which continue the OCS “canon”, whereas texts reflecting vernacular usage largely remained out of focus. Vernacular speech may be assumed to have propelled the newer futures based on *WANT* and *HAVE*. Thus, the impression that these newer constructions and PFV.PRS co-existed over such an amount of time as means of future marking most probably simply reflects an overlay of conservative OCS-traditions with much more innovative vernacular speech (see also Section 3.2.2).¹⁶

As concerns North Slavic, Borkovskij (1949: 145) points out that during the 12th–16th century PFV.PRS was the predominant choice (88%) for reference to future events in East Slavic secular texts. Similarly, Mišina (2018: 124) summarizes findings on Old East Slavic and notes that in the earliest attested period (11th–14th centuries) the most common way of referring to the future was by using PFV.PRS, that is, just as in OCS (see Section 3.1.2). Provided these conclusions can be generalized for South and North Slavic, respectively, they are indicative of a beginning split between North and South Slavic in the 13th–14th centuries.

3.2 Grammaticalization: *BECOME* vs. *WANT*

Caveats concerning discrepancies between vernacular speech and OCS-influenced writings are probably appropriate for all constructions that were grammaticalized into futures in Slavic. As a rule, the constructions in which lexical source expressions of emergent future markers and the form of the lexical verb combined were ancient minor patterns. This applies in particular to the combination of *BECOME* (**bqd-*) with the *l*-form (originally an active anterior participle), whereas the combination of *BECOME* with the infinitive was one of the really few innovations (see Section 3.2.1). In this construction, *l*-form and infinitive must have co-existed for some time in different areas, but data is too scarce and inconclusive, also as for aspect restrictions of the lexical verb and the issue whether and when **bqd-* + *l*-participle turned from a future perfect into a “simple” future. Finally, we also cannot decide to which extent this construction was used in parallel with future reference marked by PFV.PRS (cf. Křížková 1960: 95–96, 156–158, Whaley 2000: 26–33).

In South Slavic, this parallelism was probably reduced with the consolidation of the ‘want’-based future (see Section 3.2.2). However, a complex syntactic and functional distribution of the latter with the *bud*-construction and PFV.PRS continues to be characteristic of different Central South Slavic (CentSS) varieties

¹⁶I am grateful to Jasmina Grković-Mejdžor for having drawn my attention to this fallacy.

until today, in which *bud*-constructions are employed like a future perfect in certain embedded clauses.¹⁷ Regardless of meaning, contemporary Slovene has *bô*-only with the *l*-form. Likewise, adjacent Croatian Kajkavian dialects have *bud*- (or *bô/bu*-) with the *l*-form. For some of them this auxiliary is also occasionally noted with the infinitive (see fn. 17), but this combination might probably be explained through interference with Štokavian dialects or (likewise Štokavian-based) Standard Croatian.¹⁸ As for North Slavic, only Polish (with contiguous regions) and Kashubian still optionally use the *l*-form (of ipfv. stems) with the future auxiliary (*będ*-),¹⁹ in all other North Slavic languages it only occurs with the infinitive.

3.2.1 BECOME

The Slavic root **bqd-* is cognate to the existential-copular verb *by-ti* ‘be’ and can be considered a nasal extension of a PIE root (< PIE **bhū-n-d(h)-*); cf. Werner (1996), Grković-Mejdžor (2012: 95). Probably, **bqd-* originally was inchoative,²⁰ although by the time of its first written attestations it might have lost its inchoative flavour (Krämer 2005: 75–81). Nonetheless, an inchoative feature might have played a role in its reinterpretation as a future marker (see below).

The development of the BECOME-future must have started after the split of Common Slavic into North and South Slavic (i.e. more or less by the 9th century AD), as it took different paths within individual languages (Whaley 2000: 25, 73). This shows up in two respects: first, whether aspect restricts its use; second, whether **bqd-* combines with the *l*-form (originally an active anteriority participle) or the infinitive. Initially, *bqd-* only combined with the *l*-participle as a future perfect (a.k.a. *Futur II* or *futurum exactum*), without restrictions of aspect (see 16).

¹⁷Cf. Kovačević (2008) on Serbian, Vuković (2014) for Croatian complex sentences. Cf. also Andersen (2006b: 20) for diatopic variation in the Kajkavian, Lisac (2009) for diatopic variation in the Čakavian region. Cf. also Lukežić (2015: 348–349) for a brief summary on Kajkavian, Čakavian and Štokavian and Grković-Mejdžor (2019: 11–18) for a concise overview of the diachronic development and diatopic distribution of future marking in South Slavic.

¹⁸I am obliged for this suggestion to an anonymous reviewer.

¹⁹There are no conceivable functional motivations of this variation in Polish (Łaziński 2006: 317–321, Błaszczak et al. 2014: 184–187). In Kashubian the distribution of both variants rather seems to be diatopic, possibly with stronger Polish influence (for *bqd-* + INF) in the southern part (Andersen 2006b: 141–142).

²⁰Cf. Whaley (2000: 23–24). For a recent concise overview of positions cf. Pen'kova (2019b: 570–571).

(16) Old Church Slavonic

i ašte grěx-y bŏd-etъ sŏtvori-l-ъ otdad-ętъ
 PTC if sin-ACC.PL FUT-3SG make[PFV]-LPT-SG.M surrender[PFV]-PRS.3PL
sę emu
 REFL 3SG.M.DAT

‘Even if he commits [lit. **will have committed**] sins, they will devote themselves to him.’

(Jak 5:15; Večerka 1993: 180)

However, already in OCS this temporal function sometimes oscillated with an epistemic, or even inferential, interpretation (Lépissier 1960, Ivanova-Mirčeva 1962: 145–147). The subsequent earliest attested stages of Slavic languages already reveal considerable variation.²¹

3.2.1.1 South Slavic

In South Slavic, its middle Section, the Štokavian-Čakavian area, is most interesting. For this area, the most comprehensive account to date is Grković-Mejdžor (2012). She reports (2012: 95–97) that the original construction *bud-* + *l*-form was extremely rare in its western (i.e. the Čakavian) part,²² while in the eastern (Štokavian) part it was attested in the 14th century. It could still have an anterior function, like a future perfect, but it also occurred in hypothetical conditionals and in contexts with past reference. Thus, the anterior function ceased to be the core function, but it might have been the functional pivot which explains why, in some of the early documents, *bud-* + *l*-form is also occasionally attested as an epistemic downtoner or reportive marker. Otherwise, the construction simply turned into a future.

In the western part, instead of the *l*-form, we find *bud-* in combination with the infinitive from the end of the 14th century (see also fn. 22). Grković-Mejdžor (2012: 99) calls it a Štokavian-Čakavian innovation. The construction first appeared in dependent clauses, where, in the majority of cases, it marked precedence to another situation. In independent clauses *bud-* + infinitive first appeared in Čakavian (from the second half of the 14th century), before it spread into the remaining (Štokavian) territory, i.e. from west to east within that area, but not

²¹For a brief overview cf. Arkadiev & Wiemer (2020: 197–199). Reliable corpus-based surveys and insights are provided by Pen’kova (2016; 2017; 2018; 2019a; 2019b; 2020).

²²However, according to an anonymous reviewer, Lukežić (2015: 350) cites *bud-* + *l*-participle (alongside with the infinitive!) for sources dating back to the 13th century. I am unable to check this (see fn. 18).

beyond it: over the entire history of Balkan Slavic *bqd-/bud-* + infinitive is unattested (Ivanova-Mirčeva 1962: 144).

Consequently, in early Čakavian and Štokavian varieties (14th–15th centuries) we observe a somewhat unclear diatopic distribution of *bud-* + infinitive (innovative) and *bud-* + *l*-form (inherited). Nonetheless, at least Grković-Mejdžor's account suggests that the innovative pattern from the west and the older pattern from the east "ran into one another", causing different "mixtures" in various pockets, as this shows up today in some dialects (see above). In both patterns, stems of either aspect occurred, although for the new pattern (*bud-* + infinitive) ipfv. stems were 2–3 times more frequent than pfv. ones (Grković-Mejdžor 2012: 97). This was practically inverse to the proportions of pfv. vs ipfv. stems in the original pattern *bud-* + *l*-form (see below). It therefore seems that in South Slavic for either pattern there never was any restriction as for aspect. However, for various historical and contemporary local varieties it needs to be checked to what extent PFV.PRS was/is likewise widespread to mark future reference (also in main clauses) and, thus, whether the distribution of *bud-/bô-* was/is less symmetric for the aspect of the lexical verb on account of PFV.PRS being used as an alternative to *bud-/bô-* with pfv. stems.

We are left with the question how the original pattern *bud-* + *l*-form "survived" in Slovene, i.e. the northwest corner of South Slavic. Slovene remained unaffected by the "Balkanized" type of future, which, based on successors of want-verbs, was an innovation from the southeast (see Section 3.2.2), but this does not explain why in Slovene the pattern *bô-* + *l*-form has persisted as the only one. Unfortunately, the earliest Slovene texts (*Freisinger Denkmäler*, 10th–11th centuries) show no evidence of **bqd-* at all (Whaley 2000: 27–28), nor are there any attestations of **bqd-* + infinitive in the history of Slovene (Andersen 2006b: 19).

3.2.1.2 North Slavic

Concerning the development of the 'become'-based future in West Slavic, the most comprehensive and thorough account, based on a large comparative body of evidence, is still Křížková (1960), regardless of criticism (see Section 3.2.1.3). Within North Slavic, *bqd-/bud-* most probably started to consolidate as a future marker in Bohemia (13th century). Křížková (1960: 92 et passim) emphasizes that *bud-* + infinitive figures in Old Czech texts from the end of the 13th c. as a largely established construction, which outnumbered constructions with *jmati* 'have' and *chtíti* 'want' (+ infinitive of either aspect). The latter occurred much more rarely in futurate contexts and rather retained their modal or volitional

functions. The pattern *bud-* + infinitive also superseded verbs with ingressive prefixes, which in the earliest attested layers of Czech served to mark “durative” future (see Section 4). Compare, e.g., *ktož vz-bydlí.PRS.3SG v stanu tvém* ‘who starts to live (> will live) in your camp’, later “replaced” by *bud-e.FUT.3SG bydliti.IPFV.INF* (Křížková 1960: 92–94).

Further Křížková reports that the Old Czech texts provide almost no evidence of a *bud-* + *l*-form, and only in some of the few attested cases could this pattern be interpreted as a future perfect.²³ In contrast to East Slavic (see below), no modal functions were attested (Pen’kova 2016: 481). Moreover, the small empirical basis does not allow us to build any plausible link to *bud-* + infinitive, all the more as the few *l*-forms predominantly built on pfv. stems, while right from the start infinitives with *bud-* exclusively belonged to ipfv. stems. Obviously, these were two different constructions with complementary choice of aspect in the lexical verb (1960: 95–96). Shortly after, the construction with the *l*-form disappeared altogether.

In the Sorbian language area, the construction *bud-* + *l*-form has not been attested at all, while *bud-* + infinitive (with either aspect) started occurring from 1500 (Andersen 2006b: 16).

As for Old Polish, the situation basically differs only with respect to the fate of the *l*-form. In the earliest attestations (mid-late 14th c.) *będ-* + *l*-form is rare, the infinitive prevails, but *l*-forms already occur exclusively from ipfv. stems²⁴ and not as a future perfect (Křížková 1960: 106, 156); see (17):

- (17) *a będzi-e krolowa-l w domu Jakobowym na wieki*
 and FUT-3SG reign[IPFV]-LF-(SG.M) in house of Jacob on ages
 ‘And he **will** reign in the house of Jacob for ages.’
 (Ew_Zam_Mat[1_2r]_33; from Błaszczak 2019: 187)

In the mid-15th century, the *l*-form reappears (Křížková 1960: 106). Most probably, this “renaissance” of the *l*-form in the ipfv. future of Polish does not directly continue the CS construction *będ-* + *l*-participle (see example 16 from OCS). In the earliest extant texts, infinitival complements clearly prevail (approx. 90% of

²³ A slightly larger number of *bud-* + *l*-form can be found only in the *Knihy rožmberské* (1360), but even in these legal texts there are 8 attestations with *l*-form (against 11 with infinitive), from which 7 are from pfv. stems. This corresponds to a higher number of cases of *bud-* + *l*-participle in early East Slavic law texts (e.g., the *rus’kaja pravda*) (see below).

²⁴ Sporadically *l*-forms of pfv. stems appear in the 16th century, but only in translations from Ruthenian texts (e.g., from the *Statuty Litewskie*), e.g. *będzie zakupił* ‘he will have bought’, *będą dali* ‘they will have given’ (Křížková 1960: 106). These could probably better be interpreted as epistemic judgments (see below).

all attestations). Only in the 16th century does the frequency of *l*-forms increase again, but it is secular texts which forward their spread in the periphrastic future. It seems as if the *l*-form in this construction reappeared after a period of oblivion, and for that time the distribution (*l*-form vs infinitive) resembled register variation (Křížková 1960: 104, Błaszczak 2019: 208–213). Subsequently this variation was levelled out (see fn. 19).

Let us turn to East Slavic. In general, within Slavic, this was the area with the strongest propensity for ingressive verbs (in their present tense forms: *učbn-*, *počbn-* and *jm-* ‘begin’, etc.) to serve as quasi-auxiliaries of future (with ipfv. infinitives) until the late 16th century (cf. Pen’kova 2019b for a corpus-based evaluation). This was in contrast to South Slavic, where such verbs hardly played any role as candidates of future markers (Křížková 1960: 166, 174–175, Ivanova-Mirčeva 1962, Grković-Mejdžor 2012). In turn, WANT- and HAVE-based periphrases (with infinitives) marking future events in early East Slavic texts have been considered as reflecting Church Slavonic (i.e. eastern South Slavic) influence in written language (Andersen 2006b: 27; 2006a: 72–73, 81).

In turn, the construction *bud-* + *l*-form is well-attested in the earliest East Slavic texts (11th–12th century) and can be considered original. The oldest documents show *bud-* + *l*-form with stems of either aspect, although pfv. stems prevail. It was more typical of administrative language. It was not restricted to conditional or other dependent clauses, and only rarely did it function as a “simple” future, but also its function as a future perfect has been disputed. In the Old Novgorod dialect *bud-* + *l*-form lost its function as a relative tense (future perfect) early and tended towards acquiring an inferential function (Pen’kova 2019a).²⁵ In northern Ruthenian²⁶ documents of the 14th–16th century, *bud-* + *l*-form practically got restricted to contexts with suspended propositions, where it acquired dubitative and reportive functions, mainly in dependent clauses (Pen’kova 2020). Eventually, throughout East Slavic, the construction disappeared in the 16th century, leaving behind the 3sg-form *budet̃o* as an epistemic or inferential particle (Křížková 1960: 162–169, Pen’kova 2019b); see also Section 3.2.1.3.

The construction *bud-* + infinitive is attested later, and practically only with ipfv. stems. It first appears in the second half of the 14th century (in parallel

²⁵ Andersen (2006a: 76–80, 82–84) took issue with the commonplace analysis of relevant examples in the Novgorodian birchbark letters and claimed that it was wrong. However, the bulk of examples are in conditional or temporal clauses, for which one cannot clearly decide whether reference is made to a single event and, above all, from whose perspective temporal relations should be judged (the writer’s, the addressee’s, or of an independent observer?).

²⁶ *Ruthenian* refers to those parts of East Slavic which from the late 14th century came under increasing Polish influence (within the *Rzeczpospolita Obojga Narodów*) and which can be considered the soil on which later Ukrainian and Belarusian were formed.

to *imati* ‘have’ and phasal verbs) in Ruthenian texts, i.e. at the (south)western periphery of the East Slavic territory. In texts of the (north)eastern East Slavic territory (“Muscovy”) it only appears from the mid-15th c., among others (but not exclusively) in written exchange with Polish diplomats and similar official documents. Only during the 17th century does *bud-* + infinitive gain more prominence: through the first half of the 17th century it is still rare (much rarer than *uĉn-* or *stan-* ‘begin’) in the administrative language, but its use increases in the second half of the 17th century, until it finally ousts *uĉn-* (but not *stan-*) and becomes representative of daily speech (Křížková 1960: 173–176). Through the 17th century the spread of *bud-* was clearly supported by a Polish model, only by the 18th century did *bud-* + infinitive consistently appear in texts without Polish influence (Moser 1998: 311–330).

3.2.1.3 Where did the infinitive come from?

Let us turn to the question how the infinitive became integrated into a construction with *bud-*. Different theories have been developed, all of which require qualification.

In the first place, how could an originally inchoative verb combine with an infinitive, either before or after the inchoative feature was lost? Concerning Old Serbian (eastern Štokavian), Grković-Mejdžor (2012: 97) suggests that the infinitive introduced an expectative viewpoint, but she does not attempt to explain how the expectative (or goal-oriented) semantics of an infinitive might combine with an originally inchoative verb.

Actually, from a purely semantic point of view, the combination of an inchoative verb (inducing a change of state) with a goal-directed form (like the infinitive) does make sense. However, upon closer examination, diachronic Slavic syntax does not supply any good model for a combination of BE/BECOME + infinitive of purpose as an emergent future. Admittedly, constructions with a copula and an infinitive, with the latter contributing a meaning of purpose (due to its IE provenance), are well-attested in Slavic (and Baltic), and so are possessive constructions of the MIHI EST LIBER-type, i.e. with the possessor in the dative and the possessee in the nominative. Such constructions laid the basis for modal (deontic or circumstantial) constructions, particularly in Russian (e.g., *Mne.DAT tut nečego delat’.INF* ≈ ‘There is nothing for me to do here’), but also in historical documents, such as the famous example (18) from the *rus’kaja pravda*, the oldest kind of legal codex in East Slavic (11th century):

- (18) *takov-a pravd-a uzja-ti rusin-u*
such-NOM.SG.F right[F]-NOM.SG take[PFV]-INF Rus'.inhabitant-DAT.SG
≈ 'These are the rights a "Russian" must/can enjoy.'
(Holvoet 2003: 467, translation adapted)

However, in these constructions the "possessor" (i.e. more agent-like argument) is coded in the dative (cf. Holvoet 2003 for an analysis and the diachronic background), whereas in the *bud-* + infinitive-construction the corresponding argument appears in the nominative.²⁷ This is also admitted by Andersen (2006b: 34), who points out a construction with *bude* or *budetb* as an existential verb; these forms are encountered in the Novgorod birchbark letters (see 19). Equivalent examples are known from 13th century documents farther to the south (see 20).

- (19) *ažb dolgo bude dolgo medlę-ti prisl-i vest-e*
if long FUT.3SG long overstay-INF send-IMP.SG message-ACC.SG
'If it **happens** (for you) to stay much longer, send (me) a message.'
(Gramota no. 771, end of 13th–beginning 14th century; Zaliznjak 2004: 532)
- (20) *aže budętb tęrgova-ti smolnęnin-u sę nemęčic-emę*
if FUT.3SG trade-INF Smolensk.inhabitant-DAT.SG with German-INS.SG
'If it **happens** for an inhabitant of Smolensk to trade with a German.'
(contract, Smolensk, 1220s; Zaliznjak 2004: 532)

Despite the gloss FUT.3SG, here *bude/budętb* is to be considered an invariable form employed either like a clausal operator or as a predicate which takes a clausal complement with an infinitival head. Again, if mentioned at all, the most agent-like argument of the infinitive occurs in the dative (see 20). Such examples clearly show the use of petrified *bude/budętb* in habitual or conditional contexts, and it certainly was this use which paved the way for *budetb* as an epistemic-inferential particle after the 16th century (see Section 3.2.1.2). As for the structure, Andersen (2006b: 32, 35) only surmises that it might have arisen as "a syntactic neologism coined for subjectless verbs" (as in some hypothetical CS **bųdetb moroziti* '(it) will freeze'), which later expanded to clauses with nominative subjects. However, this is only an *ad hoc*-stipulation, so that Andersen ultimately concedes that there is no real solution to the issue.

²⁷By comparison, even in languages like German modal infinitive constructions only code the more patient-like argument in the nominative, while the potential agent normally is left out (e.g., *Das Problem ist (nicht) zu lösen*.INF 'The problem is (not) to be solved').

In sum, neither in East Slavic, nor anywhere else in Slavic, are any combinations of *bud-* + infinitive in purpose meaning attested even in the earliest documents. Apart from this, there is no reason to expect a free combination of *bud-* with infinitive to restrict itself to, or prefer, ipfv. aspect of the infinitive. However, from the earliest documents onwards, only ipfv. infinitives are attested in West Slavic (Section 3.2.1.2) and ipfv. infinitives considerably outnumbered pfv. ones in Old Serbian (Section 3.2.1.1).

Alternatively, Křížková (1960: 98) argues that *bud-* + infinitive acquired future meaning holistically, as a construction (with loss of internal transparency). However, since she also rightly discards any connection to possessive constructions (see above), one wonders how this holistic meaning and the implied loss of inchoativity of *bud-* came about. At any rate, these processes must have taken place prior to the earliest documents in Old Czech (i.e. during the early 13th century or even earlier).

Finally, an entirely different approach was chosen by Błaszczak et al. (2014). The authors decompose the construction into a pfv., possibly ingressive (‘inceptive’) operator (*będ-*) and its ipfv. “complement” (infinitive or *l*-form). Since the pfv. operator is in the present (or non-past) tense, it causes a “forward shift” of the reference time, which the authors take to be equivalent to a shift to future (2014: 190–191). They further claim that *będ-* denotes a Kimian state. Such a state does not represent an eventuality, but “a referential argument for a temporally bound property exemplification” (2014: 190). It is located in time but not in space and exists only for the needs of language communication, as in *Bill knew the answer (and that’s why he passed the exam)* (Tatevosov 2016: 63–65). Błaszczak et al. do not justify why *będ-* should be assigned this status, nor do they notice the inherent contradiction when they treat *będ-* as the pfv. head of the ASP-slot in syntactic structure and simultaneously claim it to represent a state. One might assume that *będ-* marks the beginning of a state (which would be compatible with the assumed inchoative origin of **bod-*), but this obviously is not what the authors accept, either, as it would not conform to their own examples. These examples, in turn, mostly exemplify processes (not states), which are however not distinguished by these authors. Finally, and after all, Błaszczak et al. only concentrate on the question why *będ-* exclusively combines with ipfv. stems of the lexical verb, but they do not clarify how the infinitive “entered the scene”, let alone why in all North Slavic languages except Polish and Kashubian it entirely ousted, or replaced, the *l*-form.

To my knowledge, there have been no more attempts at explaining the rise of *bud-/będ-* + infinitive on internal grounds. However, since the 1950s there

have been hot debates concerning the issue whether the *bud*-future construction might have been PAT-borrowed (in the sense of Matras & Sakel 2007) from Middle or Early New High German into Old Czech, or the other way around. Remarkably, this discussion mostly disregards this construction in South Slavic (see Section 3.2.1.1), and hardly anybody thought about the possibility of mutual influence between western edges of Slavic and some German varieties (also closer to Slovene and CentSS). As for German, the earliest attestations of *werdan* + infinitive have usually been dated between the early 11th century and the early 13th century (Westvik 2000). However, this construction was also attested with *werdan* in the past tense in the 13th-16th centuries (Krämer 2005: 75–81). Jointly, *werden* + infinitive seems to have kept an inchoative meaning until the time of its alleged influence on **bqd*- + infinitive in West Slavic. However, **bqd*- + infinitive could never occur in the past tense, and **bqd*- itself must have lost its inchoative flavour in Czech no later than during the 13th century.²⁸

Recently, Jäger (2024) has suggested that the genesis of the German *werden*-future started with the “reanalysis of originally predicative past participles in *werden*-copula constructions as infinitives”. This reanalysis was possible because the past participles of a small class of strong unaccusative verbs lacked a *ge*-prefix. These forms were homonymous with their infinitives (compare modern Germ. *Das Gebäude wird zerfallen*.PTCP *sein* ‘The building will (or: may) be fallen apart’ vs *Das Gebäude wird zerfallen*.INF ‘The building will fall apart’), so that as predicative “complements” of inchoative *werdan* ‘become’ in the present tense these participles created bridging contexts for the reanalysis of a future-oriented resultative construction as a future. According to Jäger, the reanalysis occurred in Old High German and Old Low German, i.e. until the 11th century. Thus, if her hypothesis turns out to be correct, the reanalysis should have been followed by analogical expansion, before the new BECOME+infinitive construction could have served as a model for Slavic varieties at the western edges. That is, this construction should have already spread considerably to become sufficiently established in Middle High German prior to being copied by speakers of Slavic, and one wonders why speakers of Old Slovene did not adopt this construction.

In sum, the issue whether PAT-borrowing or, alternatively, polysemy copying²⁹ played a role as a catalyst of the spread of *bqd*-/*bud*- + infinitive in Slavic is still open.

²⁸For some more details and a concise evaluation of these debates cf. Wiemer & Hansen (2012: 105–108). For surveys on the German *werden*-future cf. Hartmann (2021: Section 2) and Jäger (2024: Section 2).

²⁹For a discussion of these notions cf. Wiemer & Wälchli (2012: 26–29).

3.2.2 WANT

CS **xotěti/xătěti* ‘want’ is the predominant source of future marking in contemporary South Slavic, but its embryonic stages can be observed also in North Slavic, where, apart from marking intention and wish, it can at best be regarded as a prospective marker (Křížková 1960: 120–125, Ševeleva 2021). Remarkably, it does not seem to have ever shown any preference for the aspect of the stem of the lexical verb. In this respect, modern South Slavic languages (except Slovene) probably just continue the distribution from the onset of the grammaticalization of WANT-periphrases.

In early OCS, WANT-periphrases referring to future events have been claimed to be initially less frequent than equivalent HAVE-periphrases (Birnbaum 1958, Grković-Mejdžor 1995, also Xaralampiev 1981: 116 with further references). In the eastern part of South Slavic, this relation must have swapped slightly later, after or even during the 11th century. As for the Štokavian area, Grković-Mejdžor (2012) concludes that from the mid-13th century the present tense forms of *xtěti* ‘want’ started showing signs of phonetic erosion, cliticization (> 2P-enclitic) and context expansion to inanimate situations (with concomitant semantic bleaching). This process had already basically been finished in the 14th century; it seems to have lagged behind slightly in the Čakavian area, i.e. further to the Adriatic coast in the west (Grković-Mejdžor 2012: 88–89).

As concerns early Slavic in the Balkans, *xotěti* must have already started taking over the floor during the 11th century (Tomić 2004: 409–410, referring to Asenova 2002: 204 and Mirčev 1978: 223). Heine & Kuteva (2005: 191) report that WANT competed with HAVE and BEGIN until the 13th century, when *xošte* (> *šta*) + truncated infinitive gained priority. From the 15th century onwards the infinitive was gradually replaced by *da*-complements (with finite verbs), which then also appeared after *šta* > *šte* (Tomić 2004: 405, referring to Koneski 1967: 206 and Asenova 2002: 205). Tomić (2004: 407–408) argues that the *da*-construction bridged the transition from an inflected “modal clitic” (WANT) with infinitive towards the contemporary construction (uninflected proclitic + V_{PRS}). Table 1 (based on Asenova 2002: 214) summarizes these stages in a simplified manner with the ipfv. stem *pisa-/piš-* ‘write’ and its 2SG-form (of the present tense stem *piš-*) for the purpose of illustration.

What is remarkable about this sequence of stages is not so much the gradual phonological and categorial deterioration of the initial WANT-verb (*xošte-ši.PRS.2SG* > *šta* > *šte*), nor the loss of the infinitive (“replaced” by inflected present tense forms) in the lexical verb, but the suggested immense overlap among the future marker variants and their combinations with and without the irrealis pro-

Table 1: From WANT to future in South Slavic (here for Bulgarian)

9th–13th c. (OCS)	14th–20th c.	15th–early 20th c.	16th–21st c.	16th–21st c.
<i>xošt-eši pisa-ti</i>	<i>xošt-eš pisa</i>	<i>xošt-eš da piš-eš</i>	<i>šta > šte da piš-eš</i>	<i>šte piš-eš</i>
want-PRS.2SG + INF	want-PRS.2SG + truncated INF	want/IRR- PRS.2SG + IRR + PRS.2SG	FUT/IRR (uninfl.) + IRR +PRS.2SG	FUT (uninfl.) + PRS.2SG

clitic *da*.³⁰ Here, again, one has to be careful about the data used: most probably, these overlaps rather reflect an enormous diatopic and diaphasic variation for most, if not all, centuries; this analysis “overrides” rather innovative tendencies in vernacular speech with conservative text genres based on OCS writing traditions. The modern standard form of the future without *da* (*šte pišeš*) gained dominance only quite recently, whereas the pattern with *da* (*šte da pišeš*) has been marginalized as a device of marking assumptions (Asenova & Guentchéva 2022).

Regardless, *da-V_{PRS}* has persisted in the suppletive HAVE-pattern which dominates in the negated future (*njama da* + *V_{PRS}*). In this construction, HAVE occurs in the invariable form of PRS.3SG (see 21b and 22). In Macedonian we encounter an analogical situation³¹ (see 9).

(21) Macedonian

- a. *Nie ke= stign-eme utre.*
1PL.NOM FUT= arrive[PFV]-PRS.1PL tomorrow
‘We will arrive tomorrow.’
- b. *Nema da= te= zema-m.*
NEG.have.INVAR IRR= 2SG.ACC take[PFV]-PRS.1SG
‘I won’t take you (with me).’

³⁰Cf. Ivanova-Mirčeva (1962: 134 and passim) for details. She points out that truncated forms (*štq*.1SG, *šteši*.2SG, etc.) could be found as early as in the 13th century, while contemporary uninflected *šte* became predominant only by the 18th century (1962: 134–135).

³¹Tomić (2012: 388) remarks that the construction with negated *ke* conveys an additional “pre-suppositional component” (compare, e.g., *Ne ke te zemat so niv* ‘(I presume) they won’t take you with them’ with 21b).

In modern standard Macedonian and Bulgarian we also find the unnegated pattern HAVE + *da* + V_{PRS}, which still carries some obligational flavour; compare (10):

- (22) Macedonian
Ima da= oda-m.
have.INVAR IRR= go[IPFV]-PRS.1SG
'I have to go > I will go.'

These patterns suggest a persistent “rivalry” between WANT- and HAVE-based constructions showing relation to future marking, for which the degree of entrenchment varies. However, their exact diachronic relationship, in particular of the different HAVE-based future periphrases in the Balkans (is it a direct continuation or a cyclic phenomenon?) requires clarification (cf. Xaralampiev 1981: 119–122, Kramer 1997: 408, Mitkovska & Bužarovska 2014: 195–196). As for Old Serbian, the analysis presented by Grković-Mejdžor (2012: 93–97) even seems to indicate that here the history of predicative HAVE-constructions and their emergent development as possible futures started anew only in the 14th century, which would mean that it was entirely detached from HAVE-based future-like constructions in OCS.

Most of the aforementioned processes, and the variation, have parallels in other languages of the Balkans. We observe them in the history of the Greek WANT-future, and the diachronic competition between WANT and HAVE finds their diatopic equivalent in contemporary differences between North (Geg) and South (Tosk) Albanian. For overviews cf. Kramer (1994), Andersen (2006b: 36–38), Tomić (2006), and Wiemer et al. (2024: 51–53).

4 Future marked by prefixes on imperfective stems

In Czech (Šmilauer 1947: 152, Machek 1962: 439, Mrhačová 1993: 65–67, among many others), Slovak (Horecký 1957, Ružička & Dvonč 1966: 41), in Upper and Lower Sorbian (Michalk 1963, Scholze 2008: 194–195, 269–270, 279–280) and (to a lesser extent) in Slovene we observe a peculiar phenomenon. The prefix *po-* (in Sorbian also *s-/z-*) marks future tense of certain classes of ipfv. stems. The core group has always consisted of verbs of directed motion (e.g., Cz. *pojeđu* ‘I will go/drive’, *popluji* ‘I will swim’), related causatives (e.g., *potáhnu* ‘I will pull’, *povezu* ‘I will bring (by driving)’), and inchoative verbs (e.g., *poroste* ‘s/he will grow’), occasionally also some non-motion activity verbs (e.g., *mlýn pomele* ‘the mill will grind’, Kopečný 1962: 49) and even state verbs (e.g., colloquial USorb.

změjem ‘I will have’, Scholze 2008: 194) are mentioned. The extension of these groups has remained fuzzy; diachronically it has most probably been subject to variation (cf. Šlosar 1981: 79–80, 102–103 on Czech). In the modern languages we cannot assume closed classes either. Figures adduced, e.g., for contemporary Czech vary from 20 to 115 lexical items (Kopečný 1962: 49–50, Osolsobě 2014: 137, Ševčíková & Panevová 2018: 179–180, VAGSČ 2021: 156, with further references). Before dealing with diachronic issues (Section 4.2–4.3), I will survey the contemporary stage.

4.1 Contemporary stage

First of all, ipfv. stems with *po-* marking future typically denote parametric changes, i.e. changes along scales without absolute endpoints (contrary to strictly telic verbs, which imply an absolute goal). Moreover, we are dealing with stems in particular meanings here. First, the ipfv. behaviour of *po-*prefixed future forms often does not apply to these verbs in non-physical, figurative usage, although this is not a strict rule (see below). Second, *po-*prefixation with the same ipfv. stems turns these stems into pfv. aspect if the number of arguments increases (thus, from intransitives to transitives and from transitives to ditransitives). For instance, *pojít* in the meaning ‘originate, rise’ or ‘perish’, or *povléct* in the meaning ‘cover (some surface with sth)’ have a full verbal paradigm and behave like other pfv. stems;³² compare (23–24):

(23) Czech

- a. *Půjde*^{IPFV.FUT} / *šel*^{IPFV.PST} *domov celou hodinu*.
‘He **will go** / **went** home for a whole hour.’
- b. *Myš pojde*^{Pfv.FUT} / *pošla*^{Pfv.PST} *po požití jedu *celou hodinu*.
‘After the mouse eats / ate the poison, it will die / died *for a whole hour.’

(from Ševčíková & Panevová 2018: 182, with adaptations)

(24) Czech

- a. *Kůň povleče*^{IPFV.FUT} / *vlekl*^{IPFV.PST} *kaskadéra za sebou dlouhé minuty*.
‘The horse **will drag** / **dragged** the stuntman after itself for many minutes.’

³²Cf. Bondarko (1961: 544–545), Němec (2009a: 238, 248) and most systematically Ševčíková & Panevová (2018: 181–186).

- b. *Povleče*^{PFV.FUT} / *povlékla*^{PFV.PST} *postel (čerstvým prádlem) *dlouhé minuty.*
 ‘She **will cover** / **covered** the bed (with fresh linen) *for many minutes.’

(from Ševčíková & Panevová 2018: 182, with adaptations)

Basically, the same applies to other languages of the “western hemisphere” of Slavic, e.g. Slovene (Dickey 2018: 375).

In modern Czech, this kind of lexical split causes grammatical homonymy of *po*-prefixed forms, as can be demonstrated with the following examples (in 25a) and their past equivalents (in 25b taken from the Czech National Corpus (<https://www.korpus.cz/>, subcorpus SYNv5):

(25) intransitive, parametric change

- a. *Ještě dva týdny se Tour povalila*^{IPFV.FUT} *napříč Francií.*
 ‘For two more weeks, the Tour **will roll** across France.’
 b. *Dva týdny se Tour valila*^{IPFV.PST} (**poválí*) *napříč Francií.*
 ‘For two weeks, the Tour **rolled** across France.’

(from Ševčíková & Panevová 2018: 184)

(26) transitive, telic

- a. *Prasata plot povalí*^{PFV.FUT} ³³ *nebo se podhrabou.*
 ‘Pigs **will knock down** the fence or dig under it.’
 b. transitive, telic
Prasata plot povalila^{PFV.PST} (**dlouhý okamžik.*)
 ‘Pigs **knocked down** the fence (*for a long moment).’

(from Ševčíková & Panevová 2018: 184)

Note that, in the given meanings, the ipfv. stems (in 23a, 24a, 25a–25b) do not have pfv. counterparts, i.e. **pošel* ‘went’, **povlékl* ‘dragged’, **povalil* ‘rolled’ do not exist.³⁴ They are thus *imperfectiva tantum* lexemes.

This brings us to the question why, in the first place, a prefix attached to the present tense of simplex stems should mark the future of ipfv. aspect, and not the present tense of pfv. aspect (which then, via hypoanalysis, yields future). As for contemporary Czech, Kopečný (1962: 47–50) provides the following arguments:

³³Of course, like non-past forms of pfv. stems in general, *poválí* might also be used in dispositional or circumstantial meanings (see Section 3.1.2, 3.1.4 and below).

³⁴By contrast, the pfv. “homonyms” usually have ipfv. counterparts, e.g. *povléct*^{pfv} – *povlékat*^{ipfv} ‘cover (surface with sth)’, *poválit*^{pfv} – *valit*^{ipfv} ‘knock down’; however, ipfv. *pocházet* ‘die, perish’ for pfv. *pojít* is considered obsolete.

- (a) The *po*-prefix only occurs with the present tense form, not with the past tense, the infinitive or any other paradigmatic forms of the stem (except the imperative; see below).
- (b) The forms with *po*- are not used in the narrative present (unlike present tense forms of pfv. stems), nor do they usually occur in modal functions. What Kopečný (and others) mean by ‘modal’ is the use of non-past forms to mark dispositional or circumstantial meanings; these are regularly associated with meanings that in Slavic linguistics have been referred to under the label of ‘inactual’ or ‘gnomic’ present (see Section 3.1.2). Thus, Nerad (1940) finds that *po*-prefixed ipfv. verbs with future meaning never have this modal meaning. Other scholars, like Bondarko (1961: 529), Kopečný (1962: 47) and Machek (1962: 439–441), are less categorical in this regard (see below).
- (c) The *po*-prefixed forms can also be used for future actions with unlimited duration; thus, not only, e.g., *ty to poneseš dvě hodiny* ‘you(SG) (will) carry this for two hours’, but also *ty to poneseš stále* ‘you(SG) (will) carry this permanently’ (cf. also Galton 1976: 45); see also (23a, 24a). Ševčíková & Panevová (2018: 181) accept compatibility with relevant adverbials as one of two reliable criteria allowing to determine the aspect (see 23–26), the other criterion being the synonymy of *po*-prefixed present tense forms with unprefixed past tense forms of the same stem (see 25a–25b against 26a–26b).

As for (a), *po*- is also widely attested with the imperative of the same ipfv. stems, but here *po*- bears the implication that the action is to be performed towards and/or jointly with the speaker (e.g., Cz. *pojd!* ‘come (with me)!’, *polez!* ‘climb (here, to my place)!’).³⁵ Machek (1962: 438–439) argues that the prefix *po*- in non-past and in imperative forms derives from different etyma; as indirect proof he points out presumably different cognates in Baltic and earlier Indo-European.³⁶

As for (b), Machek (1962: 439–441) argues that *po*-, before it came to mark the future of ipfv. stems, might have been used to mark dispositions, or inclinations,

³⁵Imperatives from the corresponding unprefixed stems are neutral in this respect (Bondarko 1961: 541–542). This suggests that, with the imperative, *po*- has retained some spatial, or deictic, features that predate, or interfere with, the early stages of the rise of the stem-derivational aspect opposition (as suggested by Machek 1962: 438–439).

³⁶Compare *par*- ‘home’ (e.g., Lith. *par-grįžti* ‘return home’), from **pār*, vs *pa*- ‘approaching from below’ (e.g., Lith. *pa-si-rašyti* ‘sign (a paper)’), compare Germ. *unter-schreiben* from **upo*. Thus, *pa*- in imperatives would derive from **pār*, while in non-past forms it would continue **upo*.

of the subject argument (see Section 4.3). In his opinion, this is still reflected in modern Czech; see (27–28), which he interprets as remnants of an older layer. Note that *podělat* in (28) is not a motion verb, nor does *ponesou* in (27) clearly denote directed motion:

- (27) *Do práce budu chodit^{IPFV} pěšky tak dlouho, dokud mne nohy **ponesou**^{IPFV}.*
 ‘I will walk to work as long as my legs **can / will** carry me.’
- (28) *Sám toho **nepodělám**^{PFV}.*
 ‘I **can’t / won’t** do it alone.’

Machek points out parallels in Baltic; compare, for instance, Lith. *jis net ne-pa-eina* ‘he **won’t** even go / he is **unable to go** (even a bit)’ or *ji pa-kelia bet kokį sunkų lagaminą* ‘she is **able to lift** any heavy suitcase’.³⁷ Machek’s observation is important, since a diachronic link between these modal (or “timeless”) functions and the future function is very likely, more precisely: the future function might have emerged via abduction from the knowledge about lawful and, thus, predictable behaviour of individuals or of situations (see Section 3.1.2).

However, in today’s Czech, the dispositional function of *po*-prefixed ipfv. verbs is rarely noted. It now appears to be rather associated with a few verb stems prefixed with *u-* (e.g., *Karel unese 200 kg* ‘Karel is **able to lift** 200 kg’; cf. Mrhačová 1993: 62), but these stems are commonly classified as pfv., since they are incompatible with unlimited duration.³⁸

In addition to Kopečný’s three aforementioned arguments supporting the claim that we are dealing with the future of ipfv. stems, we have to take into account that most of the relevant ipfv. stems can also form their future with *bud-*. However, these stems show this “periphrastic” way of expressing ipfv. future (much) more rarely, and they vary considerably among each other as for the likelihood with which we may observe the periphrastic future. Some few stems, like *jít* ‘go’, are not attested with *bud-* at all, while others show larger figures; however the bias is always clearly in favour of prefixation with *po-* (cf., e.g., Kopečný 1962: 47–48, VAGSČ 2021: 156, Saicová Římalová 2023). It is an open question whether this dual way of marking ipfv. future tends toward functional differentiation. It has been claimed that *bud-* is more likely if the motion verb is used figuratively (e.g., *já mu budu lézt do zadku?!* ‘Me **crawl** up his ass?!’,

³⁷ According to Machek (1962: 441), this *pa-* derives from yet another etymon, namely IE **per-* (> Baltic *par-*).

³⁸ Lexicographic practice seems to treat them as *perfectiva tantum*. Thus, for instance, *unést* does have an ipfv. counterpart *unášet*, but only in the meaning ‘carry away’, not in the slightly different meaning ‘lift’ with a dispositional “touch”. I am grateful to Jana Kocková for consultation on these issues.

cited by Galton 1976: 45 with reference to Poldauf 1948), but there are also figurative uses with *po*-prefix (e.g. *Tato ulice **ponese** jméno XY* ‘This street **will** bear the name XY’, Galton 1976: 45). Native speakers’ judgments in this regard diverge considerably. Likewise, the analysis of corpus data offered by Ševčíková & Panevová (2018: 181) and by Saicová Římalová (2023) does not indicate any clear-cut distribution of *po*- vs *bud*- along the lines of, e.g., a differentiation of physical (directed) motion vs figurative use. Saicová Římalová (2023) concludes that, at least, the meaning of physical motion is “more reluctant to accept the periphrastic form” and “[d]ouble forms are more typical for figurative meanings than physical ones”.

Finally, among the ipfv. simplex stems for which the expression of future with the *po*-prefix is considered marginal (against a dominant *bud*-future) we find quite a few that are also attested in the infinitive and other forms of the regular paradigm (e.g., Cz. *vládnout* ‘rule, have control over’, *růst* ‘grow’, *kvést* ‘blossom’; Saicová Římalová, p.c.). Similarly, we find *po*-prefixed infinitives, for instance *postáti* ‘stand’ and *poležeti* ‘lie’, in Old Czech (Bondarko 1961: 529, fn. 9); for these, however, the question arises whether they were not better characterized as pfv. (see Section 4.3). Now that our survey has brought us to this question, we may turn to the diachronic situation.

4.2 Diachronic background

There have been different approaches to the ipfv. *po*-future in Czech (and some other Slavic languages) from a diachronic perspective (for partial surveys cf. Mrhačová 1993: 60–64, Bláha 2008: 26–34). They have different implications for the question how well established the PFV:IPFV opposition already had been by the 14th–15th centuries, and how much this hinged on the prefixation of verb stems. For instance, according to Lamprecht et al. (1986), in Old Czech certain prefixes with an ingressive meaning (*vz*-, *po*-),³⁹ when attached to stems with present tense inflection, gradually turned their ingressive feature into a future meaning (*begin* [to stem_{prs}] > *will* [stem_{prs}] = FUT [stem]). Under this premise, the future function of these prefixes was not dependent on a pre-existing perfectivity opposition (marked by an opposition of simplex vs prefixed stems).

Most scholars, however, base their assumptions (mostly implicitly) on the premise that the perfectivity opposition had already been stable to a sufficient extent in Old Czech times (i.e. by the 14th century). Thus, Bondarko (1961) shows that forms like *půjdu* ‘I (will) go’, *ponesu* ‘I (will) carry’ in Old Czech behaved like pfv. stems, and, till the end of the 14th century, these stems were also encountered in various types of habitual or timeless (“gnomic”) present:

³⁹For examples with *vz*- in Old Czech cf. Trávníček (1923: 192).

- (29) *Člověk vždy na vše sáhne^{PFV.PRS}, dobrého spíše **poběhne**^{PFV.PRS}, ve zlem jsá, pak se nehne.*

‘Man always reaches for everything, he rather **runs** from [= avoids] the good, being unright he nonetheless doesn’t bow.’

(Alx. V 160^{db}, 43)

- (30) *Sto jich jednu se zdi zabijí^{PFV.PRS}, a však se vždy ke zdi hrabí; pobitých dvě stě **povleku**^{PFV.PRS}, avšak zavše pod zed **poteku**^{PFV.PRS}.*

‘He knocks off 101 of them from the wall, and yet he always rummages around the wall; I **carry away** 200 of the knocked ones, but always I **fall** (back) under the wall.’

(Alx. V 167^{bß}, 68; Bondarko 1961: 532)

Moreover, Bondarko (1961: 533) argues that all alleged examples of “durative” (i.e. progressive) use also allow for an ingressive reading; he adds the caveat that often the available contexts do not allow for disambiguation among these readings (Bondarko 1961: 534).⁴⁰ Bondarko, however, does not consider the possibility of dispositional readings. Thus, compare (31) from the late 14th century, in which the adverbial *bez ustánie* ‘without interruption’ seems to force a progressive reading, with (32), which allows for either progressive or ingressive reading – in either case regardless of deictic time reference. In other contexts, *po*-prefixed verbs of directed motion take part in describing future-oriented single events; see (33):

- (31) *svatý kříž, jenž jest dřevo života těm všem, ješto jej přijímají, že **pobyěhnu**^{PFV.PRS} vesele a **poneffu**^{PFV.PRS} bez ustánie kříž svůj.*

‘The holy cross, which is the tree of life for everybody who accepts it, I (will) **run** cheerfully and **carry** my cross without interruption.’

(Matoušova evangelie, 287^b 14; Bondarko 1961: 533)

- (32) *do města svého... nemohu přijíti, leč **se tak povezu**^{PFV.PRS} se pánem Ježíšem.*

‘I cannot come to my place, but I (will) **take a ride** with my Lord Jesus.’

(ChelčPost 245b; Němec 2009b: 274)

- (33) ***Pójdem**^{PFV.PRS} pod tuto horu, dětem a skotu odpočinem^{PFV.PRS} a snad se tu s túhú minem^{PFV.PRS}.*

‘I **will go** under this hill, I will give a rest to the children and the cattle, and perhaps here I will pass away with longing.’

(Dal. L 3a, 20; Bondarko 1961: 533)

⁴⁰Cf. also Němec (2009b: 274–275), who however argues for a progressive (“durative”) reading.

In turn, Machek (1962: 440, fn. 14) considers (31) to exemplify a dispositional reading. In fact, divergent interpretations of these (and many other) examples often cannot be disambiguated by the available context. One may even suspect that, at least in some cases, such “rivalling” readings are more or less equally likely since the *po*-prefixed present tense forms, allowing for any of these readings, oscillated between them.

4.2.1 Comparison with Latin

A comparison of *po*-prefixed present tense forms as translational equivalents of the Latin future does not clarify the actional character, either: in many cases the ingressive semantics of *po*- + present tense translates the Latin future of *coepio* ‘I begin’, e.g. *puojde* as equivalent of *coeperit*^{FUT} *proficisci* ‘he will (start to) go (against)’. However, we also find *po*- + present tense as a translation of Latin future or present unspecified for actionality. For instance, *podme duch jeho a potekú vody* ‘his spirit **will blow** and the waters **will flow**’ (Psalm 147) translating Lat. *flabit*^{FUT} ...*fluent*^{PRS}, or *pomyšl’u za hříech svůj* ‘I **will consider** as my sin’ for Lat. *cogitabo*^{FUT} ‘I will think, consider’ may be interpreted either as processes, or states, or as their incipient phases. On this basis, Šlosar (1981: 78–79) concludes that this behaviour ambiguous to (or oscillating as for?) actionality precludes a clear decision as for the aspect membership of the Old Czech stems (*po-dúti* ‘blow’, *po-téci* ‘flow’, *po-mysliti* ‘think’); see also Šlosar (1981: 102) with regard to the prefix *vz(e)*-.

4.2.2 Comparison with Old Church Slavonic

Po- with motion verbs in the function of marking future (but not pfv. aspect) has already been pointed out for OCS, e.g. by Němec (2009b: 274), who provides the following example:⁴¹

(34) Old Church Slavonic

i milostv tvoě poženetъ mję vsję dъni života moego.

‘And your love **drives / will drive** me every day of my life.’

(Ps22:6)

However, again, many (if not most of the) examples adduced in the literature need not unequivocally be interpreted as marking a process in the future, they may also be read as ingresses.⁴² Šlosar (1981: 77) claims that *po*- as marker of

⁴¹More examples are discussed in Dostál (1954: 283, 299).

⁴²On the ingressive function of prefixes in OCS and Old East Slavic cf. Eckhoff (2017).

ipfv. future in Old Czech increased in relation to OCS; however, he also adduces examples from OCS sources, e.g. with activity verbs, which correspond to *bud-* forms in later sources of Old Czech. Compare, for instance, *sěmę moě porabota-jetъ jemu* ‘my kin/tribe **will work** for him’, *poleštę* ‘I will fly’ in OCS, for which we find *bude slůžiti* ‘will serve’ and *létati budu* ‘I will fly’, respectively, in Old Czech sources (Šlosar 1981: 78). We may, thus, suspect that, apart from a proper account of actionality, we have to distinguish between the (token) frequency of *po-*prefixed stems of (presumably) ipfv. aspect on text level and the productivity in relation to verb stems. Especially the latter appears to have been quite unsteady.

4.3 Parallel changes concerning past tense forms

As pointed out by Trávníček (1923: 190–191), Bondarko (1961: 529, fn. 9), and Šlosar (1981) for Old Czech, ipfv. stems which could mark future by prefixing their present tense forms with *po-* (or some other prefix) were also attested with the same prefix in infinitives and other paradigmatic forms. This obviously changed after the 14th century, when these forms, with the exclusion of the present tense and the imperative, gradually disappeared; by the 16th century they became exceptional (Bondarko 1961: 532–533). Concomitantly, *po-* eventually lost not only its ingressive (i.e. temporal) function, but also its original spatial meanings (‘on a surface’ or ‘from’ = ablative meaning; see 29),⁴³ whereas the imperative forms survived because they retained their semantic specialization (see fn. 35). All this together caused the paradigmatic isolation of the *po-*prefixed present tense forms, which could then be more easily associated with the paradigm of the respective simplex stems belonging to ipfv. aspect and marking activities or states.

The loss of the ingressive function was an important factor contributing to the paradigmatic isolation of the *po-*prefixed present tense forms, and this loss (or lack of further development) should also be considered in connection with the past tense. Past tense forms of the same stems, but without a prefix, are well attested in contexts in which they gained ingressive meaning; for instance

(35) Old Czech

a vstav šel jest po něm.

‘He then stood up and **went** after him / **followed** him.’

(BiblLit Mc2:14; cited from Němec 2009b: 277)

⁴³On the original meanings of *po-* cf. Němec (1958: 62–63; 2009a), also Šlosar 1981: 75–77.

That is, the ingressive phase of some activity was usually only induced from the context, but not marked by a prefix that also would have enhanced the chance of this stem being classified as a pfv. one.

Bondarko (1961: 542–543) shows that lacking prefixation in the case of ingressivity was typical of Old Czech (for both past and non-past forms), that is, simplex stems also clearly prevailed over their prefixed derivatives (which could then easily be interpreted as pfv. stems) in this function. See (36) for a sequence of events (*vzvědě*.AOR.3SG ‘learnt’ > *jěde*.PRS.3SG ‘goes, rides’) and (37) with a contrast between *pojedu* and *jěde*: both forms refer to the initiation of movement, but the prefixed form in the quote (*pojedu*.1SG) can easily be understood as a commissive speech act. It can thereby evoke an immediate future, whereas the latter form (*jěde*.3SG) pairs up with an (already obsolete) aorist-form (*poče*.AOR.3SG) to mark a narrative sequence.

(36) Old Czech

*Ciesar Bedřich poslední snem zapovědě. Jakž to král český **vzvědě**, k dvoru **jěde**.*

‘Emperor Friedrich announced the latter in the assembly. As the Czech king **learnt** about this, he went [lit. **goes**] to the court.’

(Dal 75^a, 139)

(37) Old Czech

„***Pojedu** já proti otci svému, snad to přivedu k konci dobrému.“ Tak i **jěde** proti otci svému i **poče** dobývati milosti svému milému.*

“‘I (will) go against my father, maybe I will lead this to a good end.’ So he went [lit. **goes**] against his father and **began** to earn love for his dear.’

(Dal 38^b, 79; cited from Bondarko 1961: 537)

Notably, simplex stems denoting directed motion – as well as other simplex stems denoting activities or states – usually did not mark ingressivity by prefixes in earlier stages of other Slavic languages, either (cf. Bondarko 1961: 537–538, Dickey 2011: 185–186).⁴⁴

In general, the integration of atelic stems (and of stems denoting punctual events) into the aspect system (via consistent derivation) was belated by a couple of centuries even in those Slavic languages that can be considered “most progressive” in this respect, i.e. in the eastern half of Slavic (cf. Bermel 1997 for

⁴⁴Only in the 17th century did the ingressive and other phases of atelic activities start being marked more and more consistently by (pfv.) prefixed derivatives in the eastern half of Slavic and in Polish (Dickey 2011).

East Slavic). In fact, after the 16th century the eastern part of Slavic (incl. Polish and Bulgarian) considerably extended the use of *po-* as a productive device for deriving delimitatives (= pfv. activities) and as an ingressive marker of directed motion.⁴⁵ After all, Czech and other languages on the western edges of the Slavic speaking territory are remarkably different from Slavic languages in the eastern half (East Slavic, Bulgarian, but also Polish). In the eastern half, ingressivity in (narrative) past contexts is consistently marked with prefixed pfv. equivalents of the respective ipfv. stems, whereas in the western group this explicit marking is largely absent (cf. Ivančev 1961, Berger 2016). Even more remarkably, the areal range of this latter group is more or less coextensive with the use of prefixed ipfv. verbs to mark future, as has been noted for Slovak, Sorbian and (although to a lesser degree) for Slovene and SerBoCroatian (cf. Dickey 2011: 178–180).

4.4 Summary of the presumable scenario

Table 2 schematically summarizes the presumable diachronic changes that affected *po*-prefixed present tense forms of certain semantic verb groups (with verbs denoting directed motion as the core). It basically reproduces the view expounded in Němec (2009b: 277). His “story” can be retold in the following way: in Old Czech the aspect opposition (PFV:IPFV) had already been well established, and prefixes served to derive pfv. stems from ipfv. simplex stems. However, verbs of motion created a problem since their non-past (“present”) tense forms occurred in contexts in which the prefix could be interpreted as marking either ingressivity or duration (i.e. progressive or, rarely, stative meaning); in certain contexts either interpretation was possible. Ingressive readings favoured the categorization of the prefixed form as representing pfv. aspect, whereas progressive readings favoured its classification as ipfv. aspect. In this regard, the prefixed non-past forms of these stems were underspecified. Simultaneously, up to a certain time, the same prefixed stems also occurred in the past tense, but, contrary to the non-past forms, the prefix here clearly marked ingressivity, so that the stem could easily be assigned pfv. aspect. With the simplex stem, in turn, actionality was underspecified, as it was with the non-past forms of the respective prefixed

⁴⁵Cf. Dickey (2011; 2018: 375–376). In her study based on the TOROT treebank, Eckhoff (2020 [2018]) found that prefixation of atelic ipfv. stems with *po-* had already been quite productive in OCS and Old East Slavic. In other words, prefixation as a means of perfectivizing simplex stems had already advanced to an extent that this derivational pattern could occur without a concomitant telicizing effect (see Section 3.1.1). However, it obviously needed some more centuries for this pattern to become productive for the derivation of pfv. stems with delimitative meanings (compare Russ. *po-sidet* ‘sit a while’).

stem. This led to paradigmatic asymmetry. Subsequently, this asymmetry was “reshuffled” in favour of treating the prefixed non-past forms as ipfv. (with future reference), probably because unambiguously ingressive contexts proved to be in the minority. Concomitantly, the prefixed past tense forms of these verbs vanished (unless in other than motion meanings), so that the prefixed non-past forms could be integrated into the tense paradigm of the unprefixed forms (for past, *šel jsem* ‘I went’, and present, *jdu* ‘I am going’) as their future. In this way, the biaspectual behaviour of *pójdu* > *půjdu* was abandoned and *jít* ‘go’ became an *imperfectivum tantum*. Table 2 sketches these changes with the 1sg-forms of *jít(i)* ‘go’, which was the most frequent motion verb in Old Czech.

Table 2: Reinterpretation of paradigmatically isolated *po*-prefixed present tense

<i>jít(i)</i> ‘go’	Past	Present	Future
Old Czech (14 th –15 th century)	PFV: <i>pošel jsem</i> IPFV: <i>šel jsem</i>	—— IPFV: <i>jdu</i>	PFV/IPFV: <i>pójdu</i> ^a
Intermediate stage	PFV: <i>pošel jsem</i> IPFV: <i>šel jsem</i>	—— IPFV: <i>jdu</i>	—— PFV/IPFV: <i>pójdu</i>
Modern Czech	IPFV: <i>šel jsem</i>	IPFV: <i>jdu</i>	IPFV: <i>půjdu</i>

^a Periphrastic **budu jíti* is unattested and probably never existed.

In addition to Němec’s considerations, one has to account for the rise of *bud-* as a future auxiliary with ipfv. stems, and as such *bud-* (+ infinitive) started interfering with *po*-prefixed ipfv. stems, although quite unevenly, depending on the verb lexeme (see Section 4.1). To a large extent, Němec’s argument is based on *jít(i)* ‘go’ and the absence of periphrastic **bud- jít(i)* in the history of Czech (see fn. a). This fact precluded any potential morphological opposition (*pójdu* vs **budu jíti*) just for this particular verb (with direct motion meaning); however, for many other items such an opposition was possible, since “doublets” of *po*-prefixation and *bud-* + infinitive already existed in Old Czech. Although Šlosar (1981: 79–80, 102–103), on the basis of the old text corpus and older grammars, argues that the number of ipfv. verbs expressing their future with *po-* (or another prefix) had already diminished in older periods of Czech (see Section 4.2.2), other findings summarized in Section 4.1 suggest that the duality of future marked by present tense with *po-* vs *bud-* (+ infinitive) is evident for a quite large and open class of verbs and that it has probably been pervading the entire history of Czech.

In sum, as concerns the approaches referred to in Section 4.1, most researchers assume (often implicitly) that the perfectivity opposition must already have been stable to a sufficient extent by the 14th century.⁴⁶ Regardless of this assumption (and how it was argued for), all scholars appear to agree that a former ingressive meaning of the relevant prefixes turned into a function of tense, which is by itself insensitive to aspect. Whether the compatibility with progressive contexts (e.g., with durative time adverbials) is an ancient feature (connected to the individual “lexical prehistory” of the prefixes) or only became possible after these prefixes were reinterpreted as marking future tense of ipfv. simplex stems, is of secondary concern.

5 Chronological relations and areal distribution

The differentiation of emergent future auxiliaries in various regions of Slavic is already discernible at the time of the first written documents after early OCS, i.e. from the 11th century onward. In the eastern part of South Slavic (= Balkan Slavic), WANT superseded its strongest “rival” HAVE, although HAVE-based patterns have remained important until today; BECOME did not play any role, quite to the contrary of the western part of Slavic, both in the North and the South. In turn, the eastern part of North Slavic (= East Slavic) for a couple of centuries showed preference for BEGIN-based periphrases (with WANT-based patterns probably imported from OCS), before BECOME spread there as well and became the dominant or exclusive auxiliary of futures with ipfv. stems.

The employment of prefixes to mark the future of ipfv. simplex stems (and not to derive a pfv. stem) looks quite exotic, also from a Slavic point of view. If the reconstruction sketched in Section 4 is correct, this could mean two things. Either, by the 14th century, a stem-derivational opposition of pfv. vs ipfv. aspect had already become sufficiently stable in general, but then in the western parts of the Slavic-speaking territory this system turned out less consistent for some groups of verb lexemes that denoted parametric changes or atelic activities. Such an interpretation is called for if we assume a reinterpretation of the relevant prefixes in the paradigm of the relevant ipfv. stems (see Section 4.4). Or, alternatively, we may argue that the use of *po-* and other prefixes as markers of ipfv. future reflects an archaic trait which, as it were, survived the first stages in the development of

⁴⁶The argument in Němec (2009b) does not contradict Bondarko’s (1961) argument. After all, the considerations of both scholars are based on the idea that the *po-*prefixed forms lost their ingressive function and that the isolated *po-*prefixed present tense forms were integrated for symmetry reasons into the paradigm of the ipfv. stem.

the stem-derivational aspect system and which also manifested itself earlier in OCS⁴⁷ (see Section 4.2.2).

At present, we cannot decide between these alternative explanations, but they do not even exclude one another. Verb stems denoting unbounded activities represent actionality groups that are more “vulnerable” within a perfectivity system whose onset and initial stages were certainly triggered by distinctions related to telicity; into such a system lexemes denoting unbounded activities were integrated only at a later stage (see Section 3.1). Their integration could fade away, or get stuck at an early stage, and this is what evidently happened in Czech, as part of a “cluster” of languages of the “Slavic west” – in contrast to the eastern part, in which prefixes are much more freely employed for the derivation of pfv. delimitatives and ingressesives of activity verbs. Scholars like Dickey have made a plausible case that contact with German was the most likely factor that weakened and hampered the use of prefixes for the derivation of pfv. stems with lexemes denoting unbounded activities (see Section 4.3). Dickey (2011) calls this ‘replica preservation’. What was preserved (or, maybe, rather restored) was the lack of an explicit differentiation between temporally limited and unlimited activities by otherwise well-developed derivational means (i.e. prefixes). However, in light of data discussed in Section 4.2–4.3, this “preservation” was obviously preceded by a period (14th–15th centuries) of rather free alternation between simplex (ipfv.) and prefixed (pfv.) stems for ingressivity (both in past and non-past tense) as well as for future-related and habitual or gnomic situations (with non-deontic modal implicatures).

When we now resume the “story” about the BECOME-future, more precisely: about the pattern BECOME + infinitive, in North Slavic, we notice that the areas from which this pattern most probably spread overlap strikingly with the areal cluster in which prefixation had been showing some perceivable tendency to mark the future of ipfv. stems. More so, even the period of these processes largely coincides: we are dealing here with the 14th–16th centuries. The innovative pattern *bud-* + infinitive became prominent first in Bohemian (Czech), then in Polish, from where it spread into western East Slavic (Ruthenian) and finally into Russian (15th–18th century; see also the map in Whaley 2000: 74). Properly speaking, this spread from Bohemia toward the northeast meant that a minor pattern⁴⁸ was supported and amplified by inner-Slavic contact from the (south)west of North Slavic, and that it was mainly characteristic of written (pre-standard)

⁴⁷That *po-* for ipfv. future might be an archaism was assumed by Némec (2009b) and by Galton (1976: 48). However, Némec also at least implicitly advocated for “paradigmatic reshuffling”, which is inherent to the first alternative (see the previous footnote).

⁴⁸Such patterns were also assumed by Andersen (2006b: 30) for prehistoric dialect zones all over from today’s Serbia to Danzig. He however also admits that this scattered spread remains mysterious.

varieties. This interpretation is also consistent with the observation that North Slavic belongs to a larger area stretching from German to northeastern Europe in which futures are either lacking or are only weakly developed on the basis of BECOME (Wiemer 2020: 276, following Dahl 2000). This makes BECOME areally salient as a source of future auxiliaries. Contact with Middle and Early New High German can be considered a plausible catalyst of the gradual entrenchment of inchoative *bqd-/bud-* in Bohemia and, thus, in all North Slavic, although this issue continues to create some problems, first of all because of the provenance of the infinitive (see Section 3.2.1.3).

Moreover, the innovative pattern *bud-* + infinitive is also attested in South Slavic, but only in its “middle section” (i.e. Štokavian-Čakavian). Within that region, it spread from west to east, more or less at the same time when in the eastern part the WANT-future developed and spread from east to west, concomitantly with the stepwise loss of the infinitive. The pattern *bud-* + infinitive has never been attested in Slovene, i.e. the northwest periphery, for which influence of German (apart from Italian) varieties was certainly more likely than on the Čakavian Adriatic coast. We cannot know whether this lack of attestation in Slovene simply arises from lack of data about its earliest manifestations in texts.

Anyway, the distribution of *bud-* + infinitive in South Slavic looks strange: its area of use is separated from German-speaking territory by Slovene, which, in turn, preserves the earlier pattern *bô-* (< **bqd-*) + *l*-form (simultaneously losing its future perfect function) and is isolated in this respect. One wonders whether there might have been a stable connection to West Slavic, namely Bohemia and Moravia despite the separation by the Magyars (since approx. 900 AD). This would not explain the preservation of the isolated *bud-* + *l*-form pattern in Slovene, but it would fit the western cluster of Slavic languages (Czech, Slovak, Sorbian, Slovene), which have shown some tendency to use prefixes as markers of the future with ipfv. stems. This cluster corresponds to an earlier continuum between what subsequently became Bohemian, Moravian and Kajkavian dialect zones; this cluster must have existed prior to the arrival of the Magyars in the Pannonian plain, and to the exclusion of the Lekhitic territory farther to the northeast (from where Polish and Kashubian developed). This cluster was also pointed out by Andersen (2006b: 25), but without an account of prefixes as markers of future with ipfv. simplex stems. One thus wonders whether the parallelism of these two processes was simply a coincidence. What can be said with more confidence is that, to a large extent, different dominant future markers (BECOME vs WANT) developed due to divergent contact-induced processes in different parts of the Slavic-speaking area and of inner-Slavic continua after that time.

How does all this relate to the hypoanalysis of PFV.PRS to pfv. future in North Slavic, and to its non-occurrence in South Slavic? First of all, since this kind of hypoanalysis was based on the stem-derivational aspect opposition, this opposition must have been strong enough by the time when among looser combinations with lexical source expressions (WANT, BECOME, HAVE, BEGIN, TAKE) areally biased preferences started acquiring shape.

Since Common Slavic had no dedicated contrasts in the non-past domain, future reference had been a natural part of the meaning range of non-past (“present”) tense, regardless of aspect, prior to the emergence of futures based on auxiliaries. However, PFV.PRS was driven into the irrealis domain (“inactual present”) by the aspect (IPFV:PFV) opposition alone. Subsequently, future readings must have increasingly outnumbered irrealis readings in North Slavic, but irrealis readings did not vanish; instead, this usage domain of PFV.PRS is still shared with South Slavic. Here, in turn, the rise of dedicated future auxiliaries also introduced a morphological contrast with pfv. stems, which is alien to North Slavic; however, remnants of genuine future uses of PFV.PRS can be found especially in the northwestern part of South Slavic to this day (cf. Grković-Mejdžor 2019: 11).

Consequently, irrealis uses have remained the large domain in which PFV.PRS continues to appear both in North and South Slavic. In this respect, nothing has changed since CS times: the non-future uses, considered “residual” for North Slavic, are just as original as are uses of PFV.PRS to refer to future events. What obviously has changed is the salience of future reference for PFV.PRS. Therefore, an explanation is required for (at least) two issues: first, how can this shift in salience be captured, also within a more overarching framework of language change? Second, should the divergent development of PFV.PRS in North and in South Slavic be explained by differences in the development of the stem-derivational aspect system?

While the first question requires another methodological take on the topic, there is no reason to give an affirmative answer to the second question. For all we know about inner-Slavic differences in the consistency with which stem-derivational aspect has been developing, the eastern half appears to be more consistent in marking aspect with stems of all actionality classes, i.e. lexical expansion is more advanced, and the respective process seems to have begun more or less in the 16th century (see Section 4.3). After all, we identified the use of prefixes marking the future with ipfv. stems that denote activities and parametric changes as conditioned by a system that existed prior to this process. As for the centuries preceding this process, there is no reason to assume any remarkable differences in the development of the aspect system, either, despite the fact that a restriction of BECOME + infinitive to ipfv. stems applying from the first documents has

been reported for North Slavic but not for South Slavic languages (see Section 3.2.1). Moreover, we are left with the finding that the “future-by-prefixes” pattern, jointly with the initial spread zones of *BECOME* + infinitive, cluster in the west of Slavic and cut the North-South divide. That is, they are orthogonal with regard to the North-South split. The lack of aspect restrictions for the *WANT*-future can be explained rather straightforwardly from the source construction, but the restriction to *ipfv.* aspect for the *BECOME*-future in the North cannot. Nor can this restriction be explained by the most plausible “external catalyst” of *BECOME* + infinitive in West Slavic, namely German.

6 Conclusions

The study of the development of futures in Slavic languages shows that a clear-cut distinction between grammaticalization and hypoanalysis as mechanisms of change not only makes sense, but that such a distinction is even necessary in order to see their entanglement and relative chronology as well as the divergent consequences this has for the formation of areal clusters and clines within the whole language group. Croft’s notion of hypoanalysis has here been adopted with two slight, but important amendments for the case under study: first, we need to apply it to paradigmatic oppositions, regardless of their morphosyntactic realization; second, the default future function of *PFV.PRS* in North Slavic results not so much from a shift towards a new function that previously was only implied (i.e. future), but this function was always possible, before it turned into the most prominent one. It remains to model this mechanism within a more comprehensive framework of linguistic changes.

Concomitantly, we should correctly describe the stem-derivational nature of the Slavic perfectivity opposition in order to understand the fundamental role which this technique plays in the grammatical outfit of Slavic languages in general, and for the verbal system in the non-past domain in particular. With this in mind, *PFV.PRS* turns out to be an extremely persistent feature of *all* Slavic languages, inasmuch as *PFV.PRS* in all of them continues to serve the same functions in the irrealis domain as it most probably has done from Common Slavic times, when the perfectivity opposition had established itself firmly enough. Thus, in this sense, its future function is just an areally restricted by-product that has not ousted other functions.

Finally, the North-South split of future marking leaves open issues. First of all, there is still no satisfying explanation for why the pattern *BECOME* + infinitive is restricted to *ipfv.* stems in North Slavic, and why this obviously has been so

through the entire attested period, i.e. since the 13th century. Simultaneously, we cannot say with certainty where, when and how the infinitive entered this pattern, in particular whether contact with German varieties provided the decisive external trigger. However, we observe that the initial spread zones of BECOME + infinitive perspicuously overlap with a “western cluster” of those Slavic languages in which prefixes have been used to mark the future of ipfv. simplex stems. I hope to have supplied sufficient arguments that this overlap is not just incidental but testifies to a common period (13th–16th centuries) during which these seemingly unrelated patterns of future marking of ipfv. stems developed. In the western cluster, the BECOME-based future suppressed the prefixation pattern, but at least in Czech it has never ceased to be productive for the relevant actionality classes.

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Chapter 9

Hypoanalysis and aspectual diversification: The Armenian future in diachrony

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Old Armenian does not exhibit a separate future tense; future events are commonly expressed by means of modal forms. In Middle Armenian, the historic subjunctive is lost as part of the process of replacing the synthetic present with an analytical, originally progressive form; the historic present indicative takes on subjunctive functions. Later in this period, a future develops by hypoanalysis and, secondarily, univerbation of the former present indicative, whilst an innovative periphrastic form supersedes it; the subjunctive persists.

In Modern Eastern Armenian, the new future takes on additional, secondary functions; other, analytical future periphrases have arisen, increasing the aspectual differentiation of the language in analogy to other tenses; not all are equally productive. The synthetic future might be considered an aspectually unmarked form.

This paper outlines in as much detail as currently possible the development of the future from Old to Modern Armenian, with particular attention to grammaticalisation and hypoanalysis processes. Furthermore, a corpus-based sketch of the future periphrases in Modern Eastern Armenian is given with a view to proposing a basic diachrony of and motivation for their development.



1 Introduction

As an Indo-European language with a respectable number of linguistic idiosyncrasies, both inherited and borrowed,¹ the oldest variety of Armenian, that is Classical or Old Armenian, has received a considerable amount of attention from historical and comparative linguists as concerns most domains, viz. phonetics and phonology, morphology, syntax, semantics, etc.² at least since the latter part of the 19th century; the linguistic study of modern varieties has similarly not been neglected. By far the longest phase of the history of the Armenian language, however, remains regrettably understudied: the Middle Armenian period, stretching from, at the latest, the 12th century to the 18th century and beyond, has received comparatively little attention whilst, at the same time, being the period of greatest diversification.

In turn, this research gap means that perspectives on the diachrony of Armenian as a whole are limited, at least to some extent, by the *tabula rasa* that is the Middle Armenian phase. In order to help remedy this situation, this chapter seeks to delineate and explain the diachronic development of the future from proto-Armenian to its modern varieties. Putting its history in a typological and comparative context, the chapter makes two principal claims: firstly, that the synthetic *k*-future in Modern Eastern Armenian (MEA) is the result of two separate instances of hypoanalysis in the Middle Armenian phase; and secondly, that the aspectual diversification of future reference by means of periphrases is the result of analogical paradigmatisation – even though perhaps overstated by current grammars when compared to actual use patterns.

Section 2 outlines the point of departure for the development of future expression in Armenian – that is, its absence in Old Armenian – and compares it to the situation in other early Indo-European languages. Thereafter, the two main changes in the Middle Armenian period, that is first the creation of a periphrastic present and its later reanalysis as a future, will be discussed in Section 3. Section 4,

¹Olsen (1999: iv) famously described Armenian as the “horror chamber of historical phonology” owing to its treatment of the cluster PIE *dw- as Arm. *erk-* (e.g. in the word for ‘two’, Arm. *erku* < *dwóh₁) or the metathesis in the PIE cluster *-br- to Arm. *-lb-* (e.g. in *elbayr* ‘brother’ as compared to Gk. *phrátēr*, Lat. *frāter* < PIE *b^hréh₂tér). For the question of borrowing, see the various loans – lexical, phonological, morphological, and syntactic – discussed in Meyer (2023a).

²See, e.g., the volumes by Meillet (1936), Godel (1975), Klingenschmitt (1982) or Schmitt (2007), which outline the development of Old Armenian from Proto-Indo-European in detail, and the etymological dictionary of Martirosyan (2010). The most recent, if compendious, discussion can be found in Klein et al. (2017: 1028–1165)’s *Handbook of Comparative and Historical Indo-European Linguistics*.

in turn, outlines the current situation in Modern Eastern Armenian and provides some insights into the aspectual diversification in the latter variety.

2 The future in Proto- and Classical Armenian

The oldest attested variety of Armenian, sources for which date back to the 5th century CE,³ does not conceive of the future as a grammatical category. While it distinguishes a present tense from two distinct past tenses – the imperfective preterite and the perfective aorist – and a stative-resultative periphrastic perfect,⁴ future reference can only be expressed by means of polysemous forms. As examples (1–2) and (3–4) illustrate, this is done either by means of the subjunctive, which has a variety of other, related functions,⁵ or by means of a periphrasis composed of the deontic verbal adjective in *-loc*‘ and a form of the copula.⁶

- (1) *erkink’ ew erkir anc’c’en*
 heaven.NOM.PL and earth.NOM.SG pass.3PL.SBJV
 ‘Heaven and earth shall pass’ (Mt. 24:35)
- (2) *ew es and jez ibrew z=mi i jēñj*
 CONJ 1SG.NOM with 2PL.ACC like OBJ=ONE.ACC.SG among 2PL.ABL
xndrec’ic’ z=jer ōgut=n
 seek.1SG.AOR.SBJV OBJ=2PL.POSS advantage.ACC.SG=DET
 ‘And I, too, like one of you, shall seek your advantage’ (Agat’angelos Cl.9; Yegavian 2003b)
- (3) *mi omn i jēñj matneloc’ ē z=is*
 one INDF.NOM.SG from 2PL.ABL betray.VBADJ be.3SG.PRS OBJ=1SG.ACC
 ‘One among you will betray me’ (Mk. 14:18)

³While the earliest epigraphic evidence of Armenian dates to the 5th century CE (Stone 1990), the literary attestations post-date this period by far, with the earliest manuscript dating to the 9th century.

⁴For a discussion of the diachronic semantics of the perfect, cf. Meyer (2024).

⁵They can, for instance, express doubt or uncertainty in a variety of subordinate clauses (Jensen 1959: 200, 205–207).

⁶The etymology of this verbal adjective is not clearly understood. Godel (1975: 129) plausibly proposes that it is composed of the formans **-lo-*, which is also used to form the infinitive and past participle, and a second, derivative suffix **-sko-* as found in Germanic and Slavic; the functional parallels are, however, less clear.

- (4) *zi ew duk' ibrew olormut'eamb K'ristosi arjakeloc'*
 for also 2PL.NOM like mercy.INS.SG Christ.GEN.SG release.VBADJ
ek' i kapanoc'=d, ew mek' ... gnaloc' emk' i
 be.2PL.PRS from chain.ABL.PL=DET and 1PL.NOM go.VBADJ be.1PL.PRS to
bnik ašxarh=n mer
 natural land.ACC.SG=DET 1PL.POSS
 'For you, by the mercy of Christ, shall be released from your chains, and
 we ... shall return to our homeland' (Łazar P'arpec'i III.50.1; Yegavian
 2003b)

While this polysemy between deontic or potential uses of the forms on the one hand and future uses on the other has the disadvantage of allowing for a certain degree of ambiguity, it is neither typologically, historically, nor comparatively uncommon.

No future category can be reconstructed for Proto-Indo-European, for instance. Those early languages like Sanskrit, Greek or Latin which do attest a synthetic future and thus a distinct grammatical category have developed it individually by reanalysis of a formerly subjunctival or desiderative form or by means of univerbation of a previous periphrastic construction.⁷ Others, like Armenian, rely on the subjunctive and its polysemy to express futurity; one such example is Tokharian (5).⁸

- (5) *nāš nu ce(smā)k āyāntu p_ukāk puskāsyo*
 1SG.NOM.M and DEM.ACC.PL.M.EMPH bone.ACC.PL.M all.PERL sinew.INS.PL.M
kaśal malkam=ām
 together join.1SG.SBJV.ACT=OBJ.PL
 'And I will join the bones completely together with the sinews'
 (Tokharian A; A 11 b5–6)

Next to the subjunctive, Tokharian, among other Indo-European languages, can employ the present tense effectively as a non-past, thus encompassing future events; at times, adverbs and serial verb constructions are used to eluci-

⁷For a discussion of the desiderative or subjunctival origin of the Greek sigmatic future, see Willi (2018: 441–452). The Latin *b*-future appears to be a univerbated form of the verbal root and the aorist subjunctive of *b^huH- 'to become' (Weiss 2009: 415); the original form and function of the verbal root is, however, as yet unclear.

⁸Cf. Malzahn (2010: 265–266).

date this futurity and distinguish it from the present,⁹ without being sufficiently widespread or grammaticalised, however, as to count as a separate tense category.

- (6) *URRAM ŠERAM kuiš ammuk EGIR-anda LUGAL-uš kišari*
in-the-future REL.NOM 1SG.DAT after king-NOM become.3SG.PRS.MID
‘Whoever shall become king after me’ (Hittite; KBo 3.1 ii 40)
- (7) *saei bigitiþ saiwala seina,*
REL.NOM.SG.M find.3SG.PRS.IND soul.ACC.SG.F 3SG.POSS.ACC.F
fraqisteiþ izai
lose.3SG.PRS.IND 3SG.DAT.F
‘Whoever finds his soul, will lose it’ (Gothic; Mt. 10:39 [Argenteus])
- (8) *nebo i zemlę mimo idetŭ*
heaven.NOM.SG and earth.NOM.PL past go.3SG.PRS
‘Heaven and earth shall pass’ (Old Church Slavonic; Mt. 24:35 [Marianus])

The above examples illustrate such uses for Hittite (6) as a language that separated early from the Indo-European core;¹⁰ and for Gothic (7) and Old Church Slavonic (8) as languages with a later attestation.¹¹

Even beyond the context of Indo-European languages alone, synthetic futures are by no means commonplace. In Dahl & Velupillai (2013)’s typological study, just under 50% of the 222 studied languages exhibited an inflectional future category; for Bybee (1985: 156), it was 44% in a sample of 50. Historically speaking, however, even these inflectional futures are often the result of grammaticalised, and thus univerbated, periphrases originally expressing volition, obligation, or motion (Bybee et al. 1994: 159);¹² the Romance languages demonstrate neatly that

⁹Hittite, for instance, can use *kāša* ‘at the point of’ for the immediate future or *URRAM ŠERAM* ‘in the future’ as in (6) below (Hoffner & Melchert 2008: 324). Gothic can use *wairþan* ‘to get to be’ as a prophetic future, or periphrases with *du-ginnan* ‘to begin’, *haban* ‘to have’, or *skulan* ‘to owe, be obliged to’ (Kleyner 2015). Old Church Slavonic equally can use *xotěti* ‘to want’ or *imamŭ* ‘to have’ to express futurity (Lunt 2001).

¹⁰On the position of the Anatolian languages, see most recently (Kloekhorst 2022).

¹¹On Gothic, cf. Miller (2019: 176 n. 1) with literature; for Old Church Slavonic, cf. Lunt (2001: 154–155).

¹²For an overview of such developments in other language families, see, e.g., Abdel-Hafiz (2017) for Arabic and the Nile-Nubian languages; for Yucatecan Maya, cf. Lehmann (2017: 213–229); for Bantu, cf. Batibo (2005) and Botne (1998) for a future in Central Bantu based on a verb meaning ‘to say’.

this process of grammaticalisation can be cyclical: the development of the deontic collocation of the type *cantāre habeō* ‘I have to sing’ into a future expression in Late Latin overlapped with the loss of the synthetic future (type *cantābō*), resulting in the eventual grammaticalisation and standardisation of the (previously periphrastic) forms as the new ‘standard’ future of the type Fr. *chanterai*, It. *cantarerò*, Sp. *cantaré* ‘I shall sing’; these, in turn, compete in modern varieties with motion-related periphrases of the type Fr. *aller chanter*, Sp. *ir a cantar* ‘going to sing’.¹³

Old Armenian therefore is, so to speak, in good company in not (yet) having a morphologised future, but equally exhibits periphrases which might potentially undergo such a process. As it turns out, however, this is not the pathway that the Armenian language will follow; as the next section explains, neither the (synchronic) subjunctive nor the deontic periphrasis will serve as the basis for a later future tense. Rather, in Middle Armenian, two cycles of grammaticalisation are required before the crystallisation of a separate future category. These are not dissimilar to the Romance cycles just mentioned, but centre around the present rather than the future tense, with the latter’s creation being a by-product.

3 Developments in Middle Armenian

In the more or less six centuries of the Middle Armenian period, the Armenian language has undergone significant changes, both in terms of its phonology and morphology as well as its general unity. Owing to the dearth of research in this area, some of these processes can only be judged by their outcomes at present, rather than by a detailed analysis of the changes itself. In terms of diversity, at the beginning of the 20th century, Ačaṙean (1911) describes 41 distinct varieties in three groups.¹⁴ Not all of these varieties have undergone the same changes: the western so-called *gə*-varieties, for instance, have lost the threefold distinction of plosives in voiced (/b/, /d/, /g/), voiceless (/p/, /t/, /k/), and unvoiced aspirated (/p^h/, /t^h/, /k^h/) attested for Old Armenian and, with some phonetic variation, retained in the Eastern, *um*-varieties (including Modern Eastern Armenian, the standard literary variety); the Western varieties only retain a distinction between voiced and voiceless aspirate.¹⁵

¹³Italian does not have such a periphrasis. Similarly, both German and English have a fully grammaticalised periphrastic future, but only English has developed a secondary, motion-based future in *going to*.

¹⁴An English translation with a commentary of this magnum opus is due to be published (Dolatian 2024).

¹⁵The old voiceless aspirate plosives retain their articulation, whereas the former voiced plosives fuse with them (/b/, /d/, /g/ > /p^h/, /t^h/, /k^h/); the old unvoiced plosives are voiced (/p/, /t/, /k/ > /b/, /d/, /g/).

This phonological diversification together with other changes seems to belie the notion of Middle Armenian as a unified whole representing all of high- and late-medieval and early modern Armenian. This complex situation is not helped by the fact that research on Middle Armenian to date has focussed largely on phonological, lexical, and dialectal questions as well as language contact,¹⁶ with very little in terms of morphological or morphosyntactic research.¹⁷ As a result, the standard work of reference remains Karst (1901), which, alas, gives only limited detail on verbal morphosyntax in general, is not primarily concerned with diachronic developments, and in general only discusses one particular variety of Middle Armenian – the variety of Cilicia. For the present purpose, Middle Armenian is thus treated as a whole, all the while acknowledging that this is a simplification.

To properly understand the diachrony of future expression in Middle Armenian, therefore, the most promising approach would seem to be triangulation: to take what is known for certain in Old Armenian and the modern varieties in order to reconstruct the most plausible trajectory of the changes involved, including eventual forks in the path. For that purpose, it is worth observing changes not only in expressions of futurity but also, as already alluded to above, in the present tense system.

Table 1 summarises the situation: Old Armenian uses a simple synthetic indicative form to express the present tense, whereas the future is marked by (aorist) subjunctive forms or a deontic periphrasis. The modern standard varieties (Modern Eastern Armenian, MEA, and Modern Western Armenian, MWA) use different formations to express future and present, respectively.¹⁸ In MEA, the present is formed periphrastically with a verbal adjective in *-um* and a finite form of the copula; the (unmarked) future is formed by adding the future marking morph *k-* to what is historically the Old Armenian present tense and synchronically the subjunctive. In MWA, by contrast, it is the present which is formed by adding the marker *kə* (or *k'*) to the historical present / synchronic subjunctive;¹⁹ the future

¹⁶For an overview with plentiful literature, cf. van Lint (forthcoming); the question of Middle Armenian and its dialectal variety is treated, amongst others, in Ĵahukyan (1993), Mkrtč'yan (2018), Muradyan (1975), Weitenberg (1996) and Weitenberg (2017).

¹⁷Notable exceptions are Ant'osyan (1972), Ant'osyan (1975); for more general linguistic grammatical overviews, cf. Ałayan (1972), Hovsep'yan (1975), Ĵahukyan (1964), Ĵahukyan (1975) and Łazaryan (1960).

¹⁸The more fine-grained, mostly phonological differences that are found in the present and future formations in the individual dialects will not be discussed here.

¹⁹These morphs are pronounced as /g(ə)/ in MWA; in this paper, however, Armenian is consistently transcribed according to Hübschmann-Meillet-Benveniste as the convention used in most subject-specific publications and journals, e.g. the *Revue des Études Arméniennes*.

is formed by means of a historically deontic periphrasis with *piti* (/bidi/) and the historical present / synchronic subjunctive.

Table 1: Future and present in Old and Modern Armenian

	Old Armenian	Middle Armenian	Modern Armenian	
PRS	<i>berem</i> 'I carry'	↘ ?	<i>berum em</i>	MEA
			'I (am) carry(ing)'	
			↗ <i>kə berem</i>	MWA
			'I (am) carry(ing)'	
FUT	[<i>beric</i> / <i>bereloc</i> 'em] ['I shall carry']	↗	<i>kberem</i>	MEA
			↘ 'I shall carry'	
			<i>piti berem</i>	MWA
			'I shall carry'	

These various formations allow for a certain number of observations of commonalities and differences to be made; they are, in no particular order:

- the Old Armenian future periphrases do not have modern cognates;
- the Old Armenian present formation is continued in the modern varieties (as a subjunctive) and involved in both present (MWA) and future formation (MEA, MWA);
- the marker *k-* / *kə* is used in both modern varieties, but for different purposes – the future in MEA, the present in MWA;
- both modern varieties have grammaticalised one tense on the basis of a periphrasis not found in that form in the other variety (MEA present; MWA future).

In terms of diachronic expectations, it therefore stands to reason that, in the first place, the Old Armenian futuristic periphrases fell out of use (or, perhaps, were never sufficiently frequent to cross the grammaticalisation threshold) and the subjunctive was lost. Secondly, the use of forms resembling the Old Armenian present indicative as subjunctives in the modern varieties suggests a shift in that direction in Middle Armenian; such developments are, of course, well-attested

and typologically unproblematic (cf. Haspelmath 1998). Concomitant with the reanalysis of the old present indicative as an innovative subjunctive, a new present indicative must have arisen to motivate this change to begin with. This innovative present must have involved the predecessor of the marker *k-* / *kə*, on the assumption of economy of change.²⁰ Western Armenian varieties would thus be expected to have retained this constellation, and innovated a deontic future periphrasis, whereas further change ensued in the Eastern Armenian varieties. There, a second round of reanalysis must have resulted in the reinterpretation of the *k*-present as a future under competition from a new present formation.

These are, of course, only working hypotheses based on a reasoned comparison of the earliest and recent varieties of Armenian. They could and should, in the long run, be verified and (dis-)proved on the basis of detailed corpus studies. In the absence of such a corpus, however, the following sections provide relevant illustrative examples from Middle Armenian, which have to suffice in order to show that the progress outlined above is, at the very least, plausible and to provide more details of the mechanisms of change.

3.1 Early Middle Armenian

The first question arising concerns the fate of the Old Armenian subjunctive. As a result of its relatively small functional load besides the expression of the future and due to a number of morphophonological idiosyncrasies,²¹ both the present and aorist forms fell out of use already towards the end of the Old Armenian period.²² Their functions are assumed by the present indicative, as witnessed by the virtual identity (barring phonological change) of modern Armenian present subjunctives and old present indicatives;²³ this inevitably increased the functional load of that form significantly. A clear indication of this replacement can

²⁰If there was a *k*-present in Middle Armenian, only one change is required to explain the current distribution: the Middle Armenian present becomes an MEA future. Otherwise, if a different grammatical value is attributed to this form, multiple changes, namely for both modern varieties, would be required.

²¹Cf. Karst (1901: 299); Weitenberg (1993); Vaux (1995).

²²A reviewer inquires whether the loss of the old subjunctive might be the result of the advent of the *ku*-present and the subsequent reanalysis of the old present as the subjunctive. Without a detailed corpus study, this cannot be excluded *a priori*. Two issues may speak against it, however: the lack of co-occurrence of all three forms (old subjunctive, old present, and *ku*-presents) in early Middle Armenian texts as might be expected in a transitional period; and the complete loss of the old subjunctive instead of its relegation to a more specific function, e.g. as a future.

²³Cf. Ačařean (1911) *passim* and the relevant commentaries by Dolatian (2024).

be found in the expression of commands or exhortations, which in Old Armenian require the subjunctive (9–10) but in Middle Armenian are taken over by the present indicative and a particle (11–12).

- (9) *ari gnasc'uk' ar kaysr=n*
 valiant go.2PL.AOR.SBJV to emperor.ACC.SG=DET
 'Let us go valiantly to the Emperor' (P'awstos Buzand IV.10; Yegavian 2003a)
- (10) *iwrak'anč'iwrok' z=iwr kin kalc'i*
 everyone OBJ=3SG.POSS woman.ACC.SG have.3SG.AOR.SBJV
 'Let everyone have his own wife' (1Cor. 7:2)
- (11) *t'ē jmerñ lini, tak' ĵur t'ol xmē*
 if winter.NOM.SG become.3SG.PRS warm water.ACC.SG PTC drink.3SG.PRS
 'If it is winter, let him drink warm water.' (Mxit'ar Herac'i, *Ĵermanc' mxitarut'iwn* 20.14)
- (12) *t'ol mašim i y=ays hog=s ew xist*
 PTC be-consumed.1SG.PRS into into=DET care=DET and firm
hənanam
 grow-old.1SG.PRS
 'Let me be consumed by this care and grow old [in] firm [belief]' (Frik, *Diwan* 38.86)

The exhortation or command signified by the subjunctive forms in Old Armenian can no longer be expressed by such forms in Middle Armenian, as they are identical to the indicative;²⁴ the necessary functional clarification is added by the particle *t'ol*.²⁵ In other instances yet, old structures or collocations are lost owing to the increased formal ambiguity; such is the case, for instance, in purposes clauses introduced by *zi* 'so that', where Old Armenian requires a subjunctive (13). In Middle Armenian, this use of *zi* is no longer possible.²⁶

²⁴Karst (1901: 299–300) mentions the existence of a subjunctive periphrasis consisting of the indicative forms of the verb *kam* 'remain', the conjunction *u* 'and', and a finite lexical verb (e.g. *et'ē kenay u gay* 'if he comes'), but this periphrasis is short-lived and never fully grammaticalised.

²⁵This particle derives from *t'olum* 'to let, allow', of which it is a, by then fossilised, imperative. Such a construction is already attested in Old Armenian, e.g. *t'ol ekesc'ē dma uraxut'iwn* 'let happiness come for him' (Agat'. 105); note, however, that here a subjunctive is still required.

²⁶The conjunction remains in active use, however, in its causal function, meaning 'for'.

- (13) *na ew episkoposac' =n žoloveal, palatans matuc'anēin*
 even bishop.GEN.PL=DET gather.CVB supplication.ACC.PL offer.3PL.PST
zi mi anp'oyt' arasc'ē z=korstenē iwroy
 so-that NEG careless do.3SG.AOR.SBJV about=loss.ABL.SG 3SG.POSS.GEN
vičaki=n
 domain.GEN.SG=DET
 'Even the bishops gathered and sent supplications lest he act indifferent concerning the loss of his domain' (Movsēs Xorenac'i III.29; Yegavian 2003b)

Next to these subjunctival functions *sensu stricto*, the old present also occasionally takes on futuristic functions, as indicated by (14).

- (14) *p'ark' Astucoy ver arak'em ew biwr hazar het*
 glory god.GEN.SG up send.1SG.IND and hundred thousand time
gohanam, or inj eret mitk' u hogi, or
 thank.1SG.IND COMP 1SG.DAT give.3SG.AOR mind and soul REL.NOM.SG
z=ays baners ku səgam.
 OBJ=DEM.ACC.SG thing.ACC.PL PROG.IND mourn.1SG
 'I shall send up high the glory of God and thank [him] a hundred thousand times | that he gave me a mind and a soul, who I am lamenting these things' (Frik, *Diwan* 2.46–7)

This functional overload – one form for present indicative, subjunctive, and future – proved to be an ideal breeding ground for other, innovative forms to intervene and take on specific functions from the old present. This is not to suggest, of course, that there is necessarily a causal relationship between functional overload and the *creation* of new forms in general. In the case of Middle Armenian, the innovation comes in the guise of the particle *ku*, likely originally a progressive marker.²⁷ This periphrasis does not take on subjunctival roles,²⁸ but expresses mainly the present (15–16); from Mxit'ar Herac'i (12th c.) onward it is indeed the standard expression of the present tense.

²⁷ Vaux (1995) argues convincingly for a progressive reading of this particle post-standardisation, since it does not occur with a number of high-frequency stative verbs; Karst (1901: 307–309) rightly remarks, however, that the (tentative) derivation of *ku* from *kay u* 'remains and' and its general distribution strongly suggest that the particle originally signified mainly the indicative.

²⁸ But see fn. 24 above and compare the origin of the present particle as mentioned in 27, a certain likeness between which cannot be denied.

- (15) *menk' ekel enk' u ku hrawirenk' z=k'ez*
 1PL.NOM come.PTCP be.1PL.PRS and PROG.IND invite.1PL.PRS OBJ=2SG.ACC
or gas sur i darpas=n
 COMP go.2SG.PRS.SBJV immediately to court.ACC.SG=DEF
 'We have come and invite you to go to court immediately' (Smbat Sparapet, *Antiok'i asizner* 13.13ff.)
- (16) *ew erb imanas or ōr k'an z=ōr*
 and when(ever) notice.2SG.SBJV COMP day after OBJ=day
erak=n ku tkaranay, apa z=hac'=n
 vein.NOM.SG=DET PROG.IND weaken.3SG.PRS then OBJ=bread.ACC.SG=DET
i gini=n t'ac', ew tur
 in wine.ACC.SG=DET soak.2SG.AOR.IMV and give.2SG.AOR.IMV
 'And when(ever) you notice that the vein weakens day after day, soak the bread in wine and give [it to him]' (Mxit'ar Herac'i, *Ĵermanc' mxitarut'iwn* 20.17)

Even if its origins (and later use in MWA) suggest a progressive reading, the periphrasis with *ku* at least in early Middle Armenian can also express futurity and should thus, perhaps, be interpreted as a more general non-past (17); another, more specific future periphrases of a volutative nature (*kamim* + INF) exists as well, but has not been standardised at this time (18).²⁹

- (17) *es ku asem u ku c'uc'anem alani*
 1SG.NOM PROG.PRS.IND say.1SG and PROG.PRS.IND demonstrate.1SG clearly
u yerewan
 and evidently
 'I will speak [at court] and will demonstrate clearly and evidently...'
 (Smbat Sparapet, *Antiok'i asizner* 27.11)
- (18) *ew mek=n i šoyt ew i yuš žoŋovel=n*
 and one.NOM.SG=DET in quick and in slow gather.INF=DET
lini niwt'i=n, or borbosel
 become.3SG.PRS substance.GEN.SG=DET REL.NOM.SG mould.INF
kami
 wish.3SG.PRS
 'One [reason] is the gathering, quickly or slowly, of a substance, which will grow mouldy' (Mxit'ar Herac'i, *Ĵermanc' mxitarut'iwn* 24.70)

²⁹The deontic periphrasis behind the later MWA future (*piti* + SBJV) exists already in Middle Armenian, but there still has only a deontic meaning (Karst 1901: 306).

The early Middle Armenian period is, as has been illustrated, a period of transition in morphosyntax, with a certain degree of functional overlap at various points in time between the old synthetic present forms, which gradually lose their role of expressing the present indicative in favour of reflecting the subjunctive and the future, and the new present periphrasis in *ku*, which also can have futuric readings. The specific value of this particle *ku* is difficult to pin down; while Vaux (1995: 137) believably argues for an original progressive meaning, at the latest by the time of the standardisation of this periphrasis, it had lost the specific meaning and only signified indicative.³⁰

As will become evident from the changes ensuing in later Middle Armenian, it is after the set of developments discussed here that Eastern and Western Armenian varieties stopped developing their verbal system in parallel (or indeed together as one); Modern Western Armenian retains a verbal system closer to the one described thus far, whereas the Eastern varieties undergo further developments, which are outlined in the following section.

3.2 Later Middle Armenian

The developments in this later Middle Armenian phase are most clearly visible from their outcomes in the various varieties of Armenian described by Ačaṙean (1911) and already referenced above. In the Western varieties, a variant of the *ku*-present persists, albeit often coupled with phonetic reduction of the particle *ku* to *k'* or *ka*.

In the Eastern varieties, which Ačaṙean (1911) refers to as *-um*-varieties because of their innovation of a periphrastic present ending in *-um*, the situation is evidently different. Here, the old *ku*-present has adopted a futuric function, with all present functions being taken over by an innovative, originally locative periphrasis. Once more, owing to the present unavailability of a corpus of Middle Armenian texts, it is difficult to establish when this shift took place. Its firm establishment in Tiflis-dialect poetry by the middle of the 18th century, however, suggests this as a reasonable *terminus ante quem*, given that poetry – with the metrical constraints imposed upon it – tends to allow for archaising language. When the Georgian-Armenian *ašut* Sayat'-Nova uses innovative forms, therefore, it can be assumed that they reflect a firmly established linguistic norm (19–20).

³⁰This effectively overrides the by then inherently subjunctival function of the finite verb form. A temporal assignation of present, future, or non-past is not straightforwardly possible, since the past indicative is also formed with this particle (e.g. *ku xmēr* 'he drank'; Smbat Sparapet, *Taregirk* 650) and since both present and future events can be expressed as shown above.

- (19) *T'e xam hak'nis, t'e zar hak'nis, ku šinis*
 whether simple wear.2SG.SBJV whether gold wear.2SG.SBJV FUT make.2SG
lumaš, nazani.
 fine darling
 'Whether you wear simple [things] or gold, you make it fine, [my]
 darling' (Sayat'-Nova 26.3; Meyer 2023b)
- (20) *Łalam=ən jerin č'=ē kangnum, mat'*
 pen.NOM.SG=DET hand.LOC.SG NEG=be.3SG.PRS rest.PTCP mate
šinec'ir natašk'ari=n
 make.2SG.AOR artist.DAT.SG=DET
 'The pen does not rest in [his] hand, you have set the artist mate'
 (Sayat'-Nova 26.15)

By contrast, at roughly the same time, the poet Pałtasar Dpir, who writes in the Istanbul variety, employs the same *ku*-forms to denote the present tense (21), while at the same time retaining older, unmarked presents for similar purposes (22); it is unclear from the context whether the different forms were in free variation, represented an aspectual contrast (e.g. dynamic vs habitual), or whether this particular variety of Middle Armenian is informed by the poet's educational aspirations.³¹

- (21) *ōtari=n lezu=n aweli c'aw*
 foreign.GEN.SG=DET tongue.NOM.SG=DET too pain.NOM.SG
eler, z=is ku talē, ku talē
 become.3SG.PST OBJ=1SG.ACC IND burn.3SG.PRS PRS burn.3SG
 'The foreign tongue became too painful, it burns me, it burns' (Pałtasar Dpir, *Poems* 1.2)
- (22) *Aregakn cagē y=erknic' i y=erkir, [...] erb*
 sun.NOM.SG shine.3SG.PRS from=sky.ABL.PL on to=earth.ACC.SG when
yišem z=k'ez, linim orpēs z=moxir
 remember.1SG.PRS OBJ=2SG.ACC become.1SG.PRS like OBJ=ash.ACC.SG
 'The sun shines from the sky unto earth, [...] when I think of you, I burn up like ashes' (Pałtasar Dpir, *Poems* 2.16–18)

³¹Pałtasar sought to revitalise Old Armenian as a literary language; it cannot, therefore, be excluded that he should have been influenced by that variety's morphosyntax.

While there is a certain inherent logic in the notion that varieties which undergo further systemic changes should show greater stabilisation and differentiation of the forms that are the basis for said changes — in this instance, the Eastern *-um*-varieties — it must be kept in mind that at least more highly educated Armenian speakers were likely diglossic. Next to their local, spoken varieties existed a more or less standardised (and more or less archaic or archaising) written variety, following Old Armenian morphosyntax and retaining its orthographic conventions, which was still used in the Church and for official documents. In some cases, it therefore remains uncertain to what extent the varietal texts in existence reflect purely the spoken language, and to what extent the diglossic situation has permitted the old, written standard to influence (or be influenced by) the more modern variety.

Returning to the question of Middle Armenian developments in the Western varieties, the question of the future remains to be discussed. As already mentioned, these varieties retain the *ku*-forms in their early Middle Armenian function as presents alone and had to innovate a separate form for the future. The periphrasis that eventually prevailed is built on a historically deontic expression with *piti* (23).³²

- (23) *ayn ordin* *or* *kə calrē* *ir*
 DEM SON.NOM.SG=DET REL.NOM.SG IND mock.3SG.PRS 3SG.POSS.REFL
hayr=ə, *noyn* *vičaki=n* *piti matnui ōr mə.*
 father.ACC.SG=DET same.NOM.SG fate.DAT.SG=DET FUT face.3sg day INDF
 ‘The son who mocks his own father will one day face the same fate.’
 (Dawit‘ Xaç‘konc‘, *Dun piti əllas baroyagēt, tntesagēt, ōrēnsagēt* (1907), 79)

While this example post-dates the Middle Armenian period by some time, it illustrates most clearly the key difference between Modern Western and Eastern Armenian. While the former retains the present innovated in early Middle Armenian and innovates a periphrastic future, the latter undergoes a further development (in later Middle Armenian), retaining *ku*-forms created earlier in their secondary, future function and innovating a present form instead; at a later stage, this future periphrasis is unverbated. In both varieties, the subjunctive is the most stable form, retained in more or less the same form since early Middle Armenian (and itself based on the Old Armenian present indicative). Notably, the

³²The form MWA *piti* is derived from OArm. *pētk* ‘necessary’. In Modern Eastern Armenian both *piti* and *petk* are still used to express necessity (debitive), e.g. MEA *petk* ‘ē *štapenk* ‘we must hurry’; Dum-Tragut (2009: 263–271) notes that, in spoken Eastern Armenian, the form *piti* is more common.

subjunctive form is also the basis for some of the innovative periphrases (*ku*-presents and *ku*-futures as well as the *piti*-future). Table 2 provides a diagrammatic overview of these developments.

Table 2: Future and present from Old to Middle and Modern Armenian. Cells shaded in dark grey indicate forms no longer employed in this function after said time period; cells in light grey indicate innovations.

	PRS	SBJV	FUT
OArm.	berem	beric [‘]	bereloc [‘] em
Early MArm.	berem ku berem	berem	berem ku berem
[dialect split]			
Late MArm.	berum em	berem	ku berem
MEA	berum em	berem	kberem
MWA	kə berem	berem	piti berem

As the table suggests, while both varieties have undergone very similar processes of retention and innovation, using different form-function pairs as their basis, it seems clear that, at least as morphosyntax is concerned, there was a significant varietal split in the Middle Armenian period when Eastern and Western varieties ceased to develop along the same lines; it is, however, at present not possible to identify a probable time period for this split owing to the insufficient availability of data and research already mentioned above.

3.3 Hypoanalysis

Quite apart from the particular diachrony of the present, subjunctive and future between Old and Modern Armenian, the developments outlined above are also of interest from a typological perspective in that they illustrate the interplay of “regular” grammaticalisation processes and other diachronic mechanisms of change. The concept of hypoanalysis in particular, as defined by (Croft 2000: 126–127), is of interest in this context:

In HYPOANALYSIS, the listener reanalyzes a contextual semantic/functional property as an inherent property of the syntactic unit. In the reanalysis, the inherent property of the context [...] is then attributed to the syntactic unit, and so the syntactic unit in question gains a new meaning or function.

The keyword in this definition is the notion of contextuality. Where certain functions or semantics are originally caused by contextual clues such as adverbs, noun phrases or prepositional phrases,³³ hypoanalysed forms do not require these clues any more; in logical terms, $A \& B \rightarrow C$ is simplified to $A \rightarrow C$, where A is the form in question, B the original context clue, and C the interpretation.

Hypoanalysis cannot, logically, apply in cases of complete innovation or creation *ex nihilo*, as contextual ambiguities are not at play here. So, when early Middle Armenian creates the *ku*-present, or late Middle Armenian the *-um*-present, or Modern Western Armenian the *piti*-future, these are all clear instances of grammaticalisation: original periphrases with specific semantics (progressive, deontic) are reanalysed, semantically bleached in the process, and retain only a functional value as well as being obligatorified for the relevant category (present, future). While other common processes associated with grammaticalisation are not as evidently at play here (phonetic reduction, decategorialisation), the concerned forms fit the “grammaticalisation cline” bill rather well, since progressives are known to turn into (imperfective) presents,³⁴ and deontic expressions into futures.

By contrast, a hypoanalytical process is at play in the functional specification of the old present as a subjunctive in early Middle Armenian. There, a set of forms sets off being functionally ambiguous (present, subjunctive, future) as a result of the loss of previous, more specific formations (the Old Armenian subjunctive and future periphrasis). As mentioned in §3.1 above, this ambiguity could be contextually reduced by means of appropriate adverbs for the future, subjunctions, or periphrases for the subjunctive (e.g. *t’ol* + finite verb), whereas otherwise unmarked forms were interpreted as presents.

As early Middle Armenian and later Modern Western Armenian clearly indicate, the *ku*-periphrasis over time grammaticalised as the standard present tense. For a time, however, there must have been a meaningful difference between these periphrastic forms and the old presents; given the occasional future use of the periphrasis in early Middle Armenian and its later grammaticalisation in late Middle Armenian as a future tense, the particular semantics of *ku* may have been neither purely indicative, as Karst (1901) holds, nor originally progressive as per Vaux (1995).

³³In the present context, that is futurity, examples like *tomorrow*, *next week*, or *in the coming days* come to mind.

³⁴Cf., e.g., Deo (2015).

For a time, then, the *ku*-present and old present co-existed, both referring (in potentially aspectually different ways) to the present indicative. This contrast is eliminated with the grammaticalisation of the *ku*-present as an aspectually unmarked form. The reduction of the old present to subjunctival (and future) function is, however, not simply a side-effect, since nothing in principle would have prevented the maintenance of an aspectual differentiation. Rather, a separate but concomitant hypoanalysis process must be assumed, chiefly because the contextually unmarked reading of the old present shifted to the subjunctive, which previously required contextual cues.

The cyclicity of grammaticalisation phases comes to the fore when a very similar process, the grammaticalisation of the *-um*-present in late Middle Armenian, causes or coincides with the hypoanalysis of the *ku*-present as a future. This development suggests that the *ku*-form had retained a contextual future meaning in the first round of grammaticalisation (now decontextualised as its only function); likewise, this suggests that the old present gradually lost not only its present but also future meaning in the first round of hypoanalysis.

But why is it necessary – or at least preferable – to separate the two processes, that is the grammaticalisation of a periphrasis as a new tense and the specialisation of an older form by hypoanalysis, rather than seeing the latter as a side-effect of the former? Firstly, innovative forms can in principle lead to three fates for older homosemous forms: loss, maintenance, or specialisation. The Romance future, originally a deontic periphrasis consisting of an infinitive + *have*, led to the loss of the established Latin *b*-future; the same future forms as well as the English *will*-future have been maintained for some time despite the coexistence and near-homosemy with periphrases of the *going-to*-type; or the original forms can retain a specialised meaning, usually assuming pre-existing polysemy, as is the case in the Armenian example discussed here.

The notion of the side-effect could, of course, be maintained by adding polysemy as a conditioning factor. The data presented here are not enough to suggest that polysemy is a necessary condition for this side-effect specialisation, but taken together with the developments presented in §4 below, it becomes clear that it isn't a sufficient condition. For in Modern Eastern Armenian, the *k*-future is polysemous, but retains as its most elementary function the expression of an (aspectually unmarked) future, whereas a plethora of aspectually diverse future periphrases exist concurrently. This, then, is the second point: specialisation is not automatic, even in very similar circumstances.

The observations made here therefore complicate somewhat the explanations of the Armenian data as proposed by Haspelmath (1998) and Bybee et al. (1994),

without however invalidating them. Instead, it is simply argued that the processes described there, that is the gain of futures and subjunctives from old presents, may not be automatic developments, conditioned or not, but separate, concomitant processes complementary to the grammaticalisation of innovative forms. In the case of the Armenian data, the simplest assumption would seem to be that this process was one of hypoanalysis, given the pre-existing polysemy of the old forms in both rounds.

4 Diversification in Modern Eastern Armenian

After the slightly more theoretical consideration of the diachronic processes involved in the establishment of the Modern Armenian future and other basic tenses as described above, the situation in Modern Eastern Armenian in particular deserves closer attention.³⁵ There, next to the *k(u)*-future already laid out in some detail, a set of other, more or less grammaticalised periphrases with future reference have established themselves. This section aims to briefly describe the form and function of these periphrases and to propose some first elements of a diachrony, without however entering into the finer details of the various constructions' semantics or complementary distribution.

These future periphrases occur in the context of a number of aspectual contrasts that pervade the Armenian verbal system, for which Dum-Tragut (2009: 193–199) recognises the following pairs:

- perfect vs imperfective – e.g. perfective *grec'i* 'I wrote' vs imperfective *grum ei* 'I was writing';
- habitual-iterative vs processual – both expressed by the present or imperfect tense and differentiated adverbially;
- actional vs stative – the former being expressed by durative periphrases with the participle in *-lis*.³⁶

³⁵In most contexts, including the present one, Modern Eastern Armenian is perhaps best understood as the more or less standardised, literary variety of the Eastern or *-um*-dialects. This literary variety began to take shape in the middle of the 19th century; Xaç'atur Abovyan's *Verk' Hayastani* ('Wounds of Armenia', 1841) is traditionally considered the first piece of Modern Eastern Armenian literature.

³⁶This explanation of the Armenian TAME system is both incomplete and somewhat self-contradictory: the *-lis*-participles, for instance, are processual rather than durative; the stative-resultative participle notion, as expressed by the *-ac*-participle is not integrated; and expressions of past evidentiality are not clearly integrated (Kozintseva 2000). For the present purpose, however, this does not pose an insurmountable problem.

These aspectually marked future periphrases occur alongside the historic *k*-future, which has been retained.³⁷ This synthetic tense, it would appear, is aspectually unmarked and next to its future use can also relate other notions such as generalised statements (24) and potentiality (25).

- (24) *k'ami=n* *k-p'č'i* *anjrew=ə* *k-tela* *bayc'*
 wind.NOM=DET FUT-blow.3SG rain.NOM=DET FUT-fall.3SG CONJ
leṛ=ə *k-mna* *kangun.*
 mountain.NOM=DET FUT-stay.3SG standing
 'The wind blows, the rain falls, but the mountain remains standing.'
 (Dum-Tragut 2009: 254–255)

- (25) *ays hin patmut'yun=ə* *deṛ k-lses* *gyul-i* *tarec mardkanc'ic'*
 DEM old story.ACC=DET still FUT-hear.2SG village-GEN aged man.ABL.PL
 'This old story you may still hear from old village folk' (Dum-Tragut 2009: 255)

The periphrastic future expressions, too, can be constructed with forms of the *k*-future, marked on the auxiliary verb *linem* 'to become', and with the aid of participles (here called *k*-periphrases). Parallel forms are, however, also available formed with the present tense of the auxiliary *em* 'to be', the dynamic participle in *-lu* and other participles (here called *-lu*-periphrases). Table 3 gives an overview of these forms.³⁸

As the terminology and glosses suggest already, these periphrases refer to immediate and/or planned actions (dynamic), resulting states (stative),³⁹ and continuous actions (processual). Examples (26–30) illustrate these different uses briefly.

- (26) *Lewon, varə* ... *meladrakan t't'i=d* *mej hastat im*
 PN tomorrow accusatory letter.DAT.SG-2POSS in surely 1SG.POSS
anun=ə *grelu* *es,* *č'-ē?*
 name.ACC.SG=DET write.PTCP.DYN be.2SG.PRS, NEG-be.3SG.PRS
 'Lewon, tomorrow [...] you are surely going to mention my name in your accusatory letter, aren't you?' (*Lur č'ka*, Gurgen Xanĵyan, 2016) –
 dynamic *-lu*-periphrasis

³⁷Modern grammatical writing in Armenian refers to these forms as the *conditional mood* (Arm. *paymanakan elanak*) owing to its secondary, non-futuristic uses. For the present purpose, the term *k*-future will be used for ease of comprehension.

³⁸The terms *dynamic*, *stative*, and *processual* used here reflect the terminology in Dum-Tragut (2009).

³⁹These forms are usually reserved for position verbs.

Table 3: Overview of future expressions in Modern Eastern Armenian

	<i>-lu</i> -periphrases	<i>k</i> -periphrases
dynamic	<i>berelu em</i> 'I shall carry'	<i>berelu klinem</i> 'I shall be about to carry'
stative	<i>paṙkac em linelu</i> 'I shall be lying down'	<i>paṙkac klinem</i> 'I shall be lying down'
processual	<i>gnalis em linelu</i> 'I shall be going'	<i>gnalis klinem</i> 'I shall be going'

- (27) *erewum ē du stipvac es*
 seem.PRS.IMPV be.3SG.PRS 2SG.NOM force.PASS.PTCP.RES be.2SG.PRS
linelu aystel arazel mi k'ani žam, et'e oč' amboļj
 become.PTCP.DYN here graze.INF for-some-time if NEG all
gišer=ə
 evening=DET
 'It seems you'll be forced to graze here for some time, if not the whole evening.' (tr. of *The Headless Horseman*, Mayne Reed, 1958) – stative *-lu*-periphrasis
- (28) *Sranic' heto erb Erewanum arjanner teladrelu*
 DEM.ABL.SG after when PN.LOC.SG statue.ACC.PL erect.PTCP.DYN
k-linenk', k-haskanank', or da šat patasxanatu
 FUT-become.1PL FUT-understand.1PL COMP DEM.NOM.SG very responsible
u nurb gorc ē
 and subtle work.NOM.SG be.3SG.PRS
 'After this, when we will be set to erect statues in Erewan, we will understand that this is a very responsible and subtle task' (*Arāwōt*, 01.10.2002) – dynamic *k*-periphrasis
- (29) *Aynpes or, šat ban kaxvac k-lini ayd*
 thus COMP many think.NOM.SG depend.PTCP.RES FUT-become.3SG DEM
k'vearkut'yan ardyunk'ic'
 vote.GEN.SG outcome.ABL.SG
 'Thus a lot will depend on the outcome of that vote.' (*Arāwōt*, 01.06.2004)
 – stative *k*-periphrasis

- (30) *erb kardalis k-lines, ušadrut‘yun darju*
 when read.PTCP.PROC become.2SG attention.ACC.SG turn.2SG.IMV
Trimalk‘ioni mecark‘i vra
 PN.GEN.SG banquet.GEN.SG upon
 ‘When you will be reading, turn your attention to the *Cena Trimalchionis*
 (Henrik Senkewič‘, *Yo Ert‘as*, 1983) – processual *k*-periphrasis

This aspectual richness of expression, surprising in view of the paucity and simplicity of future formations in later Middle Armenian, deserves to be studied in its own right as regards its use and distribution. For the present purpose, that is an overview of the diachrony of Armenian futures, a glimpse at the development of these periphrases over time will have to suffice. With the help of the *Eastern Armenian National Corpus* (EANC),⁴⁰ the figures of absolute occurrences in the whole corpus of these periphrases and their first attestations have been gathered in Table 4.

Table 4: Occurrences and first attestations of future periphrases in
 Modern Eastern Armenian
 The figures in parentheses indicate the number of texts in which the
 periphrasis occurs; if two years are cited, there is only one occurrence
 attested for the first year.

Type	- <i>lu</i> -periphrases	Year	<i>k</i> -periphrases	Year
dynamic (- <i>lu</i>)	71,300 (7,400)	1860	40 (39)	1870/1955
stative (- <i>ac</i>)	1,289 (1,033)	1958	8,662 (3,262)	1841
processual (- <i>lis</i>)	2 (2)	2005	103 (66)	1870/1950

On a purely numerical basis, the data indicate that three of the periphrases (dynamic *k*-, processual -*lu*-, and processual *k*-) are marginal occurrences and comparatively recent creations. The reason for the low occurrence of processual futures is, perhaps, self-explanatory: they are aspectually very precise but describe an action that has not yet happened and, therefore, cannot be predicted – or need not be specified to such a degree of precision. The same is true, *mutatis mutandis*, for the dynamic *k*-periphrasis which describes a planned or imminent action in the future and is similarly aspectually over-specific for common use.

⁴⁰The corpus, accessible at <http://eanc.net>, contains more than 110,000,000 tokens across written discourse (prose, poetry; fictional, non-fictional), oral discourse and press archives and includes content from the middle of the 19th century onwards.

That leaves three more frequent periphrases. The most common one – the dynamic *-lu*-periphrasis (*berelu em*) – is also the morphologically simplest and in all likelihood the best analogy to the pairing of synthetic futures and *going-to*-periphrases found in Romance and Germanic. Alongside it, the stative *k*-periphrasis is attested similarly early, even though only about 11% as often; this may be related to its aspectually more specific function and the smaller number of verbs with which it is appropriate. Finally, the stative *-lu*-periphrasis is a more recent innovation and, perhaps for that reason, again not attested as widely as the corresponding *k*-periphrasis. All of these figures contrast with the quasi-ubiquitous simple *k*-future (*kberem*), which is attested 630'680 times over 9'290 texts.⁴¹

In diachronic terms, then, and taking into account the data above, it seems most plausible to assume that next to the simple *k*-future, the stative *k*-periphrasis developed early to allow for the expression of the aspectual contrast of action vs state/result; at a similar time, the dynamic *-lu*-periphrasis arose, perhaps to distinguish a different kinds of futurity (planned vs definite; immediate vs remote; etc.). The other, less frequent and later developed periphrases, are analogical extensions: the stative *-lu*-periphrasis creates a parallel form to the stative *k*-periphrases; the processual periphrases extend present-tense formations (type *gnalis em*) to the future context. The fact that the processual *-lu*-periphrasis in particular occurs exclusively in grammar books raises the question to what extent all three of these less-common periphrases are the result of paradigmatisation without grammaticalisation.

It goes without saying that the present sketch of the future expressions in Modern Eastern Armenian does not do justice to its complexity, both in synchronic and in diachronic terms. It does serve, however, to give a brief overview of these formations and to raise questions for the future, in particular the following: Is the development of the aspectual specificity – hinted at in the future periphrases and more developed in other tenses – a purely internal development or, at least in part, informed by or modelled on similar distinctions in contact languages?

5 Conclusions and perspectives

The aim of this paper was to describe the diachrony of future expressions in Armenian, from its oldest attestations to the present varieties of the language.

⁴¹These figures are, to a certain extent, a simplification, since the *k*-periphrases can (but do not always) have additional modal connotations which have not been teased out here; cf. Kozintseva (2005). This distinction is, however, likely not relevant for the diachrony discussed here.

Beginning with a polysemous subjunctive, the Armenian language innovated a separate future category only in its Medieval stages, after the broader dialect split into an Eastern and Western variety had taken place. The Western varieties separately grammaticalised an originally deontic periphrasis as their future tense, whereas the Eastern varieties repurposed a present periphrasis. In Modern Eastern Armenian, this same future tense is maintained; at the same time, more aspectually specific periphrases have been created – some more grammaticalised than others – to analogously extend to the future the aspectual specificity Armenian exhibits in other tenses.

Over the course of its development, future expressions have had regular and close interactions both with the present tense and the subjunctive. In particular, it has been argued that on two occasions, the Armenian tense/mood categories have undergone a pair of related but separate grammaticalisation and hypoanalysis processes – first for the Old Armenian present turning into a subjunctive; then the late Middle Armenian present turning into a future. Different processes, namely analogical extension and paradigmatisation, are at play in the development of the future periphrases in Modern Eastern Armenian.

Apart from illustrating the development of the future in Armenian and thereby providing a detailed example of a fairly complex set of innovations, this paper has sought to complicate to a certain extent the well-known accounts of developments in Armenian cited in the linguistic literature as well as the relationship between hypoanalysis and grammaticalisation. Yet, as has been pointed out, much work remains to be done, in particular as regards the Middle Armenian data, to gain a complete understanding of the changes happening there and their potential interactions. Likewise, the system of future expressions in Modern Armenian deserves more attention and should prove a fruitful ground for research into grammaticalisation and related processes as well as into the diachrony of the Armenian aspect system.

Abbreviations

1	first person	M	masculine
2	second person	MEA	Modern Eastern Armenian
3	third person	MWA	Modern Western Armenian
ABL	ablative	NOM	nominative
ACC	accusative	OBJ	object(-marker)
AOR	aorist	PERL	perlative
COMP	complementiser	PL	plural
CONJ	conjunction	PN	proper noun
DAT	dative	POSS	possessive
DEM	demonstrative pronoun	PROC	processual
DET	determiner	PROG	progressive
DYN	dynamic	PRS	present
EMPH	emphatic	PST	past
F	feminine	PTC	particle
FUT	future	PTCP	participle
GEN	genitive	REFL	reflexive
IMV	imperative	REL	relative pronoun
INDF	indefinite	RES	resultative
INF	infinitive	SBJV	subjunctive
INS	instrumental	SG	singular
LOC	locative	VBADJ	verbal adjective

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Chapter 10

Grammaticalization of future markers in Korean

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Contemporary Korean has a number of forms that make the future time reference. Their grammatical statuses are variable, e.g., an adnominalizing suffix, modal auxiliary, sentence-ender, etc. Unlike the cross-linguistically attested patterns, Korean futures involve two grammatical formants that inherently involve futurity, i.e., suffixes indicating ‘(at) a later time’ and ‘mode, manner, purpose’, which are primarily responsible for the emergence of future markers. In structural terms, the omissibility of sentential arguments and agglutinating morphology play an important role, because they make the grammatical status of a construction undergoing grammaticalization ambiguous and consequently promotes functional reinterpretation and structural reanalysis. A review of grammaticalization scenarios with respect to the grammaticalization parameters shows that grammaticalized future markers exhibit variable degrees of conformity to them.

1 Introduction

Whether Korean has future tense markers or not remains controversial. Early studies propose a tripartite system of distinguishing the past, the present, and the future (Joo 1910; Park 1972[1935]; Choe 1937; Hong 1947; Yoo 1956; among others), but later studies, notably from Ko (1965) and onward (see also Nam 1972; Sohn 1975; Sung 1976; Suh 1976, 1977; Mun 2005, 2009; among others), tend to subscribe to the binary tense distinction (the past and the non-past), placing the future in the category of modality, based on the fact that the alleged future markers signal such modal notions as conjecture, inference, intention, etc. However, there is still



a body of literature that subscribes to the tripartite tense distinction, e.g., Choi (1995, 1996), Hong (1987, 2008), Min (1991, 1998), among others.

The categorial nomenclature notwithstanding, Korean does have a number of markers that signal the ‘futurity’ of an event or state of affairs, hence ‘future markers’ from a broad perspective. These markers exhibit different historical depths, degrees of grammaticalization, levels of currency, and areas of specialization. This chapter addresses the development of those markers from a grammaticalization perspective.

Korean is spoken by 81.7 million people (Eberhard et al. 2023) mostly in and around the Korean peninsula. It was formerly regarded as an Altaic language as the eminent Finnish Altaic linguist Gustaf Ramstedt proposed in 1939 (Ramstedt 1997[1939]), but in the modern linguistic research tradition, the Altaic hypothesis is contested (see Kim 1992; Sohn 1999; Song 2005, among others, for discussion of the issue). *Ethnologue* (26th ed., Eberhard et al. 2023) lists Korean as a Koreanic language along with Jejueo, a language variety spoken in Jeju Island. Recently, it is frequently discussed under the geographically motivated classificatory term ‘Transeurasian languages’ (Johanson & Robbeets 2010), often along with Japanese, since the two languages share many structural characteristics (Narrog & Rhee 2013; Narrog et al. 2018; Robbeets et al. 2021; see Tian et al. 2022 and Robbeets et al. 2022 for controversy).

The historical depth of texts written in *Hangeul* (or *Hankul*), the Korean alphabet, goes back to the 15th century CE (1443). The invention of this alphabetic-featural script is a landmark event that divides Early Middle Korean (EMiK; the 10th c. through the mid-15th c.) and Late Middle Korean (LMiK; the mid-15th c. through the 16th c.), and a large body of texts has been electronically archived through government-led projects. Prior to the invention of *Hangeul*, several different writing systems were invented using Chinese characters for their meaning (semantogram) or sound (phonogram) to represent Korean words or grammatical morphemes.

Typologically, Korean is an agglutinating, head-final, SOV language. In line with the SOV word order characteristics in morphology, it is suffixal and postpositional. Multiple suffixes and postpositions may occupy slots in a relatively fixed order. Argument NPs are often omitted and NPs may occur without their postpositional particles, especially in speech, and interpretive flexibility thus created often affects or effects grammaticalization. In terms of verbal morphology, finite verbs are marked for tense, aspect, mood, and modality, whereas non-finite verbs are often marked with ‘converb’-markers (*converbialia*, Ramstedt 1903), the suffixes whose main function is adverbializing a non-finite verb (Haspelmath 1995). Finite verbs occurring sentence-finally are further marked by a sentence-

type marker, such as declarative, imperative, interrogative, hortative, exclamative, etc., all of which forming a complex paradigm further modulated by the degree of honorification and politeness, commonly known as *hwakyey* ‘speech level’ (Rhee 2020).

This chapter is organized as follows: Section 2 addresses the development in form and function of the future markers; Section 3 discusses morphosyntactic and conceptual motivations and reviews the grammaticalization scenarios in view of grammaticalization parameters; and Section 4 summarizes the findings and concludes the chapter.

2 Future time reference in Korean

In LMiK, the notion of future was carried by several different morphosyntactic devices, such as monosyllabic *-li-* and the periphrastic forms involving the mode converb-marker *-key* (e.g., *-key taoy-*, *-key ha-*, and *-kuys-*). However, these markers did not develop their tense-aspectual scopes, and they either survived in limited usages or completely disappeared, in Present-day Korean (PDK; the 21st c.).

PDK has several future (and future-related) markers. Historically, they belong to three developmental trajectories, which form layers of different historical depths, i.e., *-li-*, *-l.ke(s).i-*, and *-keyss-*. Among these, the oldest form is *-li-*, attested in Old Korean (OK; 1st c. through early-10th c.). On the other hand, the other two are more recent innovations; *-l.ke(s).i-* developed from the 16th century (Lee 1997: 64) and *-keyss-* developed in the early 19th century (Huh 1987; Rhee 1996). We discuss those three classes of future markers focusing on their historical development in form and function, in the following.

2.1 The *-li*-Class futures

2.1.1 The future adnominalizer *-l*

The first future marker is *-l*, a future adnominalizer, known as the ‘prospective adnominalizer’ (PROS.ADN) (Rhee 2012b), of which the major function is to transform a verbal or clausal constituent into a modifier of a noun phrase, hence the label ‘adnominalizer’. This function is illustrated in the following:

- (1) a. 올 사람
 o-l *salam*
 come-FUT person
 ‘a person who will come’

- b. 살 옷
sa-l *os*
 buy-FUT clothes
 ‘clothes that (I) will buy’

Examples in (1) shows that *-l* heads a constituent, which modifies the following noun (note that Korean is head-final), e.g. *salam* ‘person’ in (1a) and *os* ‘clothes’ in (1b). Sometimes, structurally complex constituents involving the future *-l* becomes fully unverbated, e.g., *mek-ul kes* [eat-FUT thing] ‘food’ (< ‘thing to eat’); *ip-ul kes* [wear-FUT thing] ‘clothing’ (note that *-l* has a phonologically controlled allomorph *-ul*, which occurs after a coda). As is obvious from the examples above, *-l* is not a finite-verb morphology; it is used regardless of the speech time, as shown in (2), in which *-l* occurs in the sentences marked by the past tense.

- (2) a. 올 사람은 나 왔다.
o-l salam-un ta w-ass-ta
 come-FUT person-TOP all come-PST-DEC
 ‘Everyone who would come came.’
 (PDK, constructed)
- b. 그것은 내가 살 옷이었다.
kukes-un nay-ka sa-l os-i-ess-ta
 it-TOP I-NOM buy-FUT clothes-BE-PST-DEC
 ‘It was the clothes that I would buy.’
 (PDK, constructed)

The examples in (2), in which *-l* occurs in the past-marked sentences, clearly shows that the future *-l* is not a regular future-tense marker, which, by definition, is based on the speech time.¹ Thus, *-l* is a prospective marker (as ‘prospective adnominalizer’, Rhee 2008), one belonging to the ‘relative tenses’ (McCawley 1971: 111; Partee 1973: 609; Comrie 1985; Dahl 1985; Michaelis 2006: 220; Hengeveld 2011), ‘secondary tenses’ (Diewald 2011; Lyons 1977: 689), ‘relational tenses’ (Hopper & Traugott 2003: 152), and ‘anaphoric tenses’ (Lo Cascio 1986: 198), in contrast with the absolute tenses anchored in the moment of speaking. Relative tenses are also called ‘compound tenses’ because these markers are, in general, morphosyntactically and/or semantically complex, by referring back to times denoted by previous temporal expressions (e.g., English *will have done*, *had been doing*, etc.).

¹For this reason, Rhee (2008) describes *-l* 'futuristic' and exclusively uses the label 'prospective adnominalizer', avoiding 'future'.

In this regard, the Korean future *-l* is qualitatively different in that it does not accompany a reference time expression. The future *-l*, by virtue of marking a later point in time, may also be conceived as a ‘posterior’ tense (Jespersen 1924: 257; Bull 1963: 14; Kamp & Rohrer 1983: 254–257), as in the English posterior future ‘will be about to sing’ (Binnick 2001: 559).²

Historically, the future-marking *-l* had functioned as a nominalizer (e.g., gerundivizer, historically *-lh*, with a glottal stop) with futuristic meaning (‘prospective nominalizer’), before it acquired the adnominalizer function (Kim 1957; Lee 1972; Lee 1997; among others). This function survived as late as 1577 (Rhee 2008: 240–242) but has since become completely defunct; its sole function is a prospective adnominalizer (modifier) in Modern Korean. Its formal vestiges are found in certain sentence-enders, e.g., *-lkka* (futuristic question), *-lla* (apprehensional), *-lseyla* (apprehensional), in which the futuristic morpheme *-l* is completely fused (cf. for discussion of interrogative markers involving *-l* in OK and EMiK, see Ko 2011: 88–90, 101–127; for discussion on *-l* in apprehensionals, see Yae et al. 2023; Rhee & Kuteva, in press). It has been noted that, crosslinguistically, the secondary grammaticalization from a nominalizer to a modifier (such as relativizer, complementizer, etc.) is fairly common (Yap et al. 2010: 77; Yap & Wang 2011: 66).

2.1.2 The future *-li-*

The source of the future marker *-li-* involves *-l* (prospective nominalizer/ adnominalizer) addressed above. When *-l* was combined with the defective noun *i*, and the copula *i-*, a new portmanteau form *-li-* emerged (Lee 1997: 56–59; Park 1998: 163). As *-l* had the prospective function, *-li-* naturally inherited this function of marking futurity. Its use is exemplified in the following excerpts:³

- (3) a. 天心을 이루오리라.
CHENSIM-ul ilwu-o-li-la
 providence-ACC accomplish-HON-FUT-DEC
 ‘I will fulfill the Providence (Heavens’ will).’
 (LMiK, 1445 *Yongpiechenka* 108)

²The morpheme *-l* is multifunctional, and an anonymous reviewer directs the author’s attention to Lim (2009), who lists the irrealis-marking usage among its diverse functions.

³Some historical data in Korean has mixed script of Chinese characters and *hangeul*, and a general practice in romanization of such texts is to use upper-case for Chinese characters following the modern Korean pronunciation. Since old Korean alphabets are typographically incompatible, Korean scripts are not presented if the excerpts contain such characters.

- b. 내일이면 고향에 다다르리라.
nayil-i-myen kohyang-ey tatalu-li-la
 tomorrow-BE-COND hometown-AT reach-FUT-DEC
 ‘I will arrive at (my) hometown by tomorrow.’
 (MoK, *Wulimalsaym* ‘li’)
- c. 나는 가리라, 곧 가리라 이니스프리섬으로.
na-nun ka-li-la, kot ka-li-la inisuphuli.sem-ulo
 I-TOP go-FUT-DEC soon go-FUT-DEC Innisfree.island-DIR
 ‘lit. I will go, To Innisfree Isle (I) will soon go.’
 ‘org. I will arise and go now, and go to Innisfree.’
 (MoK translation, William B. Yeats, ‘The Lake Isle of Innisfree’)

Examples in (3) show that *-li-* is functioning as a future marker. In each example *-li-* is followed by other verbal morphologies, e.g., a sentence ender, and, thus, the future marker *-li-* is one that occupies a slot known as the penult ender position. Note that the clause containing the second occurrence of *-li-* in (3c) is in the inverted word order for poetic effect. Even though *-li-* was the primary future marker prior to MoK, it is now archaic or poetic.

In terms of functions, the future *-l-* signals intention or determination, as in (3a) and (3c) or conjecture as in (3b). However, as functional classification largely depends on the variable granularity, other functions related to these overarching notions such as prediction, possibility, ability, obligation, etc. have been also marked by *-li-*.⁴

2.1.3 The future sentence-ender *-li*

The next future marker is the sentence-ender *-li*, nearly identical in form and function with the previously addressed *-li-* above. Its use is exemplified in the following:

- (4) a. Declarative
YENG CWU-z alph-Ay naynay pwuskuli-li
 wise.king-GEN front-AT forever be.ashamed-FUT.DEC
 ‘(The unwise king) will be ashamed forever in the face of a wise king.’
 (LMiK, 1447 *Yongpiechenka* 16)

⁴Lee (1997: 32–35) also reports that in LMiK, the future *-li-*, in combination with a negative adverb *mot* ‘not’ or a negative verb *mal-* ‘stop; do not’, marked prohibition, premonition, etc. (see Lee 1997: 35).

b. Interrogative

CENGSAISEONG-skuy nw-i ani o-ZAVA-li

national.hero-DAT who-NOM not come-HON-FUT.Q

‘Who would not follow this hero who defended the nation?’

(LMiK, 1447 *Yongpiechenka* 99)

This future marker *-li* is different from the previously addressed *-li-* in that it is the sentence-ender (i.e., ultimate verbal morphology), whereas the latter is the one occurring in the penult verbal morphology. This is a unique situation because tense markers occur in the penult position and the ultimate position is reserved for sentence-type markers. The sentence-ending future markers occur only in declarative and rhetorical interrogative sentence types, as exemplified in (4) above. Since it is directly derived from the previous *-li-*, the future sentence-ender *-li* has the same functions, i.e., notably conjecture and intention, as well as other future-related functions.

This marker occurs very infrequently in the historical data, i.e., only in three sources that are written in the poetic style, i.e., *Yongpiechenka* (1447), *Welinchenkangcikok* (1447), and *Welinsekpo* (1447) (Yeom 1998: 20). In MoK, the use of this future sentence-ender *-li* is still restricted to poetic genres. For this reason, this future-marking sentence ender has not been discussed in earnest (Ko 1990 and Yeom 1998 are notable exceptions). The development of this unique marker results from a rather simple process. Sentence-final particles formerly attached to the tense-marking *-li-* were omitted, presumably for poetic effects (e.g., to match the number of syllables with the counterpart stanza, cf. Yeom 1998: 20–21), and, consequently, the suffix was formally reanalyzed and functionally reinterpreted as a future-marking sentence-ender. For the lack of verbal morphologies that normally follow the tense marker, Ko (1990: 265) and Ryu (1992: 422) regard this ender as an instance of the *panmal* ‘half-talk’ style, i.e., one without honorification or politeness markers.

2.2 The future *-l.ke(s).i-*

The second group of future markers consists of two historically related forms, *-l.kes.i-* and *-l.key*; the latter has diverged from the former. The future *-l.kes.i-* developed from around the 16th century (Lee 1997: 64) and is productively used in MoK.

2.2.1 The future *-l.kes.i-*

The future tense marker *-l.kes.i-* is a periphrasis involving the prospective (i.e., future-marking) adnominalizer *-l* (see Section 2.1.1 above), the formal noun (i.e., semantically vacuous) *kes* ‘thing’, and the copula *i-* ‘be’.⁵ This complex construction has become fully univerbated through structural reanalysis and functional reinterpretation (see Section 3.1 below for more discussion). The future *-l.kes.i-* is exemplified in the following:

- (5) a. 버스가 곧 올 것이다.
pesu-ka kot o-l.kes.i-ta
 bus-NOM soon come-FUT-DEC
 ‘The bus will come soon.’

(Lee & Lee 2010: 500)

A brief note on the formal noun *kes* is in order. *Kes* is among the most frequently used nominalizers in MoK. It started its life as a full-fledged noun denoting ‘thing, skin, surface’ (Hong 1983; Rhee 2011) but became a defective noun from LMiK (Huh 1983, Jeong 1989; Lee 1988), labeled as such because it is so semantically weak that it cannot be used independently (also called ‘formal noun’). Even though it is semantically bleached, syntactically it still functions as the head noun of a nominalized string (Rhee 2011: 395), and thus requires that the modifying constituent be headed by an adnominalizer (such as the prospective adnominalizer *-l* here).

As the complex periphrasis *-l.kes.i-* becomes univerbated into a single grammatical unit of signaling future, its function also develops into the conceptual domain marked by its earlier, and more strongly entrenched, competitor *-li-*. Thus, *-l.kes.i-* marks conjecture, intention, determination, possibility, prediction, obligation, etc. (see Section 2.1.2 above), largely depending on the context (in particular, the sentential subject) and the speech situation, as exemplified below (note that *-ul* is an allomorph of *-l*):

⁵The lexical origin of the noun *kes* is the noun *kach/kas/kes* ‘skin, fur, surface, appearance, thing’ (Hong 1983; Rhee 2008: 248), which have been completely bleached. As an anonymous reviewer points out, the noun *kes* in *-l.kes.i-* may be replaced by other nouns, e.g., *seym* ‘plan’, *sayngkak* ‘thought’, etc., which form predicative constructions known as ‘mermaid’ constructions (Tsunoda 2013; Kim 2013; Rhee 2020: 598–599). Since these nouns still retain their lexical meaning, the mermaid constructions are regarded as syntactic constructions, unlike *-l.kes.i-*, which is a relatively ‘pure’ future marker. The future meaning in the mermaid constructions is derived from the future adnominalizer *-l*, discussed in Section 2.1.1.

- (6) a. prediction, conjecture, conviction

내일도 추을 것이다.

nayil-to chwuw-ul.kes.i-ta

tomorrow-ALSO be.cold-FUT-DEC

‘Tomorrow will be cold, too.’

(Lee & Lee 2010: 500)

- b. intention, determination

나는 꼭 성공할 것이다.

na-nun kkok sengkongha-l.kes.i-ta

I-TOP by.all.means succeed-FUT-DEC

‘I will succeed by all means.’

(Lee & Lee 2010: 501)

In MoK, *kes* [kʌs] itself has undergone phonological erosion and is often used in the form of *ke* [kʌ] (note that *-l.ke(s).i-* is only one of many other lexical and grammatical forms *kes* participates in). Thus, the future marker *-l.kes.i-* has its phonologically reduced variant *-l.ke.i-*, from which the copula *i-* may also be eroded due to the V-V sequence, which triggers deletion of one vowel or modification of the vowel quality. This process results in *-l.ke-*, and thus the verb forms in (6) may also be *chwuw-ul.ke-ta* ‘will be cold’ and *sengkongha-l.ke-ta* ‘will succeed’, respectively. There are no functional differences between these fuller and eroded variants, other than that the reduced variant is more closely associated with the colloquial styles.

2.2.2 The future *-l.key*

The future *-l.key* originates from the reduced form of the future *-l.kes.i-*, described above. Its usage is exemplified below:

- (7) a. promissive

내가 맛있는 것 사 줄게.

nay-ka masissnun kes sa-cwu-l.key

I-NOM delicious thing buy-BEN-PROM

‘I will buy something delicious for you.’

(Lee & Lee 2010: 501)

- b. polite command/request

여기 침대에 누우실게요.

yeki chimtay-ey nwuw-si-l.key-yo

here bed-AT lie.down-HON-IMP-POL

‘Please lie on your back on the bed over here.’

(Koo & Rhee 2013: 489)

As shown in (7), *-l.key* is the sentence ender, i.e., occurring at the ultimate position (note that politeness marker *-yo* is not considered in slot assignment), thus different from *-l.kes.i-* occurring at the penult position, described in the previous section.

When the reduced form is further marked by the ‘informal’ speech level, ‘universal’ speech-act sentence-ender *-e/a*, the form contains a series of vowels as in *-l.ke.i-a* /l.kʌ.i-a/, a situation highly susceptible to vowel change in Korean, thus resulting in *-l.key* /l.ke/. This shows that the fusional process obliterates the word boundary as well as morpheme boundaries. The earliest attestations of *-lkey* (e.g., 17th c. *Pakssipwuincen* movable type ed.) carried the same function as *-l.kes.i-*, i.e., conjecture, prediction, intention, etc. (cf. (Koo & Rhee 2013: 490)), but the reduced form gradually diverged, and from around the turn of the 20th century, it became increasingly specializing as a promissive marker (‘I promise that...’) (e.g., 1911 *Moktanhwa*; cf. Koo & Rhee 2013: 490).

In the case of promissive, as in (7a), since the speaker is also the agent of the proposed act, the promissive sentence is compatible with the first person subject only (either singular or plural). A Korean idiosyncrasy is that the understood subject may freely be omitted as is the case with other sentential constituents. Naturally, in promissive sentences, the implicit subject is the speaker, and the verb cannot be marked with honorification, because self-honorification is not acceptable in any situation. Therefore, when the predicate is marked with honorification, the understood sentential subject cannot be the speaker. When such forms are encountered in a sentence in which a putative promissive marker *-l.key* is used, as in (7b), in a situation where a nurse is giving an instruction to a patient, the addressee will make a pragmatic inference that since it cannot be the promissive sentence, it should be some other speech act. In other words, due to the omissibility of the sentential subject, the only signal that forces the addressee to interpret the sentence as an imperative is the presence of the honorification marker. Thus, the functional reinterpretation of the promissive as the imperative was actualized through pragmatic inference. Through this pragmatic inference, new functions of request (a milder form of imperative) or a command (a stronger form of imperative) emerged in PDK.

2.3 The future *-keyss-*

In PDK there are two primary futurity markers, *-l.kes.i-* and *-keyss-*. As shown in Section 2.2, *-l.kes.i-* etymologically involves the noun *kes* ‘thing’. The latter, *-keyss-*, which developed in the early 19th century (Huh 1987; Rhee 1996), historically involves a number of morphemes (see below), especially, the mode converb-

marker *-key* ‘in such a manner that’, largely denoting mode, manner, purpose, etc. Its uses are illustrated in the following examples taken from Lee & Lee (2010: 67) with modification:

- (8) a. immediate future
 잠시 후에 세 시가 되겠습니다.
camsi hwu-ey sey si-ka toy-keyss-supnita
 moment after-AT 3 hour-NOM become-FUT-DFR.DEC
 ‘It will be 3 o’clock momentarily.’
- b. conjecture
 어서 가자, 학교에 늦겠다.
ese ka-ca, hakkyo-ey nuc-keyss-ta
 fast go-HORT school-TO be.late-FUT-DEC
 ‘Let’s hurry up; we will/might be late for school.’
- c. possibility
 나도 그만큼은 먹겠다.
na-to ku.mankhum-un mek-keyss-ta
 I-ALSO that.much-TOP eat-FUT-DEC
 ‘(I think) I, too, can eat that much. (It is possible that I...)’
- d. intention
 내일은 미국으로 떠나겠어.
nayil-un mikwuk-ulo ttena-keyss-e
 tomorrow-TOP America-DIR leave-FUT-END
 ‘I will leave for America tomorrow.’

Example (8a) marks the immediate future, thus a type of the ‘metrical tense’ (Chung & Timberlake 1985: 207ff; Comrie 1985: 83; Dahl 1985: 120ff; Binnick 2001: 560, among others). Even though the functions of *-l.kes.i-* and *-keyss-* overlap to a great extent, *-keyss-* tends to specialize in immediate future (see Section 3.2 for discussion on its conceptual motivation). Example (8b) makes reference to conjecture as to the future event. Since conjecture can also make reference to the past event, *-keyss-* is not restricted to future events (e.g., ‘They must have been late for school yesterday.’), in which case the future marker *-keyss-* follows the past tense marker *-ess-*. Example (8c) signals possibility but it may also mark its closely related concept ‘ability’, depending on the locus of the enabling force, i.e., within a sentient being (ability) or in the situation (possibility). Example (8d) marks the intention, typically the speaker’s in declarative and the addressee’s in interrogative.

Even though the examples in (8) show the common functions of the future tense *-keyss-*, there are a number of other functions that may be subsumed in those categories at varying degrees, e.g., ability, hypothetical willingness, current state as futuristic mode, evaluative, predestination, among others (Rhee 1996: 109–130).

The development of *-keyss-* (and its dialectal form *-kes(s)-*) has been studied by a number of researchers, from as early as Underwood (1890), Choe (1989 [1929]), Park (1972[1935]), and Ramstedt (1997[1939]) (see also Lee 1947; Na 1953; Shin 1972; Suh 1977; Gim 1981; Rhee 1996; Lee 1997, 2006). The seemingly simple form *-keyss-* originates from a construction involving multiple morphemes, i.e., the mode converb-marker *-key* ‘in such a manner that’, the light verb *ha-* ‘be/do/say/make’, the sequential converb-marker *-e* ‘and’, the existence verb *iss-* ‘exist’, even though many of them are not visible in the current form *-keyss-*. The long process can be briefly described below.

The first is the emergence of the resultative/perfect/past marker *-ess-*, developed from the converb-marker, also known as the ‘consolidating connective’ *-e/a* (Koo 1987), and the verb of existence *-is(s)*, through a long process from the 15th c. The consolidating connective, largely translatable as ‘and’, connects two sequentially related events and makes one holistic event (e.g., *pel-e-mek* ‘earn’ and ‘eat’ > ‘make a living’; *kuw-e-mek* ‘roast’ and ‘eat’ > ‘eat roasted meat’; etc.). By virtue of the presence of the existence verb in its historical origin, *-ess-* tends to imply a strong relevance of a completed event to the present time (Note that it was a resultative marker before it developed into the perfective/past).

The second is the emergence of the periphrastic causative *-key.ha-*, developed from the mode converb-marker *-key* and the light verb *ha-* ‘do/be/say/make’. The converb-marker *-key* is inherently futuristic because it refers to a resultant state of an action or event, e.g., *aphu-key* [ache-MOD] ‘in such a manner that one would feel pain; painfully’, *cwuk-key* [die-MOD] ‘in such a manner that one would die; leading to death’, ‘fatally’ etc. The verb *ha-* is a semantically bleached verb (thus the ‘light’ verb) with various general meanings. The combination of the ‘mode’ and ‘make’, thus ‘make x in such a manner that y will’, led to the development of the causative.

The finally derived sense from prospective causality, definiteness, and current relevance is the standard notion of futurity, in that the speaker is making a judgment with respect to the future, based on the current definite situation which is a result of a completed event. Therefore, the notion ‘conjecture’ is the most basic one in this marker.

3 Discussion

In the preceding section, we have seen the development of various future markers, which can be categorized in three groups, one related to the prospective adnominalizer *-l* (i.e., *-l*, *-li-*, and *-li*), one involving the defective nominal *kes* ‘thing’ (i.e., *-l.kes.i-* and *-l.key*), and one involving the mode converb-marker *-key* (i.e., *-keyss*). Now we turn to a discussion of morphosyntactic motivation (Section 3.1), conceptual motivation (Section 3.2), and grammaticalization parameters (Section 3.3).

3.1 Morphosyntactic motivation

There are a number of prominent factors of the development of the future in terms of morphosyntactic motivation. Among them are having a verb as the last component of the construction, the omissibility of arguments, and agglutinating morphology.

We can see that the copula is involved in the development of most future markers, e.g., *-li-*, *-li*, *-l.kes.i-*, *-l.ke.i-*, and *-l.key*. In these markers, the last element *i-* originates from the copula *i-*, even though it is not directly visible in *-l.key* (see Section 2.2.2). The copula in the source constructions of these markers enables the preceding nominals/nominalizers (i.e., *l* and *kes*) to function as equative or identificational predicates by being a part thereof. For instance, we noted that the future marker *-l.kes.i-* developed from a complex source construction involving the prospective adnominalizer *-l*, the formal noun *kes* ‘thing’, and the copula *i-* ‘be’, which became fully unverbated through structural reanalysis and functional reinterpretation. This is exemplified in the following:

(9) 그가 올 것이다.

a. source structure

ku-ka o-l kes-i-ta
 he-NOM come-PROS.ADN thing-BE-DEC
 ‘lit. It is (a thing) that he comes at a later time.’

b. grammaticalized structure

ku-ka o-l.kes.i-ta
 he-NOM come-FUT-DEC
 ‘He will come.’

The structural reanalysis of (9a) into (9b) involves a number of noteworthy aspects. As is obvious from the translation of (9a), the entity being equated with

‘thing’ in the predicate is not the sentential subject ‘he’ (note that *kes* ‘thing’ is not used to refer to a person in Korean). The sentential subject is not explicit, i.e., the source structure of (9a) is a subject-less sentence, an idiosyncrasy of Korean syntax. The implicit subject is ‘the current situation’, i.e., the current situation is such that ‘he comes at a later time’. What aspects of the current situation warrants the conjecture are not explicit and thus unclear – they are only vaguely referred to as a ‘thing’, a perfect reference as its semantics is nearly vacuous (thus called a formal/defective noun, functioning as a nominalizer). Thus, the absence of concrete semantic content in *kes* ‘thing’, the propinquity between the prospective adnominalizer *-l* and *kes*, the copula following *kes*, and the presence of a contentful verb preceding all these components, among others, created a perfect syntagmatic condition for the reanalysis of the complex construction *-l.kes.i* as an auxiliary (note that the syntactically designated position of an auxiliary is between the main verb and the sentence-type marker like the declarative *-ta* in 9).

It is also noteworthy that the reanalysis illustrated in (9) involves ‘syntactic upgrading’ (Heine et al. 1991: 169–170; Ramm 2008; Rhee 2012a: 297–298), i.e., the formerly subordinating clause (‘he comes at a later time’) becomes the main clause with the future-tense interpretation (‘he will come’). Reanalysis of a former subordinate clause into a main clause typically occurs with ‘insubordination’ (Evans 2007), i.e., main clause ellipsis, which is responsible for many instances of grammaticalization in Korean, but the type of reanalysis with the future marker described here is actualized when the subject is implicit and its predicate is fused with the predicate of the subordinating clause.

Another type of morphosyntactic motivation is with reference to the future *-keyss-*. As alluded to in Section 2.3, it developed from the string, *-key-ha-e-iss-* [mode-make-and-exist], which developed into [CAUS-PERF]. This is illustrated below:

(10) 곧 비가 오게 하여 있어요.

- a. *kot pi-ka o-key-ha-e-iss-e-yo*
soon rain-NOM come-IN.SUCH.A.MANNER-MAKE-AND-EXIST-END-POL
‘It (the situation) is such a manner as to make the rain come soon and exists.’
- b. *kot pi-ka o-key.ha-ess-e-yo*
soon rain-NOM come-CAUS-PERF-END-POL
‘It has made the rain come soon.’

- c. *kot pi-ka o-keyss-e-yo*
soon rain-NOM come-FUT-END-POL
‘Rain will come soon.’

One caveat is that the morphemic breakdown along with glosses and interpretations in the examples is grossly oversimplified, glossing over the complexity of the development; it is presented as such for the sake of convenience, hence the awkwardness of the interpretation. For instance, there is no instance in historical data that is comparable to (10a), simply because the construction did not string together these exponents all at once at one point of time. In other words, there were always differential degrees of semantic cohesion and formal fusion between the components, which also changed over time in the course of grammaticalization (for detailed description of the processes, see Rhee 1996: 109–130 and Lee 1997: 64–82, among others).

Simplicity of data presentation notwithstanding, the syntagmatic condition of *-keyss-* is remarkably similar to that of *-l.kes.i-*, in that the construction immediately follows a verb and is followed by a sentence-ender, a condition perfect for the exponent to be reanalyzed as an auxiliary. The implicit subject, one that takes the serialized verbs *ha-* ‘make’ and *iss-* ‘exist’ as its predicate, is something vague, such as ‘situation’, and does not surface in the sentence. As the two verbs became invisible due to formal fusion, the verb of the subordinating clause was reanalyzed as the main verb, and accordingly the fused exponent was reanalyzed as the auxiliary, once again an instance of syntactic upgrading.

From the exposition above, it is evident that the syntactic idiosyncrasy in Korean that sentential constituents (the subject in the present case) can be freely omitted has played an important role in the grammaticalization of the future markers. In structural terms, the fact that the last element of the construction is a verb (e.g., *i-* ‘be’ and *iss-* ‘exist’ above) is important because the verb occurs sentence-finally, marked by the sentence-type marker (also known as the sentence-ender). The slot immediately before the sentence-type marker is for TAM, including the future auxiliary. Also significant is the typological characteristic of agglutinating morphology because agglutination and fusion obliterate the morphemic boundary, make the exponent’s grammatical status ambiguous, and consequently promotes functional reinterpretation and structural reanalysis.

3.2 Conceptual motivation

In crosslinguistic studies, the future is known to develop from diverse conceptual sources. For instance, the lexicon by Kuteva et al. (2019) lists Change-of-state,

‘come to’, ‘go to’, ‘love’, B-Necessity, D-Necessity, H-Possessive, ‘take’, ‘then’, ‘tomorrow’, Venitive, and ‘want’, as the commonly attested sources of future markers. In their seminal work, Bybee et al. (1994: 251–280) show that lexemes related to desire (‘want’, ‘like’), ability (‘be able to’), obligation (‘owe’), movement verbs (‘come’, ‘go’), temporal adverbs (‘then’, ‘thereafter’, ‘afterward’, ‘just now’, ‘soon’), possession (‘have’, ‘get’, ‘obtain’, ‘catch’) as well as other semantically general verbs (e.g., ‘be’, ‘become’, ‘try to’, ‘need’, ‘owe’, ‘get’, ‘do’, ‘make’, ‘look for’, ‘intend’), etc. develop into the future across languages.

The Korean paths, however, are difficult to reconcile with these known cross-linguistic patterns. Even though the Korean verb *ha-* ‘make, do’ may be close to the general verbs in the inventory, and the copula *i-* ‘be’ is listed in it as well, we noted in the above that they do not make a notable contribution to the development. They are similar to syntactic placeholders rather than semantic contributors. Far more important contributors are the prospective adnominalizer *-l* (e.g., ‘book to read’), which is already inherently futuristic by virtue of indexing a later point in time, and the mode converb-marker *-key* (e.g., ‘painfully’), which inherently signals the effect- or goal-orientation (Rhee 1996). As briefly alluded to in Section 2.3, the converb suffix *-key* marks ‘manner’, ‘mode’ or ‘purpose’. The manner/mode of a situation or the purpose of an action has direct implication to the future because they are the bridge between one situation and its future situation or between one action and its future resultant situation. For this reason, the future signaled by *-keyss-*, as compared to *-l.kes.i-* or *-li-*, is close to the present time, hence, the immediate future. Even though the relation between manner/purpose and future is almost universal, such relation implied by *-key* in Korean is so prominent that (Ramstedt 1997[1939]: 91) calls the verbs connected by the morpheme *-key* ‘converbium futuri’, a position tantamount to classifying *-key* as a full-fledged future marker.

All the conceptual contributors to the development of Korean future markers can be shown as a diagram in Figure 1, which is necessarily brief.

3.3 Grammaticalization parameters

The grammaticalization scenarios of Korean future markers can be reviewed with respect to the four grammaticalization parameters suggested in Kuteva et al. (2019), i.e., desemanticization, extension, decategorialization, and erosion.

3.3.1 Desemanticization

Desemanticization refers to the loss of meaning content. Even though it is difficult to compare the source (or earlier) and target meanings, a rough generalization from the foregoing exposition can be presented as in Table 1.

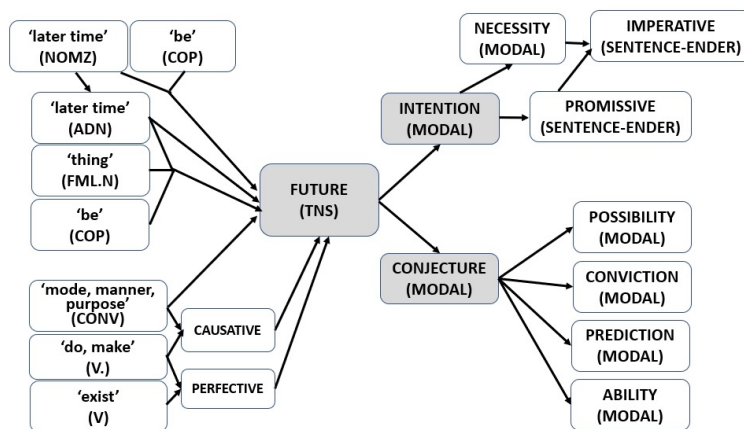


Figure 1: Conceptual contributors to Korean future markers

Table 1: Desemanticization in grammaticalization of future markers

Form	Source meaning (literal)	Target meaning	Desemanticization
-l	'later time'	Future	Not prominent
-li-	'be (at) a later time'	Future	Not prominent
-li	'be (at) a later time'	Future	Not prominent
-l.ke(s).i-	'it is the thing that P at a later time'	Future	'thing' disappeared
-l.key	'it is the thing that P at a later time'	'I promise that...' 'Please do x'	'thing' disappeared
-keyss-	'X is in the mode of Y and exists'	Future	'mode' bleached, 'exist' bleached

Table 1 shows that desemanticization in the *-li*- Class futures is not prominent, because they started out (at least from the earliest available data) as a word/morpheme denoting a later time (either as a nominalizer or adnominalizer). The words denoting ‘thing’ (*kes*) and ‘exist’ (*iss-*) as well as the converb-marker with the ‘mode’ meaning (*-key*) have their source meanings either completely absent or partially bleached.

3.3.2 Extension

Extension as a grammaticalization parameter refers to the use of a form in new contexts. A comparison of the use contexts of the source structure to those of the current, future usage can be presented as Table 2.

Table 2: Extension in grammaticalization of future markers

Form	Source context	Target context	Extension
<i>-l</i>	Verbal morphology (Nominalizer)	Verbal morphology forming a pre-nominal modifier	No extension (context shift)
<i>-li-</i>	Verbal morphology for predicate-formation (cf. <i>-l-i-</i>)	Verbal morphology, modal suffix	No extension (context shift)
<i>-li</i>	Verbal morphology, modal suffix	Verbal morphology, sentence ender	No extension (context shift)
<i>-l.ke(s).i-</i>	Verbal morphology involving ‘a thing’	Verbal morphology without referential restriction	Extension
<i>-l.key</i>	Verbal morphology involving ‘a thing’	Verbal morphology without referential restriction	Extension
<i>-keyss-</i>	Verbal morphology involving causative and perfective meaning	Verbal morphology with no causative/perfective meaning	Extension/Shift

As shown in Table 2, the future markers involving the lexeme *kes* ‘thing’ (i.e., *-l.ke(s).i-* and *-l.key*) have undergone extension in that the future does not have to involve reference to an entity. Similarly, *-keyss-*, which contains elements denoting causative and perfective in earlier stages, can be used without such restrictive meanings. Extension, by definition, means an increase in use contexts and, thus, implies the availability of earlier usage at a later stage. However, the future markers involving *-l* (i.e., *-l*, *-li-*, and *-li*) do not retain earlier usage, and thus exhibit usage context shift, rather than extension.

3.3.3 Decategorialization

Decategorialization refers to the loss in morphosyntactic properties characteristic of lexical or other less grammaticalized forms. Korean future markers display decategorialization at varying degrees, as shown in Table 3.

Table 3: Decategorialization in grammaticalization of future markers

Form	Source category (feature)	Target category (feature)	Decategorialization
<i>-l</i>	Nominalizer, verbal suffix	Adnominalizer, verbal suffix	Decategorialization Loss of argument function
<i>-li-</i>	(periphrastic) (cf. <i>-l-i-</i>)	Modal auxiliary	Not determinable
<i>-li</i>	Modal auxiliary	Modal suffix (sentence ender)	Decategorialization Loss of ability to be followed by a sentence-ender
<i>-l.ke(s).i-</i>	(periphrastic)	Modal auxiliary	Not determinable
<i>-l.key</i>	(periphrastic)	Modal suffix (sentence ender)	Not determinable
<i>-keyss-</i>	(periphrastic)	Modal auxiliary	Not determinable

The presence/absence or degrees of decategorialization should be based on a single, major (or non-minor) category lexeme. However, as shown in Table 3, most future markers are periphrastic at the source stage, and, thus, decategorialization is largely indeterminable with these forms. Though trivial in extent, some level of decategorialization is observed with *-l* and *-li*. As we discussed in Section 2.1.1, *-l* was a nominalizer, which later became an adnominalizer. As a nominalizer it can form a nominal constituent, which can function as a verbal argument. As it becomes an adnominalizer (i.e., modifier), such a nominal function is lost, an instance of decategorialization. The sentence-ender *-li*, occupying the ultimate slot, is a development from the modal auxiliary *-li-*, occupying the penult slot. Since modal auxiliaries are suffixes that are further marked by sentence-enders (i.e., sentence-type markers), by becoming a sentence-ender, *-li* has lost such earlier categorial function.

3.3.4 Erosion

Erosion refers to the reduction of phonological volume of a grammaticalizing form. Korean future markers are eroded at varying degrees, as shown in Table 4.

Table 4: Erosion in grammaticalization of future markers

Form	Source form	Target form	Erosion
<i>-l</i>	/l/	/l/	No erosion
<i>-li-</i>	/li/	/li/	No erosion
<i>-li</i>	/li/	/li/	No erosion
<i>-l.ke(s).i-</i>	/l.kʌs.i/	/l.kʌs.i/ /l.kʌ.i/	No erosion Eroded (moderate)
<i>-l.key</i>	/l.kʌs.i.a/	/l.ke/	Eroded (extensive)
<i>-keyss-</i>	/ke.ha.ʌ.is/	/kes/ /kʌs/ (dialectal)	Eroded (extensive) Eroded (extensive)

It should be noted that there is a discrepancy between the character-based romanization and phonemic representation of the markers. From Table 4, it is evident that the erosion is variable. The eroded forms are only those that involve the noun *kes* ‘thing’ /kʌs/ and the mode-marking converb connective *-key* /ke/.

From the foregoing exposition, it is evident that the grammaticalization scenarios of Korean future markers exhibit differing degrees of conformity to the grammaticalization parameters, proposed by Kuteva et al. (2019). Some of them are indeterminable because of multiple forms participating in the source construction.

4 Summary and conclusion

This chapter addressed the grammaticalization of Korean future markers. Contemporary Korean has six forms that make the future time reference, i.e., *-l*, *-li-*, *-li*, *-l.ke(s).i-*, *-l.key*, and *-keyss-*. Their grammatical statuses are variable from an adnominalizing suffix (*-l*), to a modal auxiliary (*-li-*, *-l.ke(s).i-*, *-keyss-*), and to sentence-ender (*-li*, *-l.key*).

Unlike the crosslinguistically attested patterns of grammaticalization sources, e.g., lexemes denoting desire, ability, obligation, movement, time, possession, among others, the development of Korean futures is largely due to two grammatical formants that inherently involve futurity, i.e., suffixes indicating ‘(at) a later time’ (*-l*) and ‘mode, manner, purpose’ (*-key*).

In structural terms, a number of factors contribute to the development of future markers, but, most notably, omissibility of sentential arguments and agglutinating morphology play an important role, because they make the grammatical status of a construction undergoing grammaticalization ambiguous (i.e., through omission and fusion), and consequently promotes functional reinterpretation and structural reanalysis.

A review of grammaticalization scenarios with respect to the grammaticalization parameters, i.e., desemanticization, extension, decategorialization, and erosion, shows that future markers exhibit variable degrees of conformity to them in the course of their grammaticalization.

Abbreviations

ACC	accusative	IMP	imperative
ADN	adnominalizer	LMIK	Late Middle Korean
BEN	benefactive	MOD	Mode-converb
CAU	causative	MOK	Modern Korean
COND	conditional	N	noun
CONV	converb-marker	NOM	nominative
COP	copula	NOMZ	nominalizer
DAT	dative	PDK	Present-Day Korean
DEC	declarative	PERF	perfective
DFR	deferential	POL	polite
DIR	directional	PROM	promissive
END	sentence-ender	PROS	prospective
FML	formal	PST	past
FUT	future	Q	question/interrogative
GEN	genitive	TNS	tense
HON	honorific	TOP	topic
HORT	hortative	V	verb

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Chapter 11

Grammaticalization of future markers in Thai

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Contemporary Thai has a number of forms that mark the notion of futurity. The primary future tense marker is *cà*, whose lexical origin, though not yet established, is likely the verb *càk* 'know'. Other markers are used either alone or in combination with *cà(k)*. Their morphosyntactic development reveals some influence of the typological and idiosyncratic features of Thai, such as pragmatic orientation, verb serialization, and preference for polylexemic units in lexicalization and grammaticalization, among others. Their semantic developments are also well motivated in terms of conceptual relatedness between the source forms and the grammaticalized forms. A review of grammaticalization scenarios with reference to grammaticalization parameters, e.g., desemanticization, extension, decategorialization, and erosion, shows that the Thai future markers have been grammaticalized to varying degrees.

1 Introduction

Living in the spatio-temporal dimension, all humans supposedly have linguistic means, be they lexical or grammatical, of expressing future time reference.¹ As observed in the literature, Thai does not have an inflectional future, as a general characteristic of an isolating language, but rather has a construction involving the auxiliary *cà* combining with a verb to indicate future time reference (Dahl 1985: 173–174).² Thai also has a handful of other markers, with variable degrees

¹For instance, Dahl (1985: 108–109) notes that even in the languages without a future tense marker, there are ways of signaling future time reference.

²Thai linguists use a few different romanization systems. We use the Thai IPA transcription.



of grammaticalization, that signal that an event or state of affairs will occur or exist at a future time. Their developmental patterns exhibit certain commonalities with other languages as well as peculiarities due to the typological or idiosyncratic features of Thai. This chapter describes the means of future time reference in Thai from a grammaticalization perspective. The data sources include the Thai National Corpus (TNC), dictionaries, lexica, online resources, and reference grammars.³

Thai is a Kra-Dai language (Ostapirat 2000; Eberhard et al. 2022), also known as Tai-Kadai (Diller et al. 2008), and is mostly spoken by the people in Thailand and its vicinity. Thai is an isolating language. Word order is SVO and phrases are head-initial. Thai does not have case marking morphology, but it has a complex tone system (for typological descriptions, see Noss 1954, 1964; Abramson 1962; Haas 1964; Anthony et al. 1968; Warotamasikkhadit 1972; Tingsabadh & Abramson 1999; Campbell 2000; among others).

This chapter is organized as follows: Section 2 describes and exemplifies a few forms indicating future time reference, i.e., *càk* (Section 2.1), *cà* (Section 2.2), adverbials (Section 2.3), and particles (Section 2.4); Section 3 addresses some prominent issues with respect to grammaticalization of future time reference, focusing on the morphosyntactic motivation (Section 3.1), the conceptual motivation (Section 3.2), and grammaticalization parameters (Section 3.3); and Section 4 summarizes the findings and concludes the chapter.

2 Future time reference in Thai

Some Thai reference grammars explicitly state that Thai has no future tense markers, largely because their use is not mandatory and future time reference can be inferred from a temporal adverb, an aspect marker, or from the context (Iwasaki & Ingkaphirom 2005: 123; Thiengburanathum 2013: 159). However, most reference grammars acknowledge the presence of future markers (Noss 1964: 165; Campbell & Shaweevongse 1972: 26; Setthapun 1992: 24; Smyth 2002: 67; among others). The most commonly cited future marker is *cà*, but there are other forms as well. The notion of futurity signaled by certain words is often interpreted as a lexical meaning rather than a grammatical function. This is the case because of categorial fluidity, resistance to formal change, as well as the prevalence of the polyfunctionality of forms across lexical and grammatical domains, which are characteristics typical of isolating languages.

³The Thai National Corpus (TNC) is an online searchable, 33.4-million-word corpus, developed by Chulalongkorn University.

However, if we adopt a dynamic perspective such as the one in grammaticalization theory, Thai indeed has a number markers of futurity. They include *càk*, *cà*, *cuan*, *kùap*, *thêep*, and *klây*. There are poly-lexemic forms created by means of compounding. Some authors, e.g., Supanvanich (1973), also claim that certain sentence ending particles in Thai function as markers of future time reference. In the following, we illustrate the uses of such future markers.

2.1 *Càk*

The word *càk* marks the future in Modern Thai. Evidently, it is the earlier form of *cà* (see Section 3.2 below), the primary future marker in Contemporary Thai. Following the tenets of grammaticalization theory according to which all grammatical forms originate from a lexical source, *càk* is also believed to have a lexical origin. With a regrettably shallow history of documentation (see Section 3.3), however, there has not been any serious scholarly investigation to date into its origin, and, consequently, there is no consensus regarding its lexical origin. Despite the paucity of lexical evidence, some clues do exist in language contact. For instance, the Thai lexicon was greatly influenced by Middle Chinese, and a lexicon of Middle Chinese borrowings (SEALANG 2006) includes Thai *จັก* *càk* (*jàk* in its notation). As the historical source of the Thai *càk*, this lexicon points to the Middle Chinese lexeme denoting ‘recognize (as is)’, which references Proto-Southwestern Tai **cak* ‘to recognize, know’ (Jonsson 1991), as well as Proto-Southwestern Tai **čak* and Proto-Tai **čak* (Li 1977). The same source lists the Thai verb *รู้จັก* *rúucàk* (= *rujàk*) ‘to recognize, know’ as a borrowing of the Middle Chinese lexeme 識 (*śik*) ‘to recognize, know, discern, have knowledge, etc.’, citing Manomaivibool (1975; known as ‘Prapin’), Baxter (2000), and Starostin (n.d.) (SEALANG 2006). Based on these comparative-linguistic analyses, it can be hypothesized that the future marker *càk* (and its later form *cà*, see Section 2.2 below) developed from the Middle Chinese borrowing *càk* with the meaning ‘know’.⁴ Despite *càk* being used as the cognitive verb ‘to know, under-

⁴Other possibilities from the data in SEALANG are the two homophonous lexemes (*จັก* *càk*; *jàk* in its notation), one denoting ‘weave, knit’ (Middle Chinese *čík*/*tsyik*) and the other denoting ‘split, pare bamboo’ (Proto Southwestern **cak*/**čak* and Proto-Tai **čak*), the latter being proposed in the earlier version of this paper. An anonymous reviewer raised concerns that the proposal is not substantiated by historical data. Dr. Weera Ostapirat (p.c.), a specialist in the Tai language family, also expresses his opinion that ‘split, pare bamboo’ is not a likely source and that ‘know’ is the better candidate (see Section 3.2 for conceptual motivation across languages). He further states that the native Thai verb *càak* ‘depart, leave’ is a less likely but possible candidate. At the current level of understanding, the lexical source of the future marker *cà(k)* remains inconclusive. Special thanks go to the reviewer and Dr. Weera Ostapirat for their suggestions.

stand, be aware of', it has become obsolete and only survives in the form of *rúucàk* 'to know a person, place, etc.' (thus, recognition-oriented), in which *rúu* also means 'to know a subject or piece of learned information' (thai-language.com) (thus, comprehension-oriented).⁵ The use of the reinforced *rúucàk* is exemplified in (1):⁶

(1) Modern Thai (constructed)

ผู้ชาย ที่ ผึ้ง รู้จัก ใน เมือง เชียงใหม่ ก็ มี อยู่ สอง คน
phûuchaay thîi phûng rúucàk nay muan Chiang.mây kôw mii yûu sǎw
 man that [name] know in city [name] also have be two
khon
 person

'There are two men whom Pheung knows in Chiang Mai.'

(Modern Thai, constructed)

Càk has been grammaticalized as a future auxiliary (for morphosyntactic and conceptual motivation see Section 3.1 and 3.2). As an auxiliary, first attested in the Sukhothai period (1238–1347 CE), *càk* indicates future or intention.⁷ Its use is exemplified below:

⁵It has been observed that Thai has an idiosyncratic tendency of favoring a morphological process of 'recycling', whereby synonyms or near synonyms are compounded. The multiple forms thus created tend to create 'layering' (Hopper 1991) and have different degrees of intensification or different register-specific, semantic specialization (see below for more instances). It is also common that there are no noticeable semantic differences between the multiple forms created through recycling. Some of such (near-)synonymous forms may disappear, as shown here with *càk* 'to know'.

⁶Note that Thai does not practice inter-lexical spacing, but Thai data are given with spacing for visual clarity.

⁷In the excerpt, the parts that the original Thai data do not need to be presented with interlinear morphemic glossing are presented in English translation within square brackets in the interest of space. In the English translation, the parts that are required for syntactic and semantic reasons but are only implicit in the original data are presented within parentheses. An alternative translation is presented in parentheses as well. Also for convenience, markers of future time, regardless of their grammatical status or additional semantics, are glossed as FUT(ure), unless a different label is needed for expositional clarity.

(2) *Càk* future/intention

- a. ใคร จัก ใคร่ คำ ช้าง คำ ใคร จัก ใคร่ คำ ม้า คำ ใคร จัก ใคร่ คำ เงิน คำ ทอง คำ

khray càk khây kháa cháaŋ kháa khray càk khây kháa máa
 who FUT want sell elephant sell who FUT want sell horse
kháa khray càk khây kháa ɸuan kháa thooŋ kháa
 sell who FUT want sell silver sell gold sell

‘Whoever would want to trade in elephants does so. Whoever would want to trade in horses does so. Whoever would want to trade in silver or gold does so.’

(1292, King Ramkhamhaeng’s free trade law)

- b. *Càk* future/intention

เมื่อ ไม่ รัก จัก หย่า ก็ ว่า เลิกล

mêu mây rák càk yàa kôw wâa thòet
 when not love FUT divorce so say please

‘When (you) do not love and (you) would (intend to) divorce (me), please tell (me).’

(2009, Enak Phra Khanong)

Excerpt (2a) is taken from a historical text, in which *càk* functions as a future marker with the main verb *khây* ‘want’. While *càk* can mean either future or intention (the notion of intention may have arisen from the knowledge meanings associated with the source verb), the presence of the co-occurring conative verb *khây* ‘want’ makes the intention interpretation redundant and the future interpretation contextually favored. Excerpt (2b) is taken from a Modern Thai source, part of a traditional Thai octameter poem entitled ‘The Legend of Mae Nak Phra Khanong’, which still bears many characteristic features of the classical poem. In this excerpt, *càk* can mean either future or intention. One might argue that if the phrase *càk yàa* ‘will divorce/intend to divorce’ is viewed in syntactic parallelism with the preceding *mây rák* ‘not love’, *càk* would favor the intention interpretation, i.e., as a main verb, rather than the future interpretation, i.e., as an auxiliary. However, since Thai does not require syntactic parallelism, there is no syntactically-driven interpretational favoring. The future interpretation is as much favored as in (2a), especially because the second predicate (i.e., ‘divorce’) is later in time (thus, future) relative to the first predicate (i.e., ‘not love’).

The future auxiliary *càk* can form a deontic modal auxiliary with *tôŋ* in the form of *càk-tôŋ* ‘must’, especially in formal registers, e.g., legal documents. As

briefly alluded to before, ‘recycling’ a formant for use in combination with another formant is a common lexicalization-grammaticalization strategy in Thai (Rhee & Khammee 2022, Khammee & Rhee 2022, Thirakunkovit & Rhee 2023; see also Section 3.3.4). The participating form *tôŋ* is itself the deontic modal ‘must’, which is also heterosemous (Lichtenberk 1991) with the verb *tôŋ* ‘to make contact, be exposed to, touch, agree with, etc.’

- (3) *càk-tôŋ* future/deontic necessity
 ความ ยุติธรรม จักต้อง ดำรง อยู่ ใน มวล มนุษย์
khwaam yútitham càk-tôŋ damron yuu nay muan manút
 NMLZ justice FUT maintain be in entire human
 ‘Justice will (shall, must) prevail in mankind.’
 (Modern Thai, constructed)

Excerpt (3) contains *càk-tôŋ*, a future marker also with a deontic necessity meaning. An interesting aspect is that, due to the fact that deonticity is inherently future-related, the presence of the future marker *càk* is not semantically consequential, i.e., there is no perceivable difference between *càk-tôŋ* and *tôŋ*, both denoting ‘must’. The close relationship between deontic modality (agent-oriented) and future tense has been observed in Fleischman (1982), Dahl (1985), Heine (1993), Bybee et al. (1994), among others.

2.2 Cà

In Modern Thai, the primary future marker is *cà*, which is historically an eroded form of *càk*, addressed in the previous section.⁸ The future-marking function of *cà* is first identified in the Ayutthaya Period (1351–1767). In the earliest record of *cà*, the form strongly suggests intention, thus often making the sentence ambiguous as to futurity or intention (note that the two are conceptually close, as exemplified in the development of the English future *will*, Aijmer 1985: 11–20).⁹ As indicated previously, it is quite possible that *cà*, from its earlier verbal meanings of *càk* ‘to know’, has developed into ‘to intend’, and its conceptually close ‘future’ meaning has emerged simultaneously or soon afterwards. Thus, even from the early usage, *cà* functions as a marker not only of the future but also of

⁸The older *càk* is now used with very low frequency (52 pmw in a contemporary corpus TNC) and functions as an auxiliary, marking the future tense or the subject’s intention. On the other hand, the newer and eroded *cà* occurs with very high frequency as a general future-tense marker (15,014 pmw in TNC).

⁹Soithurum (2010), conceding that the absolute lexical origin of *cà* is unavailable, asserts that its earliest recognizable meaning is ‘intend’.

intention and even desire, and other meanings such as assurance and obligation become associated with *cà*, as these notions are all closely related to futurity (cf. Dahl 1985: 103; also see below). The following is an excerpt involving the future marker *cà*:

(4) *cà* future/intention

(in a letter by the Siamese ambassador to France)

จะ ขอบ แก่ ท่าน ให้ เข้า ทูล แก่ พระมหากษัตริย์ เจ้า ผู้ ไห่ญ ...

cà khǎw kèe thân hây khâw thuun kèe phrámahākāsāt-câaw phûu

FUT beg to you:HON to enter tell to king who

yày [...] lɛɛ cà ù-sàa thamhây cháná thúkkhon nay khwaam

great and FUT attempt make win everybody in NMLZ

nóp-nǎw nán

reverence that

‘We would (intend to) beg Your Excellency to tell His Majesty [...] and (we) will (intend to) endeavor to surpass everybody else in terms of respect and veneration toward the king.’

(Siamese Documents of the 17th Century, Paris)

Excerpt (4) is taken from a letter dated from the 17th century, in which the Siamese ambassador requests the addressee to convey a message of allegiance and veneration to the French king. As seen in the excerpt, the future marker *cà*, which, in addition to being a future auxiliary, has potential as a verb, and is followed by three successive verbs *ù-sàa* ‘attempt’, *thamhây* ‘make’, and *cháná* ‘win’.¹⁰

The auxiliary *cà* denotes, in addition to future tense, various future-related modal notions, e.g., intention, promise, necessity, possibility, probability, certainty, indeterminacy, proximation, constancy, etc., either alone or through ‘univerbation’ (Lehmann 2015) with other lexemes.¹¹ These notions are so closely related that whichever is denoted by this particular instance of a *cà*-sentence largely depends on the context, and functional designation may not be exclusive. The usages of *cà* for denoting these modal notions are exemplified below:

¹⁰This is a unique situation in Thai, where multiple verbs may occur in succession asyndetically. For instance, Diller (2006: 160) notes that Thai, which he called ‘a verb-loving language’, allows such series of verbs as ‘Peel give eat’ (for ‘Peel the oranges for the guests’), ‘Tell finish not have’ (for ‘I already told you, there aren’t any’), and the like. See 3.1 for more discussion on its relevance to grammaticalization.

¹¹Since modal markers in Thai are many and varied, with tense and aspect markers omissible without affecting grammaticality, Soithurum (2010: 114) asserts that Thai is not a tense- or aspect-prominent language but a ‘mood-prominent’ language.

- (5) a. *cà* future/intention
 ฉัน จะ เข้าร่วม ประชุม ที่ เยอรมนี เดือน หน้า
chǎn cà khâwruam pràchum thii yəəramánii dʰan nâa
 I FUT attend meeting at Germany month next
 'I will (intend to) join a meeting in Germany next month.'
 (Modern Thai, constructed)
- b. *cà* future/promise/determination
 ฉัน จะ ไม่ ซื้อ ของ ร้าน นี้ อีก ต่ไป
chǎn cà mây sùu khwǎn ráan níi iik tɔ̀ɔp
 I FUT not buy thing shop this again further
 'I will (promise, am determined) not (to) buy any stuff again from this shop.'
 (Modern Thai, constructed)
- c. *cà* future/prediction
 อากาศ ที่ ปลอดโปร่ง จะ ทำให้ เรา พ้นตัว จาก ไข้ เร็ว ยิ่ง ขึ้น
aakàat thii plɔ̀ɔtpròɔŋ cà thamhây raw fɨ́ntua càak khây rew
 air that clean FUT make us recover from fever fast
yîŋ khûn
 more up
 'Clean air will (is expected to) help us recover from a fever faster.'
 (Modern Thai, constructed)
- d. *cà* future/irrealis
 ถ้าผม จะ เปลี่ยนอาชีพ ผม จะ เป็น ครู
thâa phǒm cà plian aachîp phǒm cà pen khruu
 If 1SG:M FUT change occupation 1SG:M FUT be teacher
 'If I had changed careers, I would be a teacher (now).'
 (Modern Thai, constructed)

In (5), *cà* signals that the events described belong to a point in time in the future and at the same time implies certain concepts typically associated with the future tense, as indicated above. Evidently, in (5a), the agent of the predicate with an action verb in the active voice (i.e., attend a meeting) is the sentential subject, which coincides with the speaker (i.e., 'I'), an ideal condition for invoking the speaker's intention. In (5b), the semantic-syntactic features are similar to those of (5a), with an added feature of negation of intention of any further patronage of the shop, which lends the interpretation of determination and promise. In (5c), the sentential subject is non-sentient (i.e., air), thus not amenable to any interpretation involving conative modality such as intention, promise, determination,

etc. The pragmatic inferences, based on the scientific knowledge of the causal relationship between health benefit and clean air, tend to suggest a prediction. Finally, as in (5d), if the auxiliary *cà* is used in the apodosis or protasis of a conditional sentence, especially of a hypothetical condition, it functions as an irrealis marker (cf. ‘hypothetical situation or putative action’ Noss 1964: 165; Soithurum 2010). This is in line with the observation that future is commonly associated with a prediction or extrapolation from the present state of the world (Dahl 1985: 103; Comrie 1989; Fleischman 1982: 24–31, 133–134; see also Kratzer 1991; Sarkar 1998; Sansó 2006: 243; Soithurum 2010; Aikhenwald 2010: 81; Stenzel & Gomez-Imbert 2018: 362, for individual instances).

In addition to the usages of *cà* exemplified above, *cà* also signals diverse future-related modal notions, e.g., necessity, indeterminacy, possibility, certainty, proximation, constancy, etc., through univerbation with other lexemes, which contribute additional meanings of futurity. In this context, the univerbation phenomenon, frequently observed in Thai, warrants a brief exposition. When *cà* forms a univerbated unit with another form (typically a modal), it is possible to treat the two modals in a complex form as operating separately, instead of treating the form as a single unit.¹² For instance, *cà* and *tôŋ* (see 6a below) each may be analyzed as signaling future and deontic necessity. This is a perfectly valid approach, especially considering that when *cà* is omitted from *cà-tôŋ*, there may not be loss of deontic necessity (obligation) meaning, and conversely when *tôŋ* is omitted from *cà-tôŋ*, there may not be loss of futurity meaning. Such an analytic approach, however, does not capture the insights of the researchers who acknowledge the patterns of common co-occurrence of *cà* and other modal elements, and therefore present them as more or less combined forms (even though *cà* may be optional in certain cases). For instance, Noss (1964: 117), Smyth (2002: 65–69), Iwasaki & Ingkaphirom (2005: 130–134), and Soithurum (2010: 58), among others, present certain combinations as unitized forms, either explicitly or implicitly, e.g., (*cà*) *tôŋ* ‘must’, *khuan* (*cà*) ‘should/ought’, *nâa cà* ‘should/ought’, *yàak* (*cà*) ‘want to, would like to’, *campen cà* ‘necessary to’, *kamlan cà* ‘be about to (happen)’, *kùap* (*cà*) ‘be about to (happen)’, *thêep* (*cà*) ‘be about to (happen)’, *cuan* (*cà*) ‘be about to (happen)’, *mák cà* ‘used to, would’, *yôm cà* ‘likely’, *chák cà* ‘beginning to’, *nâa cà* ‘possibly’, *khon cà* ‘probably’, *àat cà* ‘probably, perhaps’, etc. Therefore, while the treatment of such forms either analytically or holistically may be an unnecessarily granular issue, the fact that they tend to accompany other forms and that such a tendency is not applicable to all modal elements supports the practice of treating them as more or less unitized forms (note that the combined forms are represented with a hyphen between the two components).

¹²Our thanks go to the anonymous reviewer for raising this issue.

The usages of the *cà*-univerbated forms for signaling diverse future-related modal notions are exemplified in the following. The first set involves necessity as in (6):

- (6) a. *cà-tôṇ* future/necessity
 คุณ จะต้อง จ่าย ค่าปรับ ที่ สถานี ตำรวจ
khun cà-tôṇ càay khâapràp thii sathǎanii tamrùat
 you FUT pay fine at station police
 ‘You will (must, have to) pay the fine at the police station.’
 (Modern Thai, constructed)
- b. *khuan-cà* future/necessity
 เธอ เปียก ทุกคน เธอ ควรจะ พก ร่ม มา ด้วย
thəə piak mòt ləəy thəə khuan-cà phók rôm maa dûay
 you wet all already you FUT bring umbrella come together
 ‘You are all wet. You should have brought an umbrella.’
 (Modern Thai, constructed)

Example (6a) above illustrates the usage of the future auxiliary *cà-tôṇ*, which also signals necessity. The additional morpheme *tôṇ* is itself a deontic auxiliary, a heteroseme of the verb *tôṇ* ‘to touch, be exposed to, agree’ (see Excerpt 3 of *càk-tôṇ* in Section 2.1 above). Since the phonologically longer form *càk-tôṇ* and the shorter form *cà-tôṇ* carry the identical function, it can be assumed that *càk-tôṇ* and *cà-tôṇ* are variants and that the latter is the more grammaticalized of the two (see Section 3.3.4 below). Similarly, (6b) exemplifies the use of *khuan-cà*, an auxiliary for future and necessity, which denotes unfulfilled obligation. This compound incorporates the future *cà* and the necessity modal auxiliary *khuan*. Even though *cà-tôṇ* and *khuan-cà* both denote future and necessity, the former is stronger in terms of the degree of necessity than the latter, analogous to the English deontic modals *must* and *should*. Another subtlety is that *cà-tôṇ* is more strongly future-oriented than *khuan-cà*, and thus in regret contexts as in (6b), *khuan-cà* is more appropriate.

As shown above, the position of *cà* in the combined forms varies (e.g., *cà-tôṇ* and *khuan-cà*).¹³ It is generally accepted that the relative position of a form with reference to another typically carries the meaning-changing potential due to the scope effect, as in the famous contrast between *ino-rssu-anguag* [person-big-little] ‘little giant’ and *inu-ngua-rssuag* [person-little-big] ‘big midget’ in a dialect

¹³ An anonymous reviewer raised this interesting issue, which may be related to scope differences.

of Eskimo (Sadock & Olsen 1976, as cited in Bybee 1985: 96). Soithurum (2010) identifies three different structural patterns of *cà*: (i) before a verb, (ii) before or after an auxiliary, and (iii) after a complementizer (*wâa* or *thîi*), and briefly mentions, using the example of *kamlan cà*, the structural scope of *kamlan* and *cà* (Soithurum 2010: 72, 76). The combinations described in the present analysis all belong to (ii). While Soithurum's classification system is useful, it is too imprecise to differentiate the types of, e.g., *cà-tôṅ* and *khuan-cà*, and does not adequately explain scope differences. Considering that *tôṅ* and *khuan*, for example, are both full-fledged auxiliaries of necessity, and that the relative position of *cà* in *cà-tôṅ* and *khuan-cà* cannot be reversed (**tôṅ-cà* and **cà-khuan*), the fixed relative position of *cà* is a great puzzle. It is possible that this frozen order represents the syntagmatic arrangement of the forms at the time of grammaticalization (cf. Givón 1971: 413). With the current paucity of relevant data, however, the issue requires further research.

The second set involves modals denoting variable degrees of certainty.

- (7) a. *nâa-cà* future/probability ('probably')

ทุกอย่าง น่าจะ จบลง ดีด้วย ดี
thúkyàaṅ nâa-cà còp.ləṅ dūay dii
 everything FUT end by good
 'Everything should end well.'

(Soithurum 2010: 79, modified)

- b. *àat-cà* future/possibility ('possibly')

ครู อาจจะ สอน อยู่ ที่ ห้องเรียน สีแดง ตอนบ่าย พรุ่งนี้
khruu àat-cà sǎwṇ yùu thîi hōṅrian sǐdēeṅ tɔwṇbàay
 teacher FUT teach PROG at classroom red.color afternoon
phrûṅnîi
 tomorrow

'The teacher may teach in the red classroom tomorrow afternoon.'

(Modern Thai, constructed)

- c. *khon-cà* future/possibility ('possibly')

คุณตา คงจะ ปลูก พืช อยู่ หลังบ้าน เย็น นี้
khuntaa khon-cà plùuk phàk yùu lǎṅbâan yen nîi
 grandpa FUT plant vegetable at backyard evening this

'Grandpa may have planted vegetables at the backyard this evening.'

(Modern Thai, constructed)

Example (7a) illustrates the use of *nâa-cà* denoting future and probability. The added form *nâa* is a modal adverb/auxiliary denoting possibility, which is inher-

ent in future, and vice versa. The conceptual affinity may have motivated this univerbation process. As a future marker, *nâa-cà* also indicates indeterminacy or uncertainty; thus, the described event is lower on the scale of certainty than *cà-tôη* derived from E-Necessity (high probability). The marker is also used in a regret situation, i.e., in describing an event that was once possible but was missed. Example (7b), involving *àat-cà*, in which *àat* is also a modal adverb/auxiliary, indicates likelihood.¹⁴ Its degree of certainty is also lower than probability, as signaled by *cà-tôη*. Example (7c), involving *khon-cà*, in which *khon* is a modal adverb/auxiliary of possibility, still bears a semantic connection with its adjectival heteroseme *khon* ‘be permanent, fixed’. The epistemic meaning (E-Possibility) of *khon-cà* assigns the word to be used as an evidential marker, as in (7c). In this context, variability of the perceived degree of certainty as associated with the modals is evident among speakers of Thai, even conceding that the boundary between possibility and probability is often dependent on the speaker’s subjective judgment. For instance, Iwasaki & Ingkaphirom (2005: 137) state that *tôη* signals high confidence; *nâa* and *khuan* signal medium confidence (*nâa* being more ‘subjective’ than *khuan*); and *khon* and *àat* signal low confidence. An anonymous reviewer stated that *nâa*, *àat* and *khon* are widely accepted to indicate a moderate degree of possibility similar to the English *may* and *might*. The judgment may be influenced by the propositional content as well. Minor perceptual disagreements notwithstanding, *cà*-derived futurity markers may be said to signal E-Necessity and E-Possibility by virtue of involving epistemic modals in their combinations.

Another set of *cà*-derived future modals consists of those marking ‘immediate future’ (Smyth 2002: 69; ‘imminent future’ Heine et al. 2017: 10), as exemplified in the following:

- (8) a. *kamlan-cà* immediate future

เรา กำลังจะ กิน ข้าว

raw kamlan-cà kin khâaw

we FUT eat food

‘We are/were about to eat.’

(Smyth 2002: 69, modified)

- b. *cuan-cà* immediate future

เขา จะจมน้ำ แล้ว

khăw cuan-cà com nâam léew

he FUT drown water PFV

‘He is about to drown.’

(Modern Thai, constructed)

¹⁴The lexeme *àat* is also used as an adjective denoting ‘be brave, courageous, bold, be able’; it is likely, though not conclusively, a heteroseme of the modal adverb/auxiliary *àat* ‘probable’.

- c. *kùap-cà* immediate future
 ฉัน ทำ การบ้าน เกือบจะ เสร็จ แล้ว
chǎn tham kaanbâan kùap-cà sèt léew
 I do homework FUT finish PFV
 ‘I will soon finish (almost finished) my homework.’
 (Modern Thai, constructed)
- d. *thêep-cà* immediate future
 ฉัน หิว แทบจะ ตาย
chǎn hīw thêep-cà taay
 I be.hungry FUT die
 ‘I’m so hungry (I feel like) I’m going to die.’
 (Higbie & Thinsan 2002: 90, modified)
- e. *klây-cà* immediate future
 งาน นี้ ใกล้จะ เสร็จ แล้ว
ḡaan ní klây-cà sèt léew
 work this FUT finish PFV
 ‘The job is nearly done.’
 (Higbie & Thinsan 2002: 89, modified)

The examples in (8) involve future markers that also denote that the event being described will occur in the very near future. However, as Smyth (2002: 69) noted with (8a), the use of the future marker is not restricted to referring to a future time in relation to the speech time, but can also be used to refer to a later time in relation to the time of reference in the past. In the literature, the notion of immediacy is often separately labeled as ‘proximative’ or ‘prospective’ (see Heine 1994; Johanson 2017; Nevskaya 2005, 2015; Kuteva et al. 2019, for terminological issues).¹⁵ Example (8a) involves the compound modal *kamlan-cà*, in which *kamlan* is a progressive marker (note that Iwasaki & Ingkaphirom 2005: 149 label *kamlan-cà* as an ‘inceptive’ (i.e., ‘is about to do’) aspect marker). Example (8b) illustrates the use of the immediate future marker *cuan-cà*, which involves the adverb of proximity *cuan* ‘almost, nearly, close to’ (note that Iwasaki & Ingkaphirom (2005: 149) label *cuan-cà* as a ‘pre-inceptive’ (i.e., ‘is almost done’) aspect

¹⁵Johanson (2017) distinguishes prospective and proximative: prospective being an aspect marker signaling a perspective on events foreseen to occur subsequent to some reference time, and proximative being an ‘actionality’ marker referring to a pre-phase (preliminary or preparatory) of a possible event conceived of as imminent or impending, i.e., as ‘about to occur’ (as cited in Nevskaya 2015). For similar issues, see Comrie (1976: 64); Dahl (1985: 111–112); among others.

marker). The closeness in degree or space is used to denote temporal proximity. Similarly, (8c), involving the immediate future *kùap-cà*, makes use of the degree adverb/auxiliary *kùap* ‘almost, nearly, close to’. Likewise, (8d) makes use of the degree adverb *thêep* ‘almost, nearly, barely’, and (8e) makes use of *klây* ‘(be) near, beside’, a polyfunctional exponent in the adjective, adverb, and preposition word class.

Just as *cuan*, *kùap*, *thêep*, and *klây* are synonymous, so are the derived future modals *cuan-cà*, *kùap-cà*, *thêep-cà*, and *klây-cà*. There is no perceived difference in the degree of proximity in these immediate future markers. However, as compared to *kamlan-cà* in (8a), the degree of immediacy signaled by these four, (8b) through (8e), is greater.

There is another, sui generis future marker, i.e., *mák-cà*, the usage of which is exemplified below:

- (9) *mák-cà* future/habitual
 น้ำหวาน มักจะ กิน ขนม ใน รถ
Namwan mák-cà thaan khanǝm nay rót
 [name] FUT eat snack in car
 ‘Namwan would eat (always eats) snacks in the car.’
 (Modern Thai, constructed)

The future marker *mák-cà* is a compound modal involving the degree adverb *mák* ‘frequently, often’. By virtue of the involvement of the full-fledged future marker *cà*, the compound modal *mák-cà* marks the future, and by virtue of the frequency adverb *mák*, it marks the high frequency of repeated occurrence, hence a habitual future. Thus, the event being described is thought to have occurred habitually in the past, at the present time, or in the future. In Modern Thai, the semantic bleaching occurred to such an extent that *mák-cà* is also used for habitual occurrence of past events (as in the English modals *used to* or *would*). This seems to be due to the fact that the speaker’s assessment that some events will occur in the future with regularity is often based on the same events having happened with regularity in the past and in the present. If that is the case, using a future habitual marker for past habitual events is an instance of reasonable extension along the timeline (cf. the English habitual *will* and *would* based on prediction, Quirk et al. 1985: 228–229).

2.3 Adverbials

There are other, less productive, less entrenched, future markers (Supanvanich 1973), i.e., immediate future markers originating from ‘almost, close, nearly’, i.e., *cuan*, *kùap*, *thêep*, and *klây*, as exemplified below:

(10) a. Immediate future

ฝน จวน ตก แล้ว กลับบ้าน กัน

fǒn cuan tòk léew klàp bâan kan

rain FUT fall PFV return home PTCL.HORT

‘It’s going to (almost started to) rain. Let’s go back home.’

(Modern Thai, constructed)

b. เขา เกือบ/แทบ/ใกล้ จม น้ำ แล้ว

khǎw kùap/thêep/klây com náam léew

he FUT drown water PFV

‘He is about to drown.’

(Modern Thai, constructed)

The examples in (10) illustrate *cuan*, *kùap*, *thêep*, and *klây* functioning as future markers. These four lexemes are those that formed compound modals for the immediate future (see 8b through 8e above). In other words, these examples differ from the previously addressed *cuan-cà*, *kùap-cà*, *thêep-cà* and *klây-cà* only in that they do not involve the primary future marker *cà*; the futurity is marked solely by the proximity markers denoting ‘almost, nearly, close’.

One peculiarity of these adverb-based immediate future markers is that they often, but not necessarily, co-occur with *léew*, the perfective aspect particle, also functioning as a sentence ending particle of diverse discourse functions (note that 10a is bi-clausal (or bi-sentential), the first one ending with *léew*), which originated from the verb *léew* ‘finish, be done’ (Iwasaki & Ingkaphirom 2005: 149).¹⁶ The combination of the perfective *léew* with the proximity degree markers signals that something will not only occur soon but will also be completed soon, and thus creates the sense of definite and complete occurrence of the event within a short period of time.

Treating temporal adverbials as future markers, as Supanvanich (1973) does, raises an important issue that warrants mention, as an anonymous reviewer pointed out. According to the reviewer, adverbials modify actions with respect to their degree of progress, but the described actions may not be actualized. As researchers have observed, futurity is never a purely temporal concept but it necessarily includes an element of prediction or some related modal notion (Lyons 1977: 677; see also Salkie 2010: 1787). There is always an inherent uncertainty about whether an event will take place or not (Dahl 1985, 2000; Jäggi et al. 2022). Thus, a future marker does not guarantee the occurrence of a described event or

¹⁶The immediacy effect created by *léew* is evident in the fact that when the primary future marker *cà* occurs with *léew*, the sentence means that the event being described will occur very shortly.

the presence of a described state of affairs in the future; it only signals that an event or state is, unless stated otherwise, to come about at a time later than the speech time or referenced time. The detachment of a future-marked event from its actualization in reality is inherent in the notion of future, and this trait closely links the future marker to irrealis (Noss 1964; Soithurum 2010: 110). Noss (1964: 165), in his exposition of *cà*-, states that the meaning of *cà*- is ‘hypothetical situation or putative action.’ In this regard, the analysis by Iwasaki & Ingkaphirom (2005: 123) of *cà* as a marker of ‘challengeability’ (Givón 1982) is particularly noteworthy. In other words, a future event is assumed to be challengeable by virtue of not having been realized and thus may be marked by *cà*. Such an analysis can shed light on the usage of *cà* for non-temporal marking, the discussion of which is beyond the immediate scope of the present paper (see Iwasaki & Ingkaphirom 2005: 126–127 for more exposition).

The issue of whether or not the forms marking the degree of progress may be regarded as future markers is debatable. Two points are relevant to this issue: (i) the Mandarin Chinese temporal lexeme *yào* ‘immediately’ (as well as ‘want’) functions as a ‘lexical futurity’ marker (Li & Thompson 1989: 175), and (ii) the general notion of futurity may include temporal adverbials, which are lexical, as well as proximative and prospective adverbials, which are often on the boundary of lexis and grammar. In this light, the adverbials that denote temporal proximity addressed here may well be regarded, at least, as ‘peripheral’ markers of futurity.

2.4 Particles

Other linguistic forms have additionally been claimed to be future markers in Thai. For instance, Supanvanich (1973) asserts that certain sentence ending particles are future markers, e.g., *lá*, *ná*, and *lá-ná*. These future markers are subject to controversy and, even if qualified as such, are undoubtedly marginal forms. Such uses are exemplified in the following:

(11) Immediate future

ฉัน นอน ละ / ูะ / ละ ูะ

chǎn nɔɔn lá / ná / lá-ná

I sleep PTCL

‘I (will) sleep now.’

(Supanvanich 1973)

The example in (11) involves the sentence ending particles *lá*, *ná*, and *lá-ná*. The primary function of these particles, when used in any sentence, is to draw the addressee’s attention to the information being provided and indicate this information as noteworthy. The notion of futurity associated with these forms seems

to be only pragmatically inferred, i.e., not yet grammaticalized. In other words, when the example sentence (11) is spoken, and the speaker is not yet in bed, the addressee, noticing the discrepancy, reconciles it by inferring that the proposition (*Chǎn nǎon* ‘I sleep’), presented as a piece of ‘noteworthy’ information (as indicated by *lá/ná/lá-ná*), is describing a situation ‘as good as the present’, hence the immediate future. Put another way, in (11) above, there exists a gap between the speaker’s alleged claim (i.e., ‘I am sleeping’) and the hearer’s perceived reality (i.e., ‘You are not sleeping now’), which triggers the pragmatic inference (‘You will go to sleep immediately’). The motivation behind describing a future event as one that is presently occurring is conceptually straightforward, and is classifiable as an instance of hyperbole (cf. ‘That does it! I’m leaving’ Quirk et al. 1985: 215).¹⁷ This pragmatic inference seems to have triggered the functional reanalysis of the sentence ending particles as immediate future markers, or at least as forms with a future connotation. It is to be noted that this inference of future time reference is triggered by only these sentence ending particles out of numerous particles in Modern Thai. Therefore, this is different from the use of the simple present tense for future, which is widely attested across languages (cf. Quirk et al. 1985: 182–183 for English). Incidentally, the variations by means of the three sentence ending particles do not create any significant differences in meaning, but most Thai speakers understand that when the longer particle (*lá-ná*) is used instead of the shorter ones (*lá* and *ná*), the proposition carries stronger illocutionary force.

3 Discussion

In the preceding section, we have seen how Thai future markers are used. Based on the description, we will discuss a few issues involved in their development that bear theoretical significance, e.g., morphosyntactic motivation, conceptual motivation, and grammaticalization parameters.

3.1 Morphosyntactic Motivation

The morphosyntactic motivation behind the development of Thai futures comprises two primary factors, one typological, and the other idiosyncratic. The typological factor relates to pragmatic orientation and verb serialization, and the idiosyncratic factor relates to the preference of creating polylexemic units in lexicalization and grammaticalization.

¹⁷The role of hyperbole in grammaticalization has been reported in a series of works by Rhee (2003, 2009, 2022) on the Korean language.

Most functionalists hold that languages differ in terms of the degree to which they utilize pragmatic factors in their linguistic representation. For instance, Huang (2000: 276) asserts that ‘Some languages are more pragmatic than others’ (for similar positions, see Li & Thompson 1976, 1979; and on Thai, see Bamgroongraks 1987; see also O’Grady 2005, 2008, for the ‘emergentist’ approach to linguistic structure). Thai lacks many grammatical forms that are found in many languages, notably linking words. Thus, the semantic and syntactic relationships between constituents are often identified from the linguistic and situational context. This pragmatic orientation is related to the proliferation of verb serialization, in which verbs are strung together *asyndetically*. Diller (2006: 160), while illustrating an overload of verb-based detail in informal Thai, presents an example of a seemingly extreme case: ‘Go. Hurry exit go buy return come finish peel pass’ (for ‘Then hurry off and buy some. When you get back, peel them – get on with it’). The serial verb construction is so susceptible to grammaticalization that it is often regarded as the ‘seed’ of grammaticalization (cf. DeLancey 1991: 15–16). Indeed, all serial verb languages seem to develop exponents of grammatical functions out of serial verb constructions by way of reanalysis (Bynion 1985), or of pragmatics, i.e., interaction between conceptualization of world events and iconicity in the language (cf. Bruce 1988; DeLancey 1991; Lord 1995; Rhee 1996; Aikhenwald & Dixon 2006 and works therein). With the prevalence of verb serialization, Thai exhibits numerous instances of grammaticalization from serialized verbs (Vis 1978; Thepkanjana 1986; Diller 2006; Park 2011).

The development of the primary future tense marker *cà* (and its older *càk*), which originated from a verb, seems to have occurred in verb serialization. The likely lexical origin of the marker is the Sino-Thai lexeme *càk* ‘to know, recognize’, which survives in contemporary Thai as *rúucàk* ‘to know a person, place, etc.’ (see Section 2.1 above and Section 3.2 below). Thus, the source construction of the future marker is, for example, [I + know + return + home], from which the reinterpreted construction [I + will + return + home] emerged (Note that Thai is a head-initial language and the auxiliary precedes the verb).

Secondly, Thai has a strong idiosyncratic preference for the creation of polylexemic or polysyllabic units in lexicalization and grammaticalization (Note that native Thai lexemes are predominantly monosyllabic, Smyth 2002: 1). In other words, Thai often resorts to recycling a single lexeme to create similar or slightly modulated meanings by reduplicating the base form, reinforcing form and meaning by means of the juxtaposing of synonyms, and/or recruiting additional forms, such as particles, adverbs, etc. (cf. ‘polysemy strategy’ Khammee & Rhee 2022; Rhee & Khammee 2022; Thirakunkovit & Rhee 2023). This preference of ‘addition’ is also applicable to the sentence level; various sentence ending particles are

(often multiply) added to the end of the sentence, with the primary function being to soften the illocution and make the assertion or statement more polite. The speech style of ending a sentence without a sentence-final particle is stigmatized as *mây-mii-hǎaŋ-sǎaŋ* ‘having no tail of sound’ (Khammee et al. 2024).

The preference for creating polylexemic or polysyllabic units by means of the above-mentioned strategies is responsible for the creation of several *cà*-derived future markers, e.g., *cà-tôŋ*, *khuan-cà*, *nâa-cà*, *àat-cà*, *khon-cà*, *kamlan-cà*, *cuan-cà*, *khàp-cà*, *thêep-cà*, *klây-cà*, *mák-cà*, etc. (see Section 2.2 above). When multiple modals occur in a single sentence, they tend to occur next to each other in the pre-verbal slot, and thus their positional closeness, as well as the general tendency to create polylexemic, polysyllabic units, is likely to increase the bond between them (hence, *cà-tôŋ* ‘will’ (< ‘will-must’)). Similarly, adverbials often occur next to the future marker. In general, there is a small group of adverbials that occur sentence-finally (but before sentence ending particles), e.g., *samǎə* ‘always’, *bôy* ‘often’, and *rúay* ‘all the time’ (Iwasaki & Ingkaphirom 2005: 103), but others are more positionally flexible and often occur at the pre-auxiliary position. Hence, the habitual future *mák-cà* ‘will’ (< ‘frequently-will’), the immediate future *cuan-cà* ‘be about to, will’ (< ‘almost-will’), etc.

Apart from the two primary factors elaborated above, a mention is in order with respect to the immediate future markers that are heterosemous with degree adverbs, i.e., *cuan* ‘frequently’, *khàp*, *thêep*, and *klây* (see Section 2.3 above). These future markers are less established, as evident in the fact that few researchers include them in the inventory of future forms (e.g., Supanvanich 1973). The syntagmatic position of degree adverbials is flexible, but the pre-verbal position is favored, because they are modifiers of the verb or verb phrase. Thus, in this configuration, degree adverbials occupy the same position as the auxiliaries, including future markers such as *cà*, which, coupled with conceptual motivation (see below), seems to have triggered the functional reinterpretation and structural reanalysis of the degree adverbials into future auxiliaries.¹⁸

3.2 Conceptual Motivation

Semantic change in grammaticalization is not random but, rather, is motivated (‘the source determination hypothesis’ and ‘the universal path hypothesis’; Bybee et al. 1994). Conceptual motivation and the gradient nature of linguistic change lead to the formation of a conceptual network of lexemes and grammatical forms (Heine et al. 1991; Nuyts 1992; ‘semantic map’ van der Auwera & Plungian 1998; Croft 2001; Narrog 2009).

¹⁸See Section 2.2, however, for unsolved morphosyntactic positionality issues.

We have noted that the source lexeme of the Thai primary future marker *cà* is likely the Sino-Thai verb *càk* ‘to know, recognize’ (see Section 2.1 above) and that around the time when it became grammaticalized into a future auxiliary in the Sukhothai period (1238–1347 CE), its meaning became ‘to intend’ (see Section 2.2 above). Crosslinguistic research shows that the future develops from diverse sources, e.g., Change-of-state, ‘come to’, ‘go to’, ‘love’, B-Necessity, D-Necessity, H-Possessive, ‘take’, ‘then’, ‘tomorrow’, Venitive, and ‘want’ (Kuteva et al. 2019). Since B-Necessity denotes ‘intention’, the lexical source of the Thai path conforms to the crosslinguistic pattern (also the path proposed by Bybee et al. 1994). Furthermore, the lexical source ‘know’ is known to develop into PI-Possibility (Ability) and further to Habitual or Permission, or to Evidential or Experiential (Kuteva et al. 2019). Mandarin Chinese has no grammatical tense, but the verb *hui* ‘know’ can indicate futurity (Bybee et al. 1994: 265; Li & Thompson 1989: 183). Similar to this grammaticalization scenario, the Medieval Chinese verb *jie* ‘know, know of’ developed into the auxiliary of PI-Possibility (Ability), which describes a notion closely related to the future (Long et al. 2012). The Thai future marker *cà* marks Possibility (E-Possibility), closely related to PI-Possibility (Ability), as well as Necessity (i.e., E-Necessity, for Probability; D-Necessity, for Obligation; and B-Necessity, for Intention), often with the help of co-occurring forms. The states of affairs in Thai fit with the observed crosslinguistic patterns.

It is noteworthy that once a lexical form or a construction begins to mark future tense, the marker, in most cases, seems to further acquire diverse modal functions, such as Necessity, Certainty, Probability, etc., even though the focus may be selective and different. For instance, the Thai future acquired the modal function of Indeterminacy, which is likely to have derived from the characteristic of the future as inherently indeterminate and uncertain.

The adverbials of proximity denoting ‘almost, nearly’, either alone or with the primary future *cà*, have developed into future markers (see Section 2.2 and Section 2.3 above). The development must have been enabled by the general proximity (typically spatial and degree) metaphorically projected to temporal proximity. The original semantics still persists in the grammaticalized markers in that these are immediate future markers (rather than remote future or general future) and that their functions are largely restricted to marking immediate future, not extended to other modal meanings.

We also noted that some sentence ending particles have developed into future markers, albeit marginally (see Section 2.4 above). Considering that the primary function of these markers is to signal that the speaker is ‘informing’ the addressee by providing noteworthy information, we argued that the addressee’s attempt to

fill the gap between the information and reality is responsible for the emergence of the function of marking immediate future.

All these conceptual considerations and their interrelatedness may be sketched, as in Figure 1.

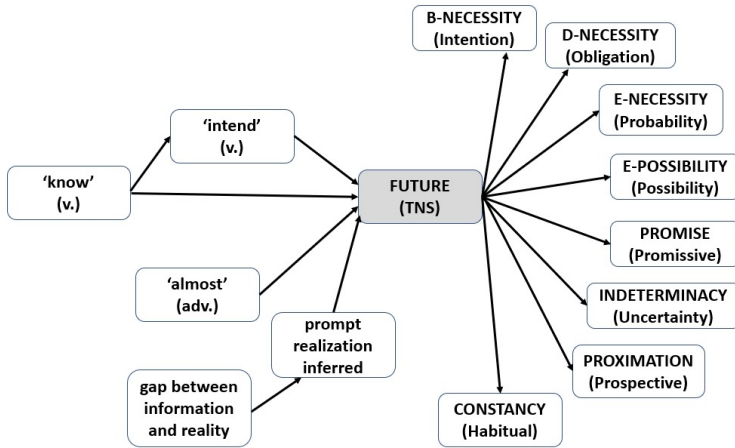


Figure 1: Conceptual network of Thai future markers

3.3 Grammaticalization Parameters

A brief mention to Thai documentation is in order before we discuss diachronic development. The documentation of the Thai language has a rather shallow historical depth. Haarmann (2012), Leppert (1992), and others state that it is possible that the early literature of the Thai people may have been written in Chinese, but no historical record of the Siamese thus far refers to this earlier literature. The Thai alphabet was established in the 13th century CE, and the earliest literature dates back to the Sukhothai period (13th to mid-14th century), chiefly surviving in the form of stone inscriptions (e.g., the King Ramkhamhaeng inscription of 1292). Classical literature dating from the Ayutthaya period (1351–1767) is written in verse but most of it was lost in the 18th century in the military conflict with Burma (Myanmar). Thus, the extant literature is recent, mostly dating back to the 19th century. One obvious implication of this shallow historical depth in grammaticalization studies is that the developmental trajectories of grammatical forms cannot be easily or conclusively established. Thus, following the tradition of researching languages with little or no historical documentation (Traugott & Heine 1991: 6), we can project diachronic grammaticalization to synchronic data

and, conversely, synchronic variations to the reconstruction diachronic developmental paths.

The relative paucity of historical data notwithstanding, we will review the grammaticalization scenarios from the available resources with respect to the four grammaticalization parameters suggested in Kuteva et al. (2019), i.e., desemanticization, extension, decategorialization, and erosion.

3.3.1 Desemanticization

Also commonly called ‘semantic bleaching’ in figurative terms, desemanticization refers to the loss in meaning content. As briefly alluded to in the preceding exposition, the source lexeme *càk* has undergone desemanticization to such an extent that most speakers of contemporary Thai are unable to recognize the connection between the verb *càk* and the future marker *cà*. The implication of cognizance, i.e., ‘know’, is nearly completely lost and its trace is only visible in the verb *rúucàk* ‘to know a person, place, etc.’ Even as a future marker, *cà* does not make a substantial semantic contribution to the *cà*-compounds. For instance, the semantic difference between the compound and non-compound is often imperceptible (e.g., *àat* ‘might’ and *àat-cà* ‘might’; *khon* ‘possibly’ and *khon-cà* ‘would’; *nâa* ‘probably’ and *nâa-cà* ‘likely’; *khuan* ‘should’ and *khuan-cà* ‘should’; *kàap* ‘almost’ and *kàap-cà* ‘almost’; etc.).

On the other hand, the meanings of general degree and spatial proximity as associated with the proximity adverbs denoting ‘almost’ have been bleached, but the bleaching is not substantial because the immediate future is still based on the notion of temporal proximity. The sentence ending particles, by virtue of having no ‘semantic’ content, cannot be discussed in this regard.

3.3.2 Extension

Also known as context generalization, extension refers to the use of a form in new contexts. With respect to the future *cà*, the source lexeme was a verb denoting the presence of knowledge, which later denoted intention, and after grammaticalization the form has become an auxiliary of the future. Through grammaticalization, the usage context has been extended because there is no more semantic constraint and the form can modify a proposition (i.e., a clause).

Similarly, the degree adverbs that, before grammaticalization, could be used as modifiers of verbal, adjectival, and predicate constituents, can now modify propositions, a clear instance of usage context extension.

The sentence ending particles that could be used as modifiers of a clause and function as a discourse operator (as markers of the speaker’s diverse stances) can

now be used as modifiers of a clause with respect to the temporal dimension of the proposition, which also constitutes an instance of usage context extension.

3.3.3 Decategorialization

Decategorialization refers to the loss in morphosyntactic properties characteristic of lexical or other less grammaticalized forms. Decategorialization typically involves, but is not restricted to, the primary categories. We will look at the source verb *càk* and the adverbs *cuan*, *kàap*, *thêep*, and *klây*.

The verb *càk*, by virtue of being a verb, could carry diverse verbal features, such as degree, quality, frequency, and manner modification, e.g., ‘fast’, ‘well’, ‘everyday’, ‘diligently’, ‘thoroughly’, etc. However, after its grammaticalization into a future marker, the form cannot be marked by such modifiers. The ability to c-select a theme as its complement, a feature of a transitive verb, has also been lost.

Similarly, the proximity adverbials, before their grammaticalization into future markers, could function as modifiers of verbs, adjectives, and predicates. After grammaticalization, such adverbial features are lost and can function only as modifiers of a clause. Thus, the grammaticalization of future markers in Thai clearly shows decategorialization.

3.3.4 Erosion

Also known as phonetic reduction, erosion refers to the loss in phonetic substance. Even though most grammaticalizing forms across languages undergo phonological reduction, often by virtue of increased frequency, Thai future markers are not eroded to a significant level, a characteristic of an isolating language (cf. Matisoff 1991; Bisang 1996, 2011; Enfield 2009; Ansaldi et al. 2018). For instance, the verb *càk* did not lose any substance in its development into the future marker *cà*. As it developed into the future marker *cà*, some minimal loss has occurred.

Similarly, the proximity adverbials have not undergone phonological erosion either, in the course of their development into future markers. Their syntagmatic properties as well as their phonological shapes (including tonal features) are retained even after the grammaticalization. The only change that occurred is their functional reinterpretation. The same is applicable to sentence ending particles: they retain all syntagmatic and phonological properties after their grammaticalization. The grammaticalization only lies in the reinterpretation of a discourse-pragmatic marker into a tense-marking modal auxiliary.

There is one noteworthy aspect in the grammaticalization of future markers in Thai, which may be extendable to Thai grammaticalization in general. It relates to the reverse of erosion, i.e., reinforcement. As indicated in Section 3.1, Thai has a strong tendency to reinforce a linguistic (particularly grammatical) form. The common strategies recruited in the realization of this inclination include reduplicating the base form, reinforcing form and meaning by means of juxtaposing synonyms, and recruiting additional forms, such as particles, adverbs, etc. (see Section 3.1 above). For instance, in their research on Thai allatives numbering 26, Rhee & Khammee (2022) report that there are many forms that recycle the base form by reinforcing it with semantically similar forms, e.g., *hây* ‘to/for’, *hây-kêe* ‘to/for’, *hây-kàp* ‘to/for’ all involving *hây* ‘to/for’; *con* ‘to’, *con-thŭŋ* ‘to’, *cùap-con* ‘to’, *con-kràthân-thŭŋ* ‘to’, *tràap-con* ‘to’, *tràap-con-thŭŋ* ‘to’, all involving *con* ‘to’, etc. (see also Khammee & Rhee 2025 for a similar analysis). Similarly, Thirakunkovit & Rhee (2023), in their research on Thai intensifiers, note that out of 23 intensifiers as many as 14 of them involve *mâak* ‘very, very much’, the primary intensifier in contemporary Thai, e.g., *mâak* ‘very’, *mâak-maay* ‘very’, *mâak-maay-kàay-kɔŋ* ‘very’, *mâak-kəən-pay* ‘very, too much’, *mâak-con-kəən-pay* ‘very, too much’, etc. These forms, though composite in makeup, are perceived as ‘univerbated’ units (Lehmann 2015) by Thai speakers. Even though further research is required to reach a solid generalization, Thai seems to favor semantic and phonological reinforcement to increase the salience, especially as a grammaticalizing form becomes semantically weak or loses novelty through frequent use.

4 Summary and Conclusion

Contemporary Thai has a number of forms that mark futurity. In addition to the primary marker *cà* and its older form *càk*, there are secondary markers of futurity, e.g., proximity adverbials denoting ‘almost’ and sentence ending particles signaling informative noteworthiness, which are used either alone or in combination with other forms. An investigation of their morphosyntactic development reveals some influence of the typological and idiosyncratic features of Thai, such as a high level of pragmatic orientation, extensive verb serialization, and preference for polylexemic units in lexicalization and grammaticalization, among others. Their semantic developments are also well motivated in terms of conceptual relatedness between the source forms and the grammaticalized forms. A review of grammaticalization scenarios with reference to grammaticalization parameters, e.g., desemanticization, extension, decategorialization, and

erosion, shows that the Thai future markers have been grammaticalized to varying degrees. One peculiarity of Thai future grammaticalization is that Thai favors semantic and phonological reinforcement to increase the salience, especially as a grammaticalizing form becomes semantically weak or loses novelty through frequent use, thus countering semantic bleaching and formal erosion.

Abbreviations

FUT	future	PFV	perfective
HON	honorific	PRS	present
HORT	hortative	PROG	progressive
M	masculine	PTCL	particle
NMLZ	nominalizer	SG	singular

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Chapter 12

West Polesian future tense constructions switch roles

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In this chapter I assess West Polesian (East Slavic) future tense constructions from historical and typological perspectives, focusing on the de-obligative (xořu robiti ‘I will (need to) do’) and de-volitive futures (majus¹(a) robiti ‘I am planning to do’), which have had the most unusual evolution of all. Against the typological expectations, the de-volitive has synchronically an obligational nuance, whereas the de-obligative expresses an intention.

First, I describe their morphological structure and how speakers use them. Then I show their cross-linguistic recurrence and presence in Slavic and other European languages. I assess the differences and similarities between the West Polesian constructions and their analogue forms or cognates and I map their distribution in Europe. In the light of this data, I try to address two closely related questions: (i) How and at which point may have this semantic change happened? (ii) Is the semantic mutation of these a side-effect of having a particularly large inventory of future tense constructions (which could have pushed some to specialize) in line with the Theory of Niches?

1 Introduction

1.1 A Quixotesque plot-twist

By the end of Miguel de Cervantes’ “Don Quixote”, there is an unexpected plot-twist. Out of contact with each other, Don Quixote, who had been always the insane character that suffered from numerous hallucinations, dies sane; whilst his loyal squire Sancho, who used to be the voice of reason, turns insane. In this



chapter, I describe an analogous situation concerning the course of two future tense constructions in West Polesian (East Slavic). Against the typological expectations, the de-obligative construction (i.e. historically based on a ‘have-to’ verb) has ended up acquiring a volitional/intentional nuance *majusj(a) robiŭ* ‘I will/intend to do’; whilst the de-volitive construction (i.e. historically based on a ‘want/will’-type of verb) has turned obligative *xofu robiŭ* ‘I will be in the need to do it’.

First, I start by presenting the main future tense constructions in West Polesian (Section 1.2), following the criteria set by Dahl (2000) to give an overview of the complexity of the set and its implications. I also discuss the sociolinguistic setting very briefly and provide some methodological remarks (Section 1.3). Second, I narrow the attention to the two future tense constructions under discussion, their morphological structure, and their uses (Section 2). Third, I look at the constructions from both an areal/typological and a diachronic perspective to assess whether they are loans from neighbors and how did they get into the system (Section 3). Finally, I present a summary of the findings and extract some conclusions (Section 4).

1.2 Overview of future tense constructions in West Polesian

West Polesian is an East Slavic variety or language spoken in southwestern Belarus (Brest), northwestern Ukraine (Volynja) and a small region on Poland’s easternmost border (Podlasie). West Polesian is primarily a language of oral communication within the community, and speakers refuse to use it outside of their trusted community members, because it is highly stigmatized. The speech community has remained fairly isolated from the outside for centuries, given the swampy topography, which made travel both inside and outside of the community very difficult. As a result, culturally and linguistically they have remained more conservative in some respects than most of their Slavic neighbors (e.g. the presence of vocatives), but this has also set the ground for innovations (e.g. pronominal numerals; postnominal and dislocated possessors), some of which I discuss in this chapter. Conversely, for the past decades, the community has been surrounded by four, genetically very closely, yet highly standardized Slavic varieties: Standard Belarusian, Polish, Russian and Standard Ukrainian.

Narrowing down the focus on the peculiarities of West Polesian grammar, future tense constructions are certainly worth studying. In the light of Dahl’s (2000) survey of future tense constructions in European languages, data from fieldwork in West Polesian future tense constructions fall into at least six of his future tense construction-families:

1. The SYNTHETIC (IMPERFECTIVE) FUTURE (cf. Dahl 2000: 319). It consists of the infinitive form of the verb plus a cliticised form of the verb ‘to have’ with **person/number** marking.

- (1) (Z10.1 00:20, own fieldwork)
ja ʃjo nɪ zna-tɪ-mu skaza-tɪ
 1SG.NOM Q.ACC.SG NEG know-INF-HAVE.1SG say-INF
 ‘I won’t be able to tell [it] (lit. I will not know to tell).’

2. The DE-OBLIGATIVE FUTURE (2000: 323).¹ It consists of a verb originally meaning ‘have to’ (in this case in the **reflexive** form) + a verb in the infinitive.

- (2) (T12.8.el 04:32, own fieldwork)
vona maj-it-sa rodɪ-tɪ v ijuɫj-e
 3SG.F.NOM have-3SG-REF deliver-INF in July-LOC.SG
 ‘She is due to (deliver in) July.’

3. The DE-VOLITIVE FUTURE (cf. Dahl 2000: 323).² It consists of a verb originally meaning ‘to want’ + a verb in the infinitive.

- (3) (Tor1.36.pr 01:51, own fieldwork)
vona xof-e kot-a pokormɪ-tɪ
 3SG.F.NOM want-3SG cat-ACC.SG feed.PERF-INF
 ‘She is going to feed the cat (lit. She’ll need to feed the cat).’

4. The ‘BECOME’-TYPE or DE-VENITIVE FUTURE (cf. Dahl 2000: 323–324). It consists of an auxiliary verb meaning ‘to become’ + a verb in infinitive.

- (4) (Tor1.53.el 03:28, own fieldwork)
jek vona stan-e [xorofo] huʃɪ-tɪ-sja, to
 COMP 3SG.NOM.F become.PRF-3SG well study-INF-REFL then
jiɹɪ da-dutj premij-u
 3SG.DAT.F give.PRF-3PL prize-ACC.SG
 ‘If/when she will start to study hard, she will get a prize.’

¹Dahl (2000: 318–319) calls this the “Balkan ‘have’ future”, although this is not the only de-obligative future-construction family in Europe. He previously mentions the “Romance inflectional future”, which is analogous to this form. However, for the sake of consistency and to avoid confusion with the rest of the terms here (i.e. not to confuse it with the inflectional or synthetic future, as in *robitimu* ‘I will do’), I use a more generic term ‘de-obligative’.

²Dahl (2000: 322) mentions the existence of another de-volitive construction-family in Northern Europe (Germanic). Yet, the West Polesian de-volitive is genetically and typologically closer to the Balkan (i.e. South Slavic) family than to the Northern European one.

5. The Slavic PERFECTIVE FUTURE (or PRESENT) (2000: 323, 326). It consists of a perfective verbal base, with **person/number**inflection as in the present tense.

- (5) (Z2.6 03:07, own fieldwork)
batju/k-a, jak ze z ja, [het-ofo], sober-u
priest-VOC how then PART 1SG.NOM this-GEN.SG gather.PRF-1SG
het-ije pir-ije?
this-ACC.PL feather-ACC.PL
'Father (Orthodox priest), how will I gather all those feathers?'

6. The Slavic COPULAR CONSTRUCTIONS (2000: 324), which consist of the auxiliary 'to be' in the future tense + a verb (in this case, in the infinitive).

- (6) (P2.5.pr 00:04, own fieldwork)
a fjo mi bud-em robi-ti na pasku?
so Q-ACC.SG 1PL.NOM BE.FUT-1PL do-INF in Easter-LOC.SG?
bud-em praznova-ti
BE.FUT-1PL celebrate-INF
'So what are we going to do for Easter? We are going to celebrate it.'

On the one hand, this should not be surprising, as, according to the cross-linguistic surveys in Bybee (1985) and Ultan (1978), future tense constructions are evolving frequently within verbal systems. Furthermore, West Polesian is on a highly diglossic sociolinguistic situation, so it is tempting to propose that the plethora of forms is poorly described code-switching or the result of extended language contact, which I try to challenge in this chapter. On the other hand, no European language appears to have more than three grammaticalized constructions for the future tense in Dahl's (2000) survey, which would make West Polesian a 'record holder' within Slavic languages, and most likely also in European languages. This represents an interesting situation from the point of view of typology, as well as historical linguistics, as this finding can assist in tracing areal influences.

1.3 Methodological remarks

Data for this chapter are the result of my own fieldwork in Belarusian Western Polesie (Brest region), between 2016–2017 as part of a larger morphological typology survey. Unfortunately, the pandemic, the migration crises with the border

and the full-scale invasion of Ukraine have made impossible to go back to collect more data. I interviewed up to 50 speakers, however, for this parameter, I focused on free-texts and elicitation sessions with 15 speakers, as the constructions displayed are not present in all the varieties surveyed. The future tense has been by far the hardest parameter to elicit. Due to its delicate semantic nature (given that it is easily replaced by other forms) and relatively low frequency in the corpus, many of the examples provided here have been obtained through direct elicitation (most often using multiple choice questionnaires and grammaticality judgments, and, less frequently, using translations or rephrasing exercises). This is a common problem: Ultan (1978: 89) also pointed out the lower saliency of the future in his cross-linguistic survey of the future tense. I have tried to be transparent about the source of all my examples. When the utterance was extracted from a free text, I have added the time stamp. In other cases, I have indicated whether it was prompted (pr) (e.g. by asking about a situation where they would be very likely to use the future tense); directly elicited (el) (e.g. by a translation task or grammaticality judgment); or overheard (ov) (and thus, not recorded) but outside of the working sessions (i.e. through participatory observation; e.g. when farming with them).

Since West Polesian is primarily an oral variety and that there is hardly any reference of a standard, I have transcribed all the examples following my ongoing analysis of West Polesian phonology using IPA. Yet, I capitalize proper nouns, which I admit is a license I have taken (despite IPA conventions).

Following the ethical standards in linguistic fieldwork, I collected consent from all participants before making the recordings. Given data protection and privacy regulations I have given pseudonyms to all participants involved. The corpus contains many recordings with sensitive data; therefore, following the common ethical guidelines, I have decided not to make them public. However, for reproducibility purposes, I invite people to email me to ask for access to the corpus. The letters of the code indicate the village where the recording was made. I am providing a list of the participants (with the names they chose to be known as) for those who opted for having their identity disclosed and get recognition for their valuable contribution to this project in the Appendix.

2 Morphology and use

2.1 The de-obligative future

2.1.1 Morphological structure

The ‘have’-type or de-obligative future construction in West Polesian is composed by the auxiliary verb *matı* ‘to have’ (synchronically) with the reflexive suffix and the infinitive. The auxiliary bears person/number and probably also tense information, whereas the infinitive encodes the lexical meaning and aspect.

- (7) *ja ma-ju-sj (po)sadı-tı kartopl-i*
 1SG.NOM have-1SG-REFL plant.PRF-INF potato-ACC.PL
 ‘I am going to plant potatoes.’

In the villages where I have worked I have only documented the reflexive form of the auxiliary (*matısj(a) + inf*) ‘to have’. Nevertheless, when I asked my language assistants, they all recognized the non-reflexive form (*matı + inf*) as grammatical. They pointed out that it is used in other villages, yet nobody managed to specify a village/variety where they use it. In any case, even though I primarily deal with the reflexive form here, the semantic base of the auxiliary is the verb ‘to have’ and so I include it in the group of de-obligative future constructions.

Concerning the morphological structure, it is remarkable that *matı* inflects for past tense (whereas other auxiliaries/synthetic futures do not) and indicates prospectivity in the past; as in (8). To give some context, the speaker was narrating how the Nazis had taken hostage her dad and other men in the village. The soldiers were about to shoot the hostages, but someone that knew her dad intervened, so suddenly someone appears with a message from a superior telling the soldiers to stop the execution.

- (8) (Tor1.25 05:31)
vosjim ʔolo'vik ma-l-ı-sja rostrılje-tı
 eight.NOM/ACC person.GRADNM have-PST-PL-REFL shoot.PRF-INF
 ‘They were about to shoot eight people.’

The auxiliary *majusj* can be used on its own as a free or unbound auxiliary, like the copular auxiliary *budu*. There are few examples of this in the corpus, but one instance is particularly worthy of mention. TL1’s daughter was supposed to water the potatoes one day, but she did not. While she was explaining to her mum why she had not been able to water them, her mother asked “*a mai/sja?*”, meaning ‘but are you going to water the potatoes?’

2.1.2 Uses and connotations

The de-obligative future tense construction has a strong sense of intentionality (in some cases it even retains some obligational flavor). After asking some speakers about the meaning of *majusj robıtı* ‘I will do’ (de-obligative), some (e.g. T2) said that it has the same meaning as *budu robıtı* (copular), and thus that there is no difference between the two. Yet after several weeks some speakers (T5, T10 and T12 i.a.) pointed out that the auxiliary *majusj* of the de-obligative is the equivalent of the Russian *sobirajus’* ‘I intend to’.

In most instances the DE-OBLIGATIVE FUTURE is inappropriate for general predictions. However, in situations such as the discussion of someone’s pregnancy, this form appears the preferred one (9a, 10a), rather than the inflectional future (9b, 10b). Therefore, there may be also some lexicalization behind the choice of one structure over another.³

(9) (T9, T12.el)

- a. *vona ma-et-sa rodı-tı v ıjunje*
3SG.NOM.F have-3SG-REFL deliver-INF in June-LOC.SG
- b. * *vona rodı-tı-me v ıjun-je*
3SG.NOM.F deliver-INF-HAVE.3SG in June-LOC.SG
‘She is due (to deliver) in June.’

(10) (T9, T12.el)

- a. *vona ma-et-sa rodı-tı dofk-u*
3SG.NOM.F have-3SG-REFL deliver-INF daughter-ACC.SG
- b. *vona rodı-tı-me dofk-u*
3SG.NOM.F deliver-INF-HAVE.3SG daughter-ACC.SG
‘She will give birth to a girl.’

According to T6, the use of this form implies a high level of intentionality, with a sense of commitment “*namjereno, vze zadumano*” ‘purposefully/intended, already planned’,⁴ whereas T9 emphasized that the de-obligative expresses more uncertainty than the perfective:

³Concerning ((9b)) and ((10b)) the SYNTHETIC form is generally good for both general predictions and intentions. In the context of discussing pregnancy and birth *rodıtıme* implies a greater certainty than the DE-OBLIGATIVE form *maetsa rodıtı*, according to my consultants. Given that people can expect, but not know for sure when they are going to give birth, the synthetic form sounded inappropriate to all three of my consultants. However, when it comes to sex of the baby, T13 argued that there can be more certainty about it (thanks to technology), thus, it is okay to speculate (using the de-obligative) about the sex of the baby, but it is also acceptable to assert it quite firmly (if we know the results from the doctor).

⁴(T6.3 01:57).

“So if it’s tomorrow, I will say “tomorrow *pojdu* ‘I will [already] go [PERFECTIVE]’ to Drahicyn, or Brest, or Pinsk... I will go tomorrow”. But if I say *majusja jixatı* ‘I’m due to go [DE-OBLIGATIVE]’ it is unknown when *pojdu* ‘I will go [PERFECTIVE]” (T9.9.el 02:32) [My translation].

According to HL1’s comments, there is a correlation between animacy and the morphological strategy used. I proposed (11a), where the object is a human (children), and she objected to it saying that “you would say *maju*, if you had an item, but with children, you say *treba~tra?a*”.

- (11) a. (HL1.el)
**ma-ju-(sja) dit-ej polozi-tı spa-tı*
 have-1PL-REFL child-GEN.PL lay.PRF-INF sleep-INF
- b. (HL1.vol) >>
minji treba dit-ej polozi-tı spa-tı
 1SG.DAT necessary child-GEN.PL lay.PRF-INF sleep-INF
 ‘I need to/will put the children to sleep.’
- (12) a. (HL1.el)
praz pjet hod-ov maj-ut-sja dostroi-tı doroh-u
 in five year-GEN.PL have-3PL-REFL build.PRF-INF road-ACC.SG
- b. (HL1.vol) >>
sobiraj-ut-sja dostroi-tı; objazan-ı dostroi-tı
 plan-3PL-REFL build.PRF-INF obliged-NOM.PL build.PRF-INF
 ‘In five years’ time they will have to have finished building the road.’

2.2 The de-volitive future

2.2.1 Morphological structure

Some of the varieties surveyed in the West (of Belarusian Western Polesie) have a de-volitive future. In these varieties the verb *xotiti* ‘to want’ has a deontic meaning, and it seems that *majusj* (de-obligative) is close in meaning to this, as it was the first response I obtained whenever I tried eliciting sentences containing *maju* / *majusj(a)*.

The de-volitive future tense is composed by an inflected form of the verb *xotiti* ‘want’+ infinitive. Most commonly the verb *xotiti* appears inflected in the present, which means that the topic-time is in the future; but less frequently the verb can

appear in the past tense if the topic-time is in the past (i.e. future-in-the-past).⁵ As with other periphrastic forms, the auxiliary encodes person, number and probably also tense information; whereas the infinitive bears the biggest semantic load (of the verb). In principle, both imperfective and perfective infinitives can appear with *xotiti*. Nevertheless, when I suggested periphrases with *xotiti* (among other constructions) with an infinitive to my main informant, Tor1, she rarely picked the imperfective infinitives with *xotiti*.

(13) (Tor1.el)

zavtra xotf-u jixa-ti v boljnits-u, bo treba operatsij-u
 tomorrow want-1SG go-INF to hospital-ACC.SG as need operation-ACC.SG
zrobi-ti
 do-INF

‘Tomorrow I will have to go to the hospital to have an operation.’

(14) (Tor1.el)

ja xotf-u zaruba-ti pivnj-a, bo vin v mena
 1SG.NOM want-1SG kill.PRF-INF rooster-ACC.SG as 3SG.NOM.M in 1SG.ACC
klu-je i moze i popad-e do dit-aj
 bite-3SG and may and attack.PRF-2SG to child-GEN.PL

‘I will need to kill the rooster as he keeps biting, and he may attack children.’

As elsewhere in East Slavic, there is also a verb *xotiti*² which means ‘to want’. It is hard to determine whether it is a completely different entry at this stage, but it is very clear that the de-volitive future (synchronically with a deontic value) originated from it.

(15) (Z10.12 00:30)

a tpera ntkto ni xotf-e za hrof-i do tibe
 and now nobody.NOM.SG NEG want-3SG for money-ACC.PL to 2SG.GEN
i-ti
 go-INF

‘So nowadays nobody wants to come [to help] you [even] for money.’

⁵The future-in-the-past is less clear for the de-volitive. I have not been able to prove whether there are any tense restrictions, but I have a strong suspicion that there may be (unfortunately my main consultant, tor1, passed away during my research so i have not been able to confirm this). I have certainly identified utterances with the verb *xotiti* in the past tense + infinitive in the corpus, although in most of them there were signs that it only had a ‘plain’ volitive meaning (i.e. no temporal or prospective value).

2.2.2 Uses and connotations

The de-volitive future has very restricted uses as a future tense periphrasis. The verb *xotiti* denotes a need, a type of obligation that is not (at least directly) imposed by external forces, such as the weather, or an institution like a bank.⁶ Note that in (13) the speaker insisted they did not want to go through an operating theatre, it was because they needed the procedure. If the verb *xotiti* is used with an externally imposed obligation it loses its deontic meaning and only has (or preserves) a volitive meaning, which as a consequence sounds unacceptable to the speakers.

- (16) (Tor1.el) [Context: A man has been outside the house for a while. He really has to get into the house quietly.]

vin xotf-e tixenko pri-ti bo djiti sp-jetj
 3SG.NOM want-3SG silently enter-INF because child-NOM.PL sleep-3PL
 ‘He will need to get in silently because the children are sleeping.’

- (17) a. (Tor1.el) [Context: There hasn’t been any rain for a while.]

**ja xotf-u poli-ti oforod, bo vs-jo*
 1SG.NOM want-1SG water-INF garden-ACC.SG because all-NOM.SG.N
posoxn-e
 dry.PRF-3SG

- b. (Tor1.vol) >>

treba (obizatilno) poli-ti oforod-a
 necessary obligatorily water-INF garden-GEN.SG
 ‘I (will) need to water the garden, or else everything will dry up.’

Thus, the de-volitive form is closer to intentions than to predictions, but it still fails to fully adjust to any of these categories.

⁶Otherwise, other forms are used; e.g. *ja kraj maju* (T6), *ja dolžn*, *minji musovo* or *minji treba*. Tor1 was also aware of this restriction when I tried eliciting different forms (Tor1.45) “*ja xotfu, fiato, f:jo to takoje, znatfit ifastni, f:jo ja [no] xotfu; fiato nado, treba platiti*” ‘*ja xotfu* [I have to] is something private, but [with your bills] you must pay [them]’.

3 Typological overview and diachrony

3.1 The de-obligative future

3.1.1 The de-obligative within East Slavic

Before continuing further, Dahl (2000: 323) suggested a possible link between the (Ukrainian) synthetic future and the Balkan de-obligative form. Since West Polesian has both forms, I distinguish the de-obligative from the synthetic *robiťi-mu*, admitting that there may be some links between the two, but their grammaticalization paths have been very different.

Not all the contemporary Slavic languages have preserved a verb for ‘to have’ (particularly, in East Slavic), but there are obligative constructions in many, which use the verb ‘to have’ ((i)miec or any of its variations); e.g. Polish *Mam coś robić* ‘I have to do something’. Neither have all the contemporary Slavic languages grammaticalised (or preserved) the verb ‘to have’ as a future tense auxiliary, but there are a few.

According to Danylenko (2011: 172) and Whaley (2000: 58) Rusyn or “Transcarpathian dialects” have a de-obligative future form, which is realised analytically, the auxiliary appearing before the infinitive. Nevertheless, Pugh (2009) does not make any reference to this form in his grammar of Rusyn. Whaley (2000: 61) briefly mentions that apparently there are de-obligative futures in North Russian dialects (more specifically she mentions the varieties spoken in the Velikij Ustjug, Grjazovec, Kadninkov and Čerepovec regions), although she does not provide any examples.

Belarusian (BEL) has the closest analogue to the West Polesian de-obligative. In both Standard Belarusian (particularly in Narkamaŭka) and West Polesian the verb (BEL) *mec’* / (WP) *matī* ‘to have’ is heavily eroded. Mazzitelli (2011) ran a corpus survey on both *mec’* and *mecca* (the reflexive form) in Contemporary Standard Belarusian texts.⁷

Given that the verbs are rare *per se*, not surprisingly, the results she obtained were that both forms are very infrequent, particularly the reflexive one: 0.05 per thousand words for *mec’*; and 0.01 per thousand words for *mecca* (2011: 180–181).⁸

⁷Lomtev (1956: 181) says that, although very rarely, constructions with *mec’* can be found in Contemporary Belarusian works and that they used to be the norm (and where widely used) in older stages of Belarusian. He claims that nowadays such constructions should be considered “dialectal”.

⁸It must be also said that the reflexive form *mecca* has a very low frequency in both standards of Belarusian (*Taraškevica* and *Narkamaŭka*), whereas the contrast in the use of *mec’* is more notorious between the standards (Mazzitelli 2011: 181).

According to Mackevič (1959: 200) in Standard Belarusian, both *mec'* and *mecca* + infinitive can be employed in the past and the present tenses.

Now, according to Mackevič (1959: 200) and Mazzitelli (2011: 182) in Belarusian, both *mec'* and *mecca* combine with perfective and imperfective infinitive forms and they do not impose semantic restrictions on the subject; i.e. they allow either animates or inanimates to be subjects of the verb. Thus, the West Polesian and Standard Belarusian de-obligative forms share many properties in common, and it is reasonable to infer a common origin.⁹

Concerning the uses and meanings of both *mec'* and *mecca* + infinitive, Mackevič (1959: 200) says that in Standard Belarusian they can have different modal meanings:

“The modal meanings vary according to the form of the verb used: *mec'* or the reflexive *mecca*. The first one expresses a stronger meaning of obligation, need. [...] Constructions with the reflexive form, *mecca*, are characterized by a modal meaning of intention, disposition to undertake the action. [...] The (modal) semantic nuances of these constructions are close to modal constructions of the type *xacec'* [‘want’], *žadac'* [‘wish’], *dumac'* [‘think’], *music'* [‘must’]” (Mackevič 1959: 200) [My translation].

On the one hand, the Belarusian form *mec'* lies between temporality and modality. Among the temporal notions it can express (mostly future), there are scheduled future (the most common), fatalistic future (especially when used in the past tense) and intentionality (Mazzitelli 2011: 182, 190).

Very rarely, Belarusian *mec'* can be used to express immediate future, but retaining a flavor of obligation, which for Lomtev (1956: 181) and Mazzitelli (2011: 184) disqualifies it as a fully grammaticalized (i.e. in their understanding, semantically neutral) future auxiliary. On the other hand, Mazzitelli (2011: 185) says that the primary function of Contemporary Belarusian *mecca* (i.e. the reflexive form) is “to express the notion of scheduled future in the past, usually with counterfactual interpretation [...], and intention”.

- (18) Standard Belarusian [See references in Mazzitelli (2011: 181)]

Ja me-l-a-sja sta-c' povar-am
1SG.NOM have-PST-F.SG-REFL become-REFL cook-INS.SG
'I should have become a cook.'

⁹I lack enough data to assert whether West Polesian imposes any animacy restriction on its subjects with de-obligatives. Nevertheless, I have observed **animacy** restrictions on the object of the de-obligatives (see Section 2.1.2, particularly (11–12)).

Notwithstanding the semantic differences between the two forms, Mazzitelli's (2011: 180–181) survey showed that *mec'* is primarily used in the present tense, whereas *mecca* (i.e. the reflexive form) almost exclusively in the past tense. That is to say, the main condition for one form or another seems to be **tense**. In contrast, it is true that West Polesian *majusj(a)* + inf appears more often in the past tense (either as a counterfactual event or an intention). Yet it is still quite frequently employed in the present/future tense (i.e. as a proper future tense auxiliary), with an added meaning of intentionality. That is to say, the West Polesian reflexive auxiliary is not **tense**-bound, and displays more traits of a canonical or grammaticalized future tense construction.

From a diachronic perspective, according to Whaley (2000: 54) “[...] the fourteenth–sixteenth centuries show widespread use of *maju* plus the infinitive, a ‘have/take’-type future similar to that found in Old Church Slavonic as well as Rus’ian Church Slavic”.¹⁰ By contrast, the reality I have found in *The Belarusian Historical Dictionary* (henceforth, HSBM)¹¹ is quite different. I have searched for all the possible variants of the verb (*mati*; *imati*, *meti*) with their respective reflexive counterparts in HSBM. For the form *mati* as ‘to own; to have’, there are only a few historical attestations of the verb and not much information is given.

As for *imati* and *imatisja*, HSBM very briefly says that both have been used as auxiliaries for the analytic future tense. The earliest instance of *imati* that it documents is from 1438, where it says it has a modal/deontic nuance. And for *imatisja*, the earliest token also comes from a fifteenth-century text (although no comments are made about any additional nuances).

Finally, the pair *meti* and *metisja* are the most prominent in the dictionary (HSBM vol. 18), and yet there is hardly any information available. Out of the many meanings for *meti*, it says that it was possible to use it as an auxiliary for future tense denoting obligation or possibility. The earliest potential example of a de-obligative future that it provides is from a text dated from 1533. Yet, whilst the deontic reading is very clear, the temporal value is very questionable. The available material for *metisja* is even more scarce. Only the 7th entry describes very concisely that it has been used as a future tense auxiliary but denoting obligation. It only cites an example from 1579 (19):

¹⁰Conversely, the copular future was marginal at that time (*ibid*).

¹¹For its abbreviation in Belarusian (*Histaryčny Sloŭnik Belaruskaj Mowy*).

- (19) Old Belarusian (AVK, 406 (XVI century) In: HSBM vol. 18 [my glossing and translation])

by-l-a *u mene* *t-aja* *prijatelk-a* ... *o*
 be-PST-F.SG in 1SG.GEN that-NOM.SG.F girlfriend-NOM.SG.F about
t-om *ne veda-l,* *esli* *by* *ee* *me-l-i-se*
 that-LOC.SG no know-PST.M COMP PART 3SG.ACC.F HAVE-PST-PL-REFL
svata-c' *ljud-i* *abo ne*
 propose-INF person-NOM.PL or no
 'I used to have a girlfriend... I didn't know whether they were going to
 offer her in marriage or not.'

Mazzitelli (2011: 186) points out that the reflexive form was already present in Old Belarusian, and it already differed from its neighbours. According to Mazzitelli's (2011) corpus study (2011: 192–193, 199), Old Belarusian *mecca* was already grammaticalized as an auxiliary; and it was available for the future and the past tenses (as in West Polesian), the future being the least common. Yet, already by the nineteenth century it was almost restricted to the past tense (*mec'* being used for the future, instead), expressing intention or mostly a counterfactual meaning (Mazzitelli 2011: 192, 193, 199). The reflexive past tense form was preserved into the twentieth century in Belarusian. The use of *mecca* did not decrease as significantly as the Belarusian *mec'* in Soviet Belarus, but it remained very much marginal.

Mazzitelli (2011) presents various theories about the grammaticalization path of the reflexive de-obligative auxiliary. According to her, one possible explanation is that the auxiliary may have emerged from one of the meanings the reflexive *mecca* had in Old Belarusian 'strive for, going to' (without an infinitive) (Mazzitelli 2011: 198). However, the theory that Mazzitelli (2011: 199–200) favors most is that Old Belarusian *meti* and *metisja* emerged almost in parallel. With time *meti* would have retained some modal value and added a value of 'external obligation', whereas *metisja* would have developed into a more neutral auxiliary to replace the gaps of *meti*, particularly in the past tense. In addition to this, Mazzitelli finds this last explanation particularly helpful, because it helps to explain why the reflexive forms are overwhelmingly more common than the non-reflexive ones in all periods of Belarusian when used with first and second persons.

In the light of this, we can say that West Polesian may have diverged from the rest of the East Slavic varieties already by the sixteenth century by preserving the reflexive form as a future tense auxiliary. So, in this respect, West Polesian is more conservative than the rest of the East Slavic clade.

3.1.2 Balkan de-obligatives

Bulgarian uses two main forms as a basis for the rest of the future forms: the form *šte* (a de-volitive which I discuss in (Section 3.2.1)¹² for the positive forms; and for the negative forms the particles *da* (“an invariable conjunctive particle”) and *njama*, which actually is “the 3rd person singular present form of the verb *njamam* ‘have not’” (Kuteva 1995: 209). Even though the Bulgarian/Macedonian verbal system is more complex than that of West Polesian and Belarusian, one of the closest similarities is the possibility of expressing a counterfactual event grammatically (Sussex & Cubberley 2006: 242). Nevertheless, besides the semantic basis and its possibility of expressing a counterfactual past event (for both imperfective and perfective verbs, see Kuteva 1995: 209–212), the differences between the West Polesian and Bulgarian/Macedonian forms are greater than their similarities. First, Bulgarian polarity restrictions of *do* do not apply to the West Polesian de-obligative. Second, Bulgarian has gone further along the grammaticalization path. As a result, Bulgarian *njama* is a semantically neutral form that can express both intentions and predictions; whereas West Polesian *majusj(a)* has a strong intentional connotation. And third, Bulgarian *njama* does not currently inflect for **person** and does not have a reflexive form, whereas the West Polesian de-obligative is particularly salient (in the Slavic family, at least) for having a reflexive form.

Finally, Dahl (2000: 323) gives examples of de-obligatives in Gheg Albanian and Romanian. Danylenko’s (2011: 164) comments suggest that the closest form (to East Slavic de-obligatives) in the Balkans is the Greek μέλλω which means ‘to intend’. However, Danylenko (2011: 164–167, 177) also believes that the emergence of the Balkan de-obligative may have a Slavic origin, but it would not be related to the forms that emerged in Middle Belarusian and Ukrainian.

3.1.3 Within the wider European context

Outside of the Balkans, the two best known de-obligative ‘future construction families’ are English *shall* (which earlier meant ‘to owe’, see Bybee 1985: 194) and the Romance simple future; e.g. [French] *aimer+ai >aimerai* ‘I will love’. In comparison to West Polesian (and East Slavic), Romance de-obligatives are exclusively employed in the future tense. That is to say, they cannot be used as a future-in-the-past; because they tend to use the conditional for that, which is also related to auxiliary *habere* (e.g. [French] *aimerais*) or most often an imperfective de-andative periphrasis. I have summarised the areas in Europe where there is a

¹²In Macedonian the cognate is *ke* (Dahl 2000: 323).

de-obligative in (Figure 1). Totally unrelated to the Balkans or Slavic, Basque has also a de-obligative future, which is particularly used in familiar speech (as an alternative to the genitive-based future). The future reading is deduced from the context; cf. (20)–(21):

- (20) Basque (own knowledge)

[Context: a friend asking another friend before ordering.]

Kafe-a eska-tu behar (al) du-zu?

coffee-DET.ABS.SG ask-INF need Q.PART have.3SG<2SG

‘Are you going to order coffee? (lit. Do you have to order coffee?)’

- (21) [Context: Doctor talking to a patient.]

Ur gehiago edan behar du-zu.

water.ABS more drink-INF need have.3SG<2SG

‘You need to drink more water.’



Figure 1: Distribution of de-obligative future in Europe. Sources (Bybee & Dahl 1989, Dahl 2000, Danylenko 2011, Mazzitelli 2011, Whaley 2000). (CC-BY-SA Kristian Roncero).

Outside Europe, Bybee (1985: 194) also mentions Kru dialects as having a de-obligative future. Hence, there are de-obligative futures in different corners of Europe and the world, with little affiliation between them (if any). In fact, according to Bybee's (1985: 194) survey, modals denoting 'obligation' "tend to develop into *epistemic markers*" and "into *futures*, as in the case of English *shall*". Ultan (1978: 118) proposed that the link between modals and their common grammaticalization into future tense markers cross-linguistically may be due "to the relative uncertainty inherent in both future events and most of the categories subsumed under the general heading of modals".

Thus, my hypothesis so far is that the East and South Slavic de-obligative future would have departed from a common material (the verb 'to have'), that due to its semantics 'naturally' grammaticalized into a future tense construction. In this process, it is noteworthy that Belarusian and especially West Polesian have developed a de-obligative future tense auxiliary based on a reflexive, which is nowhere else found in Europe, and that it expresses intention rather than obligation.

3.2 The de-volitive future

3.2.1 Within the Slavic family

There is evidence of the existence of de-volitive and de-obligative periphrases in older stages of East and Common Slavic (Danylenko 2011, Mackević 1959, Whaley 2000). Thus, since this type of construction has a long tradition in Slavic, it is not surprising to find it in other Slavic varieties.

Starting from East Slavic, Mackević (1959) vaguely mentions the existence of periphrastic constructions with the verb *xacec'* in Belarusian. However, he does not provide any examples, nor does he say in which dialects this form is used (certainly, it is not part of the current standard BEL, so he may be referring to West Polesian or Southwestern Belarusian, in general). According to Danylenko (2011: 154), Transcarpathian (i.e. Rusyn) and Eastern Ukrainian also have de-volitive periphrastic constructions based on the verb *xotity*, although they have not fully grammaticalized the verb into a future tense auxiliary yet.¹³ In this respect, West Polesian is probably not the only member of the East Slavic clade to have preserved a de-volitive form. Nonetheless, it has gone further in the path of grammaticalization than any other member and it has made the biggest switch in the semantic content of the base: from volition to obligation.¹⁴

¹³Unfortunately, Danylenko (2011) does not provide more details or examples; nor does Pugh's (2009) Rusyn grammar make any mention of it.

¹⁴HSBM (vol.37) does not record any deontic meaning of the verb *xotitr*. Levancëvič (p.c.) shared with me that other West Polesian sub-varieties (as in Malaryta and southern Byarozhy) also use *xotitr* with a deontic meaning. She considers it to be a specificity of West Polesian.

Moving further south, Bosnian-Croatian-Montenegrin-Serbian (BCMS hereafter) forms its future tense using the inflected form of the verb *hteti* / *htjeti* ‘want’ as an auxiliary, which is a cognate of West Polesian *xotiti*. When it is in a positive sentence, the auxiliary can appear either as the whole form (*hóću/hóćeš*/etc. + infinitive) or as a shortened form (*ću/češ*/etc. + infinitive). However, when the verb is negated, only the neg *ne* + the short form is possible (*nè ću* or *nèću* / *ne ćeš* or *nećeš* /etc. + infinitive; but * *ne hóću*/etc. + infinitive) (Alexander 2006: 143–144).

Gvozdanović (1995: 186, 189, 190) says that BCMS future tense (regardless of aspect) is used: “(a) for a future event, as in (22); (b) for a necessary or desired event, as in (23)”. Gvozdanović (1995: 186) shows the following examples of BCMS futures:

- (22) BCMS (Gvozdanović 1995: 186)
Sutra ć-u ići u grad.
 tomorrow will.PRS-1SG go.INF in/to city.ACC.SG
 ‘Tomorrow I shall go to the city.’

- (23) BCMS (Gvozdanović 1995: 186)
Sada ć-eš ići po to.
 now will.PRS-2SG go.INF for this.ACC.SG
 ‘Now you will go to fetch it.’

As we can appreciate from Gvozdanović’s (1995: 186) data and discussion, particularly (23), this de-volitive construction shares with its West Polesian cognate part of the necessity meaning (which is a further innovation, given the etymology of the gram). Moreover, both constructions combine with perfective as well as imperfective infinitives, which is an uncommon feature with regards to other constructions in WP (and East Slavic).¹⁵ However, the usage of this form is far more limited in West Polesian than in BCMS. BCMS allows the use of this form as a semantically neutral future form (i.e. for both intentions and predictions); whereas West Polesian de-volitive implies the existence of a need to be addressed. In exchange, West Polesian, having a morphologically richer repertoire of constructions, most frequently uses the copularform in such a context (i.e. a semantically neutral future or a prediction).

Bulgarian/Macedonian also has a future tense auxiliary which was etymologically a volitional verb, *šta* ‘will/want’, (cognate and related of BCMS

¹⁵ According to Gvozdanović (1995: 186–187, 192) the BCMS and Slovene ‘future perfects’ can be used with either perfective or imperfective verbs (though they are based on the copular form).

hteti/htjeti).¹⁶ Nonetheless, the auxiliary is almost exclusively used in positive sentences (with the vast majority of negatives it uses the de-obligative). The future (simple) is formed by the auxiliary *šte/ke*, etymologically the 3sg form of the verb *šta* (although the auxiliary itself does not inflect for **person**), and the present tense form of the main verb (Kuteva 1995: 209).

- (24) Bulgarian (Kuteva 1995: 209)
 Šte *dadeš*
 will/want(INVAR) give.PFV.2SG.PRS
 ‘You will give.’

The Bulgarian de-volitive future form is versatile with regard to **tense** and is the base of the future simple; future perfect; future-in-the-past and future perfect-in-the-past forms (Kuteva 1995: 209–212).¹⁷ Even the future simple form (as in (24)), can express “an activity which will take place [...] or will be taking place [...] after the moment of speech” or “a repeated or habitual activity either in the present or in the past” (Kuteva 1995: 209).¹⁸ Conversely, the West Polesian de-volitive is very restricted in **tense** (in contrast with other West Polesian future-tense constructions also discussed here): it can only be used in the future tense (by inflecting the auxiliary in the present tense), as in any other form it functions as a plain volitive verb (see Section 2.1).

Elsewhere in the Balkans, Dahl (2000: 323) mentions further languages with de-volitives. On the one hand Modern Greek and Gheg Albanian which, in the same vein as Bulgarian/Macedonian, have an uninflectable form of the verb ‘want’ as the semantic base of their future tense auxiliary (Duchet 1995: 257, Hedin 1995: 233).¹⁹ And on the other hand Romanian, like BCMS, has an inflected auxiliary meaning ‘want’. Having said this, we should not be surprised to find more de-volitive future constructions in the world’s languages. Since volition and intentionality are semantically very closely related, volitive verbs often develop into future tense constructions (Bybee 1985, Bybee & Dahl 1989).

Finally, even though small, it is worth noting that Resian Slavic (RES) has also a copular, de-volitive and a de-obligative future constructions (Pila 2022), which

¹⁶According to Spencer (p.c.) in contemporary BCMS that verb has been replaced by *iskam*.

¹⁷Note that in the future-in-the-past and the future-perfect-in-the-past the auxiliaries inflect for **person** as well as **tense**.

¹⁸Spencer (p.c.) pointed out to me that English *will* futures can also express generic habitual events or past events; e.g. “By 1811 Napoleon will become master of half of Europe; by 1815 he will be defeated and in exile”. He also remarked that in Bulgarian renarrated or evidential mood you also encounter future tenses.

¹⁹In Modern Greek (*tha*) and Bulgarian/Macedonian (*šte/ke*) the verb of the auxiliary has not survived as an independent volitional verb to our days.

is not only unique in South Slavic but it is also a rarity within Slavic. These constructions are not exactly interchangeable and all add some semantic nuances. In the case of the de-volitive (25), it expresses more certainty than the copular form (be + participle) (26), according to Pila (2022: 38). As for the de-obligative (27–28), Pila (2022: 41) states that it has more of a predictive function “the need of fulfilling an action, which depends on factors outside the subject of the situation, the convenience, but not necessarily an obligation strictly speaking, but rather ethical, moral or professional” (Pila 2022: 41, my translation).

- (25) Resian Pila (2022: 36, my translation and glosses).

Zütra cón det no jabulko mimo
tomorrow want-1SG give.INF one apple.ACC.SG POSS.1SG.DAT.SG
sino.
son.DAT.SG

‘Tomorrow I will (certainly, as planned) give an apple to my son.’

- (26) Resian Pila (2022: 36, my translation and glosses).

Zütra bon dala no jabulko mimo
tomorrow BE.FUT.1SG give.F.SG one apple.ACC.SG POSS.1SG.DAT.SG
sino.
son.DAT.SG

‘Tomorrow I will (probably) give an apple to my son.’

- (27) Resian Pila (2022: 45, my translation and glosses).

Se bojïn spest. Ći bon spadel, du be
REFLEX.1SG be_afraid.1SG fall.INF if BE.FUT.1SG fall.M.SG who then
ma me wzdïnot?
have.3SG 1SG.ACC lift

‘I’m afraid of falling. If I fall, who will pick me up (lit. who has to carry me)?’

- (28) Resian Pila (2022: 41, my translation and glosses).

Ko bon wbila to kököš, jä man si skühat nö
when BE.FUT.1SG kill ART chicken 1SG.NOM have.1SG REFLEX cook one
malö bröda!
little broth.GEN.SG

‘When I (will) kill the hen, I will make myself (lit. I have to cook) some broth.’

3.2.2 Within European de-volitives

Outside of the Balkan context, Dahl (2000: 322) finds a second de-volitive future construction family within the Germanic languages of the North Sea: Danish, Faroese and Norwegian Bokmål (*vil*); Frisian (*wal*); and perhaps also Yiddish (*vel*). The English future tense construction *will* is also originally a volitive verb.²⁰ According to Dahl (2000: 322), it is the most grammaticalized future tense construction of all the languages in this construction family.

Other than the semantics of the auxiliary chosen as a base for the future tense, it is unlikely that there was an areal influence of the North Sea/Germanic *will* (and its cognates) in the development of West Polesian *xoffu*. First, historically it is hard to prove (especially given that the closest Germanic language that could have been affected, High German, did not develop it). And second, there is no phonological correspondence between West Polesian *xoffu* and Germanic *will* (and cognates); whilst *xoffu* can be clearly related to the South Slavic forms like (*hó*)*ću*.

Thus, where does the West Polesian de-volitive future stand with regards to the Balkan (South Slavic) de-volitive? On the one hand, we can hypothesize that the West Polesian de-volitive is a loan or calque of South Slavic. There are a few other features in West Polesian which look like South Slavic (mostly Serbian; e.g. vocative masculines in *-u*; or the de-obligative futures).²¹ But on the other hand, all the evidence collected so far shows that the de-volitive future in West Polesian is an independent development departing from a common base that has only grammaticalized in a few modern Slavic varieties, as we have proposed for the de-obligative. According to Danylenko (2011: 162), in Common Slavic “the verbs *iměti* ‘to have’ and *xotěti* / *xotěti* ‘to want’ were used in such P[eriphrastic] C[onstruction]s as de-modal extensions rather than auxiliaries.” In any case, the final output of the contemporary semantics of *xoffu* diverges from the rest of the construction families (typological expectations).

²⁰According to *Oxford English Dictionary* (<https://oed.com>, last checked 20/03/2023) in Contemporary English the verb *will* can mean “[t]o wish, desire; sometimes with implication of intention” although it classifies it as obscure, rare and archaic.

²¹In fact, some speakers have told me that there was a wave of migration from the south (mostly around contemporary Serbia) to the lands of West Polesie around the XV century (which would be reflected in the common surnames). However, I am unaware of any scientific sources on this topic.

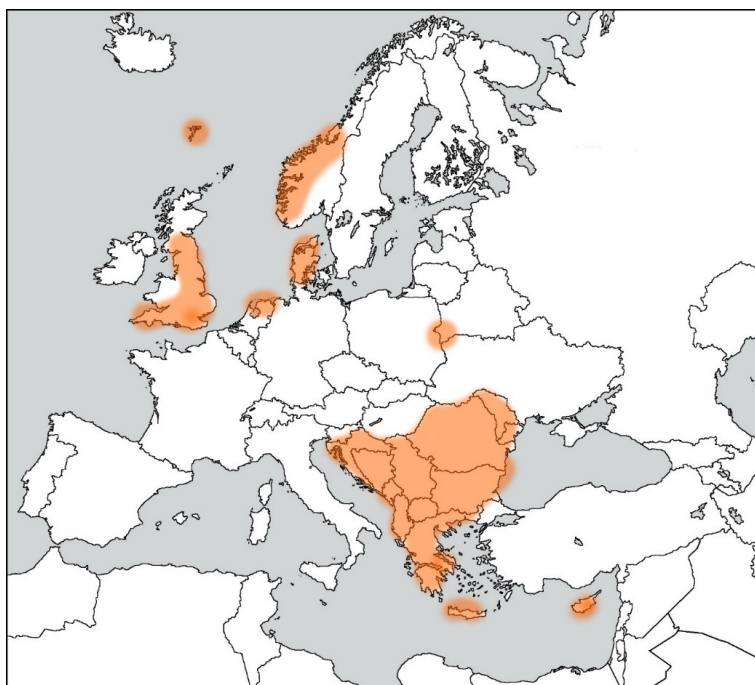


Figure 2: Distribution of de-volitive future in Europe. Sources (Dahl 2000, Danylenko 2011, Duchet 1995, Gvozdanović 1995, Kuteva 1995, Mackević 1959). (CC-BY-SA Kristian Roncero).

4 Summary and discussion

We have seen that West Polesian has several future tense constructions, some of which are further or less advanced in their grammaticalization process. For most of them, especially for the two under discussion here, they seem own creations from available materials in Slavic, as none of its neighbors (in Standard forms, except for Belarusian *mecca*) have any of them synchronically (and most of the contact with them only started about 70 years ago).

Against the general typological expectation (in Table 1 represented by English), the West Polesian de-volitive and de-obligative constructions have both retained some modal value, but they have inverted their original semantic functions. South Slavic (particularly Bulgarian and Macedonian) retains both (cognate) constructions, but it has generalized their use; that is to say, it has gone further in grammaticalization (especially if it is understood as ‘semantic bleaching’) and Resian Slavic keeps three constructions, which have developed fairly predictable nuances.

Table 1: Examples of uses of future tense constructions

	English	South Slavic	Resian	W. Polesian
Intention, Desire	<i>will</i>	(BCMS) (<i>hó</i>) <i>ću</i> ‘want’;	<i>ćon</i> ‘want’	<i>majujsj(a)</i> ‘have’
Prediction	<i>will</i>	(BG/MKD) + <i>šte/ke</i> ‘want’ - [<i>nja</i>] <i>ma</i> ‘have [not]’	<i>bon</i> ‘be’	(copular, synthetic, perfective, or become futures)
Obligation	<i>shall</i>		<i>man</i> ‘have’	<i>xofu</i> ‘want’

How did West Polesian end up in this situation? In other words, how did both constructions end up inverting their original meanings? Given then scarcity of data and the impossibility to carry out more fieldwork, I shall be cautious about any conclusion. Nevertheless, data seem to suggest that this overlapping of meanings is merely accidental, i.e. both forms were in the process of desemanticization/generalization, and they both ended up finding ‘an empty nest’ at different times in history.

According to Spencer (p.c.), English also had a similar type of ‘inversion’. The de-volitive *will* used to mean plain volition (as in *as you will*), which then became a future auxiliary. Conversely, *want* used to mean necessity or need (as in *for they shall want for nothing*), but then filled the slot vacated by *will*.

From a more traditional approach, this semantic ‘dissonance’ can be also the result of a saturated inventory of future tense constructions in West Polesian, which may have ‘forced’ certain members to specialize over time, which would be in line with Aronoff’s (2016) theory of Morphological Niches. This may explain what happened in Resian. Nevertheless, based on other observations about other morphosyntactic parameters in West Polesian, this seems to contradict the overall picture. I have gathered fine grained evidence of the same speakers using multiple endings (Roncero 2021; Roncero 2022) for the same slot, without having any additional semantic, discursive or even sociolinguistic nuance between them (Bošnjak Botica & Hržica 2016; Thornton 2011); yet conditioned by its sociolinguistic setting (cf. Dorian 2010). I shall admit though, that in some (not all) cases, language contact seemed to be one of the sources of overabundance; whereas, the neighboring varieties seem to have very limited or no effect on the development and re-semanticization of these two future tense constructions. I shall add to

this that, although marginally, the reflexive auxiliary *mecca* was already attested in Middle Belarusian, which *mutatis mutanda*, may have already the intentional nuance. As for the de-volitive, the fact that we also find traces of it in Rusyn, suggests it existed in earlier stages of East Slavic but it was not fully grammaticalized. Yet, it is still a mystery at which point in history did West Polesian de-volitive turn into de-obligative. In sum, the evolution of the de-volitive and the de-obligative future constructions in West Polesian opens further questions about grammaticalization and semantics, as well as typology and sociolinguistics.

In any case, despite the overabundance of forms, in a big part, also conditioned by language contact, the semantics seem to follow a predictable pattern. By contrast, synchronically the West Polesian de-volitive form has ended up expressing necessity, whereas the de-obligative is used to indicate intention and desire.

Abbreviations

Glossing abbreviations in this paper are based on the *Leipzig Glossing Rules* (2015 revision), with some additions.

1, 2, 3	first, second, third person (respectively)	INF	infinitive
		INS	instrumental
ABS	absolutive	LOC	locative
ACC	accusative	M	masculine
BCMS	Bosnian-Croatian- Montenegrin-Serbian	N	neuter
		NEG	negation
BEL	Standard (Literary) Belarusian	NOM	nominative
		PL	plural
DAT	dative	POSS	possessor
DET	determiner	PRF	perfective
DEM	demonstrative	PTCP	participle
ERG	ergative	PST	past
F	feminine	Q	question particle/marker
FUT	future	REFL	reflexive
GEN	genitive	RES	Resian Slavic
GRADNM	greater adnumerative	SG	singular
IMP	imperative	VOC	vocative

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Appendix A List of participants

I am providing the names of the participants who chose to have their identities disclosed. To all of those who show it here and for all the others who chose to remain anonymous, endless thanks.

Imianin [Torokan] (Drahičyn)

- T2 Mazuka, Mikalaj A.
- T5 Mazuka, Kacjaryna I.
- T6 Barysjuk, Maryja I.
- T7 Marusja
- T9 Hardzejčuk, Nadzeja I.
- T12 Sakaloŭska, Maryja P.
- Tor 1 Baba Ženja

Pare (Pinsk)

- P2 Maryja Uladzmiraŭna

Žydča (Pinsk)

- Z2 Haŭrylčuk, Nina
- Z10 Žydečkaja, Sjarafima I.

Haloŭčyci, Tolkovo (Drahičyn)

- TL1 Kanavalčyk, Vera I.
- HL1 Mancèvič, Dzina I.

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Futures of the past

Thinking of and talking about future events is a crucial part of our everyday lives. Thus, it is not surprising that in the languages of the world, a variety of markers for future time reference has developed. The goal of this volume is to take a contrastive perspective on the evolution of future constructions by bringing together scholars working on different languages and language families. In doing so, the edited volume opens up new perspectives on long-standing questions regarding the diachronic development of future constructions across languages, with implications not only for research on tense-aspect-mood systems but also for research on grammaticalization and language change more generally.